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MODULE HANDBOOK



60610300 – Computer Engineering
Bachelor's program

Submitted to **ACQUIN**
by Termiz State University
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Process



Termiz city, "Barhayot avlod"
street, house number 43



+998 76 221 74 55



info@terdu.uz



www.terdu.uz



@terdu340



@terdu340



@terdu_university



@Termizdavlatuniversiteti

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MODERN HISTORY OF UZBEKISTAN

Field Description	Description
Module Title	MODERN HISTORY OF UZBEKISTAN
Module Code	UEY1104
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Seminar) Self-Assignments: 60 hours
Frequency/Duration	I semester
Module objective	The objective of the Module Modern History of Uzbekistan is to scientifically study the historical development of the Republic of Uzbekistan during its years of independence. The Module examines the processes of state and society building, together with the substance and essence of the reforms implemented in the political, economic, social and spiritual spheres. It aims to develop in students a systematic historical thinking and an understanding of national identity. The Module fosters a sense of civic responsibility and patriotism grounded in an accurate knowledge of the nation's recent history. Students develop the skills required to evaluate and analyse historical processes impartially, using primary sources and normative legal documents. The Module further situates Uzbekistan's development within the framework of the "New Uzbekistan" strategy and the country's growing role in the international community.
Module Learning Outcomes	Knowledge: LO1. Acquires knowledge of the socio-political processes in Uzbekistan on the eve of independence. LO2. Knows the formation of the independent Republic of Uzbekistan and its historical significance. LO3. Knows Uzbekistan's unique path to independence and development. LO4. Understands the formation of the foundations of a democratic civil society in Uzbekistan and the essence of the political reforms that have been implemented. Skills:

	LO5. Understands the contribution of scholars to Uzbekistan's spiritual and cultural development during the years of independence. LO6. Is able to distinguish the social changes in the Republic of Uzbekistan. LO7. Is able to grasp the essence of events taking place in the global community. Competencies: LO8. Has experience of working independently in vacant posts appropriate to their field of higher education, and has the qualifications to pursue postgraduate studies in a selected specialisation within the relevant undergraduate programme. LO9. Is able to independently analyse and evaluate the contribution of historical events and individuals to the development of society.
Content	Lectures Introduction. The Subject, Aims and Objectives, and Theoretical-Methodological Principles of the Course "The Latest History of Uzbekistan". The subject, aims, and objectives of the course are examined. The theoretical and methodological principles underlying the study of recent history are studied. The place of the discipline within the system of historical sciences, along with its connection with other subjects, is considered. Historical Foundations of the Idea of Independence in Uzbekistan. The roots of national statehood traditions are examined. The idea of independence in the works of Jadid reformers is studied. National liberation movements of the 19th–20th centuries, along with their historical significance, are considered. Social and Political Processes in Uzbekistan on the Eve of Independence. The crisis of the Soviet totalitarian system is examined. The impact of perestroika on Uzbekistan is studied. The rise of national movements and public organizations, along with their political demands, is considered. The Establishment of the Independent Republic of Uzbekistan and Its Historical Significance. Uzbekistan's Distinctive Path to Independence and Development. The declaration of state sovereignty and independence is examined. The legal and political foundations of the new state are studied. Uzbekistan's

distinctive model of development, along with its historical significance, is considered.

The Formation of the Foundations of a Democratic, Civil Society in Uzbekistan, and the Political Reforms Implemented. The legal foundations of statehood are examined. The formation of a multi-party system and parliamentarism is studied. The stages of democratization of political power, along with the development of civil society institutions, are considered.

Economic Reforms, the Formation of Private Property, and the Development of Market Relations in Uzbekistan. The transition from a planned to a market economy is examined. Privatization and the formation of private property are studied. Structural and macroeconomic reforms, along with the development of market relations, are considered.

Social Changes in the Republic of Uzbekistan. Social policy and the social protection system are examined. Employment and living standards are studied. The development of healthcare, along with demographic changes, is considered.

Spiritual and Cultural Development in Uzbekistan During the Years of Independence. The revival of national and spiritual values is examined. The development of literature, art, and mass media is studied. The preservation of historical and cultural heritage, along with its significance, is considered.

Reforms Implemented in the Field of Education in the Republic of Uzbekistan. The reform of the general secondary education system is examined. The development of vocational and higher education is studied. The integration of science and education, along with the modernization of curricula, is considered.

Uzbekistan and the World Community. The formation of the foreign policy doctrine of independent Uzbekistan is examined. Uzbekistan's membership in international organizations is studied. Regional cooperation in Central Asia, along with its significance, is considered.

Uzbekistan and the World Community. Uzbekistan's participation in global economic and cultural exchange is examined. Bilateral and multilateral diplomatic relations are studied. The role of international cooperation in

national development, along with its outcomes, is considered.

Reflection of the Changes Taking Place in the Social, Political, Economic and Cultural Life of the New Uzbekistan in International Rankings and Indices. The methodology of international rating systems is examined. Uzbekistan's positions in global development indices are studied. The significance of ratings for further reform planning, along with their outcomes, is considered.

Reforms Being Implemented in the New Uzbekistan Under the Leadership and Initiative of President Sh.M. Mirziyoyev. The concept and stages of the "New Uzbekistan" strategy are examined. Political and judicial-legal reforms are studied. Economic and social modernization programs, along with their outcomes, are considered.

Reforms Being Implemented in the New Uzbekistan Under the Leadership and Initiative of President Sh.M. Mirziyoyev. The continuation and deepening of institutional reforms are examined. Digitalization and anti-corruption measures are studied. Regional development and public administration reforms, along with their significance, are considered.

The Socio-Political Significance of Building a People-Oriented and Humane State in the New Uzbekistan. The concept of a "people-oriented state" is examined. Mechanisms ensuring citizen participation in governance are studied. Humane policy toward vulnerable social groups, along with its socio-political significance, is considered.

Seminars

The Subject, Aims and Objectives, and Theoretical-Methodological Principles of the Course "The Most Recent History of Uzbekistan". The subject, aims, and objectives of the course are examined. The theoretical and methodological principles underlying the study of recent history are studied. The place of the discipline within the system of historical sciences, along with its connection with other subjects, is considered.

Historical Foundations of the Idea of Independence in Uzbekistan. The roots of national statehood traditions are examined. The idea of independence in the works of Jadid reformers is studied. National liberation movements of the

19th–20th centuries, along with their historical significance, are considered.

Socio-Political Processes in Uzbekistan on the Eve of Independence. The crisis of the Soviet totalitarian system is examined. The impact of perestroika on Uzbekistan is studied. The rise of national movements and public organizations, along with their political demands, is considered.

The Establishment of the Independent Republic of Uzbekistan and Its Historical Significance. Uzbekistan's Distinctive Path to Independence and Development. The declaration of state sovereignty and independence is examined. The legal and political foundations of the new state are studied. Uzbekistan's distinctive model of development, along with its historical significance, is considered.

The Formation of the Foundations of a Democratic, Civil Society in Uzbekistan, and the Political Reforms Implemented. The legal foundations of statehood are examined. The formation of a multi-party system and parliamentarism is studied. The stages of democratization of political power, along with the development of civil society institutions, are considered.

Economic Reforms, the Emergence of Private Property. The Development of Market Relations in Uzbekistan. The transition from a planned to a market economy is examined. Privatization and the emergence of private property are studied. Structural and macroeconomic reforms, along with the development of market relations, are considered.

Social Changes in the Republic of Uzbekistan. Social policy and the social protection system are examined. Employment and living standards are studied. The development of healthcare, along with demographic changes, is considered.

Spiritual and Cultural Development in Uzbekistan During the Years of Independence. The revival of national and spiritual values is examined. The development of literature, art, and mass media is studied. The preservation of historical and cultural heritage, along with its significance, is considered.

Reforms Carried Out in the Field of Education in the Republic of Uzbekistan. The reform of the general

secondary education system is examined. The development of vocational and higher education is studied. The integration of science and education, along with the modernization of curricula, is considered.

Uzbekistan and the World Community. The formation of the foreign policy doctrine of independent Uzbekistan is examined. Uzbekistan's membership in international organizations is studied. Regional cooperation in Central Asia, along with its significance, is considered.

Uzbekistan and the World Community. Uzbekistan's participation in global economic and cultural exchange is examined. Bilateral and multilateral diplomatic relations are studied. The role of international cooperation in national development, along with its outcomes, is considered.

Reflection of the Changes Taking Place in the Socio-Political, Economic and Cultural Life of the New Uzbekistan in International Ratings and Indices. The methodology of international rating systems is examined. Uzbekistan's positions in global development indices are studied. The significance of ratings for further reform planning, along with their outcomes, is considered.

Reforms Being Implemented in the New Uzbekistan Under the Leadership and Initiative of President Sh.M. Mirziyoyev. The concept and stages of the "New Uzbekistan" strategy are examined. Political and judicial-legal reforms are studied. Economic and social modernization programs, along with their outcomes, are considered.

Reforms Being Implemented in the New Uzbekistan Under the Leadership and Initiative of President Sh.M. Mirziyoyev. The continuation and deepening of institutional reforms are examined. Digitalization and anti-corruption measures are studied. Regional development and public administration reforms, along with their significance, are considered.

The Socio-Political Significance of Building a People-Oriented and Humane State in the New Uzbekistan. The concept of a "people-oriented state" is examined. Mechanisms ensuring citizen participation in governance are studied. Humane policy toward vulnerable social groups, along with its socio-political significance, is considered.

Self-study

Historical Stages of Uzbek Statehood (The History of States in the Territory of Uzbekistan).
Steps Towards State Independence.
Changes in Uzbekistan's Political System.
The Introduction of the Presidential Institution in the Republic of Uzbekistan.
Adoption of the Declaration of Independence.
Reforms Implemented in the System of State Governance During the Years of Independence.
The Principle of Separation of Powers in an Independent Uzbekistan.
Non-Governmental, Non-Profit Organisations in Uzbekistan and Their Role in Political, Social and Economic Processes.
The Essence of the New Edition of the Law on Education.
New Developments in the Republic of Karakalpakstan.
International Situation. The Role of Uzbekistan in Foreign Policy.
Brotherly Co-operative Relations Among the Central Asian Republics.
New Uzbekistan Development Strategy.
The Position of the Republic of Uzbekistan in International Ratings and Indices (On the Example of Economic Ratings).
The Position of the Republic of Uzbekistan in International Ratings and Indices (Case Study of Higher Education).
Historical Figures Who Made a Direct Contribution to Independence (Shiroq, Tomiris, Jalaluddin Manguberdi, Temur Malik, Amir Timur, and Others).
The Decline of the Soviet State.
The Establishment of the Independent Republic of Uzbekistan and Its Historical Significance.
The Introduction of the Presidency in the Republic of Uzbekistan. The Election of I.A. Karimov as President of the Republic of Uzbekistan.
The Formation of the Foundations of a Democratic Civil Society in Uzbekistan. The Role of Political Parties.
The Role of the Scientific Heritage of Central Asian Scholars in Educating High-Spirited Youth (Case Study of the Scholars' Scientific Activities).
The Activities of Mass Media in the Republic of Uzbekistan (Changes and Problems).

	The Causes of the Global Financial Crisis. The Contribution of Imam al-Bukhari and Isā al-Tirmidhi to Islam. Development of Science and Technology in the Years of Independence. The Role of Education and Upbringing in Building a New Society. The Adoption of the Law "On Education" and the "National Programme for the Training of Personnel". Economic Modernisation of the Republic of Karakalpakstan in the Years of Independence. Uzbekistan's Membership of the CIS and Shanghai Cooperation Organisations. The Accession of the Republic of Uzbekistan to the UN and Its Distinctive Place in the World Community. Strategy for the Development of the Republic of Uzbekistan for 2017–2021. The 2030 Strategy of the Republic of Uzbekistan.
Teaching Methods	The Module is delivered through interactive lectures and seminar discussions that develop students' systematic understanding of the subject matter, objectives and main stages of the discipline of Uzbekistan's most recent history. Students study primary historical sources, normative legal documents and scholarly literature, and apply historical-critical and comparative methods to analyse the processes of state and society building. Independent study assignments include the analysis of historical sources and periodical press materials, the development of chronological charts, the preparation of portrait analyses of historical figures, and work with legislative documents such as the Constitution. Project-based activities, including museum visits, systematic diagrams of political and legal development, and short video presentations on the reforms of New Uzbekistan, connect theoretical material to contemporary developments. Through this combination of guided instruction and independent research, students develop historical thinking, civic responsibility and the ability to apply historical-critical and comparative methodological principles.
Prerequisites	History of the pre-independence period The establishment of the independent Republic of Uzbekistan and its historical significance

	Social, political and economic life after independence The essence of the concept of a modernising Uzbekistan
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). Continuous assessment evaluates students' classroom participation, engagement in seminar discussions, attendance and involvement in lecture sessions, and the successful completion of independent study assignments, including analysis of historical sources and documents, work with periodical press materials, development of chronological charts of the independence period, preparation of portrait analyses on historical figures, and work with legislative documents such as the Constitution. Midterm assessment consists of written assignments, short in-class quizzes and tests, and independent project work, including a report on a visit to a historical museum or significant historical monuments, a systematic diagram of the political and legal development stages of the independence period, a short video presentation on the reforms in New Uzbekistan, proposals and analytical conclusions on the concept of "Uzbekistan's Third Renaissance," and an essay on the historical significance of independence or socio-political changes. The final assessment is a written examination consisting of theoretical questions and analytical tasks, designed to assess students' knowledge of the latest history of Uzbekistan, their skills in analyzing historical events based on cause-and-effect relationships, and their ability to independently apply historical-critical and comparative methodological principles. The timely and high-quality completion of all independent study assignments is a mandatory component of the module assessment.
Recommended Literature	1. Mirziyoyev, Sh. M. We will resolutely continue our national development path and elevate it to a new stage. Volume 1. - Tashkent: Uzbekistan, 2017. 2. Mirziyoyev Sh.M. We will build our great future together with our brave and noble people. - Tashkent: Uzbekistan, 2017. 3. Mirziyoyev Sh.M. The approval of our people is the highest assessment of our work. Volume 2. - Tashkent: Uzbekistan, 2018.

	4. Mirziyoyev Sh.M. The work of a great-minded people is great, and their life is bright and their future prosperous. Volume 3. - Tashkent: Uzbekistan, 2019. 5. Mirziyoyev Sh.M. From National Revival to National Prosperity. Volume 4. - Tashkent: Uzbekistan, 2020. 6. Mirziyoyev Sh.M. Strategy of the New Uzbekistan. Tashkent: Uzbekistan, 2021. 7. Mirziyoyev Sh.M. Let us live freely and prosperously in a New Uzbekistan. Volume 5. - Tashkent: Uzbekistan, 2023. 8. Mirziyoyev Sh.M. We will resolutely continue the course of democratic reforms based on the development strategy of the New Uzbekistan. Volume 6. - Tashkent: Uzbekistan, 2023. 9. Mirziyoyev Sh.M. People-oriented reforms serve the interests of my people. Volume 7. - Tashkent: Uzbekistan, 2023. 10. Mirziyoyev Sh.M. When we are united, we are one people; when we are together, we are our homeland. Volume 8. - Tashkent: Uzbekistan, 2024. 11. Mirziyoyev Sh.M. Strategy of the New Uzbekistan. - Tashkent: Uzbekistan, 2021. 12. I.K. G'ulomov, G.Sh. Ibadullayeva, S.F. Norbekov, X. Muzaffarov. Textbook of the Latest History of Uzbekistan (for students of all higher education specialisations). - T.: "TAMADDUN", 2025. 13. The Latest History of Uzbekistan. Textbook. / R.H.Murtazayeva and others. – Tashkent, 2024. 14. History of Uzbekistan. Edited by A.S.Sagdullayev. Volume 1. Textbook. T.: VNESHINVESTPROM, 2019. 15. N. Oblamurodov, F. To'lipov. The Latest History of Uzbekistan. Textbook. – Tashkent: Iqtisod-moliya, 2022. 16. B.J. Eshov, A.A. Odilov. History of Uzbekistan. Volume 2. Textbook. – Tashkent: Donishmand Ziyosi, 2020. 17. Khalid A. The Birth of Uzbekistan: Nation, Empire, and Revolution in the Early Soviet Period / trans. Sardor Salim. – Tashkent: Akademnashr, 2022. – 424 pp.
Responsible Lecturer	Yangiboyeva Dilnoza O'ktamovna, teacher.

RELIGION STUDIES

Field	Description
Module Title	RELIGION STUDIES
Module Code	DIN1204
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	II semester
Module objective	The objective of the Module Religion Studies is to develop in students an understanding of the essence of religion and its role in the minds of student youth. The Module surveys the world's religious landscape and examines the interconnectedness of religious and philosophical views. It addresses the phenomenon of religious fanaticism in the context of globalization, together with the negative impact of fundamentalism, extremism, and terrorism on state and societal security. Students develop theoretical knowledge of the history, doctrines, and social role of the world's major religions, including Buddhism, Christianity, and Islam. The Module further develops practical skills for combating spiritual threats and information attacks artificially linked to the name of religion. Through the study of religious organizations operating in Uzbekistan and internationally, students develop a culture of dialogue and mutual understanding among believers of different religions and non-believers.
Module Learning Outcomes	Knowledge: LO1. Understands the essence of religion, the role of religion in society and in an individual's life, the relationship between secular and religious knowledge, the interrelationship between religion and science, and the goals of religious and secular education. LO2. Explores, through the study of religious studies, the essence of various religions, the differences and commonalities among their customs and beliefs, and the basic concepts of religious studies, including God,

	prophets, humanity, destiny, the hereafter, and the essence of faith, and is able to apply this knowledge. LO3. Understands secularization processes and develops a culture of dialogue and common understanding with believers of different religions and non-believers. LO4. Has the skills to combat spiritual threats and information attacks artificially linked to the name of religion. Skills: LO5. Knows how to scientifically substantiate and express views on spiritual-national and universal human issues, and has an active worldview that adheres to the ideals of national independence. LO6. Possesses a competitive general professional preparation in line with the bachelor's degree programme. LO7. Is able to independently acquire new knowledge, organise their activity on a scientific basis, and work toward their own self-perfection. Competencies: LO8. Has experience working independently in vacant positions appropriate to their field of higher education. LO9. Tests, evaluates, optimises, and refines task results.
Content	Lectures Subject, Purpose, and Objectives of the Study of Religion. The subject, purpose, and objectives of religious studies are examined. The concept, origin, and functions of religion are studied. The significance of religious studies in understanding religious phenomena, together with the principles and methods of studying religion, is considered. The relationship of religious studies with philosophy, history, sociology, psychology, and other social sciences is also considered. National Religions. The concept and characteristics of national religions are examined. The historical development of national religions is studied, together with major national religions and their beliefs. The role of national religions in the cultural and social life of different nations is considered. The influence of national religions on traditions and customs is also considered. History and Basic Ideas of Buddhism. The origin and historical development of Buddhism are examined, together with the life and teachings of Siddhartha Gautama (Buddha). The basic beliefs and moral principles of

Buddhism are studied. Major schools and traditions of Buddhism are considered. The spread and influence of Buddhism in the world is also considered.

History and Basic Ideas of the Christian Religion. The origin and historical development of Christianity are examined, together with the life and teachings of Jesus Christ. The Holy Bible and its structure are studied. The main branches of Christianity and their characteristics are considered. The role of Christianity in world civilization is also considered.

History and Essence of Islam. The emergence and historical development of Islam are examined, together with the life and prophetic mission of Prophet Muhammad (PBUH). The fundamental beliefs and principles of Islam are studied. The Qur'an and the Sunnah as the primary sources of Islam are considered. The role of Islam in the spiritual and social life of society is also considered.

Sources of Islam, Theological Schools and Their Importance in the Present Day. The primary and secondary sources of Islam are examined, together with the development of Islamic theological schools (Kalām). Major theological schools and their principles are studied. The contribution of theological schools to Islamic thought is considered. Their significance in addressing contemporary religious issues is also considered.

Schools of Islamic Fiqh. The concept and development of Islamic jurisprudence (Fiqh) are examined, together with the sources and principles of Islamic law. The major Sunni schools of Fiqh and their founders are studied. The characteristics of each school are considered. The importance of Fiqh schools in the application of Islamic law is also considered.

The Influence of Sufi Doctrine on Eastern Culture. The concept and historical development of Sufism are examined, together with its main teachings and moral values. Prominent Sufi scholars and their works are studied. The influence of Sufi thought on Eastern literature, art, and culture is considered. The role of Sufism in spiritual and moral education is also considered.

The Contemporary Significance of Islamic Values and Enlightenment. The essence of Islamic values and enlightenment is examined. The role of Islamic values in strengthening moral education and social harmony is

studied, together with the contribution of enlightenment to preventing religious extremism and intolerance. The promotion of peace, tolerance, and mutual respect through Islamic teachings is considered. The relevance of Islamic enlightenment in modern society is also considered.

Modern Religious Movements and Sects. The concept and characteristics of modern religious movements and sects are examined, together with the factors contributing to their emergence and development. The classification and activities of contemporary religious movements are studied. Their influence on society and religious life is considered. Approaches to identifying and preventing destructive sectarian activities are also considered.

Religion in Cyberspace. The Socio-Political Risk of Missionary Work and Proselytism. The use of digital technologies in religious activities is examined, together with the spread of religious information through cyberspace. The concepts of missionary work and proselytism are studied. Their socio-political impact and legal aspects are considered. Measures to promote responsible use of online religious information are also considered.

History of Religious Fanaticism, Extremism, Fundamentalism, and Terrorism, and Their Contemporary Threat to Security. The concepts and historical development of religious fanaticism, extremism, fundamentalism, and terrorism are examined. The causes and ideological foundations of radical movements are studied. Their impact on peace, stability, and security is considered. National and international efforts to prevent religious extremism and terrorism are also considered.

Religious Organizations Operating in Uzbekistan. The legal framework governing religious organizations in Uzbekistan is examined. The types and activities of registered religious organizations are studied. The principles of freedom of conscience and religious tolerance are considered. The role of religious organizations in strengthening social harmony and interfaith cooperation is also considered.

History of the Activities of International Religious Organizations and Their Importance in Achieving the Sustainable Development Goals. The emergence and development of international religious organizations are

examined. Their humanitarian, educational, and social activities are studied. The contribution of religious organizations to peacebuilding and international cooperation is considered. Their role in supporting the achievement of the Sustainable Development Goals is also considered.

The Importance of Ensuring the Balance Between Secular and Religious Knowledge. The concepts of secular and religious knowledge are examined. The principles of balancing secular and religious education are studied. The role of balanced knowledge in promoting tolerance, social stability, and intellectual development is considered. The importance of integrating scientific and moral values in modern education is also considered.

Practical lessons

Subject, Purpose, and Tasks of the Science of Theology. The concept and scope of theology are examined, together with its subject, purpose, and tasks. Research methods in theology are studied. The relationship of theology with other sciences is considered. The significance of theology in understanding religious beliefs and values is also considered.

National Religions. The concept and characteristics of national religions are examined, together with their historical development. Major national religions and their beliefs are studied. The role of national religions in society and culture is considered. Their influence on customs and traditions is also considered.

History and Basic Ideas of Buddhism. The origin and development of Buddhism are examined, together with the life and teachings of Buddha. Basic beliefs and moral principles of Buddhism are studied. Major schools of Buddhism are considered. The spread of Buddhism throughout the world is also considered.

History and Basic Principles of the Christian Religion. The origin and historical development of Christianity are examined, together with the life and teachings of Jesus Christ. The Holy Bible and Christian beliefs are studied. Major branches of Christianity are considered. The influence of Christianity on world civilization is also considered.

History and Essence of Islam. The emergence and development of Islam are examined, together with the life

and mission of Prophet Muhammad (PBUH). The fundamental beliefs and practices of Islam are studied. The Qur'an and Sunnah as the main sources of Islam are considered. The role of Islam in society is also considered. **Sources of Islam, Theological Schools and Their Importance in the Present Day.** The primary and secondary sources of Islam are examined, together with the emergence of Islamic theological schools. Major theological schools and their doctrines are studied. Their contribution to Islamic scholarship is considered. Their importance in addressing contemporary issues is also considered.

Schools of Islamic Jurisprudence. The concept and development of Islamic jurisprudence (Fiqh) are examined, together with the sources of Islamic law. Major Sunni schools of Fiqh and their founders are studied. Distinctive features of Fiqh schools are considered. Their role in the application of Islamic law is also considered.

The Influence of Sufi Doctrine on Eastern Culture. The origin and development of Sufism are examined, together with its main ideas and spiritual values. Famous Sufi scholars and their contributions are studied. The influence of Sufism on Eastern literature, art, and culture is considered. The educational role of Sufism is also considered.

The Contemporary Significance of Islamic Values and Enlightenment. The essence of Islamic values and enlightenment is examined, together with their role in moral education and social harmony. The promotion of tolerance and mutual respect is studied. The importance of enlightenment in preventing extremism is considered. The relevance of Islamic values in modern society is also considered.

Modern Religious Movements and Sects. The concept and characteristics of modern religious movements and sects are examined, together with the factors contributing to their emergence and development. Major contemporary religious movements and their activities are studied. Their influence on society and religious life is considered. Approaches to preventing destructive sectarian activities are also considered.

Religion in Cyberspace. The Socio-Political Risk of Missionary Work and Proselytism. The role of religion in

cyberspace and digital communication is examined, together with the dissemination of religious information through online platforms. The concepts of missionary work and proselytism are studied. Their socio-political risks and legal aspects are considered. Measures to ensure responsible use of digital religious resources are also considered.

The History of Religious Fanaticism, Extremism, Fundamentalism, and Terrorism, and Their Current Threat to Security. The concepts and historical development of religious fanaticism, extremism, fundamentalism, and terrorism are examined, together with the ideological foundations of radical movements. Their impact on national and international security is studied. Measures to prevent religious extremism and terrorism are considered. The importance of promoting tolerance and peace is also considered.

Religious Organizations Operating in Uzbekistan. The legal framework governing religious organizations in Uzbekistan is examined, together with the types and activities of registered religious organizations. Principles of freedom of conscience and religious tolerance are studied. The contribution of religious organizations to social harmony is considered. State policy regarding religious organizations is also considered.

History of the Activities of International Religious Organizations and Their Importance in Achieving the Sustainable Development Goals. The emergence and development of international religious organizations are examined, together with their humanitarian, educational, and social activities. Their contribution to peacebuilding and international cooperation is studied. The role of religious organizations in achieving the Sustainable Development Goals is considered. Contemporary challenges and prospects are also considered.

The Importance of Ensuring the Balance Between Secular and Religious Knowledge. The concepts of secular and religious knowledge are examined, together with the principles of maintaining balance between secular and religious education. The role of balanced knowledge in strengthening social stability and tolerance is studied. The integration of scientific and moral values in education

	is considered. The importance of balanced education in modern society is also considered. Self-study The Ideas of Eastern Thinkers on the Influence of Islam on Human Perfection (Essay Writing). Writing an Essay on the Theme of Religion and Morality Based on Ghazzali's "Tawbah". Preparing a Presentation on the Topic "Religion and Free Thought" (Comparing Berdyaev's Ideas on Religion and Free Thought with the Ideas of Ghazzali). Preparing a Crossword on the Theme of the Protection of Human Rights, Social Equality, Justice, and Tolerance in Islam. Preparing a Test on the Topic of the Historical Experience of Interfaith Religious Tolerance in Uzbekistan. Outlining Imam Bukhari's "Al-Adab al-Mufrad" and Preparing an Essay. Preparing an Essay on Abu Hamid al-Ghazali's "Mukashafatul Qulub". Preparing a Crossword Puzzle Based on Abdurauf Fitrat's Treatise "Muxtasar Islom Tarixi". Preparing an Essay on Abdulla Avloniy's Views on the Perfect Human Being. Prepare 10 Case Questions on the Topic of the Distinctive Aspects of Ensuring Human Rights and Freedoms in Islam. Analysis of Religious Beliefs in Leo Tolstoy's "Confession". Prepare a Presentation on the Interpretation of Islamic Values in Shaykh Muhammad Sodik Muhammad Yusuf's Work "The Happy Family". Defining the Concepts of Family, Marriage, Divorce, Iddah, and Zihar in Islamic Jurisprudence and Preparing a Presentation. Prepare 10 Case Questions on the Topic of the Balance Between Secularism and Religiosity in Societal Life. Prepare an Analytical Article on the History and Content of the Cairo Declaration.
Teaching Methods	The Module is delivered through interactive lectures, seminars, case-study analysis, problem-based and inquiry-based learning, group discussions, presentations, debates, and independent learning activities. Students examine religious sources, analyze historical and contemporary religious issues, and compare different

	religious traditions. These activities develop critical thinking, analytical, and communication skills. Independent study assignments, including essays, presentations, and case-based questions, reinforce the practical application of theoretical knowledge. Digital educational resources, multimedia materials, and academic literature are used to strengthen students' understanding of theology, religious tolerance, and intercultural dialogue.
Prerequisites	Philosophy Recent History of Uzbekistan Scientific Heritage of Eastern Scholars
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, seminar discussions, presentations, case-study analysis, and the successful completion of independent study assignments, including essay writing, preparation of presentations, crossword and test development, analysis of religious sources, and comparative studies of religious doctrines and ethical values. The midterm assessment consists of written assignments, short in-class quizzes, oral presentations, and the evaluation of independent study projects, such as comparative analyses of world religions, reports on religious organizations, presentations on contemporary religious issues, and essays on Islamic values, religious tolerance, and interfaith dialogue. The final assessment is a written examination consisting of theoretical questions and analytical tasks designed to assess students' understanding of religious studies, critical thinking, analytical skills, and their ability to interpret religious concepts and contemporary issues using scientific and objective approaches. Successful completion of all independent study assignments is required as an essential component of the module assessment.

Recommended Literature	1. Muratov D., Alimova M., Karimov J. Sociology of Religion. Textbook. - Tashkent: "Navro'z" Publishing House, 2019. - 264 p. 2. Rahimdzhanov D., Ernazarov O. Introduction to the Study of Religion. Textbook. - T.: National Society of Philosophers of Uzbekistan Publishing House, 2018. - 304 p. 3. Isoqjonov R. Comparative Theology. A Textbook. - T.: OOO "Complex print", 2020. - 198 p. 4. Law of the Republic of Uzbekistan. On Freedom of Conscience and Religious Organizations. July 5, 2021, O'RQ-699. ("Xalq so'zi" newspaper, July 6, 2021, 141 (7921)). 5. Mirziyoyev, Sh. M. Strategy of the New Uzbekistan. – T.: "Uzbekiston", 2021. 464 p. 6. Majitov M. World Religions and Universal Human Values. Publishing-Printing Complex of the International Islamic Academy of Uzbekistan, Tashkent, 2019. 104 pp. 7. Guttenplan K.S., Hornsby J., Janaway C. Reading Philosophy. Blackwell Publishers Ltd. UK. 2003. P. 341. 8. Fitrat. Selected Works. - Tashkent: Ma'naviyat, 2010. - 301 p. 9. Abu Hamid al-Ghazali. Kimyo-i-Saodat // Knowledge Leading to the Happiness of Both Worlds. - Samarkand: Imam Bukhari International Center, 2019.
Responsible Lecturer	Juraev Akrom Namozovich.

PHILOSOPHY

Field	Description
Module Title	PHILOSOPHY
Module Code	FAL1204
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours

	Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	II semester
Module objective	The objective of the Module Philosophy is to form philosophical, scientific and moral knowledge in students regarding the social significance of natural and scientific knowledge and the evolution of techniques and technologies as a response to human pragmatic needs. The Module addresses the interrelationship and distinction between national values, culture and mass culture. It emphasises the importance of ensuring the harmony of practice and activity in the knowledge and understanding of the world, together with the role of correct thinking in achieving positive goals. Students acquire knowledge of the causes and forms of global corruption and of the means of combating them. The Module aims to fully demonstrate students' world knowledge and skills in the context of the reforms modernising Uzbekistan. It further seeks to instil a sense of social and professional responsibility and a desire to make a worthy contribution to the development of society.
Module Learning Outcomes	Knowledge: LO1. Based on the study of ancient and modern philosophical knowledge about the world, to organise professional activity correctly, increase social responsibility, and foster a sense of involvement in state policy, independently analyse social processes, the main ideas of Eastern and Western philosophy of the Ancient, Medieval, Modern and most recent periods; the influence of medieval Eastern philosophy on the formation of the Western Renaissance; to have an understanding of the key characteristics of philosophy in the Modern and Contemporary periods. Skills: LO2. The directions of ontology, gene ontology, bio-ontology, the social significance of medical ontology, the interrelationship and distinctions between physical and philosophical ontology, protecting the interests of individuals and society in the use of technologies in

	an information society, the role of the biosphere and noosphere in nature ontology, the transformation of the ontology of consciousness, social ecology; knowledge of and ability to apply ecological education and upbringing. Competencies: LO3. Selecting and utilising information on social networking sites to acquire methods; to make independent decisions in their professional activity, to have competitive general professional preparation in the relevant field of study, to be immune to ideologies opposing globalisation and sustainable development, and to be aware of the factors of globalisation and sustainable development, as well as cyberterrorism and cyberbullying; to have developed ideological immunity against it, to be aware of the factors of globalisation and sustainable development, and to understand the negative impact of corruption on sustainable development as a consequence of the global moral crisis; to foster a sense of involvement in the fight against it, to forecast social processes, to ensure physical and mental health, to guarantee the security of thought, to adhere to a healthy lifestyle; must possess the qualifications and skills to utilise the potential of critical thinking for the purposes of ensuring societal stability, enhancing social control and personal responsibility.
Content	Lectures The Essence of Philosophy and Its Role in the Development of Society. The concept, subject matter, objectives, and functions of philosophy are examined, together with the origin and historical development of philosophical thought. Worldview and philosophical thinking are studied in relation to the role of philosophy in shaping human consciousness and social progress. The relationship between philosophy, science, religion, and culture is considered. The significance of philosophy in solving contemporary social problems is examined. The Essence of Eastern Philosophy and Its Role in Social Development. The emergence and development of Eastern philosophical traditions are examined, including ancient Indian, Chinese, and

Central Asian philosophy. The philosophical heritage of Confucius, Laozi, Buddha, al-Farabi, Ibn Sina, al-Biruni, and other Eastern thinkers is studied. Humanism, morality, harmony, and spiritual values in Eastern philosophy are considered. The contribution of Eastern philosophy to civilization and modern society is examined.

The Main Directions of Western Philosophy and Their Influence on the Development of Science.

Ancient Greek philosophy and its representatives are examined, together with medieval philosophy, Renaissance philosophy, rationalism, empiricism, and the Enlightenment. German classical philosophy and modern philosophical movements are studied. Positivism, existentialism, pragmatism, phenomenology, and postmodernism are considered. The influence of Western philosophy on scientific methodology and technological progress is examined.

Philosophy of Being (Ontology). The concept and categories of being are examined, together with material and ideal existence. Substance, matter, space, and time are studied in relation to the relationship between existence and reality. Classical and non-classical ontological theories are considered. The philosophical interpretation of the structure and forms of existence is examined.

Philosophy of Development. The concept and principles of development are examined, together with change, movement, and evolution. Dialectical and metaphysical approaches to development are studied, along with the laws and categories of dialectics. Progress, regress, revolution, and reform are considered. Development in nature, society, and human thinking is examined.

Philosophy of Knowledge. The nature and sources of knowledge are examined, together with the subject and object of cognition. Sensory and rational knowledge are studied, along with truth, criteria of truth, and methods of verification. Scientific knowledge and its methodology are considered, together with the empirical and theoretical levels of cognition. Contemporary epistemological problems are examined.

Logic. The subject and significance of logic are examined, together with concepts, judgments, and reasoning. Deductive and inductive methods are studied, along with logical laws and principles. Argumentation and proof are considered, together with logical errors and fallacies. The role of logical thinking in scientific research and everyday life is examined.

Philosophy of Society. Society as a philosophical concept is examined, together with the origin, structure, and development of society. Social relations, institutions, and values are studied, along with civil society and the rule of law. The interaction between the individual and society is considered. Social change and modernization are examined.

Philosophy of History, Civilisation and Culture. The philosophy of historical development is examined, together with historical progress and historical consciousness. The concepts of civilization and culture are studied, along with theories of civilizational development. National and universal values are considered. Cultural heritage and intercultural dialogue in the modern world are examined.

Philosophy of Man (Philosophical Anthropology). The philosophical understanding of human nature is examined, together with the origin and essence of humanity. Consciousness, freedom, responsibility, and creativity are studied, along with human dignity and rights. Personality formation and self-realization are considered. The place of humans in nature and society is examined.

Philosophy of Values. The concept and classification of values are examined, together with moral, aesthetic, religious, political, and cultural values. Universal and national values are studied, along with value systems and social development. The formation of value orientations is considered. The role of values in personal and social life is examined.

Ethics. The subject and principles of ethics are examined, together with moral consciousness, moral norms, and virtues. Freedom, responsibility, duty, justice, and conscience are studied, along with

professional ethics and business ethics. Ethical challenges in contemporary society are considered. The importance of moral education is examined.

Aesthetics. The subject of aesthetics is examined, together with beauty, harmony, and artistic creativity. Aesthetic perception and evaluation are studied, along with art as a form of spiritual culture. Functions and types of art are considered. Aesthetic education and its importance in personality development are examined.

Philosophy of Global Processes and Sustainable Development. Globalization and global challenges are examined, together with environmental, demographic, economic, and technological problems. Sustainable development goals and principles are studied, along with ecological ethics and environmental responsibility. International cooperation and the future of humanity are considered. The philosophical foundations of sustainable development are examined.

The Nature of Corruption, Its Consequences, and the Mechanisms for Combating It. The concept and forms of corruption are examined, together with the historical and social causes of corruption. Economic, political, legal, and moral consequences of corruption are studied, along with international anti-corruption standards and conventions. National anti-corruption legislation and institutional mechanisms are considered. The role of ethics, transparency, accountability, and civil society in preventing corruption is examined.

Practical lessons

The Essence of Philosophy and Its Role in the Development of Society. The concept, functions, and significance of philosophy are discussed. Philosophical worldviews and their influence on social development are analysed. Philosophical ideas are interpreted through historical and contemporary examples.

The Essence of Eastern Philosophy and Its Role in Social Development. The major schools and representatives of Eastern philosophy are analysed. Moral and spiritual values in Eastern philosophical

traditions are discussed. Their relevance to modern society is considered.

The Main Directions of Western Philosophy and Their Influence on the Development of Science.

Major Western philosophical schools are compared and analysed. The influence of philosophical ideas on scientific methodology is discussed. Their impact on technological progress is also considered.

Philosophy of Being (Ontology). The concept of being is discussed. Ontological categories are examined in this context. Classical and modern ontological theories are compared and analysed.

Philosophy of Development. The laws and principles of development are analysed. Dialectical thinking is discussed. Its application to social and scientific processes is considered.

Philosophy of Knowledge. The sources, forms, and methods of knowledge are discussed. Scientific cognition is analysed. Truth and epistemological problems are also examined.

Logic. Practical exercises are carried out on concepts, judgments, and reasoning. Argumentation is analysed through these exercises. Logical errors are identified and logical methods are applied to problem solving.

Philosophy of Society. The structure and development of society are discussed. Social institutions and civil society are analysed. Social change is also considered.

Philosophy of History, Civilisation and Culture. Theories of historical development and civilization are analysed. Cultural diversity and national identity are discussed. Intercultural communication is also considered.

Philosophy of Man (Philosophical Anthropology). Philosophical approaches to human nature are discussed. Freedom and responsibility are examined within this context. Personality development is also considered.

Philosophy of Values. Universal, national, and personal values are analysed. Value systems are discussed. Their influence on social life is considered.

Ethics. Moral principles and ethical theories are discussed. Professional ethics is also examined. Ethical dilemmas in modern society are analysed.

Aesthetics. Beauty, art, and aesthetic perception are discussed. Artistic works are analysed from a philosophical perspective. The same works are also examined from a cultural perspective.

Philosophy of Global Processes and Sustainable Development. Globalization and sustainable development are discussed. Environmental ethics is examined in this context. Global challenges facing humanity are also considered.

The Nature of Corruption, Its Consequences, and the Mechanisms for Combating It. The causes and consequences of corruption are analysed. Anti-corruption policies and ethical standards are discussed. The role of civil society in promoting integrity is also considered.

Self-study
Classifying the Definitions of Eastern and Western Philosophers on the Essence of Philosophy.
Seneca's Moral Letters to Lucilius.
Al-Farabi's and Al-Ghazali's Views on the Science of Philosophy.
Directions and Essence of Non-Classical Ontology.
Classification of Categories in the Ontologies of Aristotle, Al-Farabi and Kant.
Family and Child-Rearing in Abdulla Fitrat's Oila.
Fitrat's Views on the 'Two Categories of Unhappy People' in Rahbari-i Najat.
Avloniy's Waves of Culture and Toffler's The Third Wave.
The Informatised Society and Its Prospects.
Importance of Artistic Works in the Aesthetic Perception of the World.
Similarities and Differences in the Cultures of the World's Peoples.
Myths of the World's Peoples.
Rumi's Tale 'The Hunter and the Falcon'.
Advantages and Disadvantages of Modern Moral Education.
Dialectics of Co-Evolution and Environmentalism.

	Behbudiy's 'Address to the Youth' and 'Address to the Esteemed Youth'. Machiavelli's The Prince. Al-Ghazali's Tawbah. Hayy ibn Yaqzan by Ibn Tufayl and Ibn Sina. The Essence of Corruption and Its Consequences. Anti-Corruption Mechanisms in Uzbekistan. Governance of Society in Nizām al-Mulk's Siyāsāt-nāma. Huntington's The Clash of Civilizations.
Teaching Methods	The module is delivered through interactive lectures, seminar discussions, case studies, inquiry-based and problem-based learning, debates, collaborative learning, presentations, and project-based activities. Students critically analyze classical and contemporary philosophical texts, compare different philosophical traditions, evaluate ethical and social issues, and develop logical reasoning and argumentation skills. Independent study includes reading philosophical works, writing analytical essays, preparing presentations, video projects, and other creative assignments. The use of digital learning resources and modern educational technologies further enhances critical thinking, communication, academic writing, and the practical application of philosophical knowledge to contemporary social and global challenges.
Prerequisites	Religion Studies Scientific Heritage of Eastern Scholars Recent History of Uzbekistan
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final assessment (50%). The continuous assessment evaluates students' classroom participation, seminar discussions, presentations, logical reasoning, analysis of philosophical texts, and the successful completion of independent study assignments, including comparative analyses of Eastern and Western philosophy, essays, presentations, video projects, crossword puzzles, and

	other research-based activities. The midterm assessment consists of written assignments, short quizzes, and the evaluation of independent study projects. Students are assessed on their understanding of philosophical concepts, major philosophical schools, ethical and social issues, logical argumentation, and the interpretation of classical and modern philosophical works. The final assessment may be conducted as either a written examination or a test examination. The written examination consists of theoretical and analytical questions covering the main topics of the module, including ontology, epistemology, logic, ethics, aesthetics, social philosophy, philosophy of values, philosophy of global processes, and anti-corruption studies. The test examination consists of multiple-choice questions designed to assess students' knowledge of philosophical concepts, theories, terminology, major philosophers, logical reasoning, ethical principles, and the application of philosophical ideas to contemporary social issues. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Shermuhamedova N. Philosophy, Textbook – Tashkent: Idris Abdurauf Publishing, 2021, p. 667. 2. Davronov Z., Shermuhamedova N, Qahharova M, Nurmatova M, Husanov B, Sultonova A. Philosophy, Study Guide – Tashkent: TMU, 2019. – 375 pp. 3. A.G. Spirkin. Philosophy: A Textbook. 2nd ed. Moscow: Gardariki, 2006. 736 pp. 4. Samatov Sh. Philosophy (a textbook). – Termez: "Surxon-nashr", 2023 – p. 340. 5. Mirziyoyev, Sh. M. The Development Strategy of the New Uzbekistan. Tashkent: O'zbekiston, 2022. – 416 pp. 6. Mirziyoyev, Sh.M. We will build a free and prosperous democratic state together. – Tashkent: Uzbekistan, 2021. – 448 p.

	8. Corbin, H. History of Islamic Philosophy (in English) — London, New York City: Kegan Paul International Ltd., 1993. — 445 pp. — ISBN 978-0-7103-0416-2
Responsible Lecturer	Mamatkulov Eldor Safarovich.

FOREIGN LANGUAGE 1,2

Field	Description
Module Title	FOREIGN LANGUAGE 1,2
Module Code	XOT1108
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (60 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	I semester
Module objective	The objective of the Module Foreign Language 1,2 is to train future specialists in non-philological fields who can use a foreign language freely in everyday life. The Module also develops the ability to use the language freely in academic and professional activities. It aims to develop students' patriotism and independent learning. Students are supported in acquiring knowledge and skills beyond classroom hours and after graduation. The Module develops students' ability to write academic articles on universal and field-related topics. It further aims to strengthen students' logical thinking.
Module Learning Outcomes	Knowledge: LO1. Demonstrate knowledge of speech competencies (reading, writing, listening and speaking), language competencies (lexical and grammatical), and socio-cultural and pragmatic competencies. LO2. Develop oral and written communication skills related to academic, professional and everyday topics. LO3. Become familiar with universal and national values and develop intercultural tolerance and interethnic friendship. LO4. Learn the terms and expressions used in academic and professional activity.

	LO5. Organize independent-study assignments according to students' academic and professional fields through clear and accurate instructions. Skills: LO6. Develop reading, writing, listening and speaking skills in an integrated manner during practical classes. LO7. Select and prioritize the relevant language skill in class according to the essence of the topic and the features of the specialization, using appropriate pedagogical methods and technologies. LO8. Conduct conversations and informal dialogues on social topics; participate in formal and informal professional discussions; manage discussions, interviews, negotiations and telephone communication. LO9. Prepare and deliver presentations or lectures on specialty-related topics; support arguments with evidence; summarize information, write and discuss articles. LO10. Develop skimming, scanning and intensive reading skills; read correspondence, emails, authentic texts, scientific and professional literature, electronic sources and press materials containing specialized terms. LO11. Produce written correspondence, messages, special reports, CVs and notes; write essays, summaries, resumes and research papers. LO12. Understand the main content of authentic materials after listening twice, identify necessary information and present it orally or in writing; understand news, reports and characters' speech in films. Competencies: LO13. Use a foreign language practically in professional contexts across speech activities; develop as a creative personality; develop skills for translating field-specific literature. LO14. Reading: work weekly throughout the academic year with texts on various general and field-specific topics (1,200-1,500 characters). LO15. Writing: write greeting cards, informal and formal letters; write summaries of texts on various topics; compose essays of 200-250 words.
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	LO16. Listening: understand radio, internet and television programs and interviews; understand complex authentic speech in familiar and unfamiliar contexts. LO17. Speaking: communicate on general, social, socio-cultural, educational, ICT-related and specialty-related topics, including daily life, global issues, education systems, scientific writing, internet technologies, professional ethics, negotiations, innovations and leading specialists in the field.
Content	Practical lessons Working in the IT Industry. Reading about meeting people is combined with listening practice on the English alphabet and pronunciation. A short piece of writing about oneself is completed, and speaking activities focus on talking about oneself. The specialty text 'Working in the IT Industry' is read and discussed to build IT-related vocabulary. The verb 'to be' in its affirmative, negative and interrogative forms is studied together with possessive pronouns. The Software Engineer. Reading about jobs in IT is combined with listening practice on common abbreviations. Writing about one's hometown is completed, and speaking activities focus on talking about one's hometown. The specialty text 'The Software Engineer' is read and discussed to build IT-related vocabulary. Verb Forms I (the verb 'to have') is studied and practised. Types of Computers. Reading about schedules is combined with listening practice on the names of places and data collection. Writing about one's daily routine is completed, and speaking activities focus on talking about family. The specialty text 'Types of Computers' is read and discussed to build IT-related vocabulary. The Present Simple tense (part 1) is introduced and practised. Inside the Computer. Reading about spelling IT acronyms is combined with listening practice on getting to know other people. Writing about problems of working on the internet is completed, and speaking activities focus on discussing computer problems with

a partner. The specialty text 'Inside the Computer' is read and discussed to build IT-related vocabulary. The Present Simple tense (part 2) is further practised.

System Software 1. Reading about computer hardware is combined with listening practice on maps of some countries. Writing a description of computer hardware is completed, and speaking activities take the form of a role-play about difficult assignments and enjoyable subjects. The specialty text 'System Software 1' is read and discussed to build IT-related vocabulary. The Present Simple tense (part 3) is further practised.

System Software 2. Reading about computer software is combined with listening practice on expressing interest and opinions. An e-mail from a software engineer is completed as a writing task, and speaking activities cover preventing security problems and weekend plans. The specialty text 'System Software 2' is read and discussed to build IT-related vocabulary. The Present Continuous tense is introduced and practised.

Programming Software. Reading about working with computers is combined with listening practice on reading numbers. Writing a description of a friend is completed, and speaking activities cover security threats and describing a nice place. The specialty text 'Programming Software' is read and discussed to build IT-related vocabulary. The Present Continuous tense (part 2) is further practised.

Application Software 1. Reading about computer usage is combined with listening practice on numbers. A company's software product index is completed as a writing task, and speaking activities focus on new developments at a software company. The specialty text 'Application Software 1' is read and discussed to build IT-related vocabulary. The structures 'there is' and 'there are' are introduced and practised.

Application Software 2. Reading about website analytics is combined with listening practice distinguishing between 'teens' and 'tens'. A formal e-mail on software developments to a company owner is

written, and speaking activities focus on software engineering projects. The specialty text 'Application Software 2' is read and discussed to build IT-related vocabulary. The Past Simple tense (part 1) is introduced and practised.

The Desktop and GUI. Reading about website development and website functions is combined with listening practice through number dictation. A troubleshooting guide is completed as a writing task, and speaking activities focus on discussing a problem with a program. The specialty text 'The Desktop and GUI' is read and discussed to build IT-related vocabulary. The Past Simple tense (part 2) is further practised.

Basic Numbers and Math. Reading about the best websites is combined with listening practice on the circulation of British newspapers. Writing about favourite websites is completed using a conversation from an earlier task, and speaking activities again focus on discussing a problem with a program. The specialty text 'Basic Numbers and Math' is read and discussed to build IT-related vocabulary. The forms 'was' and 'were' are introduced and practised.

Analyzing Numbers and Qualities. Reading about database basics is combined with listening practice on practical problem-solving. A progress report is completed as a writing task, and speaking activities involve a man talking about his job. The specialty text 'Analyzing Numbers and Qualities' is read and discussed to build IT-related vocabulary. The Past Continuous tense is introduced and practised.

Describing Change. Reading about data processing is combined with listening practice on ordinal numbers. An e-mail to a business associate is written, and speaking activities cover new software on the market and talking about oneself. The specialty text 'Describing Change' is read and discussed to build IT-related vocabulary. The Present Perfect tense ('I have done') is introduced and practised.

Presentations and Communication. Reading about data storage and back-up is combined with listening

practice through ordinal-number dictation. A presentation feedback form is completed as a writing task, and speaking activities focus on problems with one's communication style. The specialty text 'Presentations and Communication' is read and discussed to build IT-related vocabulary. The Present Perfect structure using 'have you ever' is introduced and practised.

Database Systems Benefits. Reading about education is combined with listening practice on ordinal numbers. Writing about one's university is completed, and speaking activities practise asking and telling the time. The specialty text 'Database Systems Benefits' is read and discussed to build IT-related vocabulary. The Present Perfect structure using 'how long have you been' is introduced and practised.

Information Technology. IT Department Restructuring. Reading about e-commerce companies is combined with listening practice on traffic at Britain's main airports. Writing a description of a neighbourhood is completed, and speaking activities focus on talking about one's neighbourhood. The specialty text on information technology and IT department restructuring is read and discussed to build IT-related vocabulary. The passive voice ('it is done' / 'it was done') is introduced and practised.

Introduction to Computing Systems. Reading about e-commerce features is combined with listening practice on household expenditure. Writing instructions on how to access computers is completed, and speaking activities focus on the types of computers people have. The specialty text 'Introduction to Computing Systems' is read and discussed to build IT-related vocabulary. The future forms 'will' and 'shall' are introduced and practised.

Inside the Computer (Transaction Security). Reading about transaction security is combined with listening practice on migration. A description of the transaction process is written, and speaking activities focus on taking out an old hard drive. The specialty text 'Inside the Computer' is read and discussed a second

time to build IT-related vocabulary. The structure 'to be going to' is introduced and practised.

Computing Devices. Reading about online transactions is combined with listening practice on an interview about an earthquake. Writing about devices that can be attached to computers is completed, and speaking activities focus on people's hobbies. The specialty text 'Computing Devices' is read and discussed to build IT-related vocabulary. The modal verbs 'can' and 'could' are introduced and practised.

Networking. Reading about types of network is combined with listening practice on British trade. Writing about types of network system is completed, and speaking activities focus on how a computer network helps a business. The specialty text 'Networking' is read and discussed to build IT-related vocabulary. The modal verbs 'may' and 'might' are introduced and practised.

The User Interface. Reading about networking hardware is combined with listening practice on money. A web comment about work is written, and speaking activities focus on what a user interface is. The specialty text 'The User Interface' is read and discussed to build IT-related vocabulary. The modal verb 'must' is introduced and practised.

Word Processing. Reading about talking about the past is combined with listening practice on asking about prices. A report is written using appropriate report structures, and speaking activities cover word-processing programs and university life. The specialty text 'Word Processing' is read and discussed to build IT-related vocabulary. The modal verb 'should' is introduced and practised.

Email. Network Range and Speed. Reading about modern examinations is combined with listening practice on life in London. Writing about the benefits of email for people and businesses is completed, and speaking activities discuss common problems with email. The specialty text on email, network range and speed is read and discussed to build IT-related

vocabulary. The structure 'have to' is introduced and practised.

Web Browsing. Reading about fault diagnosis is combined with listening practice on telling the time. A review is written as a writing task, and speaking activities discuss browsing the web at work. The specialty text 'Web Browsing' is read and discussed to build IT-related vocabulary. The structure 'would you like' is introduced and practised.

Images and Graphic Design. Reading about software repair is combined with listening practice on telephone conversations. Writing about editing pictures is completed, and speaking activities discuss what editing software is. The specialty text 'Images and Graphic Design' is read and discussed to build IT-related vocabulary. Adjectives are introduced and practised.

Databases vs. Spreadsheets. Reading about hardware repair is combined with listening practice on dates. Writing about old cities is completed, and speaking activities discuss the kind of information found in databases. The specialty text on databases and spreadsheets is read and discussed to build IT-related vocabulary. Antonyms and synonyms are introduced and practised.

Web Design vs. Development. Reading about customer service is combined with listening practice on survival English. Writing about shops and shopping is completed, and speaking activities discuss favourite types of websites. The specialty text on web design and development is read and discussed to build IT-related vocabulary. Linking words such as 'and', 'so', 'but' and 'because' are introduced and practised.

Desktop Publishing. Reading about IT security and safety and security solutions is combined with listening practice on university campus life. Writing about the skills needed in publishing is completed, and speaking activities discuss hobbies and interests. The specialty text 'Desktop Publishing' is read and discussed to build

	IT-related vocabulary. The phrasal verb 'look' is introduced and practised. Videoconferencing, Workstation Health and Safety. Reading about workstation health and safety is combined with listening practice on television and radio. Writing about the uses of video conferencing is completed, and speaking activities discuss how people around the world spend their free time. The specialty text 'Videoconferencing, Workstation Health and Safety' is read and discussed to build IT-related vocabulary. The quantifiers 'too', 'much', 'many', 'enough' and 'very' are introduced and practised. E-commerce. Reading about security procedures is combined with listening practice on sports. Writing about how online shopping helps businesses is completed, and speaking activities discuss personal online purchasing habits. The specialty text 'E-commerce' is read and discussed to build IT-related vocabulary. Abbreviations and abbreviated words commonly used in English and IT are introduced and practised. Self-study My Family My Weekend IT Jobs and Duties. My Flat. Popular Sports in My Country. Inside the Computer. My Biography. Popular Activities in Your Country. Types of Computers. Daily Shopping. Town and Countries. Problem Solving. My Native Town. Exam Time. Engineering Materials.
Teaching Methods	The module is delivered through a combination of lectures, practical classes, interactive discussions, and independent learning activities. Teaching methods include classroom instruction, communicative

	language teaching, pair and group work, role-plays, presentations, vocabulary and grammar exercises, reading and listening activities, writing assignments, project-based learning, and the use of digital learning resources. Independent learning activities may include preparing presentations, completing language exercises, writing essays and reports, conducting research on cultural topics, developing vocabulary portfolios, and participating in online learning tasks. These teaching methods are designed to develop students' listening, speaking, reading, and writing skills, enhance their communicative competence, and improve their ability to use the foreign language effectively in academic, professional, and everyday contexts.
Prerequisites	English language of school
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through midterm assessment (50%) and a final examination (50%). The midterm assessment consists of written assignments, short in-class quizzes, oral presentations, and evaluation of independent study projects. Independent study tasks may include preparing a presentation on a cultural topic related to English-speaking countries, creating a vocabulary portfolio, recording a video presentation on a contemporary issue, writing essays, reports and formal and informal letters, and conducting research and preparing a project on intercultural communication or language learning strategies. The final assessment is conducted as a written examination and includes theoretical questions on grammar, vocabulary, reading, and writing, together with practical tasks aimed at evaluating language proficiency, reading comprehension exercises, and writing tasks requiring students to demonstrate effective communication skills in the foreign language. The examination is designed to assess students' knowledge of the foreign language, communicative competence, analytical and critical thinking skills, and

	ability to use language accurately and appropriately in different contexts.
Recommended Literature	1. Alireza Memarzadeh. IELTS Maximiser Speaking, Writing, Reading. 2016. 2. Zhang Juan et al. Basic IELTS Reading, Listening, Writing, Speaking. Beijing Language and Culture University Press, 2013. 3. Raymond Murphy. English Grammar in Use. Elementary and Intermediate books. Cambridge University Press, 2019. 4. Constitution of the Republic of Uzbekistan. National Database of Legislation, 01.05.2023, No. 03/23/837/0241. https://lex.uz/uz/docs/-6445145 5. Labor Code of the Republic of Uzbekistan. National Database of Legislation, 29.10.2022, No. 02/22/798/0972; 12.04.2023, No. 03/23/829/0208. https://lex.uz/ru/docs/-6257288 6. Mirziyoyev Sh.M. Decree No. PF-60 dated 28 January 2022 "On the Development Strategy of New Uzbekistan for 2022-2026". 7. Materials and Man's Needs: Materials Science and Engineering. Supplementary Report of the Committee on the Survey of Materials Science and Engineering. Volume I. National Academy of Sciences, Washington, D.C., 1975. 8. William D. Callister Jr., David G. Rethwisch. Materials Science and Engineering: An Introduction. Wiley, 2018.
Responsible Lecturer	Ibodullayeva Go'zal Jo'rayevna.

CALCULUS 1,2

Field	Description
Module Title	CALCULUS 1,2
Module Code	HIS1112
Credit Points (ECTS)	12 Credits
Workload	Total workload: 360 hours Contact hours: 180 hours (90 hours Lecture, 90 hours Practical lessons) Self-Assignments: 180 hours
Frequency/Duration	I–II semester
Module objective	The objective of the Module Calculus is to develop students' knowledge of the fundamental concepts of mathematical analysis. The Module develops students' understanding of the methods of differential calculus. It further develops students' understanding of the methods of integral calculus. Students acquire skills in effectively using mathematical methods for mathematical modelling. The Module develops students' skills in applying mathematical methods to solving practical problems.
Module Learning Outcomes	Knowledge: LO1. Have an understanding of the theoretical and scientific foundations of the mathematical and natural sciences required for the engineering work education field. LO2. Be able to apply theorems on function limits and the fundamental rules of differentiation. LO3. Visualizing a function's graph using asymptotes, inflection points, and the derivative for increasing (decreasing) functions. LO4. Solve practical problems involving maxima (minima) using differentiation methods. LO5. Linear approximation; optimization methods (optimization problems, Newton's method). LO6. Application of the L'Hospital's rule when taking derivatives of certain functions, and use of differential calculus in solving problems. Skills:

	LO7. Knowledge of the concepts of elementary functions and definite integrals. LO8. Calculating integrals using substitution, simple fractions, and piecewise integration methods. LO9. Determine whether improper integrals are convergent (divergent) and evaluate convergent improper integrals. LO10. Calculating exact integrals using theorems of real analysis and Riemann sums; applying exact integrals to geometry (calculating the area of a plane figure, the volume of a solid of revolution, the length of a planar curve, and the differential of the arc length of a curve). LO11. Application of definite integrals to solving mechanics and physics problems (determining the static moments and center of mass of curved and planar shapes, calculating work), and determining whether numerical series are convergent (divergent). Competencies: LO12. Becomes competent in analyzing practical problems related to engineering and information technology and finding solutions using methods of mathematical analysis. LO13. Gains skills in mathematical modeling of processes and analysis of results by applying differential and integral calculus methods. LO14. Will be able to develop and evaluate mathematical models of complex systems using functions, series, and multivariable functions. LO15. Develops the competence to use modern information technologies to solve optimization, computational, and forecasting problems based on mathematical methods. LO16. Develops skills in independent mathematical thinking, logical analysis, and the effective application of mathematical knowledge in professional activities.
Content	Lectures The Concept of a Function; Methods of Defining Functions, Evenness and Oddness, Periodicity. The concept of a function, its domain, and its range are introduced. Methods of representing a function analytically, graphically, numerically in a table, and verbally are considered. The conditions under which

a function is even or odd and the corresponding symmetries of its graph are explained. The concept of a periodic function and its least positive period is examined.

Elementary Functions and Their Graphs. The principal elementary functions, including constant, linear, quadratic, power, exponential, logarithmic, trigonometric, and inverse trigonometric functions, are studied. Their domains, ranges, and fundamental properties are analyzed. Methods for representing their graphs in a coordinate system and performing geometric transformations of graphs are considered. The significance of elementary functions in describing mathematical and practical processes is explained.

The Concept of a Sequence of Numbers; the Limit of a Sequence of Numbers. The definition of a numerical sequence and methods of specifying it by an explicit formula or a recurrence relation are studied. The terms, general term, and principal properties of a sequence are explained. The meaning and geometric interpretation of the limit of a sequence are introduced. Convergent and divergent sequences are analyzed through representative examples.

Properties of Convergent Sequences; Calculating the Limit of Sequences of Numbers. The uniqueness, boundedness, and arithmetic properties of convergent sequences are considered. Theorems concerning the convergence of monotonic and bounded sequences are studied. Fundamental methods for calculating the limits of numerical sequences are explained. Algebraic transformations, simplification, and equivalent expressions are applied in limit calculations.

Existence of a Sequence Limit; Important Limits. Conditions ensuring the existence of the limit of a sequence and fundamental convergence theorems are examined. Methods for establishing the existence of a limit through monotonicity and boundedness are explained. Important limits widely used in mathematical analysis are introduced. Their application to the evaluation of more complicated limits of sequences and functions is considered.

Definitions of the Function Limit; Infinitely Small and Infinitely Large Functions. The definitions of the limit of a function at a point and at infinity are studied. The Cauchy and Heine definitions of a limit and the relationship between them are explained. One-sided, finite, and infinite limits are considered. The concepts of infinitesimal and infinitely large functions, together with their properties and interrelations, are examined.

Properties of Functions with Finite Limits, Important Limits, and Computing the Limit of a Function. The principal properties of functions possessing finite limits, including boundedness, preservation of sign, and uniqueness of the limit, are studied. Rules for calculating the limits of sums, differences, products, and quotients of functions are explained. Important limits are applied to trigonometric, exponential, and logarithmic functions. Methods for resolving indeterminate forms through algebraic transformations are considered.

The Concept of a Function's Continuity; Operations on Continuous Functions; Discontinuity and Its Types. Continuity of a function at a point and on an interval is studied. The preservation of continuity under arithmetic operations on continuous functions is explained. One-sided continuity and the causes of discontinuity are considered. Removable, jump, and infinite discontinuities are identified and classified.

The Concept of a Function Derivative; Its Geometric and Mechanical Meanings. The derivative is defined as the limit of the ratio of the increment of a function to the increment of its argument. Its geometric meaning as the slope of the tangent line is explained. Its physical and mechanical interpretations, including instantaneous velocity and rate of change, are considered. Fundamental differentiation rules and derivatives of elementary functions are introduced.

Calculating Derivatives of Inverse and Composite Functions. The chain rule for differentiating composite functions is studied. Procedures for differentiating expressions formed by several nested

functions are explained. The derivative of an inverse function and the conditions for its existence are considered. Derivatives of inverse trigonometric functions and their use in problem-solving are examined.

Differential of a Function; Approximate Formulas.

The differential of a function is defined as the principal linear part of its increment. The relationship between the derivative and the differential is explained. The differential of a composite function and the invariance of the differential form are considered. Differentials are applied to the approximate calculation of function values and the estimation of computational errors.

Higher-Order Derivatives and Differentials.

Second-, third-, and higher-order derivatives are introduced. Rules for calculating higher-order derivatives and their physical interpretations are explained. Higher-order differentials and procedures for their successive calculation are considered. Their applications to function analysis and the description of motion are examined.

Properties of Differentiable Functions; Taylor's Formula.

The continuity and fundamental properties of differentiable functions are studied. The principal theorems of differential calculus, including the theorems of Fermat, Rolle, Lagrange, and Cauchy, are introduced. Taylor's formula and its remainder term are explained. Taylor's formula is applied to polynomial approximation and practical calculations.

Determining the Increasing, Decreasing, and Extremum Points of Functions Using Derivatives.

The sign of the derivative is used to determine intervals on which a function increases or decreases. Critical and stationary points are introduced. Necessary and sufficient conditions for local maxima and minima are examined. First- and second-derivative tests are applied to extremum problems.

Convexity, Concavity, Inflection Point, and Asymptotes of a Function Graph.

The second derivative is used to determine intervals of convexity and concavity. The concept of an inflection point and the conditions for its existence are explained. Methods for determining vertical, horizontal, and

oblique asymptotes are considered. A systematic procedure for complete function analysis and graph construction is presented.

Basic Properties of Integrals and Integration Methods. The concepts of an antiderivative and an indefinite integral are studied. The fundamental properties of indefinite integrals and the inverse relationship between integration and differentiation are explained. A table of basic integrals is introduced. Direct integration, substitution, and integration by parts are considered.

Integration of Simple Fractions. The principal types of simple rational fractions and their integrals are studied. Methods for integrating fractions with linear or quadratic factors in the denominator are explained. Integrals of simple fractions containing repeated factors are evaluated. The importance of these formulas in integrating more complicated rational functions is demonstrated.

Integration of Rational Functions. Methods for distinguishing between proper and improper rational fractions are studied. The polynomial part of an improper rational fraction is separated. Proper rational fractions are decomposed into partial fractions, after which the resulting integrals are evaluated. Cases involving real linear and irreducible quadratic factors in the denominator are analyzed.

Integrating Some Irrational Functions. Methods for evaluating integrals containing irrational expressions are studied. Rationalizing substitutions are used to transform irrational integrals into rational form. Algebraic and trigonometric substitutions for expressions involving square roots are considered. Euler substitutions and selected special irrational integrals are examined.

Integration of Trigonometric Functions. Integrals involving trigonometric functions and their powers are studied. Trigonometric identities are used to simplify integrands. Integrals containing powers of sine and cosine, as well as tangent and cotangent, are evaluated. The universal trigonometric substitution is applied to rational expressions involving trigonometric functions.

The Concept of the Definite Integral; Properties of the Definite Integral; Table of Integrals. The definite integral is defined as the limit of integral sums. The concepts of partition, integral sum, and integrability are explained. The linearity, additivity, monotonicity, and estimation properties of the definite integral are studied. The Newton-Leibniz formula is introduced as the principal method for evaluating definite integrals.

Calculation of the Definite Integral; Approximate Calculation of the Definite Integral. Methods for evaluating definite integrals by means of antiderivatives are studied. Substitution and integration-by-parts formulas for definite integrals are explained. The need for approximate integration when an elementary antiderivative is unavailable is discussed. The rectangle, trapezoidal, and parabolic rules are considered as methods of numerical integration.

Some Applications of the Definite Integral. Methods for calculating the areas of plane regions using definite integrals are studied. Procedures for determining the volumes and surface areas of solids of revolution are explained. Problems involving the length of a plane curve and mechanical work are considered. Applications of definite integrals in geometry, mechanics, physics, and engineering are examined.

Special Functions; Calculation of Special Functions. The concept, principal properties, and methods of calculating special functions arising in mathematical analysis are studied. Attention is given to the domains of definition, characteristic properties, functional relationships, and computational procedures associated with such functions. Methods for representing and evaluating special functions and their use in solving mathematical and applied problems are considered.

Numerical Series and Its Convergence and Divergence; Positive-Term Series and Their Convergence Criteria. The concept of a numerical series, its sequence of partial sums, and the sum of a series are studied. Convergence and divergence are defined through the limiting behaviour of partial sums.

Necessary conditions for convergence are considered. Comparison, limit comparison, ratio, root, and integral criteria are applied to determine the convergence or divergence of positive-term series.

Functional Series and Their Uniform Convergence. Functional series whose terms are functions are studied. The concepts of pointwise convergence, uniform convergence, the domain of convergence, and the sum function are introduced. The distinction between pointwise and uniform convergence is explained. Criteria for uniform convergence and the conditions under which functional series may be integrated or differentiated term by term are considered.

Convergence Domain of Power Series; Functional Properties of Power Series. The concept of a power series and its domain, interval, and radius of convergence are studied. Methods for determining the radius of convergence are explained, and convergence at the endpoints of the interval is examined separately. The continuity of the sum function and the term-by-term differentiation and integration of power series are considered.

Taylor Series; Expansion of Elementary Functions into Power Series. Taylor and Maclaurin series are studied as representations of functions by power series in a neighbourhood of a given point. The coefficients of the series are determined by means of higher-order derivatives. Methods for expanding elementary functions into power series are considered. Taylor series are also applied to approximate calculations and the estimation of approximation errors.

Preliminary Information About Fourier Series. The basic concept of representing periodic functions by trigonometric series is introduced. Fourier coefficients and methods for calculating them are considered. The representation of even and odd functions by cosine and sine series is explained. Preliminary convergence conditions and the application of Fourier series to periodic processes, oscillations, and mathematical modelling are discussed.

A Function of Several Variables and Its Limit and Continuity; Derivatives and Differentials of a Function of Two Variables. Functions depending on two or more variables, their domains, ranges, and geometric representations are studied. The concepts of the limit and continuity of a multivariable function are introduced. Partial derivatives of a function of two variables and their geometric interpretations are considered. The total differential, differentiability conditions, and approximate calculation of function increments are explained.

Higher-Order Derivatives and Differentials; Gradient. Higher-order partial derivatives, mixed derivatives, and higher-order differentials of multivariable functions are studied. Conditions for the equality of mixed partial derivatives are considered. The gradient of a scalar function and its relationship with the directional derivative are explained. The geometric meaning of the gradient as the direction of the greatest rate of increase and as a normal vector to a level surface is examined.

Taylor Formula for a Function of Two Variables. Taylor's formula for approximating a function of two variables in a neighbourhood of a given point is studied. The roles of first-order and higher-order partial derivatives are explained. The second-order Taylor formula and its remainder term are considered. The formula is applied to approximate calculations and the local analysis of multivariable functions.

Extrema of a Function. The concepts of local maximum and local minimum of a multivariable function are studied. Necessary conditions for an extremum and methods for determining stationary points are considered. Sufficient conditions involving second-order partial derivatives are explained. Methods for identifying maxima, minima, and saddle points are presented, together with the basic principles of constrained extrema.

The Concept of a Multiple Integral; Properties and Calculation of Integrals. The concept of a multiple integral is introduced through the limit of integral sums over multidimensional domains. The linearity, additivity, monotonicity, and estimation properties of

multiple integrals are studied. Methods for reducing multiple integrals to iterated integrals are explained. The selection of the order of integration and the determination of integration limits are considered.

Double Integrals over Domains That Can Be Decomposed into Simple Domains. Methods for describing planar domains that are simple with respect to one of the coordinate axes are studied. Double integrals are expressed as iterated integrals with appropriate limits of integration. More complex domains are decomposed into several simple domains. Changing the order of integration as a means of simplifying calculations is also considered.

Some Applications of the Double Integral. Double integrals are applied to calculate the area of a planar region and the volume under a surface. Their use in determining the mass of a lamina with variable density, static moments, coordinates of the centre of mass, and moments of inertia is studied. Applications involving changes of variables and polar coordinates are also considered.

A Function of Three Variables, Its Partial Derivatives and Integrals. Functions of three variables, their domains, level surfaces, and partial derivatives are studied. The total differential and higher-order partial derivatives of such functions are considered. The concept of a triple integral over a spatial domain is introduced. Methods for evaluating triple integrals as iterated integrals in Cartesian, cylindrical, and spherical coordinates are explained.

Some Applications of the Triple Integral. Triple integrals are applied to calculate the volume of spatial regions and the mass of solids with variable density. Methods for determining static moments, centres of mass, and moments of inertia are studied. The use of cylindrical and spherical coordinates for simplifying integrals over symmetric spatial domains is considered.

The Concept of First-Order Curvilinear Integrals and Their Calculation. A first-order curvilinear integral is introduced as the integral of a scalar function along a curve. Parametric representations of curves and the differential of arc length are studied.

Methods for reducing a curvilinear integral to an ordinary definite integral are explained. The dependence of the integral on the curve and its independence from the direction of traversal are considered.

Some Applications of First-Kind Curvilinear Integrals. First-kind curvilinear integrals are applied to calculate the length of a curve and the mass of a wire with variable linear density. Methods for determining the static moments, centre of mass, and moments of inertia of a curved wire are considered. Applications involving scalar fields distributed along curves are also studied.

The Concept of Second-Order Curvilinear Integrals and Their Calculation. A second-order curvilinear integral is introduced as an integral with respect to coordinate differentials along an oriented curve. The influence of the curve's direction and parametrisation on the value of the integral is explained. Methods for transforming such integrals into ordinary definite integrals are considered. The interpretation of second-order curvilinear integrals in vector fields is studied.

Some Applications of Second-Order Curvilinear Integrals. Second-order curvilinear integrals are applied to calculate the work performed by a force field along a curve and the circulation of a vector field. Conservative fields and potential functions are studied. Conditions under which an integral is independent of the path are explained. The relationship between curvilinear and double integrals through Green's formula is considered.

Surface Integrals; First-Type Surface Integrals. The concept of a surface in three-dimensional space and its parametric representation is studied. The differential element of surface area is introduced. A first-type surface integral is defined as the integral of a scalar function over a surface. Methods for reducing surface integrals to double integrals by means of projections or parametric representations are explained.

Some Applications of First-Kind Surface Integrals. First-kind surface integrals are applied to

calculate surface areas and the mass of thin shells with variable surface density. Methods for determining static moments, centres of mass, and moments of inertia are considered. The calculation of the total value of physical quantities distributed over a surface is also studied.

Surface Integrals of the Second Kind; Their Calculation. Second-kind surface integrals are studied as integrals describing the flux of a vector field through an oriented surface. The orientation of a surface, unit normal vectors, and vector surface elements are explained. Methods for reducing second-kind surface integrals to double integrals are considered. The relationship between flux integrals and triple integrals through the Ostrogradsky-Gauss theorem is introduced.

Practical lessons

Examples of the Concept of a Function. Examples involving the domain and range of a function are solved. Methods of representing functions by formulas, tables, graphs, and verbal descriptions are applied. Exercises on determining the evenness, oddness, and periodicity of functions are completed.

Examples of Simple Functions and Their Graphs.

Graphs of linear, quadratic, power, exponential, logarithmic, and trigonometric functions are constructed. Transformations involving translations, stretching, compression, and reflection of graphs are applied. The fundamental properties of functions are analyzed through their graphical representations.

The Concept of a Sequence of Numbers;

Examples of the Limit of a Sequence of Numbers.

Exercises involving the general term and initial terms of numerical sequences are completed. Limits of sequences are determined by definition and by fundamental computational methods. Convergent and divergent sequences are distinguished through representative examples.

Properties of Convergent Sequences; Examples of Calculating the Limit of Sequences of Numbers.

Problems concerning boundedness, uniqueness of limits, and arithmetic properties of convergent sequences are solved. Limits are evaluated using

algebraic transformations, simplification, rationalization, and equivalent expressions. The convergence of monotonic and bounded sequences is examined.

Existence of a Sequence Limit; Examples of Important Limits. Examples concerning the existence of limits through monotonicity and boundedness are considered. Important limits of mathematical analysis are applied in calculations. Algebraic transformations are used to reduce more complicated expressions to standard limit forms.

Definitions of a Function Limit; Examples of Functions with Infinitely Small and Infinitely Large Limits. Problems involving limits at a point, one-sided limits, and limits at infinity are solved. Infinitesimal and infinitely large functions are identified, and their relationships are analyzed through examples. Elementary applications of the Cauchy and Heine definitions of a limit are considered.

Properties of Functions with Finite Limits, and Examples of Important Limits and How to Compute Them. Rules for evaluating the limits of sums, differences, products, and quotients of functions are applied. Important limits are used to evaluate trigonometric, exponential, and logarithmic expressions. Indeterminate forms are resolved through simplification, factorization, and rationalization.

Function Continuity; Operations on Continuous Functions; Examples of Function Discontinuities and Their Types. The continuity of functions at points and on intervals is examined. One-sided limits are used to identify points of discontinuity. Removable, jump, and infinite discontinuities are distinguished, and the continuity of piecewise-defined functions is analyzed.

The Concept of a Function Derivative; Examples of Derivative Applications in Geometric, Physical, and Mechanical Contexts. Derivatives are calculated from the definition, and equations of tangent lines are determined. Problems involving instantaneous velocity, acceleration, and other rates of change are solved. The geometric, physical, and

mechanical interpretations of the derivative are reinforced through practical examples.

Examples of Calculating Derivatives of Inverse and Composite Functions. Composite functions are differentiated using the chain rule. Derivatives of inverse and inverse trigonometric functions are calculated. Expressions consisting of several nested functions are differentiated through a systematic step-by-step procedure.

Functional Differential; Examples of Approximate Formulas. Differentials of functions are calculated and applied to approximate changes in function values. Numerical expressions and function values are estimated by means of differentials. Examples involving absolute and relative approximation errors are considered.

Examples of Higher-Order Derivatives and Differentials. Second-, third-, and higher-order derivatives are calculated. Higher-order derivatives of products, quotients, composite functions, and parametrically defined functions are determined. Higher-order differentials and their applications to physical problems are considered.

Properties of Differentiable Functions; Examples of the Taylor Formula. Exercises involving the conditions of Rolle's, Lagrange's, and Cauchy's theorems are completed. Functions are approximated by polynomials using Taylor and Maclaurin formulas. Approximation errors are estimated by means of the remainder term.

Examples of Determining the Increasing, Decreasing, and Extreme Values of Functions Using Derivatives. Critical and stationary points of functions are identified. Intervals of increase and decrease are determined from the sign of the derivative. First- and second-derivative tests are used to determine local maxima and minima, and practical optimization problems are solved.

Examples of Convexity, Concavity, Inflection Points, and Asymptotes of a Function Graph. Intervals of convexity and concavity are determined using the second derivative. Inflection points and vertical, horizontal, and oblique asymptotes are

calculated. Functions are fully analyzed and their graphs are constructed using domains, limits, derivatives, extrema, and asymptotes.

Examples of the Basic Properties of an Integral and Integration Methods. Antiderivatives are determined and indefinite integrals are evaluated. The linearity of integration, the table of basic integrals, direct integration, substitution, and integration by parts are applied. Obtained results are verified through differentiation.

Examples of Integrating Simple Fractions. Integrals of simple fractions with linear or irreducible quadratic factors in their denominators are evaluated. Methods for integrating fractions containing repeated factors are applied. Integrals leading to logarithmic and inverse trigonometric functions are solved.

Examples of Integrating Rational Functions. Polynomial parts are separated from improper rational fractions, and proper rational fractions are decomposed into partial fractions. Unknown coefficients are determined according to the linear and quadratic factors of the denominator. The resulting elementary integrals are evaluated and verified.

Integration of Some Irrational Functions; Examples of Integrating Trigonometric Functions. Integrals containing radicals are evaluated using rationalizing algebraic and trigonometric substitutions. Integrals involving different powers of sine, cosine, tangent, and cotangent are calculated. Trigonometric identities and the universal trigonometric substitution are applied.

The Concept of the Definite Integral; Examples of the Properties of the Definite Integral. Examples involving integral sums and the geometric interpretation of the definite integral are considered. The linearity, additivity, monotonicity, and estimation properties of definite integrals are applied. Elementary definite integrals are evaluated using the Newton-Leibniz formula.

Calculation of the Definite Integral; Examples of Approximating the Definite Integral. Definite integrals are evaluated using substitution and

integration by parts. Approximate integration is performed using rectangle, trapezoidal, and Simpson's formulas. Exact and approximate results are compared, and computational errors are analyzed.

Examples of Some Applications of the Definite Integral. Definite integrals are used to calculate areas of plane regions, volumes of solids of revolution, and lengths of curves. Practical problems involving work performed by a variable force, mass, static moments, and centres of mass are solved. Methods of definite integration are applied to problems in geometry, physics, and engineering.

Improper Integrals; Examples of Calculating Improper Integrals. Exercises involving integrals defined through limiting processes are completed. Integrals with infinite limits of integration and integrands that become unbounded at particular points are evaluated. Their convergence or divergence is determined by applying appropriate definitions, comparison methods, and limit-based calculation procedures.

Numerical Series and Its Convergence and Divergence; Examples of Positive-Term Series and Their Convergence Criteria. Numerical series, their partial sums, and the concepts of convergence and divergence are examined through practical examples. The necessary condition for convergence is verified, and geometric, harmonic, and other standard series are analyzed. Comparison, limit comparison, ratio, root, and integral tests are applied to positive-term series.

Examples of Functional Series and Their Uniform Convergence. Functional series are investigated with respect to pointwise and uniform convergence. Their domains of convergence and sum functions are determined. The Weierstrass M-test and other appropriate criteria are applied to verify uniform convergence. Examples involving term-by-term integration and differentiation are also considered.

Examples of the Convergence Domain of Power Series. The radius and interval of convergence of power series are determined using the ratio test, root

test, and the Cauchy-Hadamard formula. Convergence at the endpoints is examined separately. Exercises also involve identifying the sum function and performing term-by-term differentiation and integration within the interval of convergence.

Taylor Series; Examples of Expanding Elementary Functions in Power Series. Taylor and Maclaurin series are constructed for elementary functions. Exponential, logarithmic, trigonometric, and algebraic functions are expanded into power series around specified points. Series are applied to approximate function values, and approximation errors are estimated using the remainder term.

Examples of Elementary Information About Fourier Series. Fourier coefficients are calculated for periodic functions, and the corresponding trigonometric series are constructed. Even and odd functions are represented by cosine and sine series, respectively. Basic convergence conditions are examined, and simple periodic functions are expanded into Fourier series.

Examples of a Multivariable Function and Its Limits and Continuity. Domains and ranges of functions of two or more variables are determined, and their graphs or level curves are interpreted. Limits are evaluated along different paths to investigate their existence. Continuity at a point and over a domain is tested using the definitions and properties of multivariable functions.

Examples of Derivatives and Differentials of a Function of Two Variables. First-order partial derivatives and total differentials of functions of two variables are calculated. Differentiability conditions and the geometric meanings of partial derivatives are examined. Total differentials are applied to approximate changes in function values and to estimate computational errors.

Examples of Higher-Order Derivatives and Differentials. Second- and higher-order partial derivatives, including mixed derivatives, are calculated. Conditions for the equality of mixed partial derivatives are verified. Higher-order total

differentials, directional derivatives, and gradients are determined for various multivariable functions.

Taylor Formula for a Function of Two Variables; Examples of Function Extremums. Taylor's formula is applied to approximate functions of two variables in a neighbourhood of a given point. First- and second-order partial derivatives are used to construct Taylor polynomials. Stationary points are determined, and local maxima, local minima, and saddle points are classified using second-order derivative tests.

The Concept of a Multiple Integral; Properties of the Integral and Examples of Calculation. Multiple integrals are constructed from integral sums and evaluated as iterated integrals. Their linearity, additivity, monotonicity, and estimation properties are applied in problem-solving. Appropriate limits of integration and the order of integration are determined for two- and three-dimensional domains.

Examples of Double Integrals over a Set That Is a Union of Simple Sets. Complex planar domains are decomposed into regions that are simple with respect to one of the coordinate axes. Double integrals are expressed as sums of iterated integrals over these regions. Integration limits are determined graphically, and the order of integration is changed when this simplifies the calculation.

Examples of Some Applications of the Double Integral. Double integrals are used to calculate the areas of planar regions, volumes under surfaces, and masses of laminas with variable density. Static moments, centres of mass, and moments of inertia are determined. Problems requiring transformation to polar coordinates are also solved.

Examples of a Function of Three Variables, Its Partial Derivatives, and Its Integrals. Domains, level surfaces, partial derivatives, and total differentials of functions of three variables are examined. Triple integrals are formulated and evaluated as iterated integrals. Cartesian, cylindrical, and spherical coordinates are used according to the geometry of the integration domain.

Examples of Some Applications of the Triple Integral. Triple integrals are applied to determine the

volumes and masses of three-dimensional bodies. Static moments, coordinates of the centre of mass, and moments of inertia are calculated. Cylindrical and spherical coordinate transformations are used to simplify integrals over symmetric spatial regions.

Examples of the Concept of First-Kind Curvilinear Integrals and Their Calculation. Curves are represented parametrically, and their arc-length elements are calculated. First-kind curvilinear integrals are transformed into ordinary definite integrals and evaluated. The independence of these integrals from the orientation of the curve is verified through examples.

Examples of Some Applications of First-Kind Curvilinear Integrals. First-kind curvilinear integrals are applied to calculate curve lengths and the masses of wires with variable linear density. Static moments, centres of mass, and moments of inertia of curved wires are determined. Scalar quantities distributed along curves are also evaluated.

The Concept of Second-Order Curvilinear Integrals and Examples of Their Calculation. Second-order curvilinear integrals are formulated along oriented curves and converted into ordinary definite integrals by parametrisation. The influence of curve orientation on the sign of the integral is examined. Integrals of vector fields in the plane and in three-dimensional space are calculated.

Examples of Some Applications of Second-Order Curvilinear Integrals. Second-order curvilinear integrals are used to calculate the work performed by force fields and the circulation of vector fields. Conservative fields and potential functions are identified. Path independence is verified, and Green's theorem is applied to transform closed line integrals into double integrals.

Surface Integrals; Examples of First-Type Surface Integrals. Surfaces are described explicitly or parametrically, and their surface-area elements are determined. First-type surface integrals are transformed into double integrals over suitable projections. Integrals over planes, graphs of

functions, and parametrically defined surfaces are evaluated.

Examples of Some Applications of First-Kind Surface Integrals. First-kind surface integrals are applied to calculate surface areas and the masses of thin shells with variable surface density. Static moments, centres of mass, and moments of inertia are determined. Problems involving scalar quantities distributed over surfaces are also solved.

Examples of Second-Kind Surface Integrals. Oriented surfaces, unit normal vectors, and vector surface elements are determined. Second-kind surface integrals are formulated as flux integrals and reduced to double integrals. The effect of changing the orientation of a surface on the value of the integral is examined.

Application of First and Second Kind Surface Integrals. First-kind surface integrals are applied to scalar quantities distributed over surfaces, while second-kind surface integrals are used to calculate vector-field flux. Problems involving surface area, shell mass, fluid flow, and electromagnetic flux are solved. The Ostrogradsky-Gauss theorem is applied to relate flux through a closed surface to a triple integral over the enclosed region.

Self-study

Examples on the Concept of a Function.

Examples of Simple Functions and Their Graphs.

The Concept of a Sequence of Numbers; Examples of the Limit of a Sequence of Numbers.

Properties of Convergent Sequences; Examples of Calculating the Limit of Sequences of Numbers.

Existence of a Sequence Limit; Examples of Important Limits.

Definitions of Function Limits; Examples of Functions with Infinite Decreases and Infinite Increases.

Properties of Functions with Finite Limits and Examples of Important Limits and How to Compute the Limit of a Function.

Function Continuity; Operations on Continuous Functions; Examples of Function Discontinuities and Their Types.

	The Concept of a Function Derivative; Examples of the Derivative's Geometric and Mechanical Meanings. Examples of Calculating Derivatives of Inverse and Composite Functions. Differentiation of a Function; Examples of Approximate Formulas. Examples of Higher-Order Derivatives and Differentials. Properties of Differentiable Functions; Examples of the Taylor Formula. Examples of Determining the Increasing, Decreasing, and Extreme Values of Functions Using Derivatives. Examples of Function Graph Convexity, Concavity, Inflection Point, and Asymptotes. Indefinite Integral; Examples of the Basic Properties of Integrals and Methods of Integration. Examples of Integrating Simple Fractions. Examples of Integrating Rational Functions. Integration of Some Irrational Functions; Examples of Integrating Trigonometric Functions. The Concept of the Definite Integral; Examples of Properties of the Definite Integral. Definite Integral Calculation; Examples of Approximating the Definite Integral. Examples of Some Applications of the Definite Integral. Improper Integrals; Examples of Calculating Improper Integrals. Numerical Series and Its Convergence and Divergence; Examples of Positive-Term Series and Their Convergence Criteria. Examples of Functional Series and Their Uniform Convergence. Examples of the Convergence Domain of Power Series. Taylor Series; Examples of Expanding Elementary Functions into Power Series. Examples on the Basics of Fourier Series. Examples of a Multivariable Function and Its Limits and Continuity. Examples of Derivatives and Differentials of a Function of Two Variables.
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	Examples of Higher-Order Derivatives and Differentials. Taylor Formula for a Function of Two Variables; Examples of Extremums of a Function. The Concept of the Multiple Integral; Properties of Integrals and Examples of Calculations. Examples of Double Integrals over Sets That Can Be Decomposed into Simple Sets. Examples of Some Applications of the Double Integral. Examples of a Three-Variable Function, Its Partial Derivatives, and Its Integrals. Examples of Some Applications of the Triple Integral. Examples of First-Kind Curvilinear Integrals and the Concept of Their Calculation. Examples of Some Applications of First-Kind Curvilinear Integrals. Examples of the Concept of Second-Order Curvilinear Integrals and Their Calculation. Examples of Some Applications of Second-Order Curvilinear Integrals. Surface Integrals; Examples of First-Type Surface Integrals. Examples of Some Applications of First-Kind Surface Integrals. Examples of Second-Kind Surface Integrals. Application of First and Second Kind Surface Integrals.
Teaching Methods	The module is delivered through lectures, practical problem-solving sessions, inquiry-based and problem-based learning, mathematical modelling, computational activities, group discussions, presentations, and project-based assignments. Students solve and analyze problems involving functions, limits, derivatives, integrals, series, and multivariable functions using appropriate mathematical methods and modern mathematical software. Teacher-guided and student independent learning, together with scientific and methodological literature, are used to support learning. Digital educational resources further develop students' logical reasoning, critical thinking, computational,

	analytical, research, and mathematical modelling skills.
Prerequisites	Linear Algebra Probability and Statistics Differential Equations
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and final assessment (50%). The continuous assessment evaluates students' active participation in practical sessions and the successful completion of Teacher-Guided Independent Work. It includes mathematical problem-solving activities, computational assignments, independent tasks, analytical work, presentations, reports, and practical projects. A maximum of 8 points is awarded for participation in practical sessions, while up to 12 points is awarded for independent learning assignments. The midterm assessment consists of a theoretical assessment worth 12 points and independent work worth 18 points. The theoretical component may be conducted as a test or a written examination. When conducted in written form, it includes three theoretical questions and two practical problems. The independent work component consists of individual assignments prepared in the form of reports, analytical papers, projects, presentations, or computational tasks and submitted through the HEMIS system by the specified deadline. The final assessment may be conducted as a written examination or a test. The written examination consists of five questions selected from lecture, practical, and independent study topics, with each complete and correct answer worth 10 points. When conducted as a test, it consists of 25 questions, with each correct answer worth 2 points. The final assessment evaluates students' theoretical knowledge, mathematical problem-solving skills, computational accuracy, analytical abilities, and capacity to apply mathematical analysis methods to practical, engineering, and information technology problems. Successful and timely completion of all

	independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Alimov.Sh.O., Ashurov.R.R. Mathematical Analysis 1,2,3 vols. T. "Mumtoz so'z", 2018. 2. Fichtenholz G. M. Course of Differential and Integral Calculus, Vol. 1, 2, 3. Moscow: Fizmatlit, 2007. 3. Azlarov T. A., Mansurov X. T. Mathematical Analysis, Vol. 1, 2. Tashkent: "O'qituvchi", 1994, 1995. 4. Demidovich B. P. Collection of Problems in Mathematical Analysis. Moscow: Nauka, 1997. 5. Y. Soatov, "Higher Mathematics," Parts 1-2-3. Tashkent, 1994. 6. N. V. Bogomolov, "Practical Exercises in Mathematics," Tashkent – "O'qituvchi," 1984. 7. Tojiev Sh.I. Solving Problems in the Fundamentals of Higher Mathematics. T.: "Uzbekiston." 2002.
Responsible Lecturer	Amirkulov Chori J., Lecturer, Department of Mathematical Modeling and Computer Science, Termez State University.

PHYSICS 1,2

Field	Description
Module Title	PHYSICS 1,2
Module Code	FIZ1110
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 120 hours
Frequency/Duration	I semester
Module objective	The objective of the Module Physics 1,2 is to form fundamental concepts of classical and modern physics in students. These concepts are further internalized so that students can apply them consistently. The Module develops students' skills in applying physical phenomena and laws in their professional activities. It aims to develop scientific

	thinking in students. It further aims to lay a fundamental theoretical foundation for subsequent study. The Module also intends to develop the skills and competencies needed to apply the acquired knowledge to non-standard problems in practice.
Module Learning Outcomes	Knowledge: LO1. Will have knowledge of the fundamental laws of general physics and their formulas. LO2. Will have knowledge of the meaning, content, laws, and analysis of graphs of physical processes in general physics. LO3. Will have knowledge of the meaning of physical quantities, their units, and how to compare them. LO4. Will have knowledge of applying fundamental physical laws and processes to ecological phenomena. Skills: LO5. Understands the meaning of laws and formulas related to general physics. LO6. Knows how to apply general physics laws and formulas to non-standard problems. LO7. Applies physical laws and formulas to laboratory work and knows how to draw conclusions. LO8. Knows how to express and compare physical quantities in the SI and CGS systems of measurement. Competencies: LO9. Demonstrates the ability to identify, analyze, and solve fundamental physics problems using scientific principles, mathematical models, and logical reasoning. LO10. Performs physics experiments safely and accurately, collects and analyzes experimental data, evaluates measurement errors, and draws scientifically valid conclusions. LO11. Applies quantitative methods, mathematical equations, unit conversions (SI and CGS), and graphical analysis to interpret physical phenomena and solve practical problems. LO12. Applies the principles of mechanics, thermodynamics, electricity, and other areas of

	physics to engineering, environmental, and real-world technological applications. LO13. Communicates scientific ideas, experimental results, and technical information effectively through written reports, oral presentations, and collaborative teamwork while demonstrating ethical and professional responsibility.
Content	Lectures The Science of Physics and Its Objectives; Motion, Space, Time. The relationship of physics to other sciences is examined, together with the method of physical research. The system of physical quantities and units is studied. Forms of motion, systems of spatial coordinates, and properties of time are considered. Circular Motion of a Particle. Tangential, normal, and total accelerations in curvilinear motion are examined. The angle of rotation, angular velocity, and angular acceleration are studied. Relations between the kinematics of rotational and translational motion are considered. Dynamics of a Material Point. Inertial reference frames are examined, together with force, mass, and momentum. Newton's laws and the equations of motion are studied. Forces in nature and the concept of the centre of mass are considered. The law of conservation of momentum is examined. Angular Motion of a Rigid Body. The moment of inertia, moment of a force, and angular momentum are examined. The fundamental law of rotational dynamics is studied. The law of conservation of angular momentum is considered. Law of Conservation of Mechanical Energy. Mechanical work and power are examined, together with kinetic and potential energy. The relation of energy to work and force is studied. The law of conservation of energy in mechanics is considered. Inertial and Non-Inertial Reference Frames; Relativistic Mechanics. Inertial force is examined, together with the classical and relativistic principles of relativity. Galilean and Lorentz transformations are

studied. Relativistic transformations of physical quantities are considered.

Molecular Physics. Statistical and thermodynamic methods are examined, together with isotropic processes and the ideal gas law. Internal energy and the fundamental equation of the Generalized Law of Thermodynamics (GLT) are studied. The Maxwell distribution and the barometric formula are considered. Transition phenomena are examined.

Thermodynamics. The distribution of energy by degrees of freedom is examined. The first law of thermodynamics and its application to isothermal processes are studied. Reversible and irreversible processes, cycles, and the second law are considered. Efficiency and entropy are examined.

Electric Interaction. Electric charge and its conservation law are examined, together with Coulomb's law. The electric field and its intensity are studied, along with the induction vector and current. The Ostrogradsky–Gauss theorem and its applications are considered.

Work Done in Moving a Charge in an Electric Field. Work done in a uniform field is examined, together with the circulation of the electric field strength vector. The field potential and the energy of charges are studied. The relationship between electric field strength and potential is considered.

Dielectrics and Conductors in an Electric Field. Electron and dipole polarizations are examined, together with the polarization vector. Gauss's law is studied in relation to dielectrics. Piezoelectrics and the piezoelectric effect are considered. Conductive capacitance, capacitors, and field energy density are examined.

Electric Current. The condition for current generation, current strength, and current density are examined, together with EMF. The integral and differential forms of Ohm's and Joule-Lenz's laws are studied. Ohm's law for a closed circuit and Kirchhoff's rules are considered. Thermoelectric phenomena are examined.

Magnetic Field and the Magnetic Properties of Matter. The magnetic induction vector and current

loop are examined, together with methods for calculating magnetic fields. The Lorentz force and the Lorentz and Ampère laws are studied. The Hall effect is considered.

Biot-Savart-Laplace Law. Current loops and magnetic flux are examined, together with the circulation of the induction vector. Toroids and solenoids are studied. Molecular currents, magnetization, and para- and ferromagnetics are considered. Gauss's theorem is examined in this context.

The Phenomenon of Electromagnetic Induction. Induction EMF is examined, together with Faraday's and Lenz's laws. Alternating current and magnetic flux are studied. Inductance, self-induction, and mutual induction are considered. Magnetic field energy and Maxwell's equations are examined.

Practical lessons

Kinematics of Translational Motion. Problems related to the kinematics of translational motion are solved. Displacement, velocity, and acceleration are calculated for bodies moving along straight and curved paths. The solutions reinforce the theoretical concepts of motion, space, and time introduced in the lectures.

Kinematics of Rotational Motion of a Rigid Body. Problems on the kinematics of rotational motion are solved. Tangential, normal, and total accelerations in curvilinear motion are calculated, together with the angle of rotation, angular velocity, and angular acceleration. The relations between the kinematics of rotational and translational motion are applied to practical examples.

Dynamics of Translational Motion. Problems based on Newton's laws are solved. Force, mass, and momentum are applied to bodies in inertial reference frames. The equations of motion and the law of conservation of momentum are used to analyse practical cases.

Dynamics of Rigid Body Rotational Motion. Problems on the moment of inertia, the moment of a force, and angular momentum are solved. The fundamental law of rotational dynamics is applied to

practical examples. The law of conservation of angular momentum is used to analyse the results.

Laws of Conservation of Momentum, Angular Momentum, and Energy. Problems related to the kinetic and potential energies of a mechanical system are solved. Mechanical work and power are calculated in relation to these energies. The law of conservation of energy in mechanics is applied to verify the results.

Ideal Gas Laws. The Main Equation of the MKN. Problems concerning the distribution of gas molecules by velocity and energy are solved. The ideal gas law and the fundamental equation of the kinetic theory of gases are applied. The Maxwell distribution is used to analyse the results.

Work Done by a Gas. Heat Capacity. Problems related to the work done by a gas and its heat capacity are solved. Internal energy and the fundamental equation of the Generalized Law of Thermodynamics are applied. The results are analysed in relation to isothermal processes.

First Law of Thermodynamics. Entropy. The first law of thermodynamics is applied to isochoric and adiabatic processes. Problems related to entropy are solved. Reversible and irreversible processes, together with the efficiency of thermodynamic cycles, are analysed.

Electrostatic Field in a Vacuum. Problems on the electric field strength and potential are solved. Coulomb's law and the concept of electric charge are applied to practical examples. The Ostrogradsky–Gauss theorem is used to calculate the fields of symmetric charge distributions.

Dielectrics and Conductors in an Electric Field. Problems on the properties of dielectrics and conductors are solved. Electron and dipole polarizations and the polarization vector are applied to practical examples. Gauss's law is used to analyse the resulting fields.

Electrical Capacitance. Condensers. Problems related to the capacitance of condensers are solved. Conductive capacitance and field energy density are calculated for practical configurations. The

piezoelectric effect is discussed in relation to selected dielectric materials.

Laws of Direct Current. Kirchhoff's Rules. Circuits are analysed on the basis of Ohm's and Kirchhoff's laws. Current strength, current density, and EMF are calculated for direct-current circuits. Ohm's law for a closed circuit is applied to solve practical problems.

Magnetic Interactions. Magnetic Field in a Vacuum. Problems on calculating the simplest magnetic fields are solved. The magnetic induction vector and current loop are applied to practical examples. The Lorentz force and the Lorentz and Ampère laws are used to analyse the results.

Electromagnetic Induction. Magnetic Field of Matter. Problems on induced EMF are solved using Faraday's and Lenz's laws. The magnetic properties of matter, including magnetization and para- and ferromagnetics, are analysed. Toroids and solenoids are used as practical examples of magnetic field calculation.

Magnetic Field Energy. Problems related to magnetic field energy are solved. Inductance, self-induction, and mutual induction are applied to practical examples. Maxwell's equations are used to verify the results.

Self-study

Dynamics of Rigid Body Rotational Motion. Gyroscopes.

Motion of Bodies in the Earth's Gravitational Field. Escape Velocities.

Problem Solving in Mechanics (Kinematics and Dynamics).

Harmonic Oscillator; Damped and Forced Oscillations; Resonance.

Solving Problems on the Laws of Conservation.

Lorentz Transformations and Their Resulting Conclusions.

Transport Phenomena in Non-Equilibrium Systems. Diffusion.

Problem Solving in Molecular Physics and Thermodynamics.

Atmospheric Heat Balance and the Greenhouse Effect.

	Electric Field in Dielectrics; Piezo- and Ferroelectrics. Solving Problems in Electrostatics. Solving Problems on DC and Kirchhoff's Laws. Semiconductors and Superconductivity; Unconventional Energy Sources. Alternating Current; Resonant Circuit; Transformers. Solving Problems on Magnetic Fields and Electromagnetic Induction.
Teaching Methods	Knowledge of the fundamentals of classical mechanics, including kinematics, dynamics, and the laws of conservation, is developed through lectures and problem-solving sessions. The principles of molecular physics and thermodynamics are taught using the same approach. Electrostatics, electric current, and the physics of dielectrics and conductors are taught through lectures and laboratory-based practical work. Knowledge of magnetic fields, electromagnetic induction, and Maxwell's equations is developed through lectures and practical sessions. Skills in expressing and comparing physical quantities in the SI and CGS systems are developed through practical exercises. Competencies for applying physical laws and formulas to non-standard problems and laboratory work are developed throughout the module.
Prerequisites	Linear Algebra Differential Equations Calculus 1,2
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, attendance, laboratory performance, problem-solving exercises, seminar presentations, and the successful completion of independent study assignments. These assignments include laboratory reports, numerical problem sets, experiments on mechanics, thermodynamics, electricity and magnetism, data analysis, and short presentations on physical

	phenomena and their real-world applications. The midterm assessment consists of written assignments, short in-class quizzes, laboratory evaluations, and the assessment of independent study projects. Typical projects include investigations of oscillatory motion, energy conservation experiments, reports on electromagnetic phenomena, presentations on modern physics applications, and case studies related to renewable energy technologies. The final assessment is a written examination consisting of theoretical questions, numerical calculations, and problem-solving tasks designed to assess students' understanding of the fundamental principles of physics, analytical and quantitative reasoning, and their ability to apply physical laws to scientific and engineering problems. Successful completion of all laboratory work and independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Savelev I.V. Course of General Physics. Vol. 1, Vol. 2, Vol. 3. — M.: Astrel, 2011. 2. Sivuxin D.V. Course of General Physics. — Moscow: Fizmatlit, 2005. 3. Qosimov A.S. General Physics (textbook). — Termiz: IFRON-PRINT, 2023. 4. Qosimov A.S., Jo'rayev Sh., Qo'yliyev B. Laboratory Work in General Physics. — Tashkent, 2019, 309 p. 5. Sultonova O'.N., Begmatova D.A. et al. Collection of Problems in General Physics. — Termez: Ifron Print, 2022. 6. Tursunov I.G., Begmatov D.A. Physics (a teaching and methodological guide). — Tashkent: Excellent Polygraphy, 2020. 7. Qodirov O. Course of Physics. Part 1 (Mechanics, Molecular Physics). — Tashkent: Fan va Texnologiya, 2005. 8. Polvonov S., Daliyev X., Bozorov E., Polvonova G. Collection of Problems from General Physics. — Tashkent, 2020. 9. Ismailov M., Habibullayev P., Xaliulin M. Course in Physics. — Tashkent: O'zbekiston, 2000.

	10. Qosimov A.S. General Physics (textbook). — Termez: TerDU NMM, 2022. 11. Douglas C. Giancoli. Physics: Principles with Applications. — Pearson, 2014. 12. David Halliday, Robert Resnick, Jearl Walker. Fundamentals of Physics. — John Wiley & Sons, 2014.
Responsible Lecturer	Tursunov Alisher Isoqovich, PhD in Pedagogical Sciences, Associate Professor.

DIFFERENTIAL EQUATIONS

Field	Description
Module Title	DIFFERENTIAL EQUATIONS
Module Code	DIT1406
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 90 hours (44 hours Lecture, 46 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Differential Equations is to familiarize students with differential equations theory. Students are introduced to the principal methods used to solve differential equations. The Module develops practical skills in solving various types of differential equations. It further enables the application of mathematical models to computer engineering problems. The Module also enables the application of these mathematical models to real-world problems.
Module Learning Outcomes	Knowledge: LO1. Understand essential concepts of differential equations: order, degree, classifications, and terminology. LO2. Know the theoretical foundations: existence and uniqueness theorems, initial value problems, boundary conditions. LO3. Master solution methods for first-order equations: separable, linear, exact, homogeneous equations.

	LO4. Understand higher-order equations: characteristic equations, fundamental solutions, linear independence. Skills: LO5. Apply separation of variables and other techniques to solve first-order differential equations. LO6. Use variation of constants and undetermined coefficients methods for non-homogeneous equations. LO7. Analyze higher-order equations using characteristic equation method and reduction of order. LO8. Solve systems of differential equations and interpret solutions geometrically. Competencies: LO9. Solve practical problems involving first-order differential equations in engineering contexts. LO10. Solve and analyze higher-order differential equations with various boundary conditions. LO11. Model real-world processes (vibrations, circuits, control systems) using differential equations. LO12. Use computational tools (Maple, MATLAB) for numerical solutions and visualization. LO13. Conduct computational experiments and interpret results in context of applied mathematics.
Content	Lectures Introduction to Differential Equations. Differential equations are defined and classified, with reference to physical problems that lead to differential equations, including motion, heat transfer, population dynamics, and electrical circuits. The order and degree of an equation are introduced, together with the distinction between ordinary and partial differential equations. Homogeneous and non-homogeneous equations are distinguished. The importance of initial conditions in determining unique solutions is explained. Separable Differential Equations. Equations in which the variables can be separated are studied, together with the method of separation of variables. Integration procedures are applied to both sides of the equation. Applications to exponential growth and decay, Newton's cooling law, and chemical reactions are considered, and solutions are verified. First-order

separable equations are examined in the context of engineering problems.

Homogeneous Differential Equations.

Homogeneous equations are recognised through testing, based on their scaling properties and symmetry. The substitution method for transforming such equations into separable form is studied, together with the associated solution procedures. The geometric interpretation of solutions and solution families is considered. Real-world examples from physics and engineering are examined.

Equations Reducible to Homogeneous Form.

Coordinate transformation techniques are studied, including finding intersection points and shifting the origin. Methods for converting non-homogeneous equations to homogeneous form are examined, together with systematic solution procedures. Results are interpreted in the original coordinates. Applications in various engineering contexts are considered.

Linear First-Order Differential Equations. The standard form and structure of linear first-order differential equations are studied, together with the integrating factor method. Multiplication by an exponential factor is used to find particular solutions. Bernoulli equations and other special cases are examined. Applications to electrical circuits, mechanical systems, and chemical processes are considered.

Exact (Complete) Differential Equations. The condition for exactness is studied, together with methods for testing equations for exact form. Integrating factors for non-exact equations are examined, and exact equations are found through these methods. The potential function and its integration are considered. Applications to thermodynamics and energy conservation are examined.

Cauchy Problem and Existence-Uniqueness Theorem. Initial value problems and Cauchy conditions are studied, together with the statement of the existence and uniqueness theorem. The Picard-Lindelöf theorem and the method of successive

approximations are examined. The domain of validity of the theorem is considered. The importance of the theorem in practical applications where unique solutions are essential is explained.

Higher-Order Differential Equations. Higher-order differential equations are defined and classified by order. Initial and boundary conditions are studied, noting that the number of required conditions equals the order of the equation. Physical systems that produce higher-order equations are examined. Mechanical vibrations, electrical oscillations, and control systems are considered as examples.

Linear Higher-Order Differential Equations. The general theory and structure of linear higher-order differential equations are studied, covering both homogeneous and non-homogeneous cases. The superposition principle and the concept of linear independence are examined. The properties of the solution space are considered. Fundamental systems of solutions are studied.

Constant Coefficient Homogeneous Equations. Constant-coefficient homogeneous equations are solved using the characteristic equation method, including procedures for finding the characteristic roots. Real roots are shown to produce exponential solutions, while complex roots produce oscillatory solutions. Repeated roots and the corresponding polynomial factors are examined. Complete solution procedures are presented.

Reduction of Order Methods. Methods for lowering the order of an equation through substitution are studied. Known solutions are used to find additional solutions, including techniques for equations with a missing variable. Reduction methods for specific equation structures are examined, converting them into more manageable forms. Homogeneous equations with special structure are considered.

Variable Coefficient Equations. Special substitutions for variable-coefficient equations are studied. Euler equations and the logarithmic transformation used to solve them are examined. Solvable patterns among variable-coefficient

equations are recognised. Power series approaches for special cases are considered.

Linear Independence of Functions. The linear independence of functions is defined, together with methods for testing it. The Wronskian determinant is used to establish independence. Fundamental sets of solutions and the dimension of solution spaces are examined. The completeness of solution sets is considered.

Wronskian Determinant and Fundamental Solutions. The Wronskian determinant is defined and its calculation is studied. A non-zero Wronskian is shown to indicate the independence of solutions. The Ostrogradsky-Liouville formula and the relationship between Wronskian values at different points are examined. The application of the Wronskian to the verification of solutions is considered.

Non-Homogeneous Linear Equations. The general solution structure of non-homogeneous linear equations is studied, consisting of a homogeneous solution plus a particular solution. The variation of constants method is examined. The method of undetermined coefficients is studied for special right-hand sides. Complete solution procedures are presented.

Undetermined Coefficients Method. The method of undetermined coefficients begins with guessing the form of the particular solution and solving for its coefficients. Special cases and the modifications they require are examined. Polynomial and exponential right-hand sides are considered. Trigonometric forcing functions are also studied.

Variation of Constants (Lagrange Method). The variation of constants method is studied as a universal technique applicable to any right-hand side. Homogeneous solutions are used as the basis of the method, together with the associated integration procedures. Applications of the Wronskian within this method are examined. A systematic solution process is presented.

Higher-Order Constant Coefficient Equations. A complete systematic procedure for solving higher-order constant-coefficient equations is studied, based

on the characteristic roots and homogeneous solutions. Methods for finding the particular solution are examined. Resonance cases are considered as a special situation. Engineering applications and interpretations of the results are discussed.

Euler Equations (Equidimensional Equations).

Euler equations are studied as a special variable-coefficient form. The logarithmic substitution used to transform them into constant-coefficient form is examined. The interpretation of the resulting solutions is considered. Applications in boundary value problems are discussed.

Systems of First-Order Differential Equations. The conversion of higher-order equations into systems of first-order equations is studied, using the derivative substitution method. The matrix form representation of such systems is examined. Equilibrium points and stability are considered. Phase space concepts are introduced.

Eigenvalue Method for Linear Systems. The characteristic polynomial and eigenvalues of a linear system are studied, together with methods for calculating the corresponding eigenvectors. Solutions are constructed from the resulting eigenpairs. Real and complex eigenvalues are distinguished. Systematic procedures for applying the method are presented.

Inhomogeneous Linear Systems. Inhomogeneous linear systems with non-zero right-hand sides are studied, together with methods for finding a particular solution. The particular solution is combined with the homogeneous solution to form the complete solution. Real applications, including coupled oscillators, networks, and control systems, are considered. Complete solution procedures are presented.

Practical lessons

Basic Problem-Solving Techniques. Simple differential equations are solved using basic problem-solving techniques. The solutions are verified and represented graphically. Different solution methods are compared. The underlying structure of the solutions is examined.

Separable Equations Practice. Separable equations of increasing difficulty are solved in practice sessions. Solution curves are plotted for the resulting equations. Variations of the solution method are applied and computational verification is carried out. The accuracy of the solutions is analysed.

Homogeneous Equations Application. Problems on homogeneous equations are solved using the substitution method. Homogeneous equations are recognised among given problems. Graphical solutions are constructed and equilibrium points are analysed. The results are interpreted in the phase plane.

Reduction to Homogeneous Form. Equations that can be reduced to homogeneous form are identified. Coordinate transformations are applied, together with the corresponding solution procedures. The solutions are back-transformed into the original variables. Engineering applications of these methods are examined.

Linear Equations and Integrating Factors. Integrating factors for linear first-order equations are found and applied. The resulting solutions are verified. Circuit problems and mechanical systems are solved using this method. The physical interpretation of the results is discussed.

Bernoulli and Special Equations. Bernoulli and other special equations are recognised among given problems. Substitution techniques are applied to solve them. The applicability of the method to different equation types is examined. Real-world problems are solved using these techniques.

Exact Equations and Testing. Equations are tested for exactness, and integrating factors are found where required. Exact equations are solved using the appropriate procedures. The solutions are verified. Applications involving conservation principles are examined.

Initial Value Problems. Cauchy problems are solved by applying the given initial conditions. Solution paths are visualised graphically. Phase plane analysis is used to interpret the solutions. The uniqueness of the solutions is verified.

Implicit and Parametric Solutions. Non-explicit derivative forms are examined in practical problems. Parametric methods are applied to find solutions. Implicit solutions are interpreted in relation to the original equation. Applications of these methods are considered in various contexts.

Second and Third-Order Equations. Order-reduction techniques are applied to second- and third-order equations. Problem-solving procedures for such equations are practised. The relationship between the order of an equation and its complexity is examined. Systematic approaches to solving these equations are presented.

Real Characteristic Root Equations. Procedures for finding real characteristic roots are practised. Exponential solutions are constructed from these roots. Simple and repeated roots are handled using appropriate techniques. Independent solution sets are formed from the results.

Complex Characteristic Root Equations. Complex characteristic roots are handled using appropriate techniques. Real solutions are constructed from these roots using trigonometric functions. The amplitude and frequency of the resulting oscillations are analysed. Problems involving oscillatory systems are solved.

Independence Testing with Wronskian. The Wronskian determinant is calculated for given sets of solutions. The independence of the solutions is verified using this determinant. Fundamental systems of solutions are validated. The completeness of the solution set is examined.

Undetermined Coefficients Practice. The form of the particular solution is guessed using the method of undetermined coefficients. The unknown coefficients are then solved for. Special cases requiring modification of the method are handled. The efficiency of the method is considered in comparison with alternatives.

Variation of Constants Application. The procedures of the variation of constants method are practised as a universal solution technique. Equations with any right-hand side are handled using this

method. The Wronskian formula is applied within the solution process. The efficiency of the method is compared with the method of undetermined coefficients.

Complete Solution Procedures. The homogeneous solution of a non-homogeneous equation is found as a first step. Particular solution methods are then applied. The complete solution is assembled from these two components. Diverse engineering problems are solved using this procedure.

Resonance and Oscillating Systems. Resonance conditions are recognised in oscillating systems. Solutions that grow with time are identified as a consequence of resonance. The physical meaning of these solutions is examined. The potential danger that resonance poses to physical systems is analysed.

Euler Equation Transformation. The logarithmic substitution used for Euler equations is applied in practical problems. The transformation procedures leading to constant-coefficient form are practised. The resulting solutions are interpreted. The accuracy of the back-substitution to the original variable is verified.

System Formulation from Higher-Order Equations. Higher-order equations are converted into systems using derivative substitution. The resulting systems are constructed in standard form. The conversion procedure is applied systematically to different equations. The results are verified against the original higher-order equation.

Eigenvalue and Eigenvector Methods. Eigenvalues of a system are computed as part of the solution procedure. The corresponding eigenvectors are then calculated. Solutions are built from the resulting eigenpairs. Both complex and real eigenvalue cases are considered.

Inhomogeneous System Solutions. Particular solutions of inhomogeneous systems are found using appropriate methods. These are combined with the homogeneous solution of the system. Real system problems are solved using this approach. Complete solution procedures are presented.

	Phase Plane Analysis and Visualization. Solution curves are sketched in the phase plane for given systems. Equilibrium points are identified from these sketches. The stability of the equilibrium points is analysed. The overall behaviour of the system is visualised. Computational Verification and Exploration. Maple or MATLAB software is applied to verify analytical solutions computationally. The effect of varying parameters on the solutions is studied. The results obtained are documented systematically. Analytical and computational results are compared for consistency. Self-study Separable Equation Types and Properties. Transformation Techniques Research. Variation of Constants Thorough Study. Integrating Factor Theory and Application. Existence and Uniqueness Theorem Analysis. Higher-Order Equation Properties. All Characteristic Root Cases. Homogeneous Equation Fundamentals. Wronskian Theoretical Analysis. Non-Homogeneous Complete Treatment. Systems and Engineering Applications.
Teaching Methods	The Module is delivered through interactive lectures illustrated with real-world examples. Problem-solving sessions and practical work employing analytical and numerical methods are used to reinforce theoretical concepts. Computational experiments are carried out using Maple and MATLAB software. Group discussions and projects on modelling real systems are incorporated into the teaching process. Independent study is supported through the use of mathematical software.
Prerequisites	Physics 1,2 Calculus 1,2 Linear Algebra
Exams and Assessment	Type of examination (written, oral, project) and its weight

	The assessment consists of a written examination, practical tests, and project-based assessment. Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and final written examination (50%). Continuous Assessment evaluates classroom participation in lectures and practical sessions, problem-solving ability during hands-on work, and completion of teacher-guided independent work assignments including essays on solution methods, reports on applications, and mini-projects on modeling. Midterm Assessment consists of a written test covering theoretical knowledge of first-order and higher-order differential equations, solution methods, characteristic equations, and systems of equations. Independent work projects evaluate problem-solving skills, application understanding, and analysis capabilities. Final Examination consists of a written examination with problems requiring students to solve various types of differential equations (first-order, higher-order, systems), analyze problems, and apply appropriate solution methods. Assessment measures theoretical understanding, problem-solving skills, and ability to apply mathematical concepts to engineering applications.
Recommended Literature	1. Boyce W.E., DiPrima R.C. Elementary Differential Equations and Boundary Value Problems. 2. Edwards C.H., Penney D.E. Differential Equations and Linear Algebra. 3. Arnol'd V.I. Ordinary Differential Equations (classical reference). 4. Maple and MATLAB documentation for computational differential equations.
Responsible Lecturer	Majidova Dilnoza Bahromovna, PhD in Economics.

LINEAR ALGEBRA

Field	Description
Module Title	LINEAR ALGEBRA
Module Code	CHA1306
Credit Points (ECTS)	6 Credits

Workload	Total workload: 180 hours Contact hours: 90 hours (46 hours Lecture, 44 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Linear Algebra is to provide students with fundamental theoretical knowledge of linear algebraic models, matrix calculus, and vector spaces. The Module covers methods for solving systems of linear equations and introduces students to modern directions and problems of scientific research in this field. It aims to develop students' ability to analyze existing mathematical problems. It also develops students' ability to apply appropriate mathematical methods to solve these problems. Students learn to use their theoretical knowledge in computer engineering, computer graphics, data processing, and other related areas. The Module is further intended to develop students' independent learning and research skills, and to strengthen their logical and analytical thinking. Finally, it develops students' ability to apply linear algebra methods to practical and scientific problems.
Module Learning Outcomes	Knowledge: LO1. Thinks logically, draws correct conclusions, and improves mathematical culture through the study of linear algebra concepts, assertions and their proofs, and methods for solving practical problems, while analysing the theorems, mathematical definitions, and axioms that form the basis of linear algebra. LO2. Can explain the fundamental concepts and theorems of linear algebraic systems, matrices, and vector spaces. LO3. Understands the difference between different methods for solving systems of linear equations (Gauss, Cramer, inverse matrix) and optimal algorithms. LO4. Can identify the practical importance of matrix eigenvalues and eigenvectors in IT services, computer graphics, and data analysis.

	LO5. Searches, analyses, and processes relevant mathematical information using mathematical, comparative, and other specific methods. Skills: LO6. Applies the knowledge gained in real life. LO7. Can perform mathematical operations accurately on high-dimensional matrices, determinants, and linear operators. LO8. Models real-world engineering and IT service problems (e.g., network flows, economic models, or graphical transformations) as a system of linear equations. LO9. Can implement numerical algorithms for solving mathematical problems in a programming environment (for example, using Python's NumPy/SciPy libraries). Competencies: LO10. Independently selects and applies linear algebra methods to solve information technology problems. LO11. Works in a team on engineering projects, prepares professional technical conclusions, and justifies mathematical solutions to problems.
Content	Lectures Linear Operations on Vectors. The topic covers addition of vectors and multiplication by a scalar. Students develop the ability to perform basic linear operations. They apply these operations to vector problems. Complex Numbers. Complex numbers are defined as numbers whose components are real numbers. This topic covers the basic operations of addition and subtraction of complex numbers. It also covers multiplication and division of complex numbers. Determinants and Their Properties. Small-Order Determinants. The determinant is a numerical value associated with a square matrix. Students learn how to calculate determinants of the second and third orders. They study the main properties of these determinants. n-Order Determinants and Their Properties. This topic explains methods for calculating determinants of arbitrary order. It examines their important properties.

These properties are widely used in solving systems of linear equations and matrix problems.

Algebraic Complements and Minors. Laplace's Theorem. Minors and algebraic complements are studied as tools for simplifying the calculation of determinants. Their use in reducing determinant computations is examined. Laplace's theorem is studied as a method for expanding a determinant along a row or column.

Determinant and Additional Properties of Matrices. Inverse Matrix. Students study additional properties of determinants. They learn the concept of an inverse matrix. The inverse matrix is important for solving systems of linear equations and performing matrix operations.

Solving a System of Linear Equations. Gaussian Method. The Gaussian elimination method is used to solve systems of linear equations. It transforms the system into a simpler triangular form. This method is widely used in mathematics, computer science, and engineering.

Matrix Method for Solving a System of Linear Equations. This topic considers the use of matrix operations to solve systems of linear equations. It also considers the use of inverse matrices for this purpose. Students learn how to represent a system in matrix form and obtain its solution.

Linear Spaces. Linear Subspaces. Sum and Intersection of Subspaces. Students become familiar with the concept of linear spaces. They study the concept of subspaces. The topic explains how subspaces can be combined through their sum and intersection.

Dimension of Vector Space. Linear Dependence. Basis of Vector Space. Students study the concepts of dimension and linear dependence and independence of vectors. They study the basis of a vector space. They learn how to determine the dimension of a vector space and represent its vectors using a basis.

Linear Spaces. Linear Subspaces. Sum and Intersection of Subspaces. Students study the concepts of linear spaces and subspaces. They study

the operation of finding the sum of subspaces. They also study the operation of finding the intersection of subspaces.

Matrix Rank. Kronecker–Capelli Theorem.

Students study the concept of matrix rank. They study the Kronecker–Capelli theorem. This theorem is used to determine the existence and number of solutions of a system of linear equations.

System of Homogeneous Equations. Fundamental Solution.

Students study homogeneous systems of linear equations. They learn to determine the solutions of such systems. They also learn to construct a fundamental system of solutions.

Complex Numbers and Operations on Them.

Moivre's Formula. Students study complex numbers and basic operations on them. They study Moivre's formula. This formula is used for finding powers and roots of complex numbers.

Extracting the Square Root of a Complex Number.

Students study methods for finding the square roots of complex numbers. They learn to determine the real and imaginary parts of these roots. They also develop the ability to represent the obtained roots in algebraic form and verify the results.

Polynomials in One Variable. Horner's Scheme.

Bézout's Theorem. Students study polynomials in one variable. They learn to evaluate them using Horner's scheme. Bézout's theorem is considered for determining the remainder of polynomial division and identifying polynomial roots.

Polynomial Division and Equations. Irreducible Polynomials. Fundamental Theorem of Algebra.

Students study polynomial division and polynomial equations. They study the concept of irreducible polynomials. They also learn the fundamental theorem of algebra and its role in determining the roots of polynomial equations.

Rational Fractions and Their Expansion into Simplest Fractions.

Students study rational fractions. They learn methods for decomposing them into simpler fractions. This helps simplify algebraic

expressions and solve related mathematical problems.

Euclidean Space. Cauchy–Bunyakovsky Inequality. Orthogonalization Process. Students study Euclidean spaces and the scalar product of vectors. They study the Cauchy–Bunyakovsky inequality. They also learn the orthogonalization process for constructing mutually orthogonal vectors.

Linear, Bilinear and Quadratic Forms. Students study linear, bilinear, and quadratic forms. They study their properties and matrix representations. They learn to distinguish these forms and apply them to solve mathematical and practical problems.

Methods for Converting a Quadratic Form into Canonical Form. Students study methods for transforming quadratic forms into canonical form. They learn to simplify their structure through appropriate transformations. They also develop the ability to analyze and represent quadratic forms in a more convenient form.

Positive Definite Quadratic Forms. Law of Inertia. Students study positive definite quadratic forms. They learn to determine their definiteness. They also study the law of inertia and its application to the classification and analysis of quadratic forms.

Linear Transformations and Their Matrix. Students study linear transformations. They learn how to represent linear transformations using matrices. They also examine the relationship between a linear transformation and its matrix representation.

Image and Kernel of Linear Transformations. Invariant Subspaces. Eigenvalues and Eigenvectors. Students study the image and kernel of linear transformations. They study invariant subspaces. They also learn the concepts of eigenvalues and eigenvectors and methods for determining them.

Practical lessons

Linear Operations on Vectors. The topic covers addition of vectors and multiplication by a scalar. Students develop the ability to perform basic linear operations. They apply these operations to vector problems.

Determinants and Their Properties. Small-Order Determinants. The concept of determinants, their main properties, and methods for calculating small-order determinants are considered. Students learn to use determinants in matrix calculations. They also learn to use determinants in systems of linear equations.

n-Order Determinants and Their Properties. Methods for calculating determinants of arbitrary order are examined. Their main properties are also examined. This enables students to solve more complex problems in linear algebra.

Algebraic Complements and Minors. Laplace's Theorem. The topic introduces minors, algebraic complements, and Laplace's theorem. Students acquire the ability to expand determinants using rows and columns. They also acquire the ability to calculate determinants using rows and columns.

Determinant and Additional Properties of Matrices. Inverse Matrix. The main properties of determinants and matrices are explained. The concept of an inverse matrix is also explained. Students learn to find inverse matrices and use them in mathematical calculations.

Systems of Linear Equations. Matrices and Operations on Them. The topic focuses on systems of linear equations, matrices, and operations with them. Students learn to represent systems in matrix form. They also learn to perform basic matrix operations.

Cramer and Inverse Matrix Methods for Solving Systems of Linear Equations. Cramer's rule and the inverse matrix method are considered as important approaches to solving systems of linear equations. Students develop the ability to select an appropriate method for obtaining solutions. They also develop the ability to apply the selected method for obtaining solutions.

Permutations and Substitutions. The topic explains the concepts of permutations and substitutions. It also explains their properties. Students learn to perform transformations of elements according to specified rules.

Dimension of Vector Space. Linear Dependence. Basis of Vector Space. The concepts of dimension, linear dependence and independence, and basis are introduced. Students learn to determine the dimension of a vector space. They also learn to represent vectors through a basis.

Linear Spaces. Linear Subspaces. Sum and Intersection of Subspaces. The structure of linear spaces and subspaces is considered. Their sum and intersection are also considered. Students develop the ability to analyze relationships between subspaces and perform operations with them.

Matrix Rank. Kronecker–Capelli Theorem. Matrix rank and the Kronecker–Capelli theorem are studied as tools for analyzing systems of linear equations. Students learn to determine the existence of solutions of such systems. They also learn to determine the number of solutions of such systems.

System of Homogeneous Equations. Fundamental Solution. The properties of homogeneous systems of linear equations are examined. Their solutions are also examined. Students learn to construct a fundamental system of solutions.

Complex Numbers and Operations on Them. Moivre's Formula. The topic covers complex numbers, operations on them, and Moivre's formula. Students develop skills in performing calculations with complex numbers. They also develop skills in finding their powers and roots.

Extracting the Square Root of a Complex Number. Methods for finding the square root of a complex number are presented. Students learn to determine the real and imaginary parts of the roots. They also learn to verify the obtained results.

Polynomials in One Variable. Horner's Scheme. Bézout's Theorem. Polynomials in one variable, Horner's scheme, and Bézout's theorem are considered. Students learn to evaluate polynomials. They also learn to determine the remainder of polynomial division.

Polynomial Division and Equations. Irreducible Polynomials. Fundamental Theorem of Algebra.

	The topic covers polynomial division and polynomial equations. It also covers irreducible polynomials and the fundamental theorem of algebra. Students develop the ability to analyze polynomial equations and their roots. Rational Fractions and Their Expansion into Simplest Fractions. Methods for decomposing rational fractions into simpler fractions are studied. This enables students to simplify rational expressions. It also enables them to solve related mathematical problems. Euclidean Space. Cauchy–Bunyakovsky Inequality. Orthogonalization Process. Euclidean spaces, scalar products, and the Cauchy–Bunyakovsky inequality are considered. Orthogonalization methods are also considered. Students learn to construct and work with orthogonal systems of vectors. Linear, Bilinear and Quadratic Forms. The structure and properties of linear, bilinear, and quadratic forms are examined. Their matrix representations are also examined. Students learn to distinguish and analyze different types of algebraic forms. Methods for Converting a Quadratic Form into Canonical Form. Methods for transforming quadratic forms into canonical form are presented. Students acquire the ability to simplify quadratic forms. They also acquire the ability to analyze their structure. Positive Definite Quadratic Forms. Law of Inertia. Positive definite quadratic forms and the law of inertia are considered. Students learn to determine the definiteness of quadratic forms. They also learn to classify their properties. Linear Transformations and Their Matrix. The concept of linear transformations is introduced. Their matrix representations are also introduced. Students learn to represent transformations using matrices and analyze their basic properties. Self-study Methods for Solving Systems of Algebraic Equations.
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	Elementary Transformations of Systems of Algebraic Equations. Matrix Algebra. Second- and Third-Order Determinants and Methods for Calculating Them. Arrangements and Substitutions. Linear Spaces. Linear Dependence and Linear Independence. Dimension and Basis. Euclidean Space and Its Examples. Length of a Vector, Angle Between Vectors. Change in the Matrix of a Linear Form When the Basis Changes. Positive Definite Quadratic Forms. Condition for a Quadratic Form to Be Positive Definite. Eigenvalues and Canonical Representation of Linear Transformations.
Teaching Methods	The module is delivered through a combination of lectures, practical classes, interactive discussions, problem-solving activities, and independent learning. Teaching methods include theoretical instruction, demonstration of mathematical methods, guided problem solving, individual and group work, analytical discussions, presentations, and the use of digital mathematical tools. Practical activities focus on matrix operations, determinants, systems of linear equations, vector spaces, linear transformations, eigenvalues and eigenvectors, and quadratic forms. Independent learning activities may include studying theoretical materials, solving mathematical problems, preparing presentations, completing individual assignments, analyzing different solution methods, and conducting independent research on selected topics in linear algebra. Students may also use mathematical software and digital learning resources to verify calculations and explore selected mathematical concepts. These teaching methods are designed to develop students' theoretical knowledge and practical skills in linear algebra, and to improve their ability to apply algebraic methods to mathematical, computer engineering, computer graphics, data processing, and other related problems.

Prerequisites	Algebra, geometry
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, practical performance, problem-solving activities, short in-class quizzes, and the successful completion of independent study assignments, including matrix operations, determinant calculations, solving systems of linear equations, analysis of vector spaces, determination of matrix rank, and solving problems related to eigenvalues and eigenvectors. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as a presentation on matrix algebra, a comparative analysis of methods for solving systems of linear equations, a project on vector spaces and bases, an analysis of linear transformations, and a written report on the application of linear algebra in computer engineering, computer graphics, or data processing. The final assessment is a written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of linear algebra, their analytical and logical thinking skills, and their ability to apply mathematical methods and concepts to solve mathematical and practical problems. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Ayupov A.S.H., Omirov BA, Khudoyberdiev AX Linear algebra (textbook). "Mukhr-press" publishing house. Tashkent, 2023. -392 pages 2. Allakov I. Algebra Part I, «Surkhon-nashr» publishing house. Termiz, 2024. -332 pages. 3. Poskuryakov I.L. Collection of linear algebra. "Nauka", 2005. 4. Khojiev J.H., Feinleib A.S. Algebra and number theory course. Tashkent, "Uzbekistan", 2001.

	5. Faddeev D.K. Lektsii po algebre, M., "Nauka" 1984g. 6. Kostrykin A.I. "Vvedenie v algebru" M. 1977 7. Problems and exercises from algebra and number theory. Part II textbook, author prof. UX Narzullayev, AS Soleev, XX Ro'zimurodov. Samarkand 2021.- 201 pages. 8. Mirziyoyev Sh.M. We will build our great future together with our brave and noble people. – Tashkent: "O'zbekiston", 2017. 488 p. 9. Ziyoev Sh.M. Strategy of the New Uzbekistan. – Tashkent: Publishing House "O'zbekiston" 2021. 464 pages. 10. Gelfand I.M. Lectures on linear algebra. "High and middle school". 1964. 11. Faddeev D.K., Sominsky I.S. Zadachi po vyshey algebre, St. Petersburg, 1999. 12. Kostrykin A.I. i dr., Sbornik zadach po algebre. "Science", 1986 13. I.Allakov. Examples and problems from number theory. –T. Innovative Development Publishing House. 2020. 348p. 14. Kletenik D.V. Collection of analytical geometries. M. "FIZMATLIT", 2016. 15. Izu Waisman. Analytical Geometry. World Scientific, USA, 2007.
Responsible Lecturer	Safarov To'lqin Nazarovich; Avliyokulov Abdugafur Abduganievich

PROGRAMMING 1,2

Field	Description
Module Title	PROGRAMMING 1,2
Module Code	DAS1110
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 120 hours
Frequency/Duration	I semester

Module objective	The objective of the Module Programming 1,2 is to develop students' theoretical and practical knowledge of the fundamental concepts of programming. The Module introduces students to algorithms and data structures. Through this subject, students acquire the skills to solve problems using programming languages. They learn to develop software products and test them using programming languages. The Module also serves to develop students' logical thinking. It develops their ability to devise effective solutions to problems. The knowledge and skills acquired provide a solid foundation for future professional activities.
Module Learning Outcomes	Knowledge: LO1. Acquires theoretical knowledge of the fundamental concepts, principles and programming paradigms of programming. LO2. Develops algorithms, analyses their efficiency, and masters the main types and properties of data structures. LO3. Is familiar with the syntax of programming languages, their operators, control structures and the principles of using functions. LO4. Has knowledge of methods for working with arrays, strings, files and other data, as well as the principles of processing them. LO5. Studies methods for testing, debugging and error-troubleshooting, as well as the main approaches to optimising program code. Skills: LO6. Acquires the skills to develop various algorithms and solve practical problems using programming languages. LO7. Applies data structures, data processing and storage methods in practice. LO8. Gains practical experience in designing, creating and testing software products. LO9. Effectively uses methods for identifying, analysing and fixing errors in program code. LO10. Acquires the skills to develop software solutions using modern programming environments, libraries and tools.

	Competencies: LO11. Acquires the competence to develop effective software solutions for various practical problems using programming languages and modern software tools. LO12. Develops professional competencies in analysing algorithms and data structures, and in designing, developing and testing software products. LO13. Acquires the necessary skills to identify software errors, rectify them and improve software products. LO14. Develops competencies in the effective use of modern information and communication technologies, and in the independent acquisition of new programming technologies and their application in professional practice. LO15. Acquires the competence to develop effective software solutions for various practical problems using programming languages and modern software tools.
Content	Lectures Subject Matter, Aims and Objectives of the Discipline of Programming. Programming is introduced as a scientific and practical discipline, together with the aims, objectives and learning outcomes of the course. The importance of programming in information technology and software development, along with its history and evolution, is examined. Applications of programming in science, engineering, business and everyday life are studied. Basic concepts of programming, program development, problem solving, and programming environments and software development tools are considered. Properties of an Algorithm, Representation of an Algorithm in Words and in Graphical Form. The concept and significance of algorithms in solving computational problems are examined. Essential properties of algorithms, including finiteness, definiteness, correctness, effectiveness and generality, are studied, together with methods of algorithm design. Algorithms are represented using natural language, pseudocode and graphical

flowcharts, along with standard flowchart symbols and conventions. A comparison of different algorithm representation techniques is considered.

Linear, Branching and Iterative Algorithms.

Algorithms are classified according to control structures, distinguishing linear algorithms and the sequential execution of statements. Branching algorithms using simple and nested conditional statements are examined. Iterative algorithms, the repetitive execution of instructions, and loop structures and their practical applications are studied. A comparative analysis of linear, branching and iterative algorithms is considered through programming examples.

Classification of Programming Languages. The historical development of programming languages is examined, together with the distinction between low-level and high-level languages, including machine language and assembly language. Procedural, object-oriented, functional, scripting and visual programming languages are studied. Compiled and interpreted programming languages, as well as general-purpose and specialized programming languages, are considered. Criteria for selecting a programming language for software development are also considered.

Interactive Mode, First Program, Input and Output Operators. The programming environment is introduced, together with interactive and script execution modes. Writing, compiling and executing the first program are examined, along with program structure and syntax. Variables and constants, and standard input and output operations, are studied. Reading data from the keyboard, displaying information on the screen, formatting output, and debugging simple programs are considered.

Interactive Mode, First Program, Input and Output Operators. Interactive programming techniques are examined, together with working in the integrated development environment. Creating, editing, saving and executing programs are studied, along with input and output statements. User interaction during program execution and data input validation are

considered. Formatting output, displaying program results, and common programming errors during input and output operations are also considered.

Arithmetic and Logical Operators in the Programming Language. Arithmetic operators and their precedence are examined, together with assignment operators. Relational operators and comparison operations, as well as logical operators and Boolean expressions, are studied. The evaluation of logical conditions, and operator precedence and associativity, are considered. Practical applications of arithmetic and logical operators in programming are also considered.

Arithmetic and Logical Operators in the Programming Language. Complex arithmetic expressions are examined, together with combining arithmetic and logical operators. The short-circuit evaluation of logical expressions, compound assignment operators, and increment and decrement operations are studied. Programming problems are solved using arithmetic and logical operations. Best practices in writing expressions are considered.

Mathematical Functions and Expressions in the Programming Language. Mathematical libraries and built-in functions are examined, together with mathematical constants. Functions for calculating powers, roots, logarithms, exponentials and trigonometric values are studied. Constructing mathematical expressions and data type conversion in mathematical calculations are considered. Practical applications of mathematical functions are also considered.

Mathematical Functions and Expressions in the Programming Language. Complex mathematical expressions and nested mathematical functions are examined. Precision in numerical calculations and floating-point arithmetic are studied. Engineering and scientific problems are solved using mathematical expressions. Optimizing mathematical computations in programming is considered.

Programming Linear Processes. Sequential algorithms are programmed, together with variable declaration and initialization. The input, processing

and output stages of linear programs are examined. Simple computational programs are written, with attention to program readability and coding standards. Practical applications of linear programming structures are considered.

Programming Branching Processes. Conditional programming using if, if-else and nested conditional statements is examined, together with logical decision making in programs. Multi-way selection structures and conditional expressions are studied. Practical problems are solved using branching algorithms. Program efficiency is improved through conditional programming.

Iterative Processes and Parameter-Based Programming. The concept of iterative programming is examined, together with parameter-controlled loops. Loop initialization, continuation conditions and iteration variables are studied. Numerical and logical problems are solved using iterative algorithms that repeat computations. Performance considerations in repetitive programming are considered.

Skipping Loop Steps and Terminating Loops Prematurely. Loop control mechanisms are examined, together with the break statement and premature loop termination. The continue statement and skipping current iterations are studied, along with nested loop control. Practical applications of loop control statements are considered. Efficient looping algorithms are written using these mechanisms.

Programming Conditionally Repetitive Processes. Condition-controlled loops are examined, together with while and do-while loop structures. Parameter-controlled and condition-controlled loops are compared. Appropriate loop structures are selected for different programming problems, and infinite loops are prevented. Efficient repetitive algorithms are developed.

Practical lessons

Programming Environment. The programming environment is installed and configured. Projects are created and managed, and work is carried out with source files. Simple programs are run and tested, with familiarization with the development interface.

Print Function. The first program is written using print statements. Text, variables and expressions are displayed, and output is formatted. Multiple values and special characters are printed.

Syntax Errors. Syntax errors are identified, together with compiler and interpreter error messages. Syntax mistakes are corrected, and common programming errors are examined. Program debugging techniques are applied.

Arithmetic Operations. Arithmetic calculations are performed using arithmetic operators. Programs are written for numerical computations. Simple mathematical problems are solved.

Comments and Variables. Variables are declared and initialized, following naming conventions. Comments are used for documentation, and work is carried out with different data types. Variable values are modified during program execution.

Working with Texts. String variables and text processing are examined, together with string concatenation. The input and output of text and basic string manipulation operations are practiced. Practical exercises with textual data are completed.

Mathematical Functions. Built-in mathematical functions are used to perform calculations involving powers and roots. Calculations involving logarithms and trigonometric functions are also performed. Mathematical applications are developed.

Constructing Linear Programmes. Linear algorithms are designed and implemented. The input-processing-output model is applied. Practical computational tasks are solved using sequential programming.

Working with the Short Conditional Operator. The conditional expression, known as the ternary operator, is used. Conditional statements are simplified using this operator. Concise decision-making programs are written.

Working with Full Conditional Operators. If, if-else, and nested if statements are used. Decision-making problems are solved using these statements. Alternative solutions are compared using conditional programming.

Programming Iterative Processes. Programs are developed using loop structures, including the counting of iterations. Sequences of values are processed using these structures. Practical programming exercises involving repetition are completed.

Nesting Loops. Nested loop structures are created and used to solve multidimensional computational problems. Pattern generation is carried out using nested loops. Matrix traversal is also performed using nested loops.

The Break and Continue Operators and Their General Overview. Break and continue statements are used to control loop execution. Programming problems requiring early termination are solved using the break statement. Problems requiring skipped iterations are solved using the continue statement.

Conditional Cycle Operator. Programs are developed using condition-controlled loops. While and do-while loops are compared. Repetitive programming tasks are solved using these loop structures.

Subprograms. Functions are created and used, including their declaration and invocation. Parameters and return values are studied. Modular programming principles are applied to develop reusable program components.

Self-study

Subject Matter, Objectives and Place of Programming in the Modern IT Field.

The Concept of an Algorithm, Its Properties and Real-Life Applications.

Ways of Expressing Algorithms: Natural Language, Flowchart and Pseudocode.

Linear Algorithms and Their Application in Programming.

Solving Problems Using Branching Algorithms and Conditional Statements.

The Importance of Iterative Algorithms and Looping Operators.

	Classification of Programming Languages and Their Comparative Analysis. Programming in Interactive Mode and Writing the First Program. The Role of Input and Output Operators in Programming. Arithmetic Operations and Their Application in Programming. Solving Conditional Problems Using Logical Operations. Methods for Working with Mathematical Functions and Expressions. Recursive Processes and Parametric Loops. Use of Break and Continue in Loop Operators. Conditional Iterative Processes and Their Practical Application.
Teaching Methods	The module is delivered through interactive lectures, practical laboratory sessions, algorithmic problem-solving, coding demonstrations, pair and individual programming, project-based learning, and guided independent study. Students acquire fundamental programming knowledge through the design and implementation of algorithms, development of computer programs, debugging techniques, and program testing using modern programming environments. Practical activities emphasize computational thinking, logical reasoning, structured programming, and the application of control structures, data processing, and mathematical expressions in solving real-world problems. Independent learning, online coding platforms, digital educational resources, and collaborative discussions further strengthen analytical skills, programming competence, software development practices, and lifelong learning abilities.
Prerequisites	Calculus 1,2 Physics 1,2
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous

	assessment evaluates classroom participation, practical programming performance, laboratory exercises, coding assignments, algorithm design activities, and the successful completion of independent study tasks, including flowchart construction, pseudocode development, implementation of linear, branching and iterative algorithms, programming exercises involving arithmetic and logical operations, and debugging simple computer programs. The midterm assessment consists of written assignments, short in-class quizzes, practical coding tasks, and the evaluation of independent programming projects, such as developing simple console-based applications, implementing mathematical calculations, solving conditional and iterative programming problems, documenting algorithms using flowcharts and pseudocode, and presenting solutions to programming case studies. The final assessment is a written examination consisting of theoretical questions, algorithm design tasks, and practical programming problems designed to assess students' understanding of programming concepts, computational thinking, algorithmic problem-solving skills, and their ability to develop correct and efficient computer programs. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Narzullayev A. Fundamentals of Programming in Python: Tashkent: "Akademnashr", 2021. - 336 pp. 2. Associate Professor A. Ikramov, Tashkent State University Lecturer at Tashkent State University Q. Maxarov Fundamentals of Programming, Tashkent 2016 3. Madraximov Sh.F., Gaynazarov S.M. The C++ Programming Language. Tashkent 2008 4. Fundamentals of Programming in Python. Textbook.– Samarkand: Samarkand State University Publishing House, 2020, – 180 pages 5. ROBERT W. SEBESTA - Concepts of Programming Languages - Tenth Edition. 816 p.

	6. Daniel P. Friedman, Mitchell Wand - Essentials of Programming Languages, third edition. 2008. 433 p.
Responsible Lecturer	Tursunov Farxod Bahodir ugli, teacher.

ACADEMIC WRITING

Field	Description
Module Title	ACADEMIC WRITING
Module Code	AKY1104
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	I semester
Module objective	The objective of the Module Academic Writing is to form in students the theoretical foundations of academic writing and the norms of scientific style. The Module develops students' skills in conducting scientific research. It develops skills in creating, analysing, and editing academic texts. The Module further develops the written communicative competencies necessary for scientific and professional activity. The course is designed to develop students' competence in working with scholarly sources and adhering to the principles of academic integrity. It also develops students' ability to present research findings in a logical and coherent written form. Finally, it develops students' ability to create scholarly texts that meet the requirements of contemporary academic discourse.
Module Learning Outcomes	Knowledge: LO1. Is familiar with the fundamental rules, structures, and styles of academic writing, including the requirements of this type of writing and methods for clear and logical expression. LO2. Has an in-depth knowledge of the specialised vocabulary, grammatical constructions, and stylistic devices used in academic texts.

	LO3. Knows how to work with scholarly sources, cite correctly, compile a bibliography, and avoid plagiarism. LO4. Knows how to organise scientific research, collect and analyse data, formulate research questions, and present findings in writing. LO5. Is familiar with the various forms of academic writing (essays, research articles, scientific reports, etc.) and understands their characteristics. Skills: LO6. Expresses complex topics in writing clearly and logically, analyses the subject in depth, and thinks coherently. LO7. Writes in a scholarly and academic style, constructs clear and meaningful sentences, and uses the appropriate lexical and grammatical tools correctly. LO8. Cites sources correctly, avoids plagiarism, and lists scholarly sources in the correct format. LO9. Formulates research questions, collects and analyses data, and presents research findings in a logical and coherent written form. LO10. Edits written work, identifying and correcting grammatical, stylistic, and logical errors. Competencies: LO11. Creates academic texts independently, developing scientific articles, summaries, essays, course papers and other academic texts in accordance with the requirements of the scientific style, and correctly organising their compositional structure. LO12. Conducts research by selecting a research topic, identifying a scientific problem, formulating research questions and hypotheses, and searching for, analysing, and effectively using scholarly sources. LO13. Maintains academic integrity and handles information properly by formatting sources correctly, adhering to citation rules, avoiding plagiarism, and upholding the principles of academic ethics. LO14. Presents and edits scientific results by editing and analysing scientific texts, presenting results using
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	tables, graphs and diagrams, and drawing substantiated scientific conclusions.
Content	Lectures Key Concepts and Principles of Academic Writing. The role of academic writing in Computer Engineering and how it differs from general or creative writing is examined. Ethical standards, honesty, and intellectual responsibility in academic work are studied. A formal, precise, and professional register suitable for scientific discourse is maintained. Types of Academic Texts: Article, Report, Dissertation. The main forms of academic writing in engineering are studied, including research papers, technical reports and dissertations. The presentation of technical information is examined. Its structural foundations are also examined. Choosing a Topic and Drawing Up a Plan. A topic for a practical research project in computer engineering is chosen, together with an assessment of its relevance. Scientific resources appropriate to the chosen topic are identified. A strategy for creating a plan appropriate to the topic is developed. The logical structure of the plan is studied. Scientific Style and Language Features. Basic stylistic conventions and linguistic features required for academic and engineering writing are studied. An objective, formal register is maintained by eliminating colloquialisms, emotional expressions, and biased language. Clear technical vocabulary and correct grammar are examined. Formulation of Research Questions and Hypotheses. Specific research questions and testable hypotheses tailored to computer engineering projects are formulated. Focused research questions that address specific technical problems and gaps in the system are studied. Results are validated using measurable parameters and key evaluation indicators, together with their adaptation for research purposes. Searching for and Evaluating Scholarly Sources. Methods for finding, filtering, and evaluating reliable scientific literature for research are studied. Scopus,

Google Scholar, and other academic databases are used for this purpose. The reputation, value, and accuracy of a scientific source are examined. Research flaws are identified.

Citing Scholarly Sources and Style. Basic rules and methods for correctly citing scientific literature in technical and engineering writing are studied. Principles for formatting in-text citations and reference lists are examined. Evidence is integrated through text summarization and direct citation.

Plagiarism and Its Prevention in Academic Writing. Academic integrity, types of plagiarism, and their consequences in scientific research are studied. The causes and forms of plagiarism, including direct copying and self-plagiarism in technical writing, are examined. Methods for the effective prevention of plagiarism and the use of anti-plagiarism detection software are considered.

Analysis and Commentary in Academic Writing. Techniques for analyzing technical data are studied. Techniques for writing critical reviews in academic engineering work are also studied. Quantitative data, experimental results, and code performance metrics are examined.

Drawing Conclusions in Academic Writing. Important strategies for writing clear and impactful conclusions in technical engineering work are studied. Research findings are summarized, indicating whether the main objectives and hypotheses were met. The broader technical implications of the research and its practical applications are examined. Clear directions for future work and development are considered.

The Logical Structure of Academic Writing. Logical coherence, consistency, and structural integrity throughout academic and technical texts are studied. Linear argumentation, thematic progression, and logical flow between sections are examined. Topic sentences and paragraph structures are considered.

Editing and Revising Written Work. Systematic methods for editing and revising engineering projects to achieve professional accuracy and precision are

studied. Structural alignment by major sections is examined. Grammatical accuracy and precise technical terminology are considered. Practical methods for correcting formatting errors and self-editing are also considered.

Introduction to the Scientific Paper and Its Importance. The basic structure, purpose, and importance of scientific articles are studied. The basic architecture of a scientific article and its standard systems are examined. Publication and rigorous review are considered as means of ensuring quality, reproducibility, and continuous scientific progress.

Results and Their Presentation. Strategies for organizing and reporting experimental results and system performance in technical papers are studied. Strategies for visually presenting these results are also studied. Quantitative data, experimental findings, tables, graphs, and system architecture diagrams are presented using clear headings and labels.

The Concluding Part of Academic Writing: Analysis and Conclusion. The final sections of academic papers are studied, in which experimental results are synthesized, critically analyzed, and concluded. Data trends are analyzed and directly related to the original research questions and hypotheses. Comprehensive conclusions summarizing key technical advances and their broader engineering implications are considered.

Practical lessons

Comparing Academic and Literary Texts, Identifying Their Specific Principles. Stylistic differences between objective, formal academic prose and creative, emotional literary language are examined. The rigid, logical organization of academic articles is compared with the flexible narratives of literary works. Key linguistic features in both genres are studied.

Analysis of Article and Dissertation Texts. Structural and stylistic differences between journal articles and academic dissertations are analyzed. Macrostructural differences and stylistic variations are examined. The depth of context, breadth of literature

review, and formatting standards are studied through text analysis exercises.

Drawing Up Plans for Given Topics and Evaluating Them. Detailed plans and outlines for academic writing are developed and evaluated. Methods for developing coherent, logically structured plans for specific topics and scholarly work are studied. Criteria for evaluating project plans, based on logical development, topic coverage, and adherence to academic formatting standards, are examined.

Conduct a Stylistic Analysis of the Given Scientific Text. Basic academic language features are examined, including passive and active voice, objective tone, and barrier language. The basis for assessing stylistic quality is studied, including variety of sentence structure, conciseness, clarity, and adherence to formal standards. Stylistic flaws are identified, specific corrections are suggested, and a practical stylistic analysis of the text is conducted.

Formulate Hypotheses and Link Them to Research Questions. Testable research hypotheses are developed and aligned directly with the main research questions in technical writing. Broad research questions are transformed into concise, measurable null and alternative hypotheses. Strategies for ensuring logical alignment between research questions, variables, and hypotheses are studied.

Finding and Analysing Sources from an Academic Database. Academic databases are searched efficiently using advanced search methods, including Boolean operators and keyword strategy refinement. Filtering is applied across major databases such as IEEE Xplore, ACM Digital Library, and Scopus. Source credibility and relevance, author reputation, and citation metrics are assessed.

Learning Citation Rules and Applying Them in Practice. Citation conventions are mastered and standardized referencing styles are applied. The mechanics of academic citation styles, in-text citations, and reference list notation are studied. Rules for citing peer-reviewed journal articles,

conference proceedings, technical reports, software, and patents are examined.

Discuss Methods for Detecting Plagiarism in Given Texts and Ways to Prevent It. Plagiarism is identified through modern detection tools. Specific strategies to maintain academic integrity in engineering writing are applied. Automated and manual methods of plagiarism detection, including text matching algorithms and the identification of stylistic inconsistencies and unusual phrasing, are studied.

Analyse and Comment on the Given Text. A presented technical or scientific text is critically analyzed to assess its structural logic, argumentative force, and style. Methods for evaluating the rhetorical flow of the text and the clarity of the thesis are studied. The validity of the supporting evidence and the logical sequence between paragraphs are examined.

Analysing Various Conclusions and Formulating Them Correctly. Existing academic findings are assessed and sample findings are analyzed. Common pitfalls, such as reiterating abstracts or overgeneralizing results, are identified. Effective conclusions that overcome research limitations are formulated.

Analysis of the Logical Structure of the Text. Methods for identifying logical structures at the macro and micro levels are studied, including thematic progression and claim-argument relationships. Transitional coherence between paragraphs is examined. Analytical frameworks for identifying logical errors, broken argumentative chains, redundant assertions, and gaps in technical reasoning are applied, and text arguments are visually mapped.

Edit and Analyse the Given Text. Raw technical drafts are evaluated, and structural and stylistic errors are identified. Extensive academic editing is performed. Editing fundamentals are practiced, including line-by-line grammar improvement, terminology consistency, and eliminating redundant phrases, tracking changes, and fully revising to create a publication-ready version.

Identifying Key Aspects in the Introduction of a Scientific Paper. The main components of the introduction of a scientific article are identified. Its rhetorical structure is evaluated. The main structural elements of the introduction are identified, including establishing the broader context, identifying gaps in the existing literature, and stating the main research objectives.

Explore Methods for Presenting Results in Visual and Textual Formats. Complex data and research findings are communicated effectively in technical writing through complementary visuals and textual narratives. Principles of selecting and designing visual formats such as graphs, tables, and diagrams are studied. Clarity, proper labeling, and adherence to academic formatting standards are ensured.

Analysis of Final Conclusions in Given Texts. The effectiveness of conclusion sections in a variety of scientific and technical texts is analyzed and evaluated. A detailed rationale for deconstructing conclusions is provided, restating research objectives and synthesizing key findings. Research limitations are addressed and meaningful future work is suggested. Weak conclusion paragraphs are rewritten.

Self-study

Writing an Essay Based on the Fundamental Principles of Academic Writing

Write a Short Text in the Chosen Academic Text Type.

Independently Formulate a Topic and Outline for a Research Paper.

Write a Short Article in a Scientific Style on the Chosen Topic.

Exercises on Developing Research Questions and Hypotheses.

Find Three Reliable Sources on a Topic and Analyse Them.

Preparing a List of Sources for a Scientific Article.

Writing an Essay on the Rules of Academic Writing to Prevent Plagiarism.

	Write an Analytical Article on a Chosen Topic. Writing the Conclusion for a Scientific Article. Write a Logical Text Based on a Plan for the Topic. Editing and Revising the Initial Version of the Scientific Article. Writing the Introduction for the Text You Have Written. Present the Results of the Scientific Work Using Graphs or Diagrams. Writing the Final Conclusion and Analysis Section on the Chosen Topic.
Teaching Methods	This course utilizes interactive and practical teaching methods designed to progressively develop students' academic and research skills. Throughout the lessons, students do not merely acquire theoretical knowledge; they actively engage in selecting topics, analyzing literature, and formulating research hypotheses. Through peer review and project-based learning approaches, students enhance their ability to edit academic texts and refine scientific arguments. Consequently, students develop the competence to write research articles, visualize data effectively, and present their scientific findings in alignment with international standards.
Prerequisites	Foreign Language Uzbek Language Theory of Linguistics Textual Studies Stylistics and Editing Fundamentals Research Methodology
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment reflects the student's theoretical knowledge in the discipline, academic writing competencies, skills in creating and analysing scholarly texts, working with scientific sources, adhering to the principles of academic integrity, editing written work, and presenting research findings logically and coherently. The midterm assessment consists of written assignments, short in-class

	quizzes, and the evaluation of independent study projects, such as an annotated bibliography, a peer-reviewed article analysis, an academic outline and thesis proposal, an essay on academic integrity and plagiarism, and a draft section of a scientific manuscript. The final assessment is a written examination consisting of theoretical concepts, literature evaluation tasks, and practical academic writing problems designed to assess students' understanding of academic integrity, research methodology, structural organization, and their ability to produce clear, cohesive, and well-cited scientific texts. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Kadirova Z.Z. Academic Writing. A Textbook.– Termez: "Globe Edit", 2024. –110 pp. 2. Korotkina I.B. Academic Writing: Process, Product and Context. A Textbook for Higher Education Institutions. Moscow: Yurayt, 2021 3. Kuvshinskaya Y.M., Zevakhina N.A., Akhapkina Y.E., Gordienko E.N. Academic Writing: From Research to Text. Textbook and Practical for Higher Education Institutions. Moscow: Yurayt, 2022 4. Korotkina I.B. A Model for Teaching Academic Writing. A Textbook for Universities. Moscow: Yurayt, 2021 5. M. Aminov, A. Madvaliev, N. Mahkamov, N. Mahmudov. Business Management (Practical Guide). Tashkent, 2009. 6. Shamsimuhamedov I. Explanatory Dictionary of Diplomatic Terms. T.: "Donishmand ziyosi", 2021. 7. R.S.Samarov. New Explanatory Dictionary of Military Terms. T. "Innovatsiya-Ziyo". 2020. 8. A. Madvaliev, N. Mahkamov, Sh. Ko'chimov, Z. Mirakhmedova and others. T. A Concise Explanatory Dictionary of Social-Political Terms. "Shaffaf". 2020.
Responsible Lecturer	Z. Kadirova

DATABASE

Field	Description
Module Title	DATABASE
Module Code	MAB1306
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 90 hours (44 hours Lecture, 46 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Database is to develop students' theoretical knowledge and practical skills in database design, creation, and management. Within the course, data models and relational databases are studied. The SQL language and data processing methods are also studied. Students develop skills in the effective use of database management systems. They also develop skills in applying the principles of information storage and protection. Students acquire the competencies to use databases in the creation of modern information systems. They further acquire the ability to apply these competencies to practical problems.
Module Learning Outcomes	Knowledge: LO1. Provides information about databases, demonstrates the role of information technologies in individuals' lives and in the development of society, and reveals the possibilities of databases. LO2. Develops practical skills in creating programmes using modern programming languages. LO3. Develops practical skills in working with a computer database and on the Internet. LO4. Develops an understanding of when and for what purposes to use a database. LO5. Develops theoretical knowledge of the structure and development principles of modern information systems. LO6. Develops knowledge of the use and development of databases. Skills:

	LO7. Acquires the skills to design databases and develop their structures based on practical requirements. LO8. Applies practical skills in creating, searching, editing and managing data using the SQL language. LO9. Is qualified to work with and use modern database management systems effectively. LO10. Applies database normalisation and data integrity techniques in practice. LO11. Has skills in ensuring data security, managing user permissions, and creating backups. LO12. Creates databases for information systems in various fields, integrates them with software applications, and applies them to solve practical problems. Competencies: LO13. Possesses professional competencies in analysing, designing and creating databases for various subject areas. LO14. Acquires the competence to develop and maintain information systems using modern database management systems. LO15. Carries out professional activities to ensure the efficiency, integrity and security of databases at the enterprise level. LO16. Has the competence to process and analyse data, and to integrate it with software applications, using SQL and other relevant technologies. LO17. Demonstrates competencies in teamwork, independent decision-making and the effective resolution of problems encountered in professional practice using modern information and communication technologies.
Content	Lectures The Concept of a Database and the Capabilities of Microsoft Access. The notions of data, information, and databases are examined, together with types of database management systems. The purpose, structure, and main capabilities of Microsoft Access are studied. The advantages and limitations of desktop DBMS are considered. Basics of Creating and Managing Tables in Microsoft Access. A new database and its tables are

created, together with the study of data types and field properties. Primary keys and indexes are examined. Data in tables is entered, edited, sorted, and filtered.

Queries and Their Role in Data Processing. The purpose and types of queries, including select and action queries, are studied. Queries are created using the Design View and the Query Wizard. Criteria, calculated fields, and parameters are used in queries.

Reports and Methods of Data Presentation. The purpose and structure of reports are studied. Reports are created using the Report Wizard and Design View. Data is grouped, sorted, and summarized. Reports are formatted for printing and presentation.

Forms and Organising the User Interface. The purpose of forms as a user interface tool is examined. Forms are created using the Form Wizard and Design View, together with form controls such as text boxes, combo boxes, and buttons. Subforms and navigation between records are studied.

Relationships Between Tables and the Fundamentals of the Relational Model. The concept of the relational data model is studied. Types of relationships, including one-to-one, one-to-many, and many-to-many, are examined, together with referential integrity. The relationships diagram is built and edited.

Data Import and Export Technologies. Data is imported from Excel, text files, and other sources. Database objects are exported to various formats. External data sources are linked to Access tables.

Macros and Process Automation in Databases. The concept and purpose of macros are studied. Macros are created and edited. Routine tasks, such as opening forms, running queries, and validating data, are automated using macros.

Data Exchange and Integration with External Sources. Access databases are connected to other DBMS and applications. Methods of data exchange, including ODBC and OLE DB, are studied. Integration with Microsoft Excel and Word is examined.

Fundamentals of Database Design and Normalisation. The stages of database design, including conceptual, logical, and physical design, are

studied. Functional dependencies are examined. Normal forms, including 1NF, 2NF, and 3NF, are studied together with their purpose in eliminating data redundancy.

Artificial Intelligence Technologies and Data Analysis Methods. Basic concepts of artificial intelligence are studied. Methods of data analysis and knowledge extraction are examined. Machine learning and data mining approaches applied to databases are considered.

The Concept of Expert Systems and Their Operating Principles. The definition and structure of expert systems are studied. The knowledge base, inference engine, and user interface are examined. Areas of application of expert systems are considered.

Software and Classification of Expert Systems. Expert systems are classified by task type, including diagnostic, predictive, and planning systems. An overview of expert system development tools is presented. Expert system development shells are also examined.

Intelligent Expert Systems and Knowledge Bases. Methods of knowledge representation, including rules, frames, and semantic networks, are studied. Building and maintaining a knowledge base is examined. Inference mechanisms, including forward and backward chaining, are considered.

Database Security and Protection Methods. Threats to database security are studied. Access control and user authentication are examined. Data encryption, backup, and recovery methods are considered.

SQL Injection Attacks and Methods of Protection. The mechanism of SQL injection attacks is studied. Vulnerabilities in web applications working with databases are examined. Methods of prevention, including parameterized queries, input validation, and access restriction, are considered.

Multi-Set, Grouping and Aggregate Functions in SQL. Set operations, including UNION, INTERSECT, and EXCEPT, are studied. The GROUP BY and HAVING clauses are examined. Aggregate functions,

including COUNT, SUM, AVG, MIN, and MAX, are considered.

Database Applications and DBMSs. An overview of modern DBMSs, including MySQL, PostgreSQL, Oracle, and SQL Server, is presented. The architecture of database applications is studied. Client-server interaction models are examined.

The SELECT Operator of the SQL Language and Its Capabilities. The syntax and structure of the SELECT statement are studied. Filtering using WHERE and sorting using ORDER BY are examined. Tables are joined using different JOIN types.

Functions of the SQL Language and Their Use. Scalar functions, including string, numeric, and date/time functions, are studied. User-defined functions are examined. Functions are used within queries.

Subqueries and Their Use. The concept of subqueries is studied. Types of subqueries, including nested and correlated subqueries, are examined. Subqueries are used in SELECT, WHERE, and FROM clauses.

MySQL Database Management System and Its Installation. An overview of MySQL architecture is presented. MySQL is installed and configured. Basic administration tasks, including creating databases, users, and setting privileges, are studied.

Practical lessons

Working with the Microsoft Access Application Interface and Settings. The Access ribbon, navigation pane, and workspace are examined. Application options and the working environment are configured. A new database file is created and saved.

Working with Tables in Microsoft Access. Tables are created in Design View and Datasheet View. Field data types, properties, and primary keys are set. Data in tables is entered, edited, and validated.

Working with Queries in Microsoft Access. Select queries are built using the Query Wizard and Design View. Criteria and calculated fields are applied. Query results are sorted and filtered.

Creating Reports in Microsoft Access. Reports are generated using the Report Wizard. Data in reports is

grouped, sorted, and summarized. A report is formatted for printing and presentation.

Working with Forms in Microsoft Access. Forms are created using the Form Wizard and Design View. Controls such as text boxes, combo boxes, and buttons are added. A form is linked to its underlying table or query.

Linking Tables in Microsoft Access. Relationships between tables are established in the Relationships window. Referential integrity is enforced. Relationships are tested through linked queries and forms.

Performing Import and Export Tasks in Microsoft Access. Data is imported from Excel and text files into Access tables. Access data is exported to Excel and other formats. Data integrity is verified after import and export.

Working with Macros in Microsoft Access. Simple macros are created to automate actions. Macros are attached to form controls and events. Macro execution is tested and debugged.

Performing Import and Export Tasks in Microsoft Access. External data sources are linked to an Access database. Recurring import and export operations are automated. Data format and compatibility issues are handled.

Database Design: Constraints and Simple Forms. Field-level and table-level constraints are defined. Validation rules are applied to ensure data accuracy. Simple data-entry forms are designed based on the database structure.

Data Analysis with Artificial Intelligence. Basic AI-based methods are applied to analyze datasets. Patterns and trends in data are identified. AI-assisted analysis is compared with traditional query-based analysis.

Expert Systems and Their Operation. The structure of a simple expert system is examined. Work is carried out with a knowledge base and inference rules. The way the inference engine produces conclusions is observed.

Classification of Expert System Software. Different types of expert system tools and shells are reviewed.

Software is classified by application area, including diagnostic, advisory, and planning systems. Features of selected expert system platforms are compared.

Expert Systems with Intelligent Information. A small knowledge base is built for a chosen domain. Knowledge is represented using rules or frames. The system's reasoning is tested on sample queries.

Database Protection Methods. User accounts and access permissions are configured. Password protection and encryption options are applied. Database backup and recovery procedures are performed.

SQL Injection Attack and Its Working Process. How an SQL injection attack is carried out is demonstrated. Vulnerable points in a sample application are identified. Protective measures, such as parameterized queries and input validation, are applied.

Multiple Sets, Grouping and Aggregation. SQL queries are written using UNION, INTERSECT, and EXCEPT. GROUP BY and HAVING clauses are applied. Aggregate functions such as COUNT, SUM, AVG, MIN, and MAX are used.

Database Applications. A database is connected from an application environment. The architecture of client-server database applications is reviewed. Basic CRUD operations are performed through an application interface.

SQL – SELECT Statement Capabilities. SELECT queries are practiced with filtering, sorting, and joins. Multiple tables are combined using different JOIN types. Queries are refined with complex conditions.

Functions of the SQL Language. Built-in string, numeric, and date/time functions are used. Functions are applied within SELECT and WHERE clauses. Simple user-defined functions are created and tested.

SQL – A Query Within a Query. Nested subqueries are written within SELECT, WHERE, and FROM clauses. Correlated subqueries are used for row-by-row evaluation. Subquery results are compared with JOIN-based alternatives.

Installation and Configuration of MySQL and PHP. MySQL server and a PHP environment are installed.

	Server settings are configured and the connection is verified. The setup is tested with a simple PHP script connecting to MySQL. Managing MySQL Using PHP. PHP scripts are written to connect to a MySQL database. Basic database operations, including insert, update, delete, and select, are performed via PHP. Query results are displayed on a web page. Self-study Modern Database Systems and Their Development Trends. Creating and Managing a Database in Microsoft Access. Relational Database and Its Practical Application. Stages of Database Design and Normalisation. The SELECT Operator of the SQL Language and Data Processing Capabilities. Application of Aggregate Functions and the GROUP BY Operator in SQL. Subqueries and Constructing Complex SQL Queries. Relationships Between Tables (One-to-One, One-to-Many, Many-to-Many). Methods for Ensuring Data Security in a Database. SQL Injection Attacks and Their Defence Technologies. Creating Forms and Reports in the Database. Data Import and Export Technologies (CSV, Excel, XML). MySQL Database Management System and Its Features. Transactions and ACID Properties in a Database. Artificial Intelligence Technologies and Database Integration.
Teaching Methods	The module is delivered through interactive lectures, problem-based and case-study learning, and laboratory practicals. Hands-on database design exercises, computer simulations, seminars, group discussions, and project-based activities are also used. Students design, build, and query databases using Microsoft Access and MySQL. Through these activities they develop analytical, technical, and

	problem-solving skills. Independent learning and digital educational resources further promote critical thinking, professional communication, and the application of modern database and information security concepts.
Prerequisites	Data Structures and Algorithms Programming 1,2
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, practical database exercises, seminar presentations, and the successful completion of independent study assignments, including database design, creating and managing database tables, developing SQL queries, database normalization, creating reports and forms, and database security tasks. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, including a database design project, an SQL query portfolio, a report on Database Management Systems, a presentation on modern database technologies, and a case study on database security and data integrity. The final assessment is a written examination consisting of theoretical questions and practical problem-solving tasks, designed to assess students' understanding of database fundamentals, relational database design, SQL programming, database administration, and their ability to apply database management principles in real-world information systems. Successful completion of all independent study assignments is a mandatory and integral component of the module assessment.
Recommended Literature	1. J.D. Saidov – Database. 2022. Textbook, 164 pp. 2. Adizova Z.M., Avezov A.A., Norova F.F. – Database. A teaching and methodological guide. 2022. 99 pp.

	3. Adizova Z.M., Avezov A.A., Norova F.F. – Database. Sharq Publishing House. 2007. 198 pp. 4. B.T. Ergashev, K.Z. Abidov – Database Management Systems. Textbook. "Ilm ziyo" Publishing House. 2020 5. Serge Abiteboul, Richard Hull and Viktor Vianu: Foundations of Database Systems, Addison-Wesley, 1995. 6. Raghu Ramakrishnan and Johannes Gehrke: Database Management Systems (Third Edition). McGraw-Hill, 2003. 7. Narzullaev U.X., Qarshiev A.B., Boynazarov I.M. Database. //Textbook. Tashkent: Tafakkur Publishing House, 2013. – 192 pp. 8. A. Sattorov - Database Management System. Textbook. "Fan and Technology" Publishing House. 2006
Responsible Lecturer	Tursunov Farxod Bakhodir o'g'li.

FOUNDATIONS OF CYBERSECURITY

Field	Description
Module Title	FOUNDATIONS OF CYBERSECURITY
Module Code	KIA1406
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 90 hours (46 hours Lecture, 44 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Foundations of Cybersecurity is to develop students' theoretical knowledge and practical skills in protecting computer systems and their data from threats that compromise confidentiality, integrity and availability. The Module introduces the fundamental concepts of cybersecurity, including the principles of confidentiality, integrity and availability, and situates them within the broader context of cybercrime, cyber law and cybernetics. It covers the fundamentals of cryptography and encryption algorithms, together with

	methods for ensuring data integrity. Students examine the principal types of cyber-attacks, malicious software and network-based threats, and the access-control, identification and authentication mechanisms used to counter them. The Module further develops competence in risk analysis and risk management, and in the development and implementation of an organizational cybersecurity policy. Practical emphasis is placed on modern protection technologies such as data encryption and archiving tools, physical data protection and secure electronic commerce practices. Throughout, the primary focus is placed on malicious rather than accidental threats to information systems.
Module Learning Outcomes	Knowledge: LO1. Knows the fundamental concepts of the field of cybersecurity, and the principles of confidentiality, integrity and availability. LO2. Understands the modern concepts of cybercrime, cyber law, cybernetics and information protection. LO3. Understands the fundamentals of cryptography, symmetric and public-key cryptosystems, and methods for ensuring data integrity. LO4. Has theoretical knowledge of cyber-attack types, malicious software and network-based threats. Skills: LO5. Encrypts and decrypts data using simple and symmetric encryption algorithms. LO6. Protects against cyberattacks and uses antivirus software and security tools. LO7. Works with identification, authentication and access control tools. LO8. Applies data backup, physical protection and secure archiving methods. Competencies: LO9. Implements security settings on operating systems (Windows/Linux). LO10. Analyses incidents of cyber-attacks and develops measures to mitigate them. LO11. Develops a cybersecurity policy and implements it within the organisation.

	LO12. Applies risk analysis and risk management methodology in professional practice. LO13. Independently studies and implements modern technologies and threats in the field of cybersecurity.
Content	Lectures Basic Concepts of Cybersecurity. The definition and scope of cybersecurity as a field of study are examined. The fundamental terminology and principles underlying the protection of digital systems are studied. The relationship between cybersecurity and other areas of information technology, along with its growing importance, is considered. The Concepts of Confidentiality, Integrity and Availability. The three core principles of information security — confidentiality, integrity, and availability — are examined. The mechanisms used to ensure each of these principles in practice are studied. Their combined role in forming the foundation of a secure information system, along with real-world applications, is considered. Cybercrime, Cyber Laws and Cybernetics. The nature and classification of cybercrime are examined. The legal frameworks and regulations governing cyberspace are studied. The relationship between cybernetics as a scientific discipline and modern cybersecurity practices, along with their evolution, is considered. Cyber Security Architecture and Strategy. The principles of designing a secure system architecture are examined. The components and layers of an effective cybersecurity strategy are studied. The alignment of architectural design with organizational goals and threat models, along with best practices, is considered. Information Protection, Information Security and Its Modern Concept. The distinction between information protection and information security is examined. The evolution of information security concepts in response to technological change is studied. The modern, holistic approach to safeguarding information assets, along with its practical implications, is considered.

The Human Factor in Cybersecurity. The role of human behavior in creating and mitigating security vulnerabilities is examined. Common social engineering tactics and their psychological basis are studied. Methods for improving user awareness and organizational security culture, along with their effectiveness, are considered.

Basic Concepts of Cryptography. The fundamental principles and terminology of cryptography are examined. The historical development of cryptographic methods is studied. The role of cryptography in ensuring secure communication and data protection, along with its modern applications, is considered.

Simple Encryption Algorithms. The structure and logic of classical, simple encryption algorithms are examined. Methods of encoding and decoding messages using these algorithms are studied. Their limitations and vulnerabilities in the context of modern cryptanalysis, along with their historical significance, are considered.

Symmetric Cryptographic Algorithms. The principles of symmetric-key encryption are examined. Widely used symmetric algorithms and their operational mechanisms are studied. Their advantages, limitations, and areas of practical application, along with key management challenges, are considered.

Public-Key Cryptosystems. The concept and mathematical foundation of public-key cryptography are examined. The functioning of key pairs in encryption and digital signatures is studied. The advantages of asymmetric systems over symmetric methods, along with their practical applications, are considered.

Methods for Ensuring Data Integrity. The concept of data integrity and its importance in information systems is examined. Techniques such as hashing and checksums used to verify data integrity are studied. Their role in detecting unauthorized data modification, along with practical implementation methods, is considered.

Viruses, Trojan Horses, Worms and Rootkits. The characteristics and behavior of various types of malicious software are examined. The methods by which viruses, Trojan horses, worms, and rootkits infect and spread across systems are studied. Strategies for detecting and removing such threats, along with preventive measures, are considered.

Types of Cyberattacks and How to Protect Against Them. The classification of common cyberattack types is examined. The techniques and objectives behind each type of attack are studied. Defensive strategies and best practices for protecting systems against these threats, along with their effectiveness, are considered.

Fundamentals of the WWW and Client-Side Attacks. The basic architecture and functioning of the World Wide Web are examined. Common client-side attack methods targeting web browsers and applications are studied. Techniques for mitigating client-side vulnerabilities, along with their practical significance, are considered.

Transport Layer Security, DNS Attacks. The role and mechanisms of Transport Layer Security (TLS) in securing communications are examined. Common types of DNS-based attacks and their impact on network security are studied. Methods for strengthening transport-layer and DNS security, along with their practical application, are considered.

Network Security Issues. The fundamental challenges and vulnerabilities present in network systems are examined. Common sources of network security breaches are studied. Approaches to identifying and addressing these issues, along with their significance for organizational security, are considered.

Detecting Network-Based Threats. The methods and tools used to detect network-based threats are examined. The characteristics of common threat indicators and anomalous network behavior are studied. The role of monitoring and detection systems in early threat identification, along with their practical implementation, is considered.

Access Control Methods. The principles and models of access control are examined. Various methods of authentication and authorization used to restrict system access are studied. Their role in preventing unauthorized access and protecting sensitive resources, along with practical applications, is considered.

Risk and Risk Management. The concept of risk within the context of cybersecurity is examined. The processes of risk identification, assessment, and mitigation are studied. The role of risk management frameworks in guiding organizational security decisions, along with their practical implementation, is considered.

Data Encryption and Archiving Tools. The functions and features of common data encryption and archiving tools are examined. Methods of securely storing and compressing sensitive data are studied. Their practical application in protecting data confidentiality and integrity, along with best practices, is considered.

Cybersecurity Policy and Its Implementation. The purpose and structure of organizational cybersecurity policies are examined. The process of developing and implementing effective security policies is studied. Challenges in policy enforcement and compliance, along with strategies for successful implementation, are considered.

Physical Protection of Data. The importance of physical security measures in protecting data infrastructure is examined. Common methods of safeguarding hardware, storage devices, and facilities are studied. The integration of physical protection with broader cybersecurity strategies, along with practical applications, is considered.

Electronic Commerce Security. The specific security challenges associated with electronic commerce are examined. Methods for protecting online transactions, payment systems, and customer data are studied. The role of trust and regulatory compliance in ensuring secure e-commerce operations, along with their significance, is considered.

Practical lessons

Software Tools in Cybersecurity. The main categories of software tools used in cybersecurity are examined. The functions and practical applications of these tools in detecting and preventing threats are studied. Their role in supporting overall organizational security infrastructure, along with selection criteria, is considered.

Working on Confidentiality, Integrity and Availability. Practical methods of implementing confidentiality, integrity, and availability in information systems are examined. Techniques for testing and verifying each of these principles are studied. Their combined application in maintaining a secure and reliable system, along with common challenges, is considered.

Practical Methods of Information Protection. Applied techniques for protecting information from unauthorized access and loss are examined. The implementation of protective measures across different types of information systems is studied. Their effectiveness in real-world scenarios, along with best practices, is considered.

Scientific Analysis of the Human Factor in Cybersecurity. Methods for analyzing human behavior as a factor in security vulnerabilities are examined. Case studies and research findings on social engineering and human error are studied. Approaches to reducing human-related risks through training and policy, along with their outcomes, are considered.

Cybercrime, Cyber-Laws and Cybernetics. Practical cases and examples of cybercrime are examined. The application of cyber-laws in addressing such incidents is studied. The connection between cybernetic principles and legal frameworks in cyberspace, along with their practical relevance, is considered.

Threats, Their Types and Protection Against Them. The classification of various cybersecurity threats is examined. The characteristics and methods of operation of each threat type are studied. Practical protective measures and countermeasures against

these threats, along with their effectiveness, are considered.

Developing a Cybersecurity Strategy. The process of designing a comprehensive cybersecurity strategy is examined. Key components and stages involved in strategy development are studied. The alignment of the strategy with organizational needs and risk levels, along with practical implementation steps, is considered.

Security Policy for Mobile Devices. The specific security risks associated with mobile devices are examined. The development and implementation of mobile device security policies are studied. Practical measures for protecting data and access on mobile platforms, along with their effectiveness, are considered.

Analysis of Cybercrime Worldwide. Global trends and patterns in cybercrime are examined. Case studies of major cybercrime incidents across different regions are studied. The international response and cooperation in combating cybercrime, along with lessons learned, are considered.

Working with Symmetric Cryptographic Algorithms. The practical implementation of symmetric cryptographic algorithms is examined. Methods of encrypting and decrypting data using these algorithms are studied. Their performance, efficiency, and suitability for different applications, along with practical exercises, are considered.

Data Backup and Replication. The principles and methods of data backup are examined. Techniques for data replication to ensure availability and disaster recovery are studied. Their role in maintaining business continuity and data resilience, along with best practices, is considered.

Methods for Ensuring Data Integrity. Practical techniques for verifying and maintaining data integrity are examined. The application of hashing and checksum methods in real-world systems is studied. Their effectiveness in detecting unauthorized modifications, along with implementation challenges, is considered.

Methods of Access Control. Practical models and mechanisms of access control are examined. The implementation of role-based, discretionary, and mandatory access control methods is studied. Their application in restricting and managing user permissions, along with practical scenarios, is considered.

Working with Identification and Authentication Tools. The functions of common identification and authentication tools are examined. Practical methods of verifying user identity, including passwords, biometrics, and tokens, are studied. Their role in strengthening system security, along with implementation considerations, is considered.

Working with Physical Data Protection Methods. Practical measures for physically securing data storage and infrastructure are examined. Techniques such as access restriction, surveillance, and environmental controls are studied. Their integration with digital security measures, along with practical applications, is considered.

Encrypting Data Using Simple Encryption Algorithms. The practical application of simple encryption algorithms is examined. Step-by-step methods of encrypting and decrypting sample data are studied. The limitations of these algorithms in modern security contexts, along with their educational value, are considered.

Working with Data Encryption and Archiving Tools. The functionality of common data encryption and archiving software is examined. Practical methods of securing and compressing data using these tools are studied. Their role in protecting data confidentiality during storage and transfer, along with best practices, is considered.

Familiarisation with Network Security Issues. The common vulnerabilities and challenges present in network systems are examined. Practical methods of identifying network security weaknesses are studied. Their implications for overall system security, along with introductory protective measures, are considered.

Types of Cyberattacks and How to Protect Against Them. The classification and mechanisms of common cyberattacks are examined. Practical case studies of real-world cyberattack incidents are studied. Effective protective strategies and countermeasures against these attacks, along with their implementation, are considered.

Recommendations and Suggestions for Protecting the Operating System. Common vulnerabilities within operating systems are examined. Practical recommendations for strengthening operating system security are studied. The implementation of protective configurations and updates, along with their effectiveness, is considered.

Computer Viruses and Methods of Virus Protection. The characteristics and behavior of computer viruses are examined. Practical methods of detecting and removing viruses are studied. Preventive measures and antivirus tools used for protection, along with their effectiveness, are considered.

Electronic Commerce Security. Practical security challenges in electronic commerce platforms are examined. Methods for protecting online transactions and customer data in real systems are studied. The application of security protocols and compliance standards in e-commerce, along with their significance, is considered.

Self-study

Fundamental Concepts of Cybersecurity and Analysis of Modern Threats.

Access Control Models: A Comparison of DAC, MAC, and RBAC Systems.

DoS and DDoS Attacks: Mechanism of Operation and Protection Methods.

DNS Attacks and Link Layer Protection: Technologies and Examples.

The Human Factor in Security: Social Engineering Threats.

User Authentication and Protection Against Phishing Attacks.

	Symmetric Encryption Algorithms: AES and DES Analysis. Comparative Analysis of RSA and ECC Algorithms in Asymmetric Encryption. Practical Application of Digital Signatures and MAC Mechanisms. Differences Between Identification, Authentication and Authorisation. How the TOR Network and Anonymity Protocols Work. Operating System Security: User Roles and Permission Systems. File System Security and Defence Against Memory Attacks. Analysis of Computer Viruses, Trojans, Worms and Rootkits. Attacks and Defences via Web Browsers. Types of Cybercrime and Legislation in the Republic of Uzbekistan. Information Encryption and Secure Data Archiving Methods. The Most Common Problems and Solutions in Network Security. Types of Cyberattacks: APT, Zero-Day and Ransomware Attacks. Risk Analysis and Risk Management in Information Security. Antivirus Programs: Functions and Modern Protection Methods. Payment and Transaction Encryption in E-Commerce. Developing a Secure Password Policy and Reducing Errors. Software Vulnerabilities and Attacks Exploiting Them. Security and Data Protection on Mobile Devices. Measures Against SQL Injection and XSS Attacks in Web Applications. Cryptographic Protocols: Analysis and Application of SSL, TLS, and HTTPS. Physical Security: Ensuring Security in Server Rooms.
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	User Information Security Training Programmes. Threat Detection and Monitoring Using Artificial Intelligence.
Teaching Methods	The Module is delivered through interactive lectures combined with laboratory-based practical sessions in which students work directly with software-based security tools. Practical exercises include the implementation of encryption algorithms, access-control simulations, malware analysis tasks and network threat-detection activities. Case-based and project-based learning is used to connect theoretical material to realistic scenarios, including the preparation of a cybersecurity strategy proposal and a mobile device security policy draft. Independent study is structured around a comparative analytical report on symmetric and asymmetric cryptography and an essay on global cybercrime trends, both of which are presented and discussed in class. Seminar presentations and a video presentation on a current cyberattack case are used to develop students' analytical and communication skills. Throughout the Module, this combination of guided instruction and independent investigation develops both the technical competence and the risk-based decision-making ability required in professional cybersecurity practice.
Prerequisites	Computer Networks Design of Information Systems Programming 1,2
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). Continuous assessment evaluates students' classroom participation, engagement in laboratory sessions, performance in practical exercises, seminar presentations, and the successful completion of independent study assignments, including exercises with software-based security tools, implementation of encryption algorithms, access control simulations, malware analysis tasks, and network threat detection activities. Midterm assessment consists of written

	assignments, short in-class quizzes and tests, and independent project work, including a comparative report on symmetric/asymmetric cryptography, a cybersecurity strategy proposal, a video presentation on a current cyberattack case, a mobile device security policy draft, and an essay on global cybercrime trends. The final assessment is a written examination consisting of theoretical questions and practical problem-solving tasks, designed to assess students' knowledge of cybersecurity and information protection, their analytical skills, and their ability to independently apply security principles to real-world scenarios. The timely and high-quality completion of all independent study assignments is a mandatory component of the module assessment.
Recommended Literature	1. Cybersecurity Fundamentals Study Guide. - ISACA, 2015. - 196 p. 2. Stallings W. Cryptography and Network Security: Principles and Practice. 8th ed. - Pearson, 2023. 3. Nesterov S.A. Fundamentals of Information Security. A Textbook. 3rd ed. - St. Petersburg: Лань, 2017. 4. G'aniyev S.K., G'aniyev A.A., Xudoyqulov Z.T. "Fundamentals of Cybersecurity". Textbook. — Tashkent, 2021. 5. Masalkov A.S. Features of Cybercrimes: Tools for E ssion and Protection of Information. Moscow: DMK Press, 2018.
Responsible Lecturer	Mengnarov Adham Ergashevich, teacher.

DATA STRUCTURES AND ALGORITHMS

Field	Description
Module Title	DATA STRUCTURES AND ALGORITHMS
Module Code	MTA1204
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	II semester

Module objective	The objective of the Module Data Structures and Algorithms is to teach methods for the efficient organisation, storage, processing, and utilisation of data in computer memory. The Module develops students' knowledge, skills, and abilities in selecting and creating data structures such as arrays, linked lists, stacks, and queues. It also develops these abilities for trees, graphs, and hashing structures. Students learn to use these data structures to develop efficient algorithms. As a result, students are able to optimise the speed of software. They are also able to optimise its memory consumption. Finally, students are able to optimise the overall efficiency of software.
Module Learning Outcomes	Knowledge: LO1. Provides information on data structures and algorithms, and demonstrates the role of information technology in the lives of individuals and the development of society. LO2. Develops practical skills in creating programmes using modern programming languages. LO3. Develops an understanding of when and for what purposes data structures and algorithms are used. LO4. Develops knowledge of the application and development of data structures and algorithms. LO5. Becomes familiar with methods for ensuring information security in data processing environments. Skills: LO6. Studies the theoretical foundations of using personal computers in data structures and algorithms processes and the methods for applying them to various areas of data structures. LO7. Studies methods for the effective use of data structures and algorithms in control systems. LO8. Learns the skills to work with Internet systems and data structure technologies. LO9. Studies the development trends of data structures and algorithms and their characterisation. LO10. Becomes familiar with the methods for ensuring security when working with data structures and their functional capabilities. Competencies:

	LO11. Analyses, selects and applies data structures such as arrays, linked lists, stacks, queues, trees, graphs and hash tables effectively when solving various practical problems. LO12. Programmes, tests and documents data structures and algorithms using modern programming languages and development environments. LO13. Develops the competence to formally describe practical and scientific problems, express them as algorithmic models, and design solutions based on appropriate data structures.
Content	Lectures Formally Describing Practical and Scientific Problems as Algorithmic Models. The concept of algorithmic problem solving is studied, together with the formal description of practical and scientific problems. Algorithmic modeling techniques are examined. Appropriate data structures for efficient solutions are selected. The importance of computational thinking and algorithm design is considered. Introduction to the Subject of Data Structures and Algorithms. The objectives and scope of data structures and algorithms are studied. The importance of algorithms in software development is examined. Basic concepts of data organization and processing, together with algorithm design principles, are considered. The relationship between data structures and algorithm efficiency is also considered. General Overviews of Data Structures. The concept and classification of data structures are studied, including primitive and non-primitive data structures. Static and dynamic data structures are examined. Selection criteria for data structures are considered. Applications of data structures in computer science are also considered. Recursion and Its Application in Programming. The concept and principles of recursion are studied. Recursive functions and recursive problem solving are examined. Recursion and iteration are compared, together with the advantages and limitations of recursive algorithms. Practical applications of recursion in programming are considered.

Data Search Algorithms. The concept of searching algorithms is studied, including linear search and binary search algorithms. Search efficiency and time complexity are examined. Searching techniques are compared. Practical applications of search algorithms are considered.

Data Hashing Algorithms. The concept of hashing and hash functions is studied. Hash tables and collision resolution techniques are examined, including open addressing and chaining methods. Performance analysis of hashing algorithms is considered. Applications of hashing in data retrieval are also considered.

Basic and Extended Sorting Algorithms. The importance of sorting algorithms is studied. Bubble, Selection, and Insertion Sort algorithms are examined. Merge Sort, Quick Sort, and Heap Sort algorithms are also examined. Sorting methods are compared based on time and space complexity, and their practical applications are considered.

Linear Data Structures. The concept and characteristics of linear data structures are studied, including arrays, linked lists, stacks, and queues. Operations on linear data structures are examined. Advantages and limitations of different linear structures are considered. Their applications in programming are also considered.

Linearly Linked Lists. The structure and properties of linked lists are studied, including singly, doubly, and circular linked lists. Insertion, deletion, and traversal operations are examined. Memory management in linked lists is considered. Practical applications of linked list structures are also considered.

Sequential Data Structures (Lists, Stacks, Queues). The characteristics of sequential data structures are studied, together with list implementation methods. Stack operations and applications are examined. Queue types and operations are considered. Real-world applications of sequential data structures are also considered.

Tree Data Structures, Sets, and Priority Queues. The concept and properties of tree structures are studied. Set data structures and their operations are

examined. Priority queues and their implementation are considered, together with tree traversal methods. Applications of trees and priority queues are also considered.

Binary Search Tree. The structure and properties of binary search trees are studied. Insertion, deletion, and search operations are examined. Tree balancing concepts are considered. Performance analysis of binary search trees and their practical applications in data management are also considered.

Binary Trees in a Heap Tree Structure. The concept and characteristics of heap trees are studied, including max-heap and min-heap structures. Heap construction and maintenance algorithms are examined. Heap operations and priority queue implementation are considered. Applications of heap structures are also considered.

Algorithms for Working with Graphs. The concept and representation of graphs are studied, including directed and undirected graphs. Graph data structures and operations are examined. Shortest path and minimum spanning tree algorithms are considered. Applications of graph algorithms are also considered.

Graph Traversal Algorithms. The principles of graph traversal are studied. The Depth-First Search (DFS) algorithm is examined. The Breadth-First Search (BFS) algorithm is also examined. Graph traversal techniques are compared, and their applications in problem solving are considered.

NP-Hardness. The concept of computational complexity is studied. Complexity classes P, NP, NP-Complete, and NP-Hard are examined. Polynomial-time reductions and examples of NP-Hard problems are considered. The significance of computational complexity in algorithm design is also considered.

Practical lessons

Designing Programs to Process Various Data Types. Programs for processing numeric, character, and structured data are developed. Suitable data structures are selected. Data processing algorithms are implemented, tested, and debugged. Their performance is evaluated.

Data Abstraction, Classical ADTs. Abstract data types are implemented, including stack, queue, list, and tree ADTs. Data and operations are encapsulated. ADT implementations are tested. Practical applications of data abstraction are examined.

Algorithmic Strategies: Divide and Conquer, Greediness, Backtracking, and Dynamic Programming. Divide-and-conquer algorithms are implemented. Greedy and backtracking solutions are developed. Dynamic programming algorithms are programmed. Algorithmic strategies are compared through solving practical optimization problems.

Development of Data Search Algorithms and Programmes. Efficient search algorithms are programmed and their performance is compared. Searching techniques are optimized. Search programs are tested with large datasets. Their computational complexity is analyzed.

Hashing and Dictionaries. Dictionary data structures are implemented and hash-based applications are programmed. Collision handling techniques are developed. Dictionary operations are tested. The performance of hashing methods is analyzed.

Development of Data Sorting Algorithms and Programmes. Sorting algorithms are programmed for different datasets. Algorithm efficiency is compared and sorting methods are optimized. Sorting applications are tested. Their execution time is analyzed.

Designing Algorithms and Programs for Processing Linear Data Structures. Algorithms for arrays and lists are implemented. Insertion, deletion, and update operations are programmed. Linear data processing programs are tested and their performance is compared across different approaches.

Developing Algorithms and Programs for Working with Linearly Linked Lists. Singly, doubly, and circular linked lists are programmed. Insertion and deletion algorithms are developed. Traversal and search operations are implemented. Linked list

programs are tested and their memory management is analyzed.

Designing Algorithms and Programs for Working with Stacks, Queues and Deques. Stack, queue, and deque operations are implemented, together with practical applications. Data processing algorithms are tested. The performance of sequential structures is compared. Problems are solved using abstract data types.

Designing Algorithms and Programs for Processing Tree-Like Data Structures. Tree construction and traversal algorithms are programmed. Tree operations are implemented. Tree-processing applications are tested and their performance is evaluated. Practical applications of tree structures are examined.

Algorithms for Working with Binary Trees. Binary tree operations are implemented and recursive tree algorithms are programmed. Tree traversal techniques are applied. Binary tree applications are tested and their algorithm performance is analyzed.

Algorithms for Working with Heap-Tree-Like Binary Trees. Heap construction algorithms are programmed. Insertion and deletion operations are implemented. Priority queue applications are developed. Heap-based programs are tested and their performance is analyzed.

Graphs and Graph Algorithms. Graph representations are implemented and graph processing algorithms are programmed. Practical applications of graph theory are examined. Graph-based solutions are tested and their algorithm efficiency is analyzed.

Development of Graph Traversal Algorithms. DFS and BFS algorithms are programmed and graph exploration methods are developed. Traversal algorithms are tested on different graph structures and their performance is compared. Practical problems are solved using graph traversal.

Methods for Logical Representation of a Given Graph. Adjacency matrices and adjacency lists are implemented and compared. Graph construction algorithms are programmed. Graph models are

	tested, and appropriate representations are selected for different applications. Self-study Binary Representation of Data in Computer Memory. Representation of Data Types in Programming Systems. Representation of Simple Data Types on a Machine. Data Collection Methods and Methods for Accessing Data Elements. Methods for Data Search and Algorithm Evaluation. Classification of Data Sorting Methods. Analysis of Data Sorting Algorithms. The Nature and Purpose of Data Structures: Queue, Stack and Deque. Prioritised Queuing Characteristics. Comparative Descriptions of Methods for Organising Strings in Computer Memory. List View of Data in Computer Memory. Dynamically Allocated Memory Management. Forms of Representing Graphs in Computer Memory. Tree Data Structures. Binary and Balanced Trees.
Teaching Methods	The module is delivered through interactive lectures, practical laboratory sessions, and problem-based and project-based learning. Algorithm design exercises, coding assignments, seminars, group discussions, and independent learning activities are also used. Students develop algorithms, implement and analyze data structures, and evaluate algorithm efficiency. They solve computational problems using modern programming languages and software development tools. Digital educational resources, programming environments, algorithm visualization tools, and online coding platforms are used to strengthen students' computational thinking, programming skills, analytical abilities, and problem-solving competence in data structures and algorithms.
Prerequisites	Programming 1,2 Foundations of Artificial Intelligence Database
Exams and Assessment	Type of examination (written, oral, project) and its weight

	Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, practical laboratory performance, programming exercises, algorithm design activities, coding assignments, and the successful completion of independent study tasks, including the implementation of data structures, analysis of algorithm performance, development of sorting and searching algorithms, graph and tree applications, and comparative analysis of algorithmic solutions. The midterm assessment consists of written assignments, short in-class quizzes, practical programming tasks, and the evaluation of independent study projects, such as the implementation of linear and non-linear data structures, algorithm complexity analysis, development of searching and sorting programs, graph-processing applications, and presentations on algorithm design techniques and optimization strategies. The final assessment is a written examination consisting of theoretical questions, algorithm design problems, and practical analytical tasks designed to assess students' understanding of data structures and algorithms, computational thinking, programming proficiency, analytical skills, and their ability to design efficient algorithmic solutions for real-world computing problems. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. O. R. Yusupov, I. Q. Ximmatov, E. Sh. Eshonqulov Algorithms and Data Structures. //Coursebook.Reference. Samarkand: SamSU edition. 2021, 204 pages. 2. Gaynazarov S.M., Saidov D.Y. Algorithms and Data Structures. Textbook. – T.: "Ma'rifat", 2023. 254 pages. 3. Pat Morin, Open Data Structures: An Introduction, Published by au Press, Athabasca University 1200, 10011-109 Street, Edmonton, AB T5J 3S8

	4. S.S. Qosimov, Information Technologies. Textbook. Aloqachi Publishing House, p. 370. 5. Bobojonova M.A., The Python Programming Language. A Textbook. Durdona Publishing House, Bukhara-2023, 108 pp.
Responsible Lecturer	Ibragimova Mohigul Komiljon qizi

ELECTRONICS AND CIRCUITS 1

Field	Description
Module Title	ELECTRONICS AND CIRCUITS 1
Module Code	EVS1306
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 120 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Electronics and Circuits 1 is to familiarise students with the types and characteristics of circuits and electronic devices used in information and communication technologies. The Module studies the structure and operating mechanisms of these devices. It also studies the technological and schematic characteristics of the complex devices created with their help. The Module develops students' skills to apply these devices in practice. It aims to provide students with theoretical knowledge and practical skills. The Module teaches the application of circuits and electronic devices used in information and communication technologies. Finally, it aims to foster a scientific outlook in students.
Module Learning Outcomes	Knowledge: LO1. Knows physical laws and formulae relating to electrical circuits, the physical principles of processes in semiconductor devices, amplifiers and generators, the meaning of physical quantities, their units and how to compare them, and the methods for applying the fundamental physical laws and principles of electrical circuits to electronic circuits.

	LO2. Knows the main blocks and modules of electronic devices for transmitting and receiving data, how to set up laboratory work, carry out measurements and calculate results, how to use measuring instruments correctly, and can describe the operating principles of electronic devices based on physical laws and apply them. Skills: LO3. Plans experiments, selects the circuits and components to be used, experimentally measures the main parameters and characteristics of electrical circuits, and operates electronic and measuring equipment. LO4. Has a grasp of the fundamental laws of electricity and magnetism and the skills to apply them in electronics. Competencies: LO5. Knows how to calculate currents and voltages in passive and active elements of an electrical circuit and can apply this knowledge.
Content	Lectures Introduction to the Course "Electronics and Circuits". The aims and objectives of the course are studied, together with the role and importance of electronics. The history of modern electronics, its stages of development, and its prospects are examined. Electrical engineering and electronics standards are considered. Electrical Quantities, Their Measurement Methods and Instruments. Basic electrical quantities, including charge, voltage, current, and power, are studied, together with active and passive elements. Series and parallel connections are examined. Measurement techniques and accuracy are considered, together with magnetoelectric, electromagnetic, and electrodynamic instruments. Direct Current Electrical Circuits and Their Analysis. The concepts of nodes, meshes, and loops are studied, together with principle and replacement diagrams. Ohm's, Kirchhoff's, and Joule-Lenz laws are examined. Circuit analysis based on Ohm's law is considered.

Analysis of DC Electrical Circuits. Kirchhoff's laws are studied. Loop currents, superposition, nodal potentials, and equivalent-circuit methods are examined. Resistor, inductor, and capacitor combinations are considered.

Characteristics of Electrical Circuits Under Sinusoidal Excitation. Frequency characteristics are studied, together with active and reactive resistances. Impedance, admittance, and phase are examined. AC and DC voltages are calculated in R, L, and C circuits.

Mutually Inductive Circuits. Magnetic driving force is studied, together with the laws and calculation of magnetic circuits. The structure and characteristics of transformers are examined. Resonance and oscillation circuits are considered.

Physical Fundamentals of Semiconductor Device Operation. Semiconductors and their conductivity are studied, together with band theory. Equilibrium concentration, generation, and recombination are examined. Currents in semiconductors are considered.

Contact Phenomena in Semiconductors. Equilibrium and non-equilibrium p-n junctions are studied. The voltage-current characteristics of the p-n junction and types of breakdown are examined. Metal-semiconductor and heterojunctions are considered.

Semiconductor Diodes. Rectifier diodes are studied, together with their I-V characteristics and parameters. The mathematical model of the diode is examined. Stabilistrons, varactors, Schottky, tunnel, photo- and light-emitting diodes, and optrons are considered.

Bipolar Transistors. General information about bipolar transistors is studied, together with connection circuits. n-p-n and p-n-p configurations and energy diagrams are examined. Electrode currents, static characteristics, and parameters are considered.

Multilayer Semiconductor Devices. The structure and operating principle of the diac, thyristor, and simistor are studied. The thyristor-transistor equivalent circuit is examined. Controlled rectifiers and their applications are considered.

Field-Effect Transistors (FETs). Types and designations of FETs are studied, including p-n

junction-controlled FETs. The structure of MOSFETs is examined, together with VGS and parameters. Areas of application are considered.

Integrated Microcircuits. Technological processes in the manufacture of ICs are studied, together with integrated diodes. Microcircuits based on bipolar and MOSFET technologies, and passive components, are examined. Nanoelectronics and its prospects are considered.

Analogue Electronics. The classification of electronic devices is studied, together with analogue circuit design. Amplifier stages and amplification classes are examined, together with negative feedback. Amplifier stages based on BJT and FET are considered.

Operational Amplifiers (OAs). Fundamental elements of analogue ICs are studied. The structure, parameters, and characteristics of operational amplifiers are examined. Non-inverting and inverting feedback connections, and non-linear circuits, are considered.

Practical lessons

Studying the Operation of the Diode. The structure, operating principle, and electrical properties of semiconductor diodes are investigated. Forward and reverse bias characteristics are measured. Diode applications in electronic circuits are analyzed.

Rectification of Alternating Current Using a Bridge Circuit. A full-wave bridge rectifier circuit is constructed and analyzed. Input and output voltages are measured. Rectification efficiency, ripple voltage, and filtering methods are investigated.

Study of the Operation of the Bipolar Transistor. The structure of bipolar junction transistors is investigated. Their operating principles are also investigated. Current amplification, operating regions, and transistor characteristics are analyzed.

Study of the Operation of the Field-Effect Transistor. The structure of field-effect transistors is investigated. Their operating principles are also investigated. Voltage-controlled operation, input characteristics, and switching performance are analyzed.

Study of Amplifiers Based on Field-Effect Transistors. FET amplifier circuits are constructed and analyzed. Voltage gain and frequency response are measured. Input and output impedance and amplifier performance are also measured.

Study of the Transistor Amplifier. Transistor amplifier circuits operating in different configurations are investigated. Voltage gain and current gain are analyzed. Signal distortion and frequency characteristics are also analyzed.

Study of the Transistor Switch. Transistor switching circuits are investigated. Transistor operation in cutoff and saturation regions is analyzed. Switching time and switching characteristics are measured.

Investigating the Transistor as a Functional Generator. Transistor-based oscillator and signal generator circuits are constructed and analyzed. Waveform generation and oscillation conditions are investigated. The output frequency is also investigated.

Investigation of the Characteristics and Parameters of a Stabilatron. Zener diode operation is studied. Voltage stabilization principles are also studied. Breakdown voltage, stabilization characteristics, and voltage regulation performance are measured.

Investigation of the Optron. The operating principle of optocouplers is investigated. Their characteristics are also investigated. Electrical isolation, signal transmission, and practical applications are analyzed.

Investigation of the Static Volt-Ampere Characteristics of the Bipolar Transistor in the Common-Emitter Circuit. The input characteristics of a bipolar transistor connected in the common-emitter configuration are measured and analyzed. Its output characteristics are also measured and analyzed. Operating regions and amplification properties are determined.

Investigation of the Static Voltages of the BT in the Common-Base Connection Scheme. The static operating characteristics of a bipolar transistor in the common-base configuration are investigated. Voltage

	distribution is analyzed. Current transfer characteristics are also analyzed. Investigation of the Static Characteristics of the Metal-Oxide Transistor. The static characteristics of MOS transistors are studied. The threshold voltage and drain current are measured. Transfer characteristics are measured under different operating conditions. Investigation of the Characteristics of an Induced-Channel Metal-Dielectric-Semiconductor Transistor. Enhancement-mode MOSFET characteristics are investigated. Transfer and output characteristics are analyzed. Operating regions and switching behavior are also analyzed. Investigation of Operational Amplifiers. Operational amplifier circuits are constructed and analyzed. Amplifier configurations, gain, and offset voltage are investigated. Bandwidth and practical applications are also investigated. Self-study Electrical Quantities and Their Measurement Methods (Multisim). Direct Current Circuits: Ohm's and Kirchhoff's Laws. Methods for Calculating Electrical Circuits (Current Loops, Node Potentials). Circuits Under the Action of Sinusoidal Signals: Impedance, AC and DC Characteristics. Mutually Inductive Circuits, Transformers and Resonance. Electrical Conductivity of Semiconductors and Band Theory. p-n Junction and Its Voltage-Current Characteristics. Semiconductor Diodes and Their Applications. Bipolar Transistors: Structure, Connection Circuits and V-I Characteristics. Field-Effect Transistors (FETs) and Their Characteristics. Integrated Circuits and Nanoelectronics. Analogue Electronics and Amplifier Stages. Negative Feedback in Amplifiers.
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	Operational Amplifiers and Their Parameters. Analogue Signal Converters Based on Operational Amplifiers.
Teaching Methods	The module develops students' knowledge of the theory of electrical circuits, including Ohm's, Kirchhoff's, and Joule–Lenz laws, and the analysis of electrical circuits. Calculation methods for electrical circuits under the action of constant and sinusoidal signals are taught. The structure and operating principles of semiconductor devices, including diodes, bipolar transistors, and field-effect transistors, are studied. Students develop skills in working with multilayer semiconductor devices and integrated circuits. They also develop proficiency in analogue electronics, amplifier stages, and the use of operational amplifiers.
Prerequisites	Physics 1,2 Calculus 1,2 Programming 1,2
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final assessment (50%). The continuous assessment evaluates students' classroom participation, practical laboratory performance, circuit design and simulation activities, problem-solving skills, and the successful completion of independent study assignments. These include electrical circuit analysis using Multisim, calculations based on Ohm's and Kirchhoff's laws, investigation of AC and DC circuits, analysis of semiconductor devices, transistor characteristics, operational amplifier applications, and presentations on modern developments in electronics and nanoelectronics. The midterm assessment consists of written assignments, short in-class quizzes, laboratory reports, and the evaluation of independent study projects. Students are assessed on their ability to analyze electrical and electronic circuits, interpret the operating principles of semiconductor devices, calculate circuit parameters, simulate electronic circuits, compare analogue

	electronic components, and explain the practical applications of diodes, transistors, integrated circuits, and operational amplifiers. The final assessment may be conducted as either a written examination or a test examination. The written examination consists of theoretical questions, circuit analysis, and numerical problem-solving tasks covering the main topics of the module, including electrical quantities and measurements, DC and AC circuit analysis, semiconductor physics, p-n junctions, semiconductor diodes, bipolar and field-effect transistors, thyristors, integrated circuits, analogue electronics, amplifier circuits, and operational amplifiers. The test examination consists of multiple-choice questions designed to evaluate students' theoretical knowledge, analytical thinking, circuit analysis skills, and their ability to apply the principles of electronics and circuit design to practical engineering problems. Successful completion of all practical laboratory exercises and independent study assignments is required as an essential component of the overall module assessment.
Recommended Literature	1. A.A. Tulyaganov, S.S. Parsiev, V.A. Tulyaganova, U.M. Abdullayev. Theory of Electrical Circuits (textbook). — T.: Aloqachi, 2018, 144 pp. 2. X.K. Aripov, A.M. Abdullayev, N.B. Alimova, J.T. Maxsudov, A.A. Tulyaganov, Sh.T. Toshmatov. Electronics and Circuit Design (textbook). — Tashkent: «Aloqachi», 2017, 376 pp. 3. Aripov X.K., Abdullaev A.M., Alimova N.B., Bustanov X.X., Toshmatov Sh.T. Schematics (textbook). — Tashkent: «Tafakkur bo'stoni», 2013, 447 pp. 4. Thomas F. Schubert, Jr., Ernest M. Kim. Fundamentals of Electronics Book 1: Electronic Devices and Circuit Applications. — 2014. 5. Nigmatov X. Fundamentals of Radioelectronics. — Tashkent: 'Uzbekistan', 1994. 6. Tursunov M.O., Iliyev X.M., Abdiyev U.B. Fundamentals of Radioelectronics (textbook). — Termiz: Bekshoh Print Servis, 2023, 192 pp.

	7. Multisim User Guide. — National Instruments, 2007. 8. Robert L. Boylestad. Introductory Circuit Analysis. — Pearson Education Limited, 2014, 1091 p. 9. Behzad Razavi. Fundamentals of Microelectronics. 2nd ed. — 2014, 932 p. 10. V.T. Yeremenko et al. Fundamentals of Electrical Engineering and Electronics (textbook). — Orel, 2012.
Responsible Lecturer	Tursunov Mamasobir Ochildiyeovich, PhD in Physics and Mathematics

DISCRETE STRUCTURES

Field	Description
Module Title	DISCRETE STRUCTURES
Module Code	DIT1306
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 90 hours (46 hours Lecture, 44 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Discrete Structures is to provide students of Information Systems and Technologies with the fundamental knowledge, concepts, theorems, and their proofs relating to discrete structures. The Module covers methods for solving practical problems using discrete structures. It provides students with knowledge of the theoretical foundations of computer science and programming. The Module provides the foundational knowledge necessary for mastering the specialisation. It introduces students to modern mathematical methods for solving practical problems with high quality and precision.
Module Learning Outcomes	Knowledge: LO1. Should have knowledge of set theory, relations, relational algebra, special binary relations, algebra of judgements, formal axiomatic theory and calculus of judgements.

	LO2. Should have knowledge of these functions, Post's theorem, predicate algebra, predicate algebraic formulae and their evaluation, and predicate calculus. LO3. Must have an understanding of the basic principles of combinatorics, graphs and their types, graph colouring, trees, and flows. LO4. Performing operations on sets and relations, constructing truth tables, finding normal forms, and proving theorems, must be able to determine the completeness of a system of functions, perform operations on predicates, solve combinatorial problems, apply combinatorial principles to practical problems, know graph-colouring algorithms, and use algorithms on trees, and must have the skills to apply the methods of discrete mathematics and mathematical logic to solve practical problems. LO5. Is able to apply the knowledge gained in real-life situations. Skills: LO6. Analyses independent mathematical and scientific literature and draws substantiated conclusions. LO7. Can analyse problem situations and correctly select a mathematical approach to their solution. LO8. Conduct conversations and informal dialogues on social topics; participate in formal and informal professional discussions; manage discussions, interviews, negotiations and telephone communication. LO9. Prepare and deliver presentations or lectures on specialty-related topics; support arguments with evidence; summarize information, write and discuss articles. LO10. Develop skimming, scanning and intensive reading skills; read correspondence, emails, authentic texts, scientific and professional literature, electronic sources and press materials containing specialized terms. LO11. Has knowledge of the fundamental concepts and laws of discrete structures and their application to solving practical problems.
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	LO12. Analyses logical expressions, uses Boolean algebra operations and constructs logical circuits. Competencies: LO13. Works with sets, relations and functions, and creates mathematical models based on them. LO14. Models real-world processes using the elements of graph theory (graph, tree, path, connection). LO15. Has knowledge of the fundamental concepts and laws of discrete structures and their application to the solution of practical problems.
Content	Lectures Sets and Operations on Them. The concept of sets and their properties is studied. Different ways of representing sets are examined. Set operations such as union, intersection, difference, and complement are studied. Practical problems involving set operations are solved, developing logical and analytical thinking through set theory applications. Relationships. Binary Relationships. The concept of relations between sets is studied. Binary relations and their properties are examined. Reflexive, symmetric, antisymmetric, and transitive relations are studied. Relations are represented using matrices and graphs, and problems involving binary relations are solved. Basic Elements of Combinatorics. Fundamental Combinations. The basic principles of combinatorics are studied. Permutations, arrangements, and combinations are examined. Counting principles are applied to solve practical problems. Skills in mathematical reasoning and problem solving are developed. Repetitive Combinations in Combinatorics. Combinations with repetitions are studied. Formulas and methods for counting repeated selections are examined. Combinatorial problems involving repetition are solved. Combinatorial techniques are applied to real-world situations. Propositions. Operations on Propositions. The Concept of a Formula. Propositions and their truth values are studied. Logical operations such as conjunction, disjunction, negation, implication, and

equivalence are examined. Logical formulas and their construction are studied. Logical reasoning is developed through truth tables and examples.

Equivalences of Formulae. Basic Equivalences of the Algebra of Logic. Logical equivalence and its significance are studied. Fundamental equivalence laws of Boolean algebra are examined. Logical expressions are simplified using equivalence transformations. Problems involving logical identities are solved.

Normal Forms. Perfect Disjunctive and Conjunctive Normal Forms. Normal forms of logical expressions are studied. Disjunctive normal form (DNF) and conjunctive normal form (CNF) are examined. Perfect normal forms are constructed from truth tables. Normal forms are applied in logic and computer science.

Elementary Boolean Functions. Methods of Their Representation. Important and Spurious Variables. Boolean functions and their properties are studied. Different methods of representing Boolean functions are examined. Essential and non-essential variables are identified. Practical problems involving Boolean expressions and logic circuits are solved.

Composite Functions. Expansion of a Boolean Function onto the Zhegalkin Polynomial. Monotonic Functions. The composition of Boolean functions is studied. Zhegalkin polynomial representation is examined. Monotonic Boolean functions and their properties are studied. These concepts are applied to logical and computational problems.

Closed Classes of Functions. Post's Theorem. Classes of Boolean functions and closure properties are studied. Post's theorem and its significance in mathematical logic are examined. Function systems and completeness are analyzed. Theoretical foundations of discrete mathematics are developed.

Axioms of the Calculus of Thoughts. The Deduction Theorem. Gödel's Completeness Theorem. The axiomatic foundations of logical reasoning are studied. Formal proofs and deduction systems are examined. The deduction theorem and

Gödel's completeness theorem are studied. Rigorous mathematical proof techniques are developed.

The Concept of a Predicate. Logical Operations on Predicates. The Quantifiers of Universality and Existence. Predicates and predicate logic are studied. Universal and existential quantifiers are examined. Logical operations are applied to predicates. Problems involving quantified statements and logical reasoning are solved.

Formulas of the Algebra of Predicates. Equivalence of Formulas of the Algebra of Predicates. Predicate formulas and their structure are studied. Equivalence transformations in predicate logic are examined. Logical expressions involving quantifiers are simplified. Predicate algebra is applied to mathematical reasoning.

Normal Canonical Forms of Predicate Algebra Formulae. Canonical forms in predicate logic are studied. Methods for transforming formulas into standard forms are examined. Logical expressions are analyzed using canonical representations. Formal reasoning skills are developed.

Applications of Predicate Logic. Practical applications of predicate logic in mathematics and computer science are explored. Problems are solved using formal logical methods. The role of predicate logic in artificial intelligence and databases is studied. Analytical and problem-solving skills are developed.

Terms and Formulae. The Deduction Theorem. The structure of terms and formulas in formal logic is studied. Deduction systems and proof techniques are examined. The deduction theorem is applied in logical reasoning. Exercises involving formal proofs are solved.

Functional Elements. Relay Contact Circuits. Logical functional elements and switching circuits are studied. Relay-contact circuit design is examined. Logical operations implemented through circuits are analyzed. Boolean algebra is applied to circuit analysis.

Graphs and Their Representations. Types of Graphs. The fundamentals of graph theory are studied. Graph representations such as adjacency

matrices and lists are examined. Different types of graphs are studied. Problems involving graph structures are solved.

Connectivity of Graphs. Path, Chain, Cycle. Eulerian and Hamiltonian Graphs. Graph connectivity and traversal are studied. Paths, chains, and cycles are examined. Eulerian and Hamiltonian graph concepts are studied. Graph theory is applied to practical and theoretical problems.

Graph Metrics. Trees, Networks. Graph metrics and distance concepts are studied. Trees and their properties are examined. Network structures and applications are studied. Optimization and connectivity problems in networks are solved.

Algorithms. Recursive and Recursively Enumerable Sets. The concept of algorithms and computation is studied. Recursive definitions and recursively enumerable sets are examined. Algorithmic processes and formal computation models are studied. Theoretical computer science foundations are developed.

Partially Recursive and Total Recursive Functions. Recursive function theory is studied. Partially recursive and total recursive functions are examined. Computability and algorithmic processes are studied. Recursive concepts are applied to mathematical problems.

Turing Machine. Algorithmically Unsolvable Problems. Turing machines are studied as a model of computation. Computability and decision problems are examined. Algorithmically unsolvable problems and their significance are studied. The limits of computation and formal algorithms are explored.

Practical lessons

Introduction to the Theory of Discrete Structures. Sets and Operations on Them. The fundamental concepts of discrete mathematics and discrete structures are studied. The definition and properties of sets are examined. Methods of representing sets and performing set operations such as union, intersection, difference, and complement are studied. Practical problems involving sets are solved, developing logical thinking skills.

Relationships. Binary Relationships. The concept of relations and their applications in mathematics and computer science is studied. Binary relations and their properties are examined. Reflexive, symmetric, antisymmetric, and transitive relations are studied. Relations are represented using matrices, graphs, and diagrams, and problems involving binary relations are solved.

Basic Elements of Combinatorics. Basic Combinations. The principles of combinatorial analysis are studied. Counting techniques, permutations, arrangements, and combinations are examined. Combinatorial methods are applied to solve practical and theoretical problems. Analytical and problem-solving skills are developed.

Repetitive Combinations in Combinatorics. Combinations with repetitions and their mathematical foundations are studied. Formulas for repeated selections are examined. Combinatorial problems involving repeated elements are solved. Combinatorial reasoning is applied to practical situations.

Propositions. Operations on Propositions. The Concept of a Formula. Propositions and their truth values are studied. Logical operations such as conjunction, disjunction, negation, implication, and equivalence are examined. The structure and formation of logical formulas are studied. Truth tables are constructed and logical expressions are analyzed.

Equivalences of Formulas. Basic Equivalences of the Algebra of Logic. The concept of logical equivalence is studied. Fundamental equivalence laws of propositional logic are examined. Logical formulas are simplified using equivalence transformations. Logical identities are applied in mathematical reasoning and computer science.

Normal Forms. Complete Disjunctive and Conjunctive Normal Forms. The concept of normal forms in logic is studied. Complete disjunctive normal form (CDNF) and complete conjunctive normal form (CCNF) are examined. Logical formulas are transformed into standard forms. Normal forms are applied in logical analysis and digital systems.

Elementary Boolean Functions. Methods of Their Representation. Important and Spurious Variables. Boolean functions and their characteristics are studied. Various methods of representing Boolean functions, including truth tables and algebraic expressions, are examined. Essential and non-essential variables are identified. Problems involving logical circuits and Boolean algebra are solved.

Composite Functions. Mapping a Boolean Function onto the Zhegalkin Polynomial. Monotonic Functions. The composition of Boolean functions is studied. The representation of Boolean functions using Zhegalkin polynomials is examined. Monotonic functions and their properties are studied. These concepts are applied to logical modeling and computational systems.

Functional Closed Classes. Post Theorem. Closed classes of Boolean functions and their properties are studied. Post's theorem and its role in the classification of Boolean functions are examined. Completeness and closure properties of function systems are analyzed. Theoretical foundations of logic and discrete mathematics are developed.

Axioms of the Theory of Computable Functions. The Deduction Theorem. Gödel's Completeness Theorem. The axiomatic foundations of formal logic and computability theory are studied. Proof systems and methods of logical deduction are examined. The deduction theorem and Gödel's completeness theorem are studied. Formal proof and reasoning skills are developed.

The Concept of a Predicate. Logical Operations on Predicates. The Quantifiers of Universality and Existence. The concept of predicates and predicate logic is studied. Logical operations on predicates are examined. Universal and existential quantifiers and their applications are studied. Logical problems involving quantified statements are solved.

Formulas of the Algebra of Predicates. Equivalence of Formulas of the Algebra of Predicates. Predicate formulas and their structures are studied. Equivalence transformations in predicate logic are examined. Formulas involving quantifiers

and predicates are simplified and analyzed. Predicate logic is applied to mathematical and computational problems.

Normal Canonical Forms of Predicate Algebra Formulae. Standard and canonical forms of predicate logic formulas are studied. Methods for transforming formulas into canonical representations are examined. Logical expressions are analyzed using formal techniques. Advanced logical reasoning skills are developed.

Applications of Predicate Logic. Practical applications of predicate logic in mathematics, computer science, databases, and artificial intelligence are explored. Real-world problems are solved using logical models. The importance of predicate logic in knowledge representation and reasoning systems is studied.

Functional Elements. Relay Contact Schemes. Logical functional elements and switching devices are studied. The principles of relay-contact circuits and logical design are examined. Circuit behavior is analyzed using Boolean algebra. Logic theory is applied to electrical and computer engineering systems.

Graphs and Their Representations. Types of Graphs. The basic concepts of graph theory are studied. Graph representations such as adjacency matrices and adjacency lists are examined. Different types of graphs and their properties are studied. Problems involving graph structures and relationships are solved.

Connectivity of Graphs. Route, Chain, Cycle. Eulerian and Hamiltonian Graphs. Graph connectivity and traversal methods are studied. Routes, chains, and cycles in graphs are examined. The characteristics of Eulerian and Hamiltonian graphs are studied. Graph theory is applied to practical and theoretical problems.

Graph Metrics. Trees, Networks. Graph metrics and distance measurements are studied. Trees and their structural properties are examined. Network models and applications are studied. Optimization and

	connectivity problems in graph-based systems are solved. Algorithms. Recursive and Recursively Enumerable Sets. The concept of algorithms and algorithmic processes is studied. Recursive definitions and recursively enumerable sets are examined. Computability and formal methods of problem solving are studied. Foundations in theoretical computer science are developed. Partially Recursive and Total Recursive Functions. Recursive function theory and its applications are studied. Partially recursive and total recursive functions are examined. Computability and effective procedures are studied. Recursive concepts are applied to algorithm design and analysis. Turing Machine. Algorithmically Unsolvable Problems. Turing machines are studied as a formal model of computation. Algorithmic solvability and decision problems are examined. Undecidable and algorithmically unsolvable problems are studied. The theoretical limits of computation and automated problem solving are explored. Self-study Set Theory and Operations on Sets. Relations and Binary Relations. Basic Elements of Combinatorics. Propositions and Algebra of Logic. Boolean Functions and Post Theorem. Predicate Logic. Graph Theory. Trees, Networks, and Graph Metrics. Algorithms and Computability Theory.
Teaching Methods	The module is delivered through a combination of lectures, practical classes, interactive discussions, and independent learning activities. Teaching methods include classroom instruction, communicative language teaching, pair and group work, role-plays, presentations, vocabulary and grammar exercises, reading and listening activities, writing assignments, project-based learning, and the use of digital learning resources. Independent learning activities may include preparing

	presentations, completing language exercises, writing essays and reports, conducting research on cultural topics, developing vocabulary portfolios, and participating in online learning tasks. These teaching methods are designed to develop students' listening, speaking, reading, and writing skills, enhance their communicative competence, and improve their ability to use the foreign language effectively in academic, professional, and everyday contexts.
Prerequisites	Linear Algebra Programming 1,2
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, problem-solving activities, practical exercises, seminar presentations, and the successful completion of independent study assignments, including set theory exercises, logical formula transformations, Boolean function analysis, graph theory problems, combinatorial calculations, and algorithmic modeling tasks. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as logical proof construction, graph analysis reports, Boolean algebra applications, combinatorial problem-solving projects, presentations on computability theory, and analytical essays on discrete mathematical structures. The final assessment is a written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of discrete structures, mathematical logic, graph theory, combinatorics, Boolean algebra, predicate logic, recursive functions, Turing machines, and their ability to apply theoretical concepts to computational and mathematical problems.
Recommended Literature	1. E. Mendelson, Introduction to Mathematical Logic, Sixth Edition, 2015.

	2. Richard Johnsonbaugh, Discrete Mathematics, Eighth Edition, De Paul University, Chicago, 2010. 3. Igoshin V. I., Mathematical Logic and Algorithm Theory. – M.: Academy, 2008. 4. H.T. To'rayev, I. Azizov. Mathematical Logic and Discrete Mathematics. Volumes 1 and 2. Tashkent. 2011 5. I. Allakov., Practical Exercises in Solving Examples and Problems from Discrete Mathematics and Mathematical Logic, Termiz, 2022 6. Turaev H., Azizov I., Otaqulov S., Combinatorics and Graph Theory, Tashkent, 2009 7. Yablonsky S.V., Introduction to Discrete Mathematics. – M.: Vyshaya Shkola, 2003
Responsible Lecturer	Sakiyeva Ozoda Batirovna

COMPUTER NETWORKS

Field	Description
Module Title	COMPUTER NETWORKS
Module Code	KOT1306
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 120 hours
Frequency/Duration	III semester
Module objective	The objective of the Module Computer Networks is to provide students with the principles of construction, technologies, and interconnection of open systems and network protocols. The Module covers computer network architectures and devices. It covers construction methods for local, metropolitan, and global networks. The Module addresses issues in designing networks in computer applications, including routing protocols and data routing processes. It covers flow control, buffering, and network management. The Module provides knowledge of quality of service and network troubleshooting methods. It also provides knowledge of network security. Finally, it covers modern

	computer networks, wireless sensor networks, and related technologies.
Module Learning Outcomes	Knowledge: LO1. Can explain the architecture of computer networks, their main components, data transmission protocols, and operating principles, and analyze the application areas of various network technologies. LO2. Will have practical skills in designing, configuring, setting up, and managing local and global computer networks, and will be able to use network devices effectively. LO3. Can implement IP addressing, routing, switching, and network service provisioning processes, and can evaluate the operational efficiency of the network infrastructure. LO4. Can solve practical problems using methods to ensure network security, protect information, and detect and mitigate security threats. LO5. Demonstrates competencies in developing projects based on modern network technologies, cloud services, IoT devices, and wireless communication systems, independently solving technical problems, and effectively using information and communication technologies. Skills: LO6. Will have skills in designing computer networks, selecting network topologies, and installing and configuring network devices. LO7. Will be qualified to practically implement IP addressing, subnetting, routing, and switching processes, as well as to configure network services. LO8. Demonstrates practical skills in identifying, diagnosing, monitoring, and troubleshooting network faults. LO9. Can effectively use tools to ensure network security, enforce security policies, control access, and protect data. LO10. Will have the skills to apply modern network technologies, cloud services, wireless networks, and IoT solutions in practical projects, manage network infrastructure, and independently resolve technical issues.

	Competencies: LO11. Will have the competence to analyze, design, and evaluate the efficiency of computer network architectures, protocols, and network infrastructure. LO12. Demonstrates competence in selecting, configuring, and effectively using routers, switches, wireless communication devices, and other network equipment to solve practical problems. LO13. Will have professional competence in designing, installing, configuring, managing, and providing technical support for local, global, and corporate networks. LO14. Will have professional competence in designing, installing, configuring, managing, and providing technical support for local, global, and corporate networks. LO15. Becomes competent in creating and managing network services, server technologies, cloud computing platforms, and virtual network environments.
Content	Lectures Introduction to Computer Networks. The Concept of a Network. The fundamental concepts, objectives, and functions of computer networks are introduced. The main components of a network, including end devices, intermediary devices, transmission media, and network services, are examined. Different types of network connections, communication modes, network topologies, and resource-sharing principles are discussed. The role of computer networks in information exchange, distributed processing, and modern digital infrastructure is also considered. Standard Technologies of Computer Networks. The principal standards and technologies used in computer networks are studied. Ethernet, Fast Ethernet, Gigabit Ethernet, and other commonly used wired networking technologies are examined together with their operating principles. Standardization activities of organizations such as ISO, IEEE, and ITU are considered. The role of standards in ensuring compatibility, interoperability, reliability, and effective data transmission among network devices is explained.

Network Architectures. The main principles and structural components of network architectures are studied. Centralized, distributed, peer-to-peer, client-server, and multi-tier architectures are compared. The organization of network services, communication processes, data exchange, and resource management within different architectures is examined. The advantages, limitations, scalability, reliability, and application areas of each architectural model are discussed.

OSI Reference Model. TCP/IP Protocol Stack. The seven-layer OSI reference model and the layers of the TCP/IP protocol stack are studied. The functions of each layer, including data encapsulation, addressing, routing, error control, session management, and application support, are explained. The relationship between the OSI and TCP/IP models is analyzed. The interaction of protocols at different layers and the process of transmitting data from a source device to a destination device are examined.

Local, Metropolitan, and Global Networks and Their Construction Methods. Local Area Networks, Metropolitan Area Networks, and Wide Area Networks are studied according to their geographical coverage, structure, transmission technologies, and application areas. Methods of constructing LAN, MAN, and WAN infrastructures are examined. Network devices, communication channels, connection schemes, addressing principles, and topology selection are discussed. The performance, reliability, scalability, and cost requirements of different network types are compared.

Network Layer. The functions and operating principles of the network layer are studied. Logical addressing, packet forwarding, routing, fragmentation, and internetwork communication are explained. The role of routers and routing tables in determining packet transmission paths is examined. Static and dynamic routing principles, routing metrics, and the interaction of the network layer with adjacent layers are also considered.

IPv4 and IPv6 Protocols. The structures and operating principles of IPv4 and IPv6 are studied. IPv4

addressing, subnet masks, network and host portions, subnetting, and classless addressing are explained. The limitations of IPv4 and the need for IPv6 are examined. IPv6 address formats, address types, simplified notation, autoconfiguration, security support, and transition mechanisms between IPv4 and IPv6 are discussed.

Mobile Communication Networks. Subscriber Access Networks. The architecture and operating principles of mobile communication and subscriber access networks are studied. Cellular network components, base stations, switching systems, frequency reuse, mobility management, and handover processes are examined. Wired and wireless subscriber access technologies, including xDSL, fibre-optic access, mobile broadband, and fixed wireless access, are considered. Their performance, coverage, capacity, and quality-of-service characteristics are compared.

Transport Networks. The structure, functions, and technologies of transport networks are studied. Transport networks used to carry large volumes of voice, video, and data traffic between access and core network segments are examined. Optical fibre communication, multiplexing, synchronous transmission systems, packet-based transport, and backbone technologies are discussed. Reliability, redundancy, traffic management, bandwidth allocation, and quality-of-service requirements are also considered.

Client-Server Architectures. The principles of client-server interaction and the distribution of tasks between clients and servers are studied. Two-tier, three-tier, and multi-tier client-server architectures are examined. The functions of application, database, file, web, mail, and authentication servers are discussed. Request-response communication, session management, centralized administration, scalability, load distribution, fault tolerance, and security issues are analyzed.

Wireless Sensor Networks. The structure and operating principles of wireless sensor networks are studied. Sensor nodes, gateways, communication

channels, network topologies, data collection, and routing mechanisms are examined. Energy efficiency, limited computing resources, coverage, connectivity, reliability, and self-organization are discussed. Applications of wireless sensor networks in industry, agriculture, healthcare, environmental monitoring, smart cities, and the Internet of Things are considered.

Network Security. The fundamental principles, objectives, and mechanisms of network security are studied. Confidentiality, integrity, availability, authentication, authorization, and accountability are explained. Common network threats, including unauthorized access, malware, eavesdropping, spoofing, denial-of-service attacks, and data interception, are examined. Firewalls, encryption, virtual private networks, intrusion detection and prevention systems, access control, and security policies are discussed.

Converged Networks. The concept and architecture of converged networks that integrate voice, video, and data services within a unified infrastructure are studied. The principles of packet-based service delivery, multimedia communication, and unified network management are examined. Quality of service, traffic prioritization, bandwidth management, reliability, and interoperability are discussed. The advantages and challenges of implementing converged networks in enterprises and telecommunications systems are analyzed.

Network Operating Systems. The functions, structure, and administration of network operating systems are studied. User and group management, resource sharing, file and print services, authentication, access control, directory services, and remote administration are examined. Server operating systems and network device operating systems are considered. Configuration, monitoring, maintenance, security, and fault-management procedures in network environments are discussed.

Modern Computer Networks. Current trends and emerging technologies in computer networking are studied. Software-Defined Networking, Network

Function Virtualization, cloud networking, edge computing, the Internet of Things, 5G, and next-generation mobile networks are examined. Automation, virtualization, artificial intelligence-based network management, cybersecurity, scalability, and energy efficiency are discussed. The development prospects and practical applications of modern computer networks are also considered.

Practical lessons

The Concept of a Network and Its Organization.

The fundamental concept of a computer network and the principles of its organization are examined in practice. Network components, communication channels, end devices, intermediary devices, and resource-sharing mechanisms are identified. Students analyze common network topologies and learn how network devices are logically and physically connected to establish communication.

Network Devices. The functions and operating principles of network devices, including routers, switches, hubs, modems, repeaters, access points, gateways, and network interface cards, are studied. Students identify appropriate devices for different network configurations and examine their ports and interfaces. Basic skills in connecting and configuring network equipment are developed.

Using the WWW Service. Working with Data in It.

The operating principles of the World Wide Web and the use of web browsers are studied. Students work with websites, hyperlinks, search engines, web addresses, downloads, uploads, and online forms. Methods for locating, evaluating, organizing, saving, and securely exchanging information through web services are practiced.

Creating a Local Network and Exchanging Data on It.

A local area network is created by connecting computers and network devices. Students configure basic network parameters, assign addresses, verify connectivity, and establish shared access to files, folders, printers, and other resources. Data exchange between devices is tested, and common connection problems are identified and resolved.

Working with a Wi-Fi Network. The principles of wireless network operation and Wi-Fi connectivity are studied. Students configure a wireless access point or router, connect devices to a Wi-Fi network, manage network names and passwords, and examine signal strength and coverage. Basic wireless security settings, encryption methods, and procedures for troubleshooting connection problems are practiced.

IP Addresses and Their Handling. IPv4 and IPv6 address structures and their roles in network communication are examined. Students identify network and host portions, configure static and dynamic IP addresses, work with subnet masks, default gateways, and DNS settings, and verify addressing parameters. Basic subnetting and IP connectivity tests are also performed.

Working with Protocols and Domains. The functions of common network protocols and the domain name system are studied. Students examine protocols such as HTTP, HTTPS, FTP, TCP, UDP, DNS, and DHCP and analyze their roles in data exchange. Domain names are resolved into IP addresses, and basic commands and tools are used to investigate protocol operation and domain information.

Connecting to the Internet via VPN. The concept, purpose, and operating principles of a Virtual Private Network are examined. Students configure a VPN connection using authorized settings, establish and terminate secure connections, and verify changes in network access. The use of VPN technology for encrypted communication, remote access, and the protection of transmitted data is discussed.

Emails and How to Work with Them. The structure and operating principles of electronic mail services are studied. Students create and manage email accounts, compose and send messages, attach files, organize incoming mail, and use folders, labels, and search functions. Email etiquette, protection against spam and phishing, password security, and safe handling of attachments are also examined.

Types of Network Cables. The principal types of network cables, including twisted-pair, coaxial, and

fibre-optic cables, are studied. Students compare their structure, connectors, transmission speed, distance limitations, resistance to interference, and areas of application. Basic procedures for selecting, preparing, testing, and connecting network cables are practiced.

Working with Network Protection Software. The functions of network protection software, including firewalls, antivirus programs, intrusion detection tools, and traffic-monitoring applications, are examined. Students configure basic protection settings, scan systems for threats, review security alerts, and analyze network activity. Safe software use, access control, regular updates, and incident-response procedures are practiced.

Ethernet Technology. The principles and standards of Ethernet technology are studied through practical network configuration. Students examine Ethernet frames, MAC addresses, transmission speeds, switching principles, and physical connections. Ethernet devices are connected and tested, and common problems related to cables, ports, addressing, and communication are identified.

Wired and Wireless Networks. Wired and wireless network technologies are compared according to speed, reliability, mobility, coverage, installation requirements, cost, and security. Students establish both types of connections, configure basic parameters, and test their performance. Appropriate network solutions are selected for different practical environments and user requirements.

Connecting Computers to Each Other. Methods of connecting two or more computers directly or through network devices are practiced. Students configure physical connections and network settings, assign compatible IP addresses, verify communication, and enable resource sharing. Connectivity is tested using appropriate system tools, and common configuration errors are diagnosed and corrected.

Internet Speed and Methods for Determining It. The main indicators of Internet performance, including download speed, upload speed, latency, jitter, and packet loss, are studied. Students use speed-testing

and network-diagnostic tools to measure connection quality. Test results are analyzed, and factors affecting Internet speed, such as signal strength, network load, equipment limitations, and service-provider conditions, are examined.

Self-study

History of the Development of Computer Networks and Their Role in the Modern Information Society.

Basic Components of Computer Networks and Their Principles of Interaction.

Comparative Analysis of the Advantages and Disadvantages of Various Network Topologies.

Layers of the OSI Model and Their Application in Real-World Networks.

The TCP/IP Model and Its Importance in the Modern Internet Infrastructure.

Analyzing the Similarities and Differences Between the OSI and TCP/IP Models.

International Network Standards and the Activities of Standardization Organizations.

Criteria for Selecting an Optimal Network Architecture for a Company or Educational Institution.

Stages of Development and Prospects for Ethernet Technology.

Modern Approaches to Designing Local Area Networks (LANs).

Analysis of the Construction Principles of Metropolitan (MAN) and Global (WAN) Networks.

The Practical Significance of the IPv4 Addressing System and Subnetting Methods.

Advantages of the IPv6 Protocol and a Comparative Analysis with IPv4.

Study of the Impact of Routing Protocols on Network Performance.

Advantages of Switching Technologies and VLAN Networks.

Analysis of PDH, SDH, SONET, and DWDM Technologies Used in Transport Networks.

Comparative Analysis of Client-Server and Peer-to-Peer Architectures.

The Role of the DNS Service in Internet Operations and Its Operating Principle.

	The Importance of the DHCP Service in Network Management. Organizing Email and Web Services in Corporate Networks. Exploring the Capabilities of Network Monitoring and Diagnostic Tools. Factors Threatening Network Security and Countermeasures. Encryption Technologies and Secure Data Exchange Tools. Developing an Information Security Policy for the Corporate Network. Development Trends of Mobile Communication Networks (3G, 4G, 5G). Applications of Wireless Sensor Networks and Prospects. The Role of IoT (Internet of Things) Technologies in Computer Networks. Software-Defined Networking (SDN) and Its Advantages. The OpenFlow Protocol and Its Functions in the SDN Architecture. Networks of the Future: Analysis of NDN, Cloud Networks, and AI-Based Network Management Technologies.
Teaching Methods	The module is delivered through lectures, laboratory and practical sessions, inquiry-based and problem-based learning, hands-on network configuration, analytical activities, group discussions, presentations, network simulations, and project-based assignments. Students design and configure local and wireless networks, work with routers, switches, modems, and access points, and perform practical tasks involving IP addressing, subnetting, VLANs, routing, switching, DNS, DHCP, network monitoring, diagnostics, and security. Packet Tracer, GNS3, and other simulation environments are used to develop network models and test technical solutions. Teacher-guided and student independent learning, scientific and technical literature, electronic resources, reports, presentations, and mini-projects further develop students' practical, analytical, research,

	troubleshooting, network management, and information security skills.
Prerequisites	Computer Organization Operating Systems Computer Architecture Embedded Systems
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and final assessment (50%). The continuous assessment evaluates participation in 15 practical sessions and completion of six Teacher-Guided Independent Work assignments. A maximum of 8 points is awarded for practical sessions and 12 points for independent assignments, reports, presentations, network configuration tasks, and projects. The midterm assessment includes a theoretical component worth 12 points and independent work worth 18 points. It may be conducted as a test or a written examination consisting of three theoretical questions and two practical tasks. The final assessment may be conducted as a written examination or a test. The written examination consists of five questions worth 10 points each, while the test consists of 25 questions worth 2 points each. All independent study assignments must be submitted before the final examination.
Recommended Literature	1. Computer Networks / A. Aripov, R. Alimov, et al. – Tashkent: Publication of the Ministry of Higher and Secondary Special Education of the Republic of Uzbekistan. 2. Kurose, James F. Computer networking: a top-down approach / James F. Kurose, Keith W. Ross. — 6th ed. ISBN-13: 978-0-13-285620-1 3. Z.Z. Miryusupov, J.X. Djumanov. Computer Networks. (Textbook) - T.: "Aloqachi", 2020, 144 pages. 4. Musaev M.M. "Computer Systems and Networks". Tashkent: "Aloqachi" Publishing House, 2013. 8

	chapters. 394 pages. – A textbook for higher education institutions. 5. Miryusupov Z.Z., Djumanov J.X. Computer Networks: A Textbook, named after Muhammad Al-Khwarizmi TATU. - T.: Aloqachi, 2020. - 144 p. 6. N. A. Olifer, V. G. Olifer. "Computer Networks: Principles, Technologies, Protocols." Anniversary Edition. St. Petersburg: Piter, 2020. - 1008 pp.
Responsible Lecturer	Kholliyev Faxriddin Boxodirovich, PhD in Physical and Mathematical Sciences

FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

Field	Description
Module Title	FOUNDATIONS OF ARTIFICIAL INTELLIGENCE
Module Code	SIA1406
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 120 hours
Frequency/Duration	IV semester
Module objective	The objective of the Module Foundations of Artificial Intelligence is to introduce students to the theoretical foundations, principles, and modern technologies of artificial intelligence. Through this Module, students develop skills in data analysis. They also develop skills in the use of machine learning methods. The Module develops students' skills in the creation of intelligent systems. It teaches the practical application of artificial intelligence in various fields. It also teaches methods for its effective implementation. The Module is designed to develop students' competencies in solving problems and creating innovative solutions using modern information technologies.
Module Learning Outcomes	Knowledge: LO1. Knows the basic concepts, development history, and modern directions of artificial intelligence. LO2. Will have knowledge of the structure, operating principles, and application areas of artificial intelligence systems.

	LO3. Understands machine learning algorithms, their types, and their application characteristics. LO4. Knows the theoretical foundations of artificial neural networks, deep learning, and data processing methods. LO5. Has knowledge of databases, big data, and intelligent systems for using artificial intelligence technologies. LO6. Knows the practical applications of artificial intelligence in various fields, its advantages, limitations, and development prospects. Skills: LO7. Will have the skills to select appropriate algorithms and methods for solving artificial intelligence problems and to apply them in practice. LO8. Can create, train, and evaluate machine learning models and artificial neural networks. LO9. Can apply methods for data collection, preparation, analysis, and processing in practice. LO10. Develops skills in creating intelligent programs using artificial intelligence libraries and software tools. LO11. Becomes proficient in modeling and solving practical problems in various fields using artificial intelligence technologies. Competencies: LO12. Will have practical experience in developing AI-based projects, analyzing results, and improving them. LO13. Will have the competence to analyze problems in professional activities and develop effective solutions using artificial intelligence technologies. LO14. Acquires professional competencies in designing, developing, and implementing intelligent information systems. LO15. Will be able to apply machine learning and data analysis methods in various practical fields. LO16. Will have competencies in applying innovative approaches in professional activities, working in teams, independent learning, and the regular enhancement of their knowledge. LO17. Demonstrates skills in planning, managing, and evaluating the results of artificial intelligence projects.
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Content	Lectures Artificial Intelligence: History of Development and Current State. The history and evolution of artificial intelligence are introduced, from early symbolic systems to modern AI technologies. Students explore current trends and applications in AI. They also explore future challenges in AI. Data Analysis. Students learn the fundamentals of data collection and preprocessing. They also learn visualization and statistical analysis. The topic emphasizes the importance of high-quality data for building reliable AI models. Biological Neuron and Artificial Neural Network Model. The structure and function of biological neurons are explained. They are compared with artificial neurons. Students learn the basic architecture and operation of artificial neural networks. Machine Learning Models: Linear Regression, Logistic Regression, Decision Tree. Students study supervised learning techniques used for prediction and classification tasks. The lecture covers the principles and advantages of common machine learning models. It also covers their practical applications. Unsupervised Learning: K-Means Clustering and PCA. Unsupervised learning methods for discovering hidden patterns in data are introduced. Students learn clustering with K-Means. They also learn dimensionality reduction using Principal Component Analysis (PCA). Deep Learning: Neural Network Structures. Students explore deep learning concepts, including multilayer neural networks and activation functions. They also explore backpropagation. The topic highlights the advantages of deep learning for solving complex problems. Convolutional Neural Networks (CNN) and Computer Vision. This lecture focuses on CNN architectures. Their applications in image classification, object detection, and computer vision are examined. Students learn how CNNs automatically extract visual features from images.
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Recurrent Neural Networks (RNN) and Natural Language Processing. Students study recurrent neural networks. Their ability to process sequential data is examined. Applications in natural language processing, speech recognition, and machine translation are discussed.

Reinforcement Learning: Q-Learning and Deep Q-Network (DQN). Reinforcement learning is introduced, in which intelligent agents learn through interaction with an environment. Students explore Q-learning algorithms. They also explore Deep Q-Networks for decision-making tasks.

Model Evaluation and Validation Methods. Students learn techniques for evaluating machine learning models using accuracy, precision, and recall. They also learn evaluation using F1-score, cross-validation, and confusion matrices. The importance of avoiding overfitting is also discussed.

Big Data and Cloud Technologies. The concepts of Big Data and distributed computing are explained. Cloud platforms used in AI development are also explained. Students learn how cloud technologies support large-scale data processing and AI applications.

Recommendation Systems: Collaborative Filtering and Content-Based Filtering. Students study the principles of recommendation systems used in e-commerce and online services. The lecture compares collaborative filtering and content-based filtering techniques. Practical examples of these techniques are presented.

Understanding Information Security and Attack Vectors in Artificial Intelligence Programs. Cybersecurity risks in AI systems are examined, including adversarial attacks and data poisoning. Model theft is also examined. Students learn methods for improving the security and reliability of AI applications.

Getting Started with Libraries Like TensorFlow and PyTorch. Students gain practical experience with popular deep learning frameworks, including TensorFlow and PyTorch. The lecture introduces

model creation and training using these libraries. It also introduces model testing and deployment.

Examples of Using Artificial Intelligence in Entrepreneurial and Practical Projects. Real-world AI applications in business, healthcare, and education are presented. Applications in finance, agriculture, and industry are also presented. Students learn how AI technologies can be integrated into innovative entrepreneurial and practical projects.

Practical lessons

Artificial Intelligence and the Technologies Used in It. Students become familiar with the core technologies of artificial intelligence, including machine learning and deep learning. They also become familiar with computer vision and natural language processing. The practical session introduces commonly used AI tools and development environments.

Data Analysis Using Artificial Intelligence. This practical focuses on using AI techniques to preprocess and visualize datasets. It also focuses on analyzing datasets. Students gain experience in identifying patterns and extracting meaningful insights from data.

Data Analytics Projects Using Artificial Intelligence. Students develop small AI-based data analytics projects by applying machine learning algorithms to real-world datasets. The emphasis is on data preparation and model building. Result interpretation is also emphasized.

Creating a Regression Model Using Artificial Intelligence. Students are guided through the process of building regression models for prediction tasks. They also evaluate these models. Practical exercises include model training, testing, and performance analysis.

Solving Mathematical and Economic Problems Using Artificial Intelligence. Students apply AI techniques to solve mathematical optimization problems. They also apply AI techniques to economic forecasting problems. The practical demonstrates how machine learning supports decision-making and predictive analysis.

Developing Recommendation Projects Using Artificial Intelligence. Recommendation systems using collaborative filtering techniques are introduced. Content-based filtering techniques are also introduced. Students create simple recommendation models for products, movies, or other digital content.

Working with Text and Voice Using Artificial Intelligence. Students explore AI methods for processing text data. They also explore AI methods for processing speech data. Practical tasks include text classification, speech recognition, and basic natural language processing applications.

Developing Chatbots Using Artificial Intelligence. This topic focuses on designing AI-powered chatbots. It also focuses on implementing these chatbots. The chatbots are capable of answering user questions and performing simple conversational tasks using NLP technologies.

Developing Interactive Game Projects. Students create interactive games that incorporate AI techniques such as decision-making and pathfinding. They also incorporate intelligent agents. The practical demonstrates how AI enhances user interaction and gameplay.

Developing Interactive Web Projects. This practical teaches students how to integrate AI models into web applications. Students develop interactive web interfaces that utilize machine learning services. They also utilize deep learning services in these interfaces.

A System for Providing Recommendations Based on Images and Audio via a Web Interface. Students build a web-based AI system capable of analyzing images to generate personalized recommendations. The system also analyzes audio for this purpose. The project combines computer vision, audio processing, and web technologies.

Emotion Detection Using Artificial Intelligence. AI techniques for recognizing human emotions from facial expressions are introduced. Techniques for recognizing emotions from speech or text are also introduced. Students develop simple emotion recognition models using machine learning libraries.

Development of Programs for Processing Graphical Data. Students create AI applications for image processing tasks such as object recognition. They also create applications for image classification. Feature extraction is performed using computer vision techniques.

Working with Professional Artificial Intelligence Platforms. This practical familiarizes students with professional AI platforms and frameworks such as TensorFlow and PyTorch. Google Colab and cloud-based AI services are also introduced. These platforms are used for model development and deployment.

Final Projects and Their Evaluation. Students complete an individual or team-based AI project that integrates the knowledge and skills acquired during the course. Projects are presented and evaluated. They are discussed based on technical quality, innovation, and practical implementation.

Self-study

History of Artificial Intelligence Development and Analysis of Modern Technologies.

Methods of Data Analysis Using Artificial Intelligence and Their Practical Applications.

Structure and Operating Principles of Biological and Artificial Neural Networks.

Solving Regression and Classification Problems Based on Machine Learning Algorithms.

The Practical Significance of Clustering Algorithms and Dimension Reduction Methods.

Deep Learning Technologies and Modern Neural Network Architectures.

Image Processing and Computer Vision Tasks Using Convolutional Neural Networks.

Artificial Intelligence Technologies for Natural Language Processing and Working with Audio Data.

Application of Reinforcement Learning Algorithms in Intelligent Systems and Games.

Creating Chatbots and Virtual Assistants Technologies.

Recommendation Systems and Their Application in E-Commerce and Web Applications.

	Methods for Detecting Emotions and Processing Graphical Data Using Artificial Intelligence. Capabilities of Big Data, Cloud Technologies, and Professional Artificial Intelligence Platforms. Developing Artificial Intelligence Projects Using the TensorFlow and PyTorch Libraries. Prospects for Using Artificial Intelligence Technologies in Entrepreneurial, Economic, and Innovative Projects.
Teaching Methods	The module teaches the basic concepts, principles, and stages of development of artificial intelligence. It ensures the mastery of machine learning, neural networks, and data analysis methods. Students develop skills in developing artificial intelligence algorithms and applying them to solve practical problems. The module teaches the creation of intelligent information systems and intelligent software tools and their effective use. It develops the competence to analyze and evaluate opportunities for applying artificial intelligence technologies in various fields. Finally, it develops students' competencies in innovative thinking, conducting research, and independent decision-making based on modern digital technologies.
Prerequisites	Data Structures and Algorithms Programming 1,2 Foundations of Cybersecurity
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, practical programming activities, seminar presentations, and the successful completion of independent study assignments. These assignments include data analysis exercises, machine learning model implementation, neural network simulations, AI-based programming tasks, TensorFlow or PyTorch practice activities, and short reports on current developments in artificial intelligence. The midterm assessment consists of

	written assignments, short in-class quizzes, practical programming tasks, and the evaluation of independent study projects. Typical projects include developing a regression or classification model, designing a recommendation system, creating a simple chatbot, implementing a computer vision or natural language processing application, and preparing a presentation on an emerging artificial intelligence technology or its real-world applications. The final assessment is a written examination consisting of theoretical questions, analytical case studies, and problem-solving tasks designed to assess students' understanding of the fundamental concepts of artificial intelligence, machine learning, deep learning, data analysis, and intelligent systems. The examination also evaluates students' ability to apply AI techniques to practical and real-world problems. Successful completion of all laboratory work, practical programming assignments, and independent study projects is required as an essential component of the module assessment.
Recommended Literature	1. J.SH. Bekkulov, I. Ibragimov – Fundamentals of Artificial Intelligence. Textbook. Qarshi 2023. 139 pp. 2. T.A. Khujakolova, N.T. Malikova - Artificial Intelligence. Textbook. Tashkent 2020. 218 p. 3. D. Rakhmatov – Artificial Intelligence: A Modern Approach. Textbook. "Renessans Press", Tashkent 2022 4. Types and Programming Languages by Benjamin C. Pierce ISBN: 0262162091 The MIT Press © 2002 (623 pages) 5. Essentials of Programming Languages. Friedman Mitchell Wand The MIT Press Cambridge, Massachusetts London, England ISBN 978-0-262-06279-4 (hbk. : alk. paper) QA76.7.F73 2008 6. R.G. Asadullayev - Fuzzy Logic and Neural Networks. A Textbook. Belgorod 2017. 309 pp.
Responsible Lecturer	Tursunov Farxod Bakhodir o'g'li, teacher

PROBABILITY AND STATISTICS

Field	Description
Module Title	PROBABILITY AND STATISTICS
Module Code	EVS1406
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 90 hours (46 hours Lecture, 44 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	IV semester
Module objective	The objective of the Module Probability and Statistics is to provide students with skills to mathematically model random events and processes. The Module develops students' theoretical knowledge, practical skills, and competencies in the collection, processing, analysis, and scientific interpretation of data. Students learn probability theory, random variables, and probability distributions. They also learn mathematical statistics, hypothesis testing, and correlation and regression analysis. These skills enable students to make data-driven decisions. They also enable students to assess uncertainties in software and information systems.
Module Learning Outcomes	Knowledge: LO1. Understands basic concepts of probability theory, random events, and classical, geometric and statistical definitions of probability. LO2. Knows random variables, their distribution laws, mathematical expectation, variance and other numerical characteristics. LO3. Understands properties of discrete and continuous probability distributions, Bernoulli scheme, binomial, Poisson and normal distributions. LO4. Knows basic concepts of mathematical statistics, population and sample, statistical series and indicators. LO5. Knows statistical estimation, confidence intervals, hypothesis testing, correlation and regression analysis.

	Skills: LO6. Is proficient in calculating probabilities of random events and solving probability theory problems. LO7. Determines numerical characteristics of random variables and works with probability distributions. LO8. Collects, groups, analyzes and graphically represents statistical data. LO9. Uses statistical methods to identify data relationships and conduct correlation and regression analysis. Competencies: LO10. Creates mathematical models of real processes using probabilistic and statistical methods. LO11. Draws statistical inferences, estimates parameters and tests hypotheses based on sample data. LO12. Applies probabilistic and statistical methods in solving practical problems in software engineering, computer engineering, artificial intelligence and data analysis.
Content	Lectures Space of Elementary Events. Probability Definitions. The definition and classification of probability are studied. Classical, geometric, and statistical definitions are examined, together with elementary events and event space. The importance of choosing the correct definition for the problem type is discussed. Applications in various fields of science and engineering are considered. Algebra of Events. Kolmogorov Axioms. Operations on events, including union, intersection, and complement, are studied. The Kolmogorov axioms as the foundation of probability theory are examined, together with the structure of the probability space. Properties derived from the axioms are considered. A rigorous mathematical framework for probability is established. Conditional Probability and Dependence. The definition and calculation of conditional probability are studied. Dependent and independent events are examined, together with the multiplication rule for probabilities. Bayes' theorem and the total probability

formula are considered. Applications to real problems in diagnostics and inference are discussed.

Bernoulli Scheme and Binomial Distribution. The sequence of independent trials is studied, together with the Bernoulli formula for exact success numbers. Binomial distribution properties are examined. The mean and variance of the binomial distribution are considered. Applications to quality control and reliability analysis are discussed.

Limit Theorems of Muavr-Laplace and Poisson. Local and integral limit theorems are studied. The normal distribution approximation to the binomial distribution is examined, together with the Poisson distribution as a limit case. Conditions for the applicability of these theorems are considered. Their practical importance in statistics and engineering is discussed.

Random Variable Definition and Distribution Function. The concept and classification of random variables are studied, including discrete and continuous random variables. Distribution function properties are examined, together with the cumulative distribution function. The relationship of these functions to probability measures is considered.

Random Vector and Multivariate Distributions. The concept of a random vector is studied, together with the joint distribution function. Marginal distributions are derived from the joint distribution. The independence of random variables is examined. Covariance and correlation between components are considered.

Mathematical Expectation of Random Variables. The concept and calculation methods of expectation are studied, for both discrete and continuous distributions. The linearity of expectation is examined. Expectation properties and applications are considered. Conditional expectation concepts are also considered.

Variance and Standard Deviation. The definition and calculation of variance are studied, together with its properties. Standard deviation is examined as a measure of dispersion. The relationship between

expectation and variance is considered. Variance calculations for common distributions are discussed.

Important Probability Distributions. A survey of common distributions and their properties is presented, including the binomial, geometric, Poisson, uniform, and exponential distributions. Distribution selection for modeling is examined. Relationships between distributions are considered. Real-world applications are discussed.

Multivariate Distributions in Detail. Joint distributions of multiple random variables are studied, together with their correlation and covariance structure. Copulas for modeling dependencies are examined. Marginal and conditional distributions are considered. Applications to multivariate problems are discussed.

Functions of Random Variables and Composite Formulas. Transformations of random variables are studied, together with the distribution of a function of a random variable. The method of transformations is examined. The convolution of distributions is considered. Moment-generating functions are also considered.

Higher-Order Moments and Correlation Coefficient. Skewness and kurtosis are studied, together with central moments and their significance. The definition and interpretation of the correlation coefficient are examined. Correlation properties and limitations are considered. Rank correlation methods are also considered.

Chebyshev Inequality and Theorem. The Chebyshev inequality is studied, including its statement and proof, together with the Markov inequality. Convergence in probability is examined, along with the weak law of large numbers. Their practical significance for error bounds is discussed.

Law of Large Numbers. The classical law of large numbers is studied, together with convergence concepts in probability. The convergence of empirical frequency to probability is examined. The strong law of large numbers is considered. Applications to statistical estimation are discussed.

Main Problems of Mathematical Statistics.

Population and sample concepts are studied, together with descriptive versus inferential statistics. Point and interval estimation are examined. Hypothesis testing procedures are considered. The quality of estimates and decision-making are also considered.

Empirical Distribution Function and Estimation.

The sample-based distribution function is studied, together with empirical frequency and cumulative frequency. The Glivenko-Cantelli theorem is examined. The consistency of the empirical distribution is considered. Its relationship to the population distribution is discussed.

Point Estimation Methods. The method of moments and maximum likelihood estimation are studied, together with least squares estimation. The properties of good estimators, including unbiasedness, consistency, and efficiency, are examined. A comparison of estimation methods is considered. Selection criteria for estimators are discussed.

Confidence Interval Methods. The concept and interpretation of interval estimation are studied, together with the meaning of confidence level. Construction methods for confidence intervals are examined. Precision and confidence trade-offs are considered. Interval estimation for various parameters is discussed.

Chi-Square, Student t and Fisher Distributions.

Distributions related to the normal distribution are studied, including the chi-square distribution's properties and uses. The Student t-distribution for small samples is examined. The Fisher F-distribution for variance comparison is considered. Applications to hypothesis testing are discussed.

Linear Correlation Analysis. Correlation coefficient calculation is studied, together with significance testing for correlation. Partial and conditional correlations are examined, along with correlation matrices. The visualization of relationships is considered. The interpretation of correlation results is also considered.

Statistical Hypotheses and Testing Procedures.

Hypothesis formulation and the null and alternative

hypotheses are studied. Type I and Type II errors are examined, together with significance level and power concepts. Test statistics and decision rules are considered. P-value interpretation and usage are discussed.

Practical lessons

Combinatorics and Counting Principles.

Permutations and combinations are studied, together with the multiplication principle. Binomial coefficients are examined. Applications to probability calculations are considered, and counting problems are solved systematically.

Limitations of Probability Calculation Methods.

The limitations of the classical definition of probability are studied. Geometric probability challenges and the empirical nature of statistical probability are examined. The consistency of the axiom system is considered. An appropriate definition is chosen for the given context.

Kolmogorov Axioms and Derived Properties.

Probability properties are derived from the axioms, including monotonicity and sub-additivity. The inclusion-exclusion principle is examined. Complement and intersection properties are considered. Theorems are proved rigorously.

Conditional Probability Applications. Conditional probabilities are calculated, and event independence is verified. The total probability formula is applied. Bayes' theorem is used for inference. Real problems are solved using conditional probability.

Event Independence and Pairwise Independence.

The distinction between mutual and pairwise independence is studied. Independence verification methods are examined, including Bernstein's examples. Counterintuitive independence cases are considered. The practical implications of independence assumptions are discussed.

Poisson Process and Scheme. The Poisson distribution is derived, and its applications to rare events are examined. Parameter estimation and goodness-of-fit testing are considered. Real applications in queuing and reliability are discussed.

Generating Functions and Their Properties.

Probability generating functions and characteristic functions are studied. Moment calculation using generating functions is examined. Their properties and applications are considered, together with inversion formulas.

Important Distributions in Practice. Various distributions are recognized and their parameters are estimated. Distributions are fitted to data, and the choice of distribution is justified. A comparative analysis of distribution properties is conducted. Goodness-of-fit is assessed.

Random Variables Without Expectation or Variance. The Cauchy distribution and undefined moments are studied. Heavy-tailed distributions are examined. The practical implications of non-existing moments are considered, along with risk assessment. Alternative measures to expectation are discussed.

Correlation Coefficient Applications. The strength of a relationship is measured, together with interpretation guidelines and cautions. Nonlinear relationships and outlier effects on correlation are examined. Correlation is computed and visualized.

Characteristic Functions. The properties and calculation of characteristic functions are studied. The inversion formula for density recovery is examined, together with moment calculation. The uniqueness theorem is considered. Applications to probability problems using Laplace and Fourier transforms are discussed.

Law of Large Numbers and Chebyshev Results. Convergence is verified, and error bounds are estimated. Probability bound calculations are performed. Practical sample size determination and convergence rate analysis are considered.

Central Limit Theorem and Applications. The statement and conditions of the central limit theorem are studied. The normal approximation to distributions and the sample mean distribution are examined. Applications to statistical inference are considered. The Lyapunov condition is checked for satisfaction.

Population and Sample Relationship. The definition of population in context is studied, together

with sampling schemes such as random, stratified, and cluster sampling. Sample representativeness is examined. Selection bias and its remedies are considered, along with finite population corrections.

Empirical Distribution and Measures. The empirical function is calculated, together with empirical moments and statistics. Histograms and frequency polygons are examined. Cumulative frequency is visualized, and the Glivenko theorem is verified.

Point Estimation Concepts and Methods. Unbiasedness is checked, and consistency is verified. Efficiency is compared among estimators. The method of moments and maximum likelihood estimation are applied.

Confidence Interval Construction. The properties and computation of confidence intervals are studied. The interval formula is derived, and the margin of error is calculated. Confidence level selection and its interpretation are considered. The relationship between interval width and precision is discussed.

Parameter Estimation for Distributions. Parameters are estimated for common distributions. Confidence intervals are constructed for normal distribution parameters. Intervals are also constructed for exponential and Poisson parameters. Reliability interval calculations and practical estimation procedures are considered.

Empirical Functions and Glivenko Theorem. The properties of the empirical distribution function are studied, together with its convergence to the theoretical distribution. The Kolmogorov-Smirnov statistic and the supremum distance measure are examined. Asymptotic theory is verified.

Normal-Related Distributions. Applications of the chi-square distribution are studied, together with the use of the Student t-distribution. The Fisher F-distribution for variance ratios is examined. The concept of degrees of freedom is considered. Distribution tables and statistical software are used.

Confidence Intervals for Normal Parameters. Confidence intervals for the known variance case are constructed. Intervals for the unknown variance case

	are also constructed. The trade-off between interval width and confidence level is examined. Robustness to the normality assumption and small sample behavior are considered. Variance Estimation and Confidence Intervals. The properties of sample variance are studied, together with the confidence interval for variance. The chi-square distribution is applied for this purpose. The equivalence of variance estimation methods and the interpretation of variance results are considered. Hypothesis Testing for Means and Variances. One-sample and two-sample tests are studied, together with test statistic calculation. P-value computation is examined. Decision-making procedures are considered, along with Type I and II error analysis. Self-study Combinatorial Foundations and Probability. Random Trials and Bernoulli Experiments. Random Variables and Expectations. Probability Distributions in Theory and Practice. Convergence and Limit Theorems. Mathematical Statistics Foundations. Estimation Theory and Methods. Normal Distribution and Related Topics. Hypothesis Testing Framework. Correlation and Regression Analysis. Advanced Statistical Methods.
Teaching Methods	The module is delivered through interactive lectures with practical examples and problem-solving sessions. Laboratory experiments and computational practice with software are also used. Seminars and group discussions form part of the delivery, together with project-based learning using real datasets. Independent study with statistical software is undertaken by students. Research activities analyzing actual data further develop students' practical skills.
Prerequisites	Programming 1,2 Linear Algebra Discrete Structures

Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). Continuous assessment evaluates classroom participation in practical sessions, problem-solving ability during hands-on work, and completion of teacher-guided independent work assignments including problem solutions, statistical analysis reports, and mini-projects on data analysis. Midterm assessment consists of a written test covering theoretical knowledge of probability theory, random variables, probability distributions, and basic statistics, together with independent work projects evaluating practical skills in probability calculation, data analysis, and statistical hypothesis formulation. The final examination is a written examination with problems requiring students to solve probability problems, analyze statistical data, conduct hypothesis tests, and interpret results. Assessment measures theoretical understanding, computational skills, and the ability to apply methods to practical data problems.
Recommended Literature	1. Narbayev A.B., Mamadazimov M.M. Probability and Statistics. TerDU, 2025. 2. Marsaglia G., Tsang W.W. The ziggurat method for generating random variables. Journal of Statistical Software, 2000. 3. Box G.E.P., Hunter W.G., Hunter J.S. Statistics for Experimenters. Wiley, 2005. 4. Freedman D., Pisani R., Purves R. Statistics. W.W. Norton, 2007.
Responsible Lecturer	S. Aymatov, Associate Professor at the Department of Algebra and Geometry, Termiz State University, PhD in Pedagogical Sciences

COMPUTER ORGANIZATION

Field	Description
Module Title	COMPUTER ORGANIZATION
Module Code	KAT1506
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 120 hours
Frequency/Duration	V semester
Module objective	The objective of the Module Computer Organization is to provide a scientifically grounded understanding of the fundamental principles, structure, and operation of modern computer systems. The Module examines the organization and functional components of computers, including the central processing unit, memory hierarchy, input/output systems, data representation, instruction execution, and system buses. It aims to explain how hardware components interact to execute programs and process information. The Module also develops students' ability to analyze computer architectures and evaluate their performance and efficiency. It fosters students' understanding of the relationship between computer hardware and software. The Module develops analytical and problem-solving skills. Finally, it provides the knowledge and practical competencies required for the design, analysis, and effective utilization of modern computer systems.
Module Learning Outcomes	Knowledge: LO1. Develops the necessary knowledge of the theoretical foundations of computer organization, computer architecture, and its components. LO2. Explains the operating principles of the computer's central processor, memory devices, and input-output systems. LO3. Has knowledge of instruction set architecture, data formats, and information processing processes in a computer.

	LO4. Studies the structure and performance characteristics of modern computer systems, processors, and the memory hierarchy. Skills: LO5. Has the skills to describe the computer's main and auxiliary devices and analyze their functions. LO6. Applies arithmetic-logical fundamentals of computers, number systems, and logical operations to practical problems. LO7. Analyzes the interaction processes between the processor, memory, and buses, and evaluates their efficiency. LO8. Develops practical skills in modern computer architectures, multicore, and parallel processing technologies. Competencies: LO9. Applies modern computer architectures in professional activities. LO10. Critically evaluates the technical and functional characteristics of computer systems. LO11. Applies the principles of computer organization in the design and optimization of computing systems. LO12. Applies high-performance computing systems and parallel technologies in research and practical activities. LO13. Independently studies innovations in modern computer architectures and computing technologies and acquires the skills to implement them in practice.
Content	Lectures Computer Organization and Classification. Computer organization and computer architecture are introduced, together with the concepts of computer organization, computer architecture, and computer system. The functional components of a computer system and their interaction are examined, along with the classification of computers according to purpose, performance, size, data processing capabilities, and application areas. The general structure of modern computer systems is studied, including the Von Neumann architecture and its fundamental principles, and the Harvard and modified Harvard architectures. Basic concepts of data, information, instructions, programs, and processing, together with the general

principles of operation and organization of modern computer systems, are considered.

Computer Architecture and Its Stages of Development. The historical development and evolution of computer architecture are studied, together with the main stages and generations of computer development, from first- to fifth-generation computers. The development of vacuum tube, transistor, integrated circuit, microprocessor, and modern multicore technologies is examined. The evolution from centralized systems to personal computers, distributed systems, mobile devices, cloud computing, and high-performance computing systems is considered. The development of modern architectural concepts, including parallel, distributed, mobile, embedded, and heterogeneous computer architectures, is discussed.

Arithmetic Foundations of Computer Organization. Number systems used in computer systems, including binary, decimal, octal, and hexadecimal systems, are studied, together with conversion between them. The representation of integers and real numbers is examined, including signed and unsigned representation, and one's complement and two's complement representations. Fixed-point and floating-point representations and binary arithmetic operations are considered, together with the role of the arithmetic logic unit (ALU) in computer organization. Overflow, underflow, precision, and rounding errors in computer arithmetic are discussed.

The Logical Foundations of Computer Organization. The fundamentals of Boolean algebra and mathematical logic are studied, together with logical variables and logical functions. Basic logical operations, including AND, OR, NOT, XOR, NAND, and NOR, are examined through truth tables and logical expressions, and the simplification of Boolean functions is considered. Logic gates and their implementation in digital systems, including combinational and sequential logic circuits, flip-flops, registers, and counters, are discussed. The role of

logical foundations in the design and operation of computer hardware is also considered.

Instruction Set Architecture: CISC and RISC Principles. The concept and structure of instruction set architecture (ISA) are studied, together with machine instructions and their basic components. Instruction formats and instruction execution are examined, including data transfer, arithmetic, logical, control, and input-output instructions. CISC and RISC architectures and their fundamental principles are compared. Instruction-level parallelism, the role of ISA in processor design, and examples of modern instruction set architectures are discussed.

Data Formats, Operand Addressing Modes, and Instruction Format. The representation and organization of data in computer systems are studied, including integer, floating-point, character, logical, and encoded data formats. Operand addressing modes, including immediate, direct, indirect, register, indexed, relative, and base addressing, are examined. The structure and components of machine instructions, including operation codes, operands, addressing fields, and instruction length, are considered, together with fixed-length and variable-length instruction formats. The relationship between instruction formats, memory organization, and processor performance is discussed.

Control Flow, Stack Organization, and Interrupt System. Control flow in computer programs is studied, including sequential, conditional, and unconditional branching, branch instructions, and subroutine calls. Stack organization and stack operations are examined, together with the role of the stack pointer and stack frame in procedure and function calls, parameter passing, and return mechanisms. Interrupt concepts and interrupt handling mechanisms are considered, including hardware and software interrupts, maskable and non-maskable interrupts, interrupt priority, interrupt vectors, and interrupt service routines. The role of interrupts in operating systems and input-output systems is discussed.

Internal Structure of the Processor: Registers, ALU, and Control Unit. The general organization and internal structure of a central processing unit (CPU) are studied, together with the classification of processor registers into general-purpose and special-purpose registers, including the program counter, instruction register, stack pointer, and status register. The Arithmetic Logic Unit (ALU) and its functions are examined, together with the control unit and its role in instruction execution, including hardwired and microprogrammed control units. The instruction cycle — fetch, decode, execute, and write-back stages — is considered. The interaction between the processor, memory, and input-output devices is discussed.

Modern Processors: Pipelined, Superscalar, and Multicore Architectures. Modern processor architectures and techniques for increasing computational performance are studied, including instruction pipelining, pipeline stages, and pipeline hazards, together with techniques for hazard detection and resolution. Superscalar processors and multiple instruction issue are examined, along with out-of-order execution and speculative execution. Multicore and many-core processors are considered, including shared and distributed memory in multicore systems. Parallel processing and the influence of processor architecture on system performance are discussed.

Memory Hierarchy and Cache Memory. The concept and organization of the memory hierarchy are studied, including registers, cache memory, main memory, and external storage and their characteristics. The principles of locality, temporal and spatial, are examined, together with cache memory organization and operation, including cache levels L1, L2, and L3, and direct-mapped, fully associative, and set-associative cache organizations. Cache hit and miss concepts, cache replacement, and write policies are considered. Memory access time and the influence of cache organization on computer system performance are discussed.

Main and Virtual Memory Organization. The organization and operation of main memory are

studied, including RAM and ROM technologies, and static and dynamic random-access memory. Memory addressing, address space, memory allocation, and protection are examined. The concept and principles of virtual memory are considered, including paging and segmentation, page tables and translation lookaside buffers (TLB), and page faults and demand paging. Memory management mechanisms and the role of virtual memory in modern operating systems and computer architecture are discussed.

External Storage Devices and RAID Arrays. The types and organization of external storage devices are studied, including magnetic disks, solid-state drives, optical storage, and modern storage technologies, together with the structure and operation of hard disk drives. Storage capacity, access time, data transfer rate, and reliability are examined, along with solid-state storage and flash memory. RAID technology and its principles are considered, including RAID 0, RAID 1, RAID 5, RAID 6, and RAID 10. Data redundancy, fault tolerance, performance improvement, and the application of RAID arrays in servers and data centers are discussed.

System Interfaces and Modern Buses. The concept and role of system interfaces in computer organization are studied, including communication between the processor, memory, and peripheral devices. System buses and their classification into data bus, address bus, and control bus are examined, together with bus architecture and bus arbitration, and parallel and serial interfaces. Modern interfaces and communication standards, including PCI Express, USB, SATA, and other contemporary interfaces, are considered, along with high-speed data transfer technologies. The role of interfaces and buses in the performance and scalability of modern computer systems is discussed.

Input-Output Systems, Ports, and Parallel/Heterogeneous Computing. The organization and principles of input-output (I/O) systems are studied, including I/O devices and their classification, programmed I/O, interrupt-driven I/O,

and direct memory access (DMA). I/O ports, device controllers, and communication between the CPU and peripheral devices are examined. Parallel computing concepts and types of parallelism are considered, including shared-memory and distributed-memory systems, and heterogeneous computing architectures combining CPUs, GPUs, and specialized accelerators. General-purpose GPU computing and the role of accelerators in modern computational systems are discussed.

Performance Metrics and Optimization Methods for Memory and Input-Output Systems.

Fundamental performance indicators of computer systems are studied, including processor performance, clock frequency, instruction execution time, throughput, latency, and speedup, together with benchmarking and performance evaluation methods. Memory performance metrics, including access time, bandwidth, cache hit ratio, and memory latency, are examined, along with input-output system performance and data transfer efficiency. Bottlenecks in memory and I/O systems are considered. Methods for optimizing cache usage, memory access, data transfer, and I/O operations, together with principles of system-level performance optimization, are discussed.

Practical lessons

Analysis of the Organizational Components, Key Indicators, and Modern Technical Specifications of Computers.

The main organizational components of modern computer systems are analyzed, including the processor, main memory, cache memory, storage devices, input-output devices, and system buses. The main technical specifications of computers are studied, including processor frequency, number of cores, cache capacity, memory capacity, storage capacity, data transfer rate, and power consumption. A comparative analysis of the technical characteristics of modern computer systems is conducted, and their influence on overall system performance is evaluated.

Performing Arithmetic Operations in Various Number Systems and Converting Between Them.

Arithmetic operations are performed in binary, octal, decimal, and hexadecimal number systems, and numbers are converted between different positional number systems. Addition, subtraction, multiplication, and division of binary numbers are performed. Signed number representations and two's complement arithmetic are applied. The results of arithmetic operations are analyzed, and overflow conditions are identified.

Representation of Floating-Point Numbers Based on the IEEE 754 Standard and Performing Operations on Them. The structure and representation of floating-point numbers according to the IEEE 754 standard are studied, including the sign, exponent, and mantissa fields. Decimal numbers are converted into binary floating-point representation. Single-precision and double-precision numbers are represented. Arithmetic operations are performed with floating-point numbers, and rounding, precision, overflow, and underflow errors are analyzed.

Simplifying Logical Expressions and Designing Logic Circuits Based on Boolean Algebra. The basic laws and rules of Boolean algebra are applied to simplify logical expressions, and truth tables are constructed and analyzed. Logical functions are implemented using AND, OR, NOT, NAND, NOR, and XOR gates. Combinational logic circuits are designed based on simplified Boolean expressions. Circuit operation is verified using digital logic simulation tools.

Comparative Analysis of Instruction Set Architectures (CISC and RISC) Based on Examples. A comparative study of CISC and RISC instruction set architectures is conducted. The main differences in instruction complexity, instruction length, addressing modes, register organization, memory access, and execution principles are analyzed. Practical examples of modern processor architectures are examined. The advantages and limitations of CISC and RISC approaches are evaluated in terms of code size, execution speed, power consumption, and hardware complexity.

Learning Operand Addressing Methods and Instruction Formats Through Practical Examples.

Immediate, direct, indirect, register, indexed, relative, and base addressing modes are studied and practically applied. The structure of machine instructions, including operation codes, operands, and addressing fields, is analyzed. Different instruction formats and their influence on instruction execution are examined. Instructions are practically encoded and decoded, and the relationship between addressing modes and processor performance is analyzed.

Study of the Interrupt System and Stack Organization Using Simulation Programs.

The principles of interrupt operation and interrupt handling mechanisms are studied. Hardware and software interrupts, interrupt priorities, interrupt vectors, and interrupt service routines are analyzed. Stack organization and stack operations, including PUSH and POP operations, are examined. Procedure calls, return mechanisms, and interrupt handling are simulated using appropriate programming or processor simulation environments.

Modeling the Operating Principles of the Processor's Internal Registers, ALU, and Control Unit.

The internal organization of a processor is studied and modeled. The functions of general-purpose and special-purpose registers, including the program counter, instruction register, stack pointer, and status register, are analyzed. Arithmetic and logical operations performed by the ALU are modeled, and the instruction execution cycle and the interaction between registers, the ALU, and the control unit are examined. The fetch, decode, and execute stages of instruction processing are simulated.

Comparing the Technical Specifications and Performance Evaluation of Modern Processors.

A comparative analysis of modern processors is conducted based on technical specifications, including clock frequency, number of cores and threads, cache memory capacity, architecture, thermal design power, and supported instruction sets. The characteristics of processors based on Intel,

AMD, and ARM architectures are studied. Processor performance is practically evaluated using benchmark programs. The relationship between technical specifications and actual computational performance is analyzed.

Learning the Operating Principle of Cache Memory and Its Impact on Performance Through Hands-On Experiments. The organization and operating principles of cache memory are studied. Cache levels L1, L2, and L3 and their role in reducing memory access time are analyzed, together with temporal and spatial locality. Cache hit and cache miss mechanisms are practically examined. The influence of data organization, memory access patterns, and program structure on cache performance is measured.

Analysis of Virtual Memory and Paging Mechanisms Using the Operating System's Tools. Virtual memory organization and paging mechanisms are studied. Virtual and physical address spaces, page tables, page frames, and page faults are analyzed. The translation of virtual addresses into physical addresses is investigated using operating system tools to monitor memory allocation, page usage, and virtual memory activity. The influence of paging on system performance is analyzed.

Testing the Performance of SSD/NVMe Storage Devices and Configuring RAID Arrays. The technical characteristics and performance indicators of SSD and NVMe storage devices are practically studied. Sequential and random read/write speeds, access time, and input-output operations per second are measured, and different storage technologies are compared. RAID arrays, including RAID 0, RAID 1, RAID 5, and RAID 10, are configured and practically investigated. The relationship between performance, data redundancy, and fault tolerance is analyzed.

Working with and Configuring Devices Connected via PCIe, USB, and Thunderbolt Interfaces. The operating principles and technical characteristics of modern computer interfaces are studied, including PCI Express, USB, and Thunderbolt interfaces. Data transfer rates, interface generations, bandwidth,

device compatibility, and communication protocols are analyzed. Peripheral devices are connected and configured through different interfaces. The performance and application areas of modern system interfaces are comparatively evaluated.

Learning the Basic Principles of GPU-Based Parallel Computing Using CUDA/OpenCL as Examples.

The principles of parallel computing using graphics processing units (GPUs) are introduced, together with the differences between CPU and GPU architectures. Threads, blocks, grids, and parallel execution models are studied. Simple parallel algorithms are practically implemented using CUDA or OpenCL, and CPU-based and GPU-based program execution times are compared. The advantages and limitations of GPU-based computing for large-scale computational problems are analyzed.

Measuring Memory and Disk System Performance Using Benchmarking Tools.

Methods and tools for evaluating the performance of memory and disk subsystems are studied. Memory bandwidth, latency, cache performance, disk read/write speed, access time, and input-output performance are measured using benchmarking tools. A comparative analysis of the obtained results is conducted, and system bottlenecks are identified. Conclusions and recommendations for improving memory and storage system performance are prepared.

Self-study

Structure and Architecture of a Personal Computer.

Principle of Operation of Computers and Their Types.

Working with Texts. Microsoft Word Text Editor.

Main and Auxiliary Computer Devices.

Minicomputers and Their Function.

Netbooks and Their Function.

Modern Tools for Developing Presentations.

Ways to Use Computer Technologies in Various Sectors and Industries of the National Economy.

Concept of the Instruction Set Architecture.

Advantages and Disadvantages of CISC and RISC Architectures.

Features of Modern ARM Architecture.

	Data Formats and Their Representation in Computers. Comparative Analysis of Operand Addressing Methods. Instruction Formats and Their Structure. Stack Memory and Its Operating Principles. Interrupt System and Processor Resource Management. Functions of the ALU, Registers, and Control Unit. Pipeline, Superscalar, and Multicore Processor Technologies. Memory Hierarchy and Its Impact on Performance. Cache Memory Levels and Operating Mechanisms. Development of RAM Technologies: DDR, DDR2, DDR3, DDR4, and DDR5. Virtual Memory and Paging Mechanisms. External Storage Devices: A Comparative Analysis of HDD and SSD Technologies. RAID Arrays and Their Types. PCI Express, USB, SATA, and Other Modern Interfaces. Input-Output Systems and Data Exchange Mechanisms. Parallel and Heterogeneous Computing Systems. GPU-Based Computing Technologies and Their Applications. Criteria for Evaluating Computer System Performance. Methods for Optimizing Memory and I/O Systems.
Teaching Methods	The module is delivered through interactive lectures, problem-based and inquiry-based learning, practical exercises, computer simulations, and project-based activities. Interactive lectures explain the fundamental concepts of computer organization, processor architecture, instruction systems, memory hierarchy, and input-output systems, encouraging active student participation through questions, discussions, and real-world examples. Problem-based learning enables students to analyze computational and architectural problems, apply theoretical knowledge, and develop effective solutions, while practical exercises focus on number systems, data representation, Boolean logic, instruction formats, processor components, cache

	and virtual memory, storage systems, computer interfaces, and system performance evaluation. Computer simulations and specialized software are used to model the operation of processors, memory systems, logic circuits, and parallel computing architectures, and students perform experiments, analyze technical specifications, conduct benchmarking, and interpret obtained results. Project-based learning and independent study promote the development of research, analytical, and technical skills through individual and group assignments, comparative studies, presentations, analytical reports, and mini-projects, while group discussions, seminars, and student presentations develop scientific communication, teamwork, critical thinking, and the ability to justify technical decisions. Digital educational resources, online simulations, visualization tools, and modern software environments further support independent learning and enable students to apply contemporary concepts of computer organization to practical and professional problems.
Prerequisites	Fundamentals of Informatics Programming 1,2 Discrete Structures Digital Logic and Electronics Operating Systems
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, practical performance, computer-based laboratory activities, seminar presentations, and the successful completion of independent study assignments, including number system conversion, Boolean logic and circuit analysis, instruction set and addressing mode analysis, processor architecture studies, cache memory experiments, virtual memory analysis, storage system performance testing, and computer system benchmarking. The midterm assessment

	consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as a personal computer architecture analysis, a comparative study of CISC and RISC architectures, an ARM architecture research project, a processor organization and ALU block diagram, a cache memory performance analysis, a virtual memory and paging analysis, and a comparative study of modern computer interfaces or storage technologies. The final assessment is a written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of computer organization and architecture, data representation, instruction systems, processor organization, memory hierarchy, input-output systems, storage technologies, and computer system performance, as well as their ability to apply theoretical principles to practical and analytical problems. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Patterson, D. A., & Hennessy, J. L. (2020). Computer Organization and Design RISC-V Edition: The Hardware/Software Interface (2nd ed.). Morgan Kaufmann. 2. Null, L., & Lobur, J. (2022). The Essentials of Computer Organization and Architecture (6th ed.). Jones & Bartlett Learning. 3. Stallings, W. (2022). Computer Organization and Architecture: Designing for Performance (12th ed.). Pearson. 4. Harris, D. M., & Harris, S. L. (2021). Digital Design and Computer Architecture: RISC-V Edition (2nd ed.). Morgan Kaufmann. 5. Bryant, R. E., & O'Hallaron, D. R. (2023). Computer Systems: A Programmer's Perspective (4th ed.). Pearson. 6. D.Y. Akbarov, P.F. Hasanov, X.P. Hasanov, O.P. Akhmedova, I.U. Kholimtoyeva. Mathematical Foundations of Cryptography. Textbook.–T.: "Aloqachi", 2019, 192 pages.

	7. RISC-V International. (2026). RISC-V specifications and resources. https://riscv.org 8. ACM Digital Library. (2026). Computer architecture and organization resources. https://dl.acm.org 9. IEEE Computer Society. (2026). Computer architecture publications. www.computer.org 10. Intel Corporation. (2026). Intel Architecture Documentation. https://www.intel.com 11. ARM Limited. (2026). ARM Architecture Reference Manuals. https://developer.arm.com
Responsible Lecturer	Mengnarov Adham Ergashevich

INDIVIDUAL PROJECT

Field	Description
Module Title	INDIVIDUAL PROJECT
Module Code	INL1704
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	VII semester
Module objective	The purpose of the Module Individual Project is for students to apply the theoretical knowledge and practical skills acquired in previous semesters within a single independent software project. Students identify real-world problems, analyze these problems, and develop solutions using modern technologies. The project process consists of several defined stages. The first stage involves defining the problem and the project objective. The second stage involves research and analysis. The third stage involves selecting and designing appropriate technologies. The fourth stage involves developing and testing the software. The final stage involves evaluating the results and preparing a professional presentation.
Module Learning Outcomes	Knowledge: LO1. Knows the principles, objectives, and life cycle of individual software project development, including

	project planning, requirement analysis, implementation, testing, deployment, and maintenance. LO2. Knows modern software development methodologies, project management techniques, software architecture principles, and approaches to selecting appropriate technologies, programming languages, and development environments. LO3. Understands database design principles, cloud technologies, API integration, user interface (UI) and user experience (UX) design concepts, and their application in software development projects. LO4. Knows software testing methods, debugging techniques, quality assurance principles, technical documentation standards, and software evaluation criteria used in professional software engineering. LO5. Understands current trends in software engineering, including artificial intelligence, cloud computing, innovative digital technologies, and their role in the development of modern software systems. Skills: LO6. Is able to analyze project requirements, define project objectives and scope, prepare project plans, and select appropriate software development methodologies for solving real-world problems. LO7. Possesses practical skills in designing software architecture, developing database structures, implementing software solutions, and integrating cloud services and application programming interfaces (APIs). LO8. Is able to design user-friendly interfaces, develop functional software applications, perform software testing and debugging, and improve software quality using modern development tools. LO9. Possesses practical skills in preparing technical documentation, user manuals, project reports, and professional presentations in accordance with academic and industry standards. LO10. Is able to plan, implement, monitor, evaluate, and successfully present an individual software project by applying project management principles and software engineering best practices. Competencies:
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	LO11. Possesses the competence to independently identify, analyze, and solve practical software engineering problems through the design and implementation of innovative software projects. LO12. Possesses the competence to select appropriate programming languages, software development tools, databases, cloud technologies, and APIs according to project requirements and technical specifications. LO13. Possesses the competence to design scalable, reliable, and maintainable software systems by applying software engineering principles, modern development methodologies, and quality assurance practices. LO14. Possesses the competence to evaluate software performance, identify technical limitations, optimize software solutions, and ensure compliance with functional and non-functional requirements. LO15. Possesses the competence to prepare professional technical documentation, communicate project outcomes effectively, and present software solutions to academic and professional audiences. LO16. Possesses the competence to work independently and collaboratively in software development projects, demonstrating project management, communication, leadership, and ethical professional behavior. LO17. Possesses the competence to engage in lifelong learning, adapt to emerging software technologies, apply innovative approaches in software development, and continuously improve professional knowledge and practical competencies in software engineering.
Content	Lectures Introduction and Basic Concepts. The concept, objectives, and significance of individual software projects are introduced. The role of project-based learning in developing professional competencies is examined. Types of software projects, the project life cycle, and the responsibilities of project developers are considered. Ethical considerations and professional standards in software engineering are discussed.

Project Purpose and Objectives. The purpose, objectives, scope, and expected outcomes of a software project are defined. Stakeholders and user requirements are identified, together with functional and non-functional requirements. Clear project goals are prepared. Measurable success criteria for software development are determined.

Research and Analysis. Research methodologies for software project development are studied. Literature review, problem identification, and feasibility analysis are examined. User requirements are collected and analyzed. Existing software solutions are evaluated, and appropriate approaches for project implementation are selected.

Project Planning. The principles of software project planning are studied. Project schedules are prepared, milestones are defined, resources are allocated, project duration is estimated, and potential risks are identified. Project management techniques are applied to ensure effective project execution.

Technology Selection. The criteria for selecting programming languages, frameworks, databases, development tools, and software platforms are studied. A comparative analysis of available technologies is conducted. Appropriate technologies are chosen according to project requirements, scalability, security, and maintainability.

Software Development. Software implementation using modern programming techniques is studied. Coding standards, modular programming, version control systems, and software engineering principles are applied. Software components are developed according to project specifications, ensuring code quality.

Interface Design. The principles of user interface (UI) and user experience (UX) design are studied. Intuitive, accessible, and responsive user interfaces are created. Wireframing, prototyping, and interface evaluation are examined, together with methods for improving user interaction through effective design practices.

Testing and Verification. Software testing methodologies, including unit testing, integration

testing, system testing, and user acceptance testing, are studied. Debugging techniques, error detection, and software verification and validation are examined. Software reliability, correctness, and quality are ensured before deployment.

Project Implementation. Software modules are integrated into a complete application. Database connectivity, API integration, deployment procedures, and system configuration are examined. Software implementation processes are managed while ensuring that functional and technical requirements are satisfied.

Project Monitoring and Control. Project progress is monitored throughout the software development life cycle. Project performance is measured, deviations from the project plan are identified, risks are managed, and corrective actions are implemented. Project management tools are applied to ensure timely and successful project completion.

Project Results and Impact. Software project outcomes are evaluated based on predefined objectives and performance indicators. Software quality, usability, efficiency, and practical impact are assessed. Project achievements, limitations, and opportunities for further improvement are identified.

Project Presentation. Professional project presentations and software demonstrations are prepared. Project objectives, development methodology, implementation process, testing results, and project outcomes are presented. Effective communication and presentation skills for technical audiences are developed.

Problems and Solutions. Technical, organizational, and implementation challenges encountered during software development are identified. Problem-solving strategies, debugging approaches, and optimization techniques are analyzed. Lessons learned throughout the project implementation process are discussed.

Future Trends. Current and emerging trends in software engineering are studied, including artificial intelligence, cloud computing, the Internet of Things (IoT), DevOps, cybersecurity, low-code development, and modern software architectures. Their impact on

future software projects is evaluated. Their influence on professional practice in software engineering is also considered.

Completion and Evaluation. The software project, its documentation, deployment, and evaluation are finalized. Technical reports, user documentation, and maintenance plans are prepared. Project outcomes, professional competencies acquired, and recommendations for future software development projects are reflected upon.

Practical lessons

Defining the Project Objectives and Tasks. The project problem is identified, and project objectives, scope, and expected outcomes are defined. Project tasks are prepared, functional requirements are determined, and measurable success criteria are established. A clear project proposal is developed based on real-world needs.

Conduct a SWOT Analysis. A SWOT (Strengths, Weaknesses, Opportunities, and Threats) analysis is performed for the proposed project. Internal and external factors that may influence project success are evaluated. SWOT results are used to support project planning and decision-making.

Developing the Project Plan. A comprehensive project plan is prepared, including project milestones, timelines, task scheduling, resource allocation, and risk identification. Project planning techniques are applied to ensure systematic project execution. The resulting plan supports efficient and coordinated implementation of the software project.

Research on Technology Selection. Available programming languages, frameworks, databases, cloud services, and development tools are analyzed. Alternative technologies are compared according to project requirements, scalability, security, and performance. The selection of appropriate technologies for project implementation is justified.

Software Architecture. The overall software architecture of the project is designed. System structure diagrams are developed, software modules are defined, component interactions are identified, and architectural documentation is prepared.

Software engineering principles are applied to build scalable and maintainable systems.

User Interface Design. User-friendly and responsive interfaces are designed using wireframes and prototypes. Screen layouts, navigation structures, and user interaction components are developed. Usability and accessibility are evaluated according to UI/UX design principles.

Software Development. The software project is implemented according to the approved design and requirements. Program modules are developed, and system components are integrated. Coding standards are applied, and version control tools are used throughout software development.

Conducting the Testing Process. Unit testing, integration testing, and system testing are performed to verify software functionality. Software defects are identified, documented, and corrected. Debugging techniques are applied, and software quality is evaluated before deployment.

Preparation of the Project Implementation Plan. A deployment and implementation plan is prepared for the completed software project. Installation procedures, configuration requirements, and data migration steps are defined. User support activities are established to ensure successful implementation.

Monitoring and Control Practices. Project progress is monitored by comparing planned and actual activities. Project performance is evaluated, implementation risks are identified, and project changes are managed. Corrective actions are applied to ensure successful project completion.

Evaluation of Results. Project outcomes are evaluated based on predefined objectives and performance indicators. Software functionality, reliability, usability, and overall project effectiveness are assessed. Strengths, limitations, and recommendations for future improvements are identified.

Preparing the Presentation. Professional project presentations, demonstration materials, and technical reports are prepared. The project objectives, implementation process, testing results, and achieved

	outcomes are presented. Effective communication and presentation skills are developed. Problem Solving. Technical and organizational problems encountered during project implementation are identified. Analytical and problem-solving techniques are applied to resolve software development challenges. Solutions and lessons learned throughout the project are documented. Studying Future Trends. Emerging technologies and current trends in software engineering are investigated, including artificial intelligence, cloud computing, DevOps, cybersecurity, and intelligent software systems. Their potential applications in future project development are evaluated. Their relevance to professional software engineering practice is also considered. Project Completion and Evaluation. The software project is completed by finalizing documentation, reviewing project deliverables, and conducting the final evaluation. Project achievements, acquired competencies, and opportunities for further development are reflected upon. The project is prepared for successful presentation and defense. Self-study Project Concepts and Characteristics of Individual Projects. Project Objectives, Scope, and Expected Outcomes. SWOT Analysis and Project Planning Techniques. Research Methodology and Technology Selection. Software Architecture and Project Design Principles. Programming Languages and Development Environments User Interface (UI) and User Experience (UX) Design Principles. Database Design and SQL Fundamentals. Cloud Technologies and API Integration. Software Implementation and Project Monitoring. Software Testing Methods and Debugging Techniques. Project Evaluation and Performance Assessment.
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	Technical Documentation and User Manual Preparation. Project Presentation and Defense Techniques. Future Trends in Software Engineering, Artificial Intelligence, and Innovative Technologies.
Teaching Methods	The module is delivered through interactive lectures, practical project-based sessions, independent research, case study analysis, problem-based learning, software design workshops, and guided project supervision. Students develop professional competencies by applying project planning techniques, software architecture design, database development, user interface (UI/UX) design, software implementation, testing, and project management practices throughout the software development life cycle. Practical activities emphasize analytical thinking, problem-solving, collaborative discussions, technical documentation, software testing, debugging, and project presentation skills. Independent learning, digital educational resources, modern software development tools, cloud platforms, and version control systems further enhance students' ability to design, implement, evaluate, and present innovative software solutions that address real-world challenges while promoting professional responsibility, creativity, and lifelong learning.
Prerequisites	Programming 1,2 Database Computer Networks
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final project assessment (50%). The continuous assessment evaluates students' classroom participation, practical project activities, independent research, project planning, software design exercises, and the successful completion of independent study assignments. Students are assessed on their ability to define project objectives, conduct requirement analysis, prepare project plans, perform SWOT analysis, select appropriate technologies, and

	develop software design documents. Active participation in discussions, practical sessions, and timely submission of project-related tasks are also considered in the assessment. The midterm assessment consists of written assignments, practical project tasks, progress reports, and the evaluation of project development activities. Students demonstrate their ability to design software architecture, develop user interfaces, implement software components, integrate databases and APIs, conduct software testing, prepare technical documentation, and apply project management techniques throughout the software development process. The final project assessment is based on the successful completion, presentation, and defense of the individual software project. Students are evaluated on the quality of project planning, software implementation, functionality, innovation, testing results, technical documentation, presentation skills, and their ability to justify design decisions and respond to technical questions. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Bjarne Stroustrup. Programming: Principles and Practice using C++ (Second Edition). Addison-Wesley. 2014, 1305 pages. 2. O. R. Yusupov, E. Sh. Eshonqulov. Database Management Systems. A textbook for higher education institutions. - Samarkand: SamDU Publishing House. 2023, 264 pages. 3. O. R. Yusupov, E. Sh. Eshonqulov, Yu. Sh. Yuldoshev. Fundamentals of Programming. Textbook for higher education institutions. - Samarkand: Samarkand State University Publishing House. 2023, 780 pages. 4. Stamp M. Information security: principles and practice // John Wiley & Sons, 2011, p. 606. 5. Markov A. S., Barabanov A. V., Dorofeev A. V., Tsirlov V.L. Seven Safe Information Technologies / ed. by A. S. Markov.—M.: DMK Press, 2017.— 224 pp. 6. D.Y. Akbarov, P.F. Hasanov, X.P. Hasanov, O.P. Akhmedova, I.U. Kholimtoyeva. Mathematical

	Foundations of Cryptography. Textbook.–T.: "Aloqachi", 2019, 192 pages. 7. M.X. Hakimov, S.M. Gaynazarov. Database Management Systems. 8. Manjikian M. Cybersecurity ethics: an introduction.– Routledge, 2017, 328 p. 9. Kostopoulos G. Cyberspace and cybersecurity.– CRC Press, 2017, 316 p. 10. Christen M., Gordijn B., Loi M. The Ethics of Cybersecurity.– Springer Nature, 2020.– P. 384. 11. Pande J. Introduction to Cyber Security. Uttarakhand Open University, 2017, 152 p. 12. Cybersecurity Fundamentals Study Guide, ISACA 2015, 196 p. 13. Easttom C. Computer security fundamentals.– Pearson IT Certification, 2019, 447 p.
Responsible Lecturer	Norqobilov Akram Mallayevich, teacher.

SIGNAL AND IMAGE PROCESSING

Field	Description
Module Title	SIGNAL AND IMAGE PROCESSING
Module Code	STI1506
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 90 hours (46 hours Lecture, 44 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	V semester
Module objective	The objective of the Module Signal and Image Processing is to develop students' theoretical foundations and practical skills in the digital processing of single-dimensional signals, including audio, sensor, and biomedical signals, and two-dimensional signals, namely images. Modern software tools, including MATLAB and Python, are used to solve real-world signal and image processing problems. During the course, students are introduced to the analysis of discrete signals and systems. Students study Fourier and z-transforms, together with DFT and FFT spectral analysis. Digital filter

	design is examined. Image enhancement, restoration, segmentation, and compression techniques are studied. The course also introduces modern machine learning approaches applied to signal and image processing.
Module Learning Outcomes	Knowledge: LO1. Understands the principles of analysing discrete signals and systems in the time, frequency, and z-domains. LO2. Understands the Fourier transform and knows the operating principles of DFT and FFT hardware. LO3. Explains the nature of sampling and aliasing phenomena. LO4. Knows the design methods for FIR and IIR digital filters. LO5. Understands the fundamentals of digital image generation, sampling and quantisation. LO6. Knows methods for enhancing, restoring and denoising images in the spatial and frequency domains. Skills: LO7. Can perform spectral analysis of a signal using DFT/FFT. LO8. Can design and implement FIR and IIR filters based on the given specification. LO9. Can enhance, denoise and restore digital images. LO10. Can process colour images and perform morphological operations. LO11. Applies image segmentation, edge detection and feature extraction algorithms. LO12. Independently develops a signal and image processing project using MATLAB/Python tools. Competencies: LO13. Applies digital signal and image processing techniques effectively in professional practice. LO14. Applies spectral analysis and digital filtering technologies to solve practical problems. LO15. Applies image enhancement, segmentation, morphological processing and compression algorithms in real-world projects.

	LO16. Independently develops signal and image processing projects using MATLAB/Python tools. LO17. Visualises the results of analysis, interprets them and prepares analytical reports. LO18. Independently studies the latest technologies in the field (CNN, deep learning) and applies them in scientific and professional activities.
Content	Lectures Fundamentals of Signals and Systems. Analogue and digital signals, discrete-time sequences, linear time-invariant (LTI) systems, convolution, difference equations, and stability and causality are introduced. The mathematical formulation of discrete-time signals and their classifications is presented. The system impulse response is evaluated, and bounded-input bounded-output (BIBO) stability conditions are verified. Frequency Domain: DTFT. The Discrete-Time Fourier Transform (DTFT) and its properties are studied, including symmetry properties, frequency response, and fundamental theorems. The mathematical formulation of the DTFT for aperiodic sequences is presented. Magnitude and phase spectra are analyzed, together with the shifting, differentiation, and convolution properties. z-transform. The z-transform and region of convergence (ROC) are studied, together with the inverse z-transform, system function, and pole-zero analysis. Discrete sequences are mathematically mapped into the complex z-plane. ROC properties are analyzed to determine system causality and BIBO stability using pole-zero plots. Sampling Continuous Signals. Periodic sampling, the Nyquist theorem and aliasing, signal reconstruction, and changing the sampling frequency are studied. Continuous-to-discrete signal conversion is theoretically evaluated. Anti-aliasing low-pass filtering and mathematical reconstruction using ideal interpolation functions are analyzed. Digital Filter Structures. FIR and IIR filter structures are studied, including serial, cascaded, and parallel configurations, signal-flow graphs, and multifrequency and multirate structures. Digital filters

are graphically modeled using signal-flow graphs. Coefficient sensitivity and computational complexity are analyzed in direct-form I and II realizations.

Design of Digital Filters. IIR filters, including Butterworth, Chebyshev, and elliptic filters designed via the bilinear transform, and FIR filters, designed using the window method and the Parks-McClellan algorithm, are studied, together with linear-phase conditions. Discrete filters are mathematically designed from continuous-time prototypes. Linear phase conditions and ripple trade-offs in passband and stopband specifications are evaluated.

Discrete Fourier Transform (DFT). The Discrete Fourier Transform (DFT) and its properties are studied, including circular convolution and spectral analysis using the DFT. The mathematical formulation of discrete frequency sampling for finite-length sequences is presented. Circular time-shifting properties and the relationship between linear and circular convolution are analyzed.

Fast Fourier Transform (FFT). Radix-2 and radix-4 Fast Fourier Transform (FFT) algorithms are studied, together with time and frequency decimation and computational complexity. The algorithmic implementation of Decimation-In-Time (DIT) and Decimation-In-Frequency (DIF) structures is examined. The reduction in computational complexity achieved by FFT algorithms compared with direct DFT computation is analyzed.

Random Signals and Spectral Estimation. Correlation and power spectral density, window functions, the periodogram, Welch's method, and quantization noise are studied. Discrete random processes and autocorrelation functions are statistically modeled. Non-parametric power spectrum estimation methods and the impact of finite word-length effects are evaluated.

Fundamentals of Digital Image Processing. Visual perception, image formation, discretization and quantization, and inter-pixel relationships are studied. The principles of human visual system response and illumination models are examined. Spatial sampling

grids, gray-level resolution, and topological pixel adjacency (4-, 8-, and m-connectivity) are analyzed.

Image Enhancement in the Spatial Domain. Point operations, histogram equalization, and spatial filtering, including smoothing and sharpening, are studied. Pixel intensities are directly manipulated using transformation functions. Spatial-domain convolution is performed using derivative masks, Laplacian operators, and local statistic filtering.

Frequency Domain Image Processing. The two-dimensional Fourier transform, frequency filters, including low-pass, high-pass, Butterworth, and Gaussian filters, and the convolution theorem are studied. One-dimensional Fourier theory is extended to two-dimensional spatial frequency representations. High-pass, low-pass, and band-pass spatial frequency filtering are analyzed based on the 2D convolution theorem.

Image Restoration and Denoising. Noise models, inverse and Wiener filtering, least-squares methods, and restoration from projections are studied. Image degradation mechanisms and additive noise probability density functions are mathematically modeled. Optimal minimum mean square error filters are formulated for image reconstruction.

Processing of Colour Images. Color spaces, including RGB, HSI, and CMYK, and the fundamentals of color enhancement and segmentation are studied. Trichromatic theory and color transformation matrices are analyzed. Full-color image processing techniques are applied for feature isolation and dynamic range adjustment.

Morphological Processing. Binary and grey-level morphology, including dilation, erosion, opening, and closing, and morphological algorithms are studied. Structuring elements for binary and grayscale image analysis are formulated using set theory. Boundary extraction, hit-or-miss transformations, and skeletonization are applied.

Image Segmentation and Edge Detection. Thresholding, Sobel and Canny edge detectors, the Hough transform, and the watershed algorithm are studied. Digital images are partitioned into constituent

regions based on intensity discontinuities and similarities. Geometric primitives are detected using parameter space mappings and gradient-based operators.

Feature Extraction and Image Registration.

Feature descriptors, multiresolution and pyramid representations, eigenimages, and image registration are studied. Scale-invariant keypoints, corner boundaries, and statistical spatial descriptors are extracted. Mathematical spatial alignment techniques are applied for multimodal image fusion.

Image Compression. Entropy coding, including Huffman and arithmetic coding, the discrete cosine transform (DCT), JPEG, and the fundamentals of wavelet-based and JPEG2000 compression are studied. Information theory constraints for redundant data reduction are examined. Spatial-domain and transform-domain coding standards are theoretically analyzed for bandwidth and storage optimization.

Modern Approaches in Signal and Image Processing.

Machine learning and deep learning approaches, including convolutional neural networks (CNNs), image classification, and segmentation, are studied. Trainable feature hierarchies are integrated for complex pattern recognition. Convolutional layers, pooling strategies, and semantic segmentation frameworks are structurally analyzed.

Practical lessons

Generation of Discrete Signals. Convolution and the response of an LTI system are calculated using Python or MATLAB. Basic discrete sequences, such as unit impulse, step, and sinusoidal signals, are generated. Continuous and discrete convolution are numerically computed to evaluate system behavior under different input conditions.

Analysis of the Signal Spectrum. The DTFT, DFT, and FFT are studied, together with the effect of window functions on spectral analysis. Spectral analysis is practically realized using Fast Fourier Transform algorithms. Main-lobe and side-lobe trade-offs are evaluated using Rectangular, Hamming, and Hann windowing techniques to minimize spectral leakage.

Sampling and Aliasing. The phenomenon of aliasing is observed experimentally by varying the sampling frequency. The Nyquist-Shannon sampling theorem is practically verified using continuous audio and image inputs. Reconstruction errors and Moiré patterns caused by undersampling are analyzed.

Design of FIR Filters. FIR filters are designed and implemented using the window method and the Parks-McClellan algorithm. Filter coefficients are calculated to achieve desired magnitude and phase responses. Linear-phase FIR digital filters are practically tested on noisy data streams.

IIR Filter Design. IIR filters are designed via bilinear substitution and applied to audio signal filtering. Analog Butterworth and Chebyshev filters are transformed into discrete-time IIR filters. IIR filtering is applied to real-time audio signals for noise suppression.

Image Enhancement. Image reading, histogram equalization, and point enhancement are performed. Digital images are represented and manipulated as arrays in Python or MATLAB. Contrast is expanded using global and local adaptive histogram equalization algorithms.

Image Filtering. Spatial and frequency-domain filtering are performed, including median, Gaussian, and Wiener noise removal. Spatial convolution matrices and 2D Discrete Fourier Transforms are implemented. The performance of spatial smoothing and frequency-domain noise removal filters is comparatively assessed.

Segmentation. Edge detection using the Canny detector and segmentation using thresholding and the watershed algorithm are performed. Discontinuities are detected using gradient operators and optimal edge detection techniques. Otsu thresholding and the watershed algorithm are executed for object isolation and boundary extraction.

Morphology and Characteristics. Morphological operations and image characteristics are isolated and analyzed. Structuring element matrices for dilation, erosion, opening, and closing are implemented.

Geometric shape descriptors and pixel-level region properties are extracted.

Colour Image. Color spaces are worked with, and color images are processed. Transformations between RGB, HSI, and YCbCr representations are performed. Color-based thresholding, contrast adjustment, and dynamic channel manipulation are implemented.

Image Compression. JPEG and DCT-based compression are performed, and the quality-size trade-off is assessed using PSNR and SSIM. The 8x8 block Discrete Cosine Transform, quantization matrix application, and entropy coding are implemented step by step. Objective quality metrics are calculated across different compression ratios.

Final Mini-Project. A real audio or image processing problem is solved in Python or MATLAB. An integrated processing pipeline is independently developed and deployed. The project is presented orally, code functionality is demonstrated, and technical results are defended.

Self-study

Classification of Signals: Analysis Using Audio, Biomedical (ECG/EEG) and Sensor Signals as Examples.

Convolution, Correlation and Their Application in Real-World Problems.

Comparative Analysis of DTFT and the Continuous Fourier Transform.

Verification of System Stability and Pole–Zero Location Using the z-transform.

Sampling and Aliasing: Practical Cases Using Audio and Image Examples.

Comparison of the Advantages and Disadvantages and Application Areas of FIR and IIR Filters.

Analysis of FFT Algorithms and Computational Complexity.

Window Functions, Spectral Leakage and the Trade-off Between Accuracy and Resolution.

Study of Image Formats (BMP, PNG, JPEG, TIFF) and Pixel Structure.

	Enhancing Contrast Using Histogram Methods (Equalisation, Matching). Comparison of Noise Models (Gaussian, Salt-and-Pepper, Speckle) and Denoising Methods. Colour Spaces (RGB, HSI, CIE Lab) and Their Conversion. Comparative Study of Edge Detectors and Segmentation Algorithms. JPEG Compression Stages and Quality Assessment Criteria (PSNR, SSIM). Fundamentals of CNNs and Applications of Deep Learning in Image Processing.
Teaching Methods	This course integrates problem-based learning and scaffolded instruction to bridge mathematical signal theory with practical computer engineering applications. Interactive coding workshops in MATLAB and Python enable students to implement digital filtering, spectral analysis, and computer vision algorithms in real time. Collaborative peer review and group project assignments foster critical thinking, algorithm optimization, and effective technical communication skills. Furthermore, laboratory experiments utilizing real-world audio, biomedical, and image datasets prepare students to independently design and deploy efficient data-processing solutions.
Prerequisites	Differential Equations Programming 1,2 Probability and Statistics
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and final written examination (50%). The continuous assessment results reflect the student's theoretical knowledge in the discipline, practical skills in processing discrete signals and digital images, ability to implement algorithms in MATLAB/Python, analyze spectral characteristics, apply spatial and frequency domain filters, and present technical findings logically and coherently. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as a

	signal classification presentation, a convolution and correlation analytical review, a DTFT vs CFT comparative analysis, a z-transform stability report, and a sampling and aliasing practical assignment. The final assessment is a written examination consisting of theoretical questions, algorithm design tasks, and practical signal and image processing problems designed to assess students' understanding of discrete-time systems, spectral analysis, filter design, compression mechanisms, and deep learning approaches in image processing. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Szeliski R. Computer Vision: Algorithms and Applications. 2nd ed. — Springer, 2022. szeliski.org/Book 2. Lichtman M. PySDR: A Guide to SDR and DSP using Python. — 2022 (continuously updated). pysdr.org 3. Zhang A., Lipton Z. C., Li M., Smola A. J. Dive into Deep Learning. — Cambridge University Press, 2023. d2l.ai 4. Prince S. J. D. Understanding Deep Learning. — MIT Press, 2023. udlbook.github.io/udlbook 5. Deisenroth M. P., Faisal A. A., Ong C. S. Mathematics for Machine Learning. — Cambridge University Press, 2020. mml-book.github.io
Responsible Lecturer	D. Abdualimov

COMPUTER ARCHITECTURE

Field	Description
Module Title	COMPUTER ARCHITECTURE
Module Code	KMA1606
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 120 hours
Frequency/Duration	VI semester

Module objective	The purpose of the Module Computer Architecture is to teach students the structure of computer systems, their operating principles, and the relationship between hardware and software. The Module also examines the architecture of modern computing systems. Through this course, students gain an in-depth understanding of the main components of a computer. Students study processors and their organization. Memory systems are examined in detail. Input-output devices and their operating principles are studied. The mechanisms by which these components interact are analyzed to develop a comprehensive understanding of computer architecture.
Module Learning Outcomes	Knowledge: LO1. Knows the theoretical foundations of computer architecture. LO2. Knows the structure and operating principles of the central processor. LO3. Has knowledge of the instruction set architecture (ISA). LO4. Knows the characteristics of cache memory, RAM, and external memory. LO5. Has knowledge of the operating principles of input-output devices. LO6. Knows the criteria for evaluating the performance of computer systems. Skills: LO7. Is able to analyze the components of a computer system. LO8. Explains the operation of the processor. LO9. Is able to use number systems and data encoding methods. LO10. Identifies the factors that affect computer performance. LO11. Is able to compare different architectural solutions. LO12. Is able to evaluate the capabilities of modern computing systems. Competencies:

	LO13. Is able to evaluate the performance of computing systems based on the elements of computer architecture. LO14. Is able to analyze the interaction of the processor, memory, and input-output devices. LO15. Is able to select and justify appropriate hardware for computing systems. LO16. Develops the ability to analyze the technical parameters of computer systems and formulate recommendations for increasing their efficiency. LO17. Has the competence to effectively use modern computer architectures and computing technologies in practical and professional activities.
Content	Lectures Introduction to Computer Architecture. The concepts of computer architecture and computer organization are introduced. The scope and objectives of studying computer architecture are defined. Computer architecture, referring to functional behavior, is distinguished from computer organization, referring to implementation details. The relationship between hardware design and software execution is introduced. Stages of Development of Computing Systems and the von Neumann Architecture. The historical evolution of computing systems through successive generations is traced. The principles of the von Neumann (stored-program) architecture are explained. Von Neumann and Harvard architectures are compared in terms of instruction and data handling. Representation of Data in a Computer: Number Systems and Encoding Methods. Binary, octal, decimal, and hexadecimal number systems and conversions between them are covered. Integer representation methods, including two's complement, are explained. Floating-point encoding and character encoding standards are introduced. Structure and Operating Principles of the Central Processor. The main functional blocks of the CPU, including the control unit, arithmetic-logic unit, and registers, are described. The coordination of these blocks in executing instructions is explained. The

concept of the datapath and its role in processing are introduced.

Instruction Set Architecture (ISA) and Addressing Modes. The concept of an instruction set and its role as the hardware-software interface are defined. Instruction types, including data transfer, arithmetic, and control instructions, are classified. Common addressing modes used to access operands are explained.

RISC and CISC Architectures: Features and Comparison. The design philosophy behind RISC and CISC processors is explained. Instruction complexity, execution speed, and hardware requirements are compared. Examples of processors representing each architectural approach are discussed.

Processor's Instruction Execution Cycle and Control Unit. The fetch-decode-execute cycle is examined in depth. The role of the control unit in generating control signals is explained. The coordination of instruction timing across processor components is described.

Memory Hierarchy: Registers, Cache, RAM, and External Memory. The concept of the memory hierarchy and its performance rationale are introduced. The speed, capacity, and cost of registers, cache, RAM, and secondary storage are compared. The principle of locality of reference underlying hierarchy design is explained.

Cache Memory Architecture and Its Impact on Performance. Cache organization types, including direct-mapped, set-associative, and fully associative organizations, are explained. Cache replacement and write policies are discussed. The influence of cache design choices on overall system performance is analyzed.

Input-Output Systems and Bus Architecture. The structure and function of system buses connecting components are described. I/O techniques, including programmed I/O, interrupt-driven I/O, and direct memory access (DMA), are explained. Bus arbitration and data transfer protocols are discussed.

Pipelined Processors and Their Operating Principles. The concept of instruction pipelining for improving throughput is introduced. Pipeline stages and the handling of structural, data, and control hazards are explained. Techniques for resolving pipeline stalls and hazards are discussed.

Parallel Processing and Instruction-Level Parallelism. The concept of instruction-level parallelism (ILP) and its exploitation are explained. Superscalar execution and out-of-order processing are discussed. Techniques such as branch prediction and speculative execution are introduced.

Multi-core Processors and Modern Microarchitectures. The motivation for and design of multi-core processor systems are explained. Cache coherence and inter-core communication challenges are discussed. Contemporary microarchitectural features in commercial processors are introduced.

Virtual Memory and Memory Management Mechanisms. The concept of virtual memory and address translation is explained. Paging, segmentation, and the role of the page table are described. The function of the translation lookaside buffer (TLB) and page replacement algorithms are discussed.

Modern Computer Architectures: ARM, x86-64, RISC-V, and Cloud Computing Systems. The design principles of ARM, x86-64, and RISC-V architectures are compared. Their respective application domains, including mobile, desktop/server, and open-source hardware, are discussed. The architectural considerations of cloud computing and virtualized infrastructure are introduced.

Practical lessons

Conversion Between Number Systems and Performing Binary Arithmetic Operations. Values are converted between binary, octal, decimal, and hexadecimal systems. Addition, subtraction, and multiplication in binary are performed using manual methods. Results are verified by cross-checking conversions across different bases.

Logical Elements and Performing Boolean Algebra Operations. Basic logic gates, including AND, OR, NOT, and XOR gates, and their truth tables are examined. Logical expressions are simplified using Boolean algebra laws. Simple combinational circuits are constructed from logical elements.

Analysis of the Computer's Main Components and Their Interrelationships. The function of each major hardware component within a system is identified. The flow of data and control signals between components is traced. A block diagram illustrating component interconnections is constructed.

Modeling Registers and Inter-Register Data Exchange. The structure and function of processor registers are simulated. Data transfer operations between registers are demonstrated. Timing and control signals governing register-to-register transfers are analyzed.

Analysis of Instruction Formats and Addressing Methods. The structure of fixed-length and variable-length instruction formats is examined. Operand fields, opcodes, and addressing mode indicators are identified. Various addressing modes are applied to sample instructions.

Familiarize Yourself with Assembler Instructions and Write Simple Programs. The syntax and structure of basic assembly language instructions are reviewed. Short assembly programs are written to perform simple computational tasks. Program execution is traced step by step to verify correct behavior.

Comparison of RISC and CISC Instruction Sets. Sample instruction sets from RISC and CISC processors are analyzed. Instruction count, complexity, and execution time are compared for equivalent tasks. Conclusions are drawn about trade-offs between the two design approaches.

Step-by-Step Analysis of the Processor's Instruction Execution Cycle. The fetch-decode-execute cycle is broken down for a sample instruction. Changes in registers and control signals are tracked at each stage. Timing relationships across the execution cycle are documented.

Memory Hierarchy Study and Calculation of Access Times. The average memory access time across hierarchy levels is calculated. The trade-off between memory speed, capacity, and cost is analyzed. Formulas relating hit rate and access time to overall performance are applied.

Calculating Cache Memory Performance. Cache hit and miss rates are computed for given access patterns. The average memory access time incorporating cache performance is calculated. Performance outcomes are compared across different cache mapping strategies.

Modeling of Bus Architecture and Data Transmission Processes. Data transfer across a system bus between components is simulated. Bus arbitration methods and their effect on transfer efficiency are analyzed. Data transfer rates are calculated under different bus configurations.

Calculation of the Efficiency of Pipelined Processing. Pipeline speedup and throughput are calculated for a given instruction sequence. Pipeline hazards and their impact on efficiency are identified. Techniques to resolve stalls and improve pipeline performance are applied.

Evaluating the Performance of Parallel Computing and Multicore Systems. Performance metrics for parallel task execution are calculated. Speedup and efficiency are analyzed using Amdahl's Law. Performance outcomes between single-core and multi-core configurations are compared.

Solving Virtual Memory, Paging, and Segmentation Problems. Practical problems involving page table lookups and address translation are solved. The effect of page faults on system performance is calculated. Paging and segmentation approaches to memory management are compared.

Analysis and Comparison of the Technical Specifications of ARM, x86, and RISC-V Architectures. The technical specifications and design philosophies of the ARM, x86, and RISC-V architectures are reviewed. Performance, power efficiency, and application domains are compared.

	Conclusions are drawn on architecture suitability for specific computing tasks. Self-study Development History and Generations of Computer Architecture. Distinctive Features of the von Neumann and Harvard Architectures. Architectural Solutions of Modern Computer Systems. Number Systems and Their Importance in Computer Technologies. Data Encoding and Decoding Methods. Fundamentals of Logic Gates and Digital Circuits. Components and Functions of the Central Processor. The Concept and Types of Instruction Set Architecture (ISA). Instruction Addressing Methods and Their Advantages. Comparison of RISC and CISC Architectures. The Role of ARM Architecture in Mobile Devices. Development Stages of the x86 and x86-64 Architectures. Prospects of the RISC-V Open Architecture. Registers and Their Importance in Processor Performance. Memory Hierarchy and Its Impact on Performance. Working Principle and Types of Cache Memory. Development of RAM Technologies (DDR, DDR2, DDR3, DDR4, DDR5). Virtual Memory and Its Operating Mechanism. Input-Output Systems and Their Organization. Computer Peripherals and Data Transmission Technologies. Advantages of DMA (Direct Memory Access) Technology. Pipeline Processing Technology. Superscalar Processors and Their Operating Principles. Parallel Computing Systems and Their Application Areas. Architecture and Performance of Multicore Processors.
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	Graphics Processing Units (GPUs) and Their Role in Computational Systems. Cloud Computing Systems Architecture. Processor Architectures Used in IoT (Internet of Things) Devices. Processors (TPU, NPU) for Artificial Intelligence and Machine Learning. Architecture of Quantum Computers and Future Computing Systems.
Teaching Methods	The module is delivered through interactive lectures, problem-based and inquiry-based learning, laboratory practical sessions, computer simulations, hardware analysis activities, seminars, group discussions, and project-based learning. Students analyze the architecture and performance of computer systems using simulation software and hardware analysis tools. Practical skills are developed in processor organization, memory systems, instruction set architecture, and computer performance evaluation. Independent learning and digital educational resources further promote critical thinking, technical communication, analytical thinking, and the application of modern computer architecture concepts.
Prerequisites	Computer Organization Computer Networks Programming 1,2
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, practical computer architecture exercises, seminar presentations, and the successful completion of independent study assignments, including computer architecture analysis, processor organization and instruction execution, number system conversion and data representation, memory hierarchy and cache memory analysis, instruction set architecture (ISA) exercises, input/output system and bus architecture analysis,

	and computer performance evaluation and optimization tasks. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, including a computer architecture analysis project, a report on processor architecture and memory systems, a presentation on modern computer architectures, a comparative study of RISC and CISC architectures, and a case study on multicore processors, parallel computing, and computer performance. The final assessment is a written examination consisting of theoretical questions and practical problem-solving tasks, designed to assess students' understanding of computer architecture fundamentals, processor organization, memory hierarchy, instruction set architecture (ISA), input/output systems, computer performance evaluation, and their ability to apply computer architecture principles in modern computing systems. Successful completion of all independent study assignments is a mandatory and integral component of the module assessment.
Recommended Literature	1. William Stallings. Computer Organization and Architecture: Designing for Performance. 11th Edition. – Boston: Pearson Education, 2019. – 800 p. 2. David A. Patterson, John L. Hennessy. Computer Organization and Design RISC-V Edition: The Hardware/Software Interface. 2nd Edition. – Cambridge, MA: Morgan Kaufmann, 2020. – 784 p. 3. Andrew S. Tanenbaum, Todd Austin. Structured Computer Organization. 6th Edition. – Boston: Pearson Education, 2016. – 800 p. 4. Morris Mano, Charles R. Kime, Tom Martin. Logic and Computer Design Fundamentals. 5th Edition. – Boston: Pearson Education, 2016. – 912 p. 5. John P. Hayes. Computer Architecture and Organization. 3rd Edition. – New York: McGraw-Hill Education, 2017. – 704 p. 6. David Money Harris, Sarah L. Harris. Digital Design and Computer Architecture. 2nd Edition. – Cambridge, MA: Morgan Kaufmann, 2015. – 592 p.
Responsible Lecturer	Ibragimova Mohigul Komiljonovna.

EMBEDDED SYSTEMS

Field	Description
Module Title	EMBEDDED SYSTEMS
Module Code	OPT1506
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 120 hours
Frequency/Duration	V semester
Module objective	The purpose of the Module Embedded Systems is to provide students with knowledge of embedded systems and their types. The Module examines the requirements for embedded systems. The integration of embedded systems with the control objects of their devices is studied. The Module also develops students' skills in working with system processors and development tools. Students gain practical experience in designing, programming, and testing embedded system hardware and software. The Module prepares students to apply embedded systems knowledge to solve practical engineering problems.
Module Learning Outcomes	Knowledge: LO1. Can explain the architecture, main components, and operating principles of embedded systems, and analyze their application areas. LO2. Will have practical skills in designing, configuring, and managing the hardware and software of systems built on microcontrollers and microprocessors. LO3. Can work with sensors, actuators, and external devices, and develop algorithms for collecting, processing, and managing data from them. LO4. Can solve practical problems using software development, testing, and optimization methods for embedded systems. LO5. Demonstrates competencies in developing projects based on IoT and modern embedded technologies, independently solving technical

	problems, and effectively using information and communication technologies. Skills: LO6. Will have skills in programming microcontrollers for embedded systems, loading programs onto them, and monitoring the operation process. LO7. Can assemble electronic circuits, connect sensors and actuators, and practically adjust their operation. LO8. Can apply methods for collecting data from embedded devices, processing it, and displaying the results through various interfaces. LO9. Has practical experience in creating and managing real-time systems and their software components. LO10. Can develop solutions for connecting devices to a network, organizing data exchange, and remote control based on IoT technologies, and can develop, test, debug, and technically justify the results of practical projects based on embedded systems. Competencies: LO11. Has the competence to analyze and design the hardware and software of embedded systems and to evaluate their performance. LO12. Demonstrates competence in selecting microcontrollers, microprocessors, and peripheral devices and in using them effectively to solve practical problems. LO13. Will have the competence to develop, test, optimize, and document software products for embedded systems. LO14. Acquires the competence to create automated control systems using sensors, actuators, and various communication interfaces. LO15. Will have the competence to develop and implement modern smart devices based on real-time operating systems (RTOS) and IoT technologies. LO16. Demonstrates the competence to develop engineering solutions to ensure the reliability, security, and energy efficiency of embedded systems. LO17. Will have the competencies to use modern information and communication technologies in professional activities, work in teams, independently
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	solve technical problems, and carry out innovative projects.
Content	Lectures Introduction to Embedded Systems. The concept and characteristics of embedded systems are introduced, together with their architecture and components. The role of embedded systems in modern technology is examined. Embedded systems are compared with general-purpose computer systems. Applications of embedded systems in industry and everyday life are discussed. Basic Requirements for Embedded Systems and Their Application Areas. The fundamental requirements of embedded systems are studied, including performance, reliability, and power efficiency considerations. Hardware and software constraints are examined. Application areas of embedded systems in industry, healthcare, transportation, and consumer electronics are discussed. Current development trends in embedded systems are considered. Interfaces for Connecting Embedded System Devices to the Controlled Object. The concept of hardware interfaces in embedded systems is introduced, together with communication between embedded devices and controlled objects. Analog and digital interface techniques are studied. Sensor and actuator integration methods are examined. Practical applications of interface technologies are discussed. Microcontroller Organization and Its Architecture. The architecture of microcontrollers is studied, including the CPU, memory, and peripheral components. Instruction execution and control mechanisms are examined. Microcontroller families and their characteristics are discussed. Applications of microcontrollers in embedded system design are considered. Parallel Information Processing Tools. The principles of parallel information processing are studied, together with hardware and software approaches to parallel processing. Multi-core processors and parallel architectures are examined.

Synchronization and communication techniques are discussed. Applications of parallel processing in embedded systems are considered.

Organizing the Interrupt Operation Mode in Embedded Systems. The concept and importance of interrupt mechanisms are studied, together with the types of interrupts and interrupt handling. Interrupt priority management is examined. Interrupt-driven programming techniques are discussed. Practical applications in real-time embedded systems are considered.

Hardware Means for Direct Memory Access (DMA). The concept and operation of Direct Memory Access (DMA) are studied, together with DMA controller architecture. Data transfer between peripherals and memory is examined. The advantages of DMA over CPU-controlled transfer are discussed. Applications of DMA in high-speed embedded systems are considered.

Input–Output Interfaces of Embedded Systems. The principles of input and output operations are studied, together with digital and analog input/output interfaces. Serial communication protocols, including UART, SPI, and I²C, are examined. Peripheral device integration is discussed. Applications of I/O interfaces in embedded systems are considered.

Operating Systems for Embedded Systems. The concept and functions of embedded operating systems are studied, together with resource management and task scheduling. The features of embedded operating systems are examined. Embedded operating systems are compared with general-purpose operating systems. Practical applications of embedded operating systems are discussed.

Real-Time Operating Systems (RTOS) and Their Usage Tools. The principles of real-time operating systems are studied, together with task scheduling and synchronization mechanisms. Memory and resource management in RTOS are examined. Development tools for RTOS applications are discussed. The practical implementation of real-time embedded systems is considered.

Designing Embedded Systems Software. The principles of embedded software development are studied, together with software architecture and modular programming. Programming languages and development environments are examined. Code optimization and debugging techniques are discussed. Software reliability and maintenance are considered.

Configuration and Software Testing of Microcontroller Systems. The configuration of microcontroller-based systems is studied, together with software testing methods and debugging tools. Functional and performance testing techniques are examined. Error detection and correction methods are discussed. The practical verification of embedded applications is considered.

Designing Hardware for Embedded Systems Based on the Arduino Platform. The architecture of the Arduino platform is studied, together with hardware design using Arduino boards and peripheral modules. Sensor and actuator interfacing is examined. The development of prototype embedded systems is discussed. Practical applications of Arduino in engineering projects are considered.

Embedded System Diagnostic and Monitoring Tools. The concept of embedded system diagnostics is studied, together with hardware and software monitoring techniques. Performance measurement and fault detection are examined. Debugging tools and diagnostic interfaces are discussed. Methods for ensuring system reliability and maintenance are considered.

Practical Hardware–Software Implementation of Embedded Systems Based on Modern Tools and the Use of Artificial Intelligence Elements. The integration of hardware and software components in embedded systems is studied, together with modern development platforms and design tools. The application of artificial intelligence techniques in embedded systems is examined. Intelligent control and automation solutions are discussed. The practical implementation of AI-based embedded applications is considered.

Practical lessons

Familiarization with the Application Areas of Embedded Systems and the Development Environment. The objectives and application areas of embedded systems are examined. The embedded system development environment is installed and configured. Development tools and software platforms are explored. A basic project is created and executed to practically apply embedded technologies.

Analysis of Requirements for Embedded Systems and Development of a Technical Specification.

Functional and non-functional requirements are identified. Hardware and software constraints are analyzed. Technical specifications for embedded system projects are prepared, following documentation standards and development planning. Practical requirement analysis is performed.

Creating Connection Diagrams for Sensors and Actuators to the Control Object.

The principles of connecting sensors and actuators are studied. Hardware interface diagrams are developed, and appropriate communication methods are selected. Circuit connections are verified. Sensor-based systems are practically implemented.

Study of the Microcontroller Architecture and Configuration of Its Main Modules.

The architecture of the selected microcontroller is studied. GPIO, timers, and communication peripherals are configured, together with the clock system and memory organization. Hardware modules are practically initialized. The performance of the configured components is evaluated.

Data Processing Using Parallel Processing Tools.

The principles of parallel data processing are studied, and concurrent processing techniques are implemented. Sequential and parallel processing are compared in terms of performance. Synchronization methods are examined. Practical applications of parallel processing in embedded systems are demonstrated.

Creating a Device Control Program Using Interrupts.

Interrupt sources are configured, and interrupt service routines are developed. Event-driven

programming techniques are applied. Interrupt-based applications are tested and debugged. Interrupt-controlled devices are practically implemented.

Using the DMA (Direct Memory Access) Mechanism and Analyzing Its Efficiency. DMA channels are configured, enabling data transfer between memory and peripherals without CPU intervention. DMA and processor-controlled transfer are compared in terms of performance. System efficiency is analyzed. DMA applications are practically implemented.

Organizing Data Exchange over UART, SPI, and I2C Interfaces. Serial communication interfaces are configured, and data transmission and reception are performed using UART, SPI, and I²C protocols. Communication between embedded devices and peripheral modules is established. Error detection and communication testing are conducted. Interface protocols are practically implemented.

Installing and Configuring the Operating System Environment for Embedded Systems. An embedded operating system environment is installed, and development tools and software packages are configured. System initialization and resource management are performed. Operating system functionality is verified. The development platform is practically prepared.

Creating and Managing Tasks in a Real-Time Operating System (RTOS). RTOS tasks are created and scheduled. Task synchronization and inter-task communication are implemented, together with resource allocation and priority management. Task execution is monitored. Real-time applications are practically implemented.

Software Development and Testing for Embedded Systems. Embedded software applications are developed. Program compilation and debugging techniques are applied. Functional and performance testing are conducted. Software errors are identified and corrected, and embedded applications are practically verified.

Configuration, Debugging, and Troubleshooting of Microcontroller Systems. Microcontroller

hardware and software are configured. Debugging is performed using integrated development tools. Hardware and software faults are identified. Troubleshooting techniques and system optimization are applied, and system performance is practically verified.

Designing and Programming an Embedded System Device on the Arduino Platform.

Embedded hardware is designed using Arduino, and Arduino-based applications are programmed. Sensors, actuators, and communication modules are integrated. The developed system is tested and validated. Arduino projects are practically implemented.

Working with Embedded System Diagnostics and Monitoring Tools.

Diagnostic and monitoring tools for embedded systems are used. System performance and resource utilization are measured. Hardware and software faults are detected and analyzed. Communication and processing activities are monitored, supporting the practical maintenance of embedded systems.

Developing a Smart Embedded System Design Using Artificial Intelligence Elements.

Artificial intelligence techniques are integrated into embedded systems. Intelligent hardware-software solutions are developed. Sensor data is processed using AI algorithms. Smart embedded applications are tested and evaluated, resulting in the practical implementation of AI-enabled embedded systems.

Self-study

The Concept of Embedded Systems and Their Modern Application Areas.

History and Prospects of Embedded Systems Development.

Key Requirements and Constraints for Embedded Systems.

Microcontrollers and Microprocessors.

Internal Architecture and Functional Blocks of Microcontrollers.

	Hardware Components of Embedded Systems and Their Functions. Methods for Interfacing Embedded Systems with Control Objects. Analysis of the Capabilities of Modern Hardware Platforms (Arduino, ESP32, STM32). Operating Principles of Input–Output Ports and GPIO Technology. Types of Sensors and Their Application in Embedded Systems. Operating Principles of Actuators and Execution Mechanisms. The Interrupt Mechanism and Its Advantages. Methods for Using Timers and Counters. Principle of Operation of DMA (Direct Memory Access) Technology. Comparative Analysis of UART, SPI, and I2C Interfaces. Parallel Processing Tools and Their Efficiency. Programming Languages and Tools for Embedded Systems. Capabilities of Compilers and Development Environments. General Overview of Operating Systems for Embedded Systems. Real-Time Operating Systems (RTOS) and Their Characteristics. Task Management Methods in an RTOS Environment. Process Synchronization and Resource Management. Software Optimization and Debugging. Designing Smart Devices Based on the Arduino Platform. The Role of the ESP32 Microcontroller in IoT Systems. Features and Applications of the STM32 Platform. IoT Technologies and Wireless Communication Tools. Embedded Systems Security, Diagnostics, and Monitoring. Integrating Artificial Intelligence Elements into Embedded Systems and Creating Smart Devices.
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Teaching Methods	The module is delivered through interactive lectures, practical laboratory sessions, hardware and software development exercises, project-based learning, demonstrations, simulation activities, programming assignments, seminars, group discussions, and independent learning tasks. Students develop practical skills in designing, programming, configuring, testing, and debugging embedded systems using modern microcontrollers, development platforms, sensors, actuators, and communication interfaces. Real-time operating systems are also used to reinforce these practical skills. Modern integrated development environments (IDEs), simulation software, Arduino, ESP32, and STM32 platforms, digital educational resources, and artificial intelligence technologies are used to enhance students' technical competence, problem-solving abilities, innovation skills, and professional readiness in embedded systems design and development.
Prerequisites	Computer Organization Database
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, practical programming activities, hardware configuration tasks, microcontroller programming exercises, and the successful completion of independent study assignments, including embedded system design, sensor and actuator interfacing, communication interface implementation, Arduino- and ESP32-based projects, software debugging, and technical presentations. The midterm assessment consists of written assignments, short in-class quizzes, practical laboratory tasks, oral presentations, and the evaluation of independent study projects, such as microcontroller-based application development, implementation of UART, SPI, and I²C communication, RTOS task management, embedded

	software testing, hardware–software integration, and smart embedded system design using modern development platforms. The final assessment is a written examination consisting of theoretical questions, system design tasks, and analytical problem-solving exercises designed to assess students' understanding of embedded systems architecture, microcontroller programming, hardware–software integration, communication protocols, real-time operating systems, embedded software development, and their ability to design, analyze, and implement reliable embedded solutions using modern engineering approaches. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. R.P. Abdurakhmanov. Embedded Systems. Textbook. Tashkent 2023. 2. M.E. Shukurova, X.J. Ro'ziyev, S.J. Yaxyayev, S.R. Ochilova. Embedded Systems (textbook). 3. E. A. Lee and S. A. Seshia. Introduction to Embedded Systems - A Cyber-Physical Systems Approach. Second Edition, MIT Press, 2017. 4. Making Embedded Systems: Design Patterns for Great Software. https://homel.vsb.cz/~sta048/mcu/doc/book/making_embedded_systems.pdf 5. Manuel Jiménez, Rogelio Palomera, Isidoro Couvertier. Introduction to Embedded Systems: Using Microcontrollers and the MSP430. Springer New York Heidelberg Dordrecht London. 6. Rob Toulson, Tim Wilmshurst. Fast and Effective Embedded Systems Design. Elsevier Ltd., 2017. https://homel.vsb.cz/~sta048/mcu/doc/book/fast_effective_embedded.pdf
Responsible Lecturer	Xolliyev Faxriddin Boxodirovich

OPERATING SYSTEMS

Field	Description
Module Title	OPERATING SYSTEMS
Module Code	OPT1704
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	VII semester
Module objective	The Module Operating Systems introduces the design and implementation of modern general-purpose operating systems. It focuses primarily on the operating system kernel, concentrating on concepts that lead to practical implementation. The aim of the Module is to develop theoretical and practical knowledge of system software. Students study the operating system and its applications. The architecture of operating systems is examined. The Module also develops students' understanding of the operating system as an interface between the user and the computing system.
Module Learning Outcomes	Knowledge: LO1. To have a deep understanding of the architecture of operating systems and the structure, functions and operating principles of the kernel. LO2. Knowledge of the concepts of processes and threads, process creation and management, inter-process communication (IPC), and synchronisation mechanisms. LO3. Master the allocation of central processor time, scheduling algorithms for processes, resource management and methods for ensuring operating system efficiency. LO4. Knowledge of the causes of deadlock situations, and of the methods for their detection, prevention and resolution. LO5. Knowledge of modern memory management techniques, including memory allocation, protection and swapping mechanisms, physical and virtual

memory, and the principles of organising and managing paged virtual memory systems.

Skills:

LO6. Be able to analyse and describe the architecture, components and functional capabilities of modern operating systems.

LO7. Be able to link the theoretical (abstract) concepts of operating systems with practical mechanisms and explain their operating principles using real-world systems as examples.

LO8. Be able to conduct a comparative analysis of the approaches, algorithms and management mechanisms used in different operating systems, and to evaluate their advantages and limitations.

LO9. Be able to determine the suitability of operating systems for specific practical problems and software environments, evaluate their performance, and select appropriate solutions.

LO10. Possess practical skills in designing, configuring, optimising operating system components and enhancing their operational efficiency based on user and system requirements.

Competencies:

LO11. To have the competence to analyse the architecture, services and management mechanisms of operating systems and to apply them effectively in information and communication systems.

LO12. Be competent in installing, configuring, managing, monitoring and providing technical support for operating systems on computer systems and servers.

LO13. Be competent in resolving practical issues related to managing process, memory, file system and device resources, and in optimising system performance.

LO14. Be competent in making decisions to ensure operating system security, manage user rights, protect data and enhance system reliability.

LO15. Be competent in comparative analysis of various operating systems, selecting the appropriate platform for specific practical tasks, and designing and integrating system components.

	LO16. Be competent in the effective use of modern technologies in the field of operating systems, virtualisation, cloud computing and containerisation solutions in professional practice. LO17. Be competent in identifying, diagnosing and resolving issues related to operating systems, and in ensuring the continuous and stable operation of systems.
Content	Lectures Introduction to the Subject of Operating Systems. Basic Concepts of Operating Systems. Operating systems are introduced, together with their objectives, functions, and importance in modern computing. The historical development of operating systems is examined. Basic concepts and classifications of operating systems are studied, together with the relationship between hardware, software, and operating systems. Operating System Architecture. The Operating System Kernel. The architecture of operating systems and the role of the kernel are studied. Monolithic, layered, microkernel, hybrid, and modular architectures are examined. Kernel services, system calls, and communication between hardware and software are discussed. The Role of Operating Systems in Managing Computer Devices. Device Binding in Operating Systems. Device management mechanisms are studied, including input/output systems, interrupt handling, device controllers, drivers, and resource allocation. Static and dynamic device binding techniques are examined. The role of these mechanisms in managing computer devices is discussed. Types of Operating Systems. Virtual Machines. Operating systems are classified, including batch, multiprogramming, multitasking, distributed, real-time, embedded, and network operating systems. The principles of virtualization are studied. Virtual machines, hypervisors, and their applications are examined. Processes and Threads in Operating Systems. The Concept of a Process. Processes, threads,

process states, process control blocks, context switching, and multithreading are studied. Processes and threads are compared. Their role in modern operating systems is discussed.

Process Management in Operating Systems. Scheduling Algorithms.

Process creation, termination, scheduling, and resource allocation are studied. CPU scheduling algorithms are analyzed, including First-Come-First-Served (FCFS), Shortest Job First (SJF), Priority Scheduling, Round Robin, and Multilevel Queue Scheduling. Their role in process management is discussed.

Synchronisation and Deadlocks in Operating Systems. Inter-process Communication.

Synchronization mechanisms, race conditions, semaphores, mutexes, monitors, and deadlocks are studied. Methods of deadlock prevention, avoidance, detection, and recovery are examined. The principles of inter-process communication are discussed.

User Interface in Operating Systems. Command-Based Terminals.

Graphical and command-line user interfaces are studied. Command interpreters, shell environments, terminal commands, and scripting are examined. User interaction with operating systems through these interfaces is discussed.

Memory Management in Operating Systems.

Memory allocation techniques are studied, together with memory protection and swapping mechanisms. Segmentation, paging, and fragmentation are examined. Memory management strategies used in modern operating systems are discussed.

Managing Paged Virtual Memory. Virtual memory concepts and demand paging are studied. Page replacement algorithms, page tables, and translation lookaside buffers (TLB) are examined. Memory optimization techniques are discussed.

File Systems in Operating Systems. Virtualisation of File Systems.

File organization and directory structures are studied. File allocation methods and file access mechanisms are examined. Journaling file systems and virtual file system architecture are discussed.

Security in Operating Systems. Separation, Protection and Sharing. Operating system security principles are studied, together with authentication, authorization, and access control. User privileges, data protection, and encryption are examined. Malware protection and security policies are discussed.

The Windows Operating System, Architecture and Its Significance. The architecture, components, and services of Microsoft Windows operating systems are studied. The registry and file system are examined. Process management and built-in applications of Windows are discussed.

Operating Systems in the UNIX Family. macOS. Operating Systems for Cloud Computing. UNIX and Linux operating systems are studied, together with macOS architecture. Cloud operating systems and distributed computing environments are examined. Cloud infrastructure management is discussed.

Android Mobile Operating System, Architecture and Significance. Android architecture and Linux kernel integration are studied, together with the application framework. Mobile operating systems and an overview of iOS are examined, including the application lifecycle. The role of operating systems in modern software development is discussed.

Practical lessons

Learning the Main Functions and Capabilities of the Operating System. Operating system functions are practically explored using modern operating system environments. System configuration and user management are performed. Resource monitoring is conducted.

Analysis of the Operating System Architecture and Kernel Structure. The architecture of an operating system is investigated, together with its kernel components. System services are examined. The interaction between software and hardware is analyzed.

Working with Devices and Drivers in Operating Systems. Hardware devices are practically studied, together with driver installation and configuration.

Troubleshooting techniques are applied. Device management utilities are used.

Creating Virtual Machines and Installing Operating Systems. Virtual machines are installed and configured using virtualization software. Different operating systems are practically deployed in a virtual environment. The functionality of the deployed systems is verified.

Monitoring and Analysing Processes and Threads. System utilities are used to monitor running processes and threads. CPU utilization and memory consumption are measured. Process performance is analyzed.

Practical Study of Process Management and Scheduling Algorithms. Process scheduling algorithms are simulated and analyzed. CPU scheduling performance is evaluated using practical examples. The results are compared across different scheduling strategies.

Study of Inter-process Communication and Synchronisation Mechanisms. Synchronization techniques, including semaphores and mutexes, are implemented and analyzed. Shared memory mechanisms are studied. Communication between processes is examined.

Working with Basic System Commands in a Command-Line Terminal. Command-line interfaces and shell commands are practically used. File operations and process management are performed. Scripting and system administration tasks are practiced.

Learning to Manage Memory Resources in Operating Systems. Memory allocation is practically investigated. Memory usage is monitored using system tools. Memory resources are optimized based on the observed results.

Practical Analysis of Virtual Memory and Paging Mechanisms. Paging techniques and virtual memory allocation are investigated. Page replacement algorithms are analyzed. Memory performance is evaluated based on the results.

Working with File Systems and Studying Their Structure. Files and directories are practically

managed, together with disk partitioning. File permissions and storage management are studied. File system utilities are used.

Security Mechanisms and User Rights Management. User accounts and access permissions are configured. Authentication methods are applied. Security policies and protection mechanisms are implemented.

Working with the Windows Operating System Environment. Windows administration tools are practically used, together with system configuration. Registry management is performed. Built-in Windows applications are explored.

Analysis of UNIX/Linux, macOS and Cloud Operating Systems. UNIX/Linux and macOS environments are practically compared. Command-line utilities in each environment are used. Cloud-based operating systems and remote system management are examined.

Exploring Mobile Operating Systems. Android and iOS operating systems are investigated, together with their system architecture. Application management and permissions are examined. The role of these systems in modern mobile software development is discussed.

Self-study

History of the Development of Operating Systems and Modern Operating Systems.

Architecture of Operating Systems and Types of Kernels.

Virtual Machines and Virtualisation Technologies.

Device Management and the Role of Drivers.

Concepts of Processes and Threads.

Process Scheduling Algorithms.

Inter-process Communication and Synchronisation Mechanisms.

Command-Line Interfaces (CLI) and Graphical User Interfaces (GUI).

Memory Management Techniques in Operating Systems.

Virtual Memory and Paging Technologies.

	Modern File Systems and Their Features. Data Archiving, Backup and Recovery Technologies. Information Security and Antivirus Tools in Operating Systems. Comparative Analysis of Windows, Linux and macOS. Mobile Operating Systems (Android and iOS) and Their Development Prospects.
Teaching Methods	Students' knowledge of the theoretical foundations and operating principles of operating systems is developed. Mechanisms for managing computer resources, including the processor, memory, file system, and devices, are taught. Skills are developed in analyzing the architecture, components, and functions of modern operating systems. Practical skills are developed in installing, configuring, managing, and effectively using operating systems. The operating principles of operating system components, such as processes, threads, memory management, security, and network services, are taught. Students' systems thinking is developed, building their competence in independently solving operating systems-related problems and using modern information technologies effectively.
Prerequisites	Computer Organization Database
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final test examination (50%). The continuous assessment evaluates students' classroom participation, practical laboratory performance, command-line and graphical user interface activities, operating system installation and configuration tasks, virtual machine deployment, process and memory management exercises, file system management, security configuration, and the successful completion of independent study assignments, including comparative analyses of operating system architectures, kernel types, virtualization technologies, process scheduling algorithms, memory

	management techniques, file systems, operating system security, and presentations on modern operating systems and mobile platforms. The midterm assessment consists of written assignments, short in-class quizzes, laboratory reports, and the evaluation of independent study projects. Students are assessed on their ability to analyze operating system architecture, compare process scheduling algorithms, manage memory and file systems, configure virtual machines, evaluate operating system security mechanisms, and explain the practical applications of Windows, Linux, macOS, Android, and iOS operating systems. The final assessment is conducted as a test examination consisting of multiple-choice questions covering all lecture topics of the module. The test evaluates students' knowledge of operating system architecture, kernel functions, process and thread management, synchronization, memory management, virtual memory, file systems, security mechanisms, virtualization technologies, Windows, Linux, macOS, Android, iOS, and cloud operating systems, as well as their ability to apply operating system concepts to practical computing problems. Successful completion of all practical laboratory exercises and independent study assignments is required as an essential component of the overall module assessment.
Recommended Literature	1. U.R. Khammadamov, Dj.B. Sultanov, S.S. Parsiyev, U.M. Abdullayev. Operating Systems. (A textbook). – T.: «Nihol print» OK, 2021. – 436 pp. 2. M.M.Nurullayev, U.M.Ibragimov, M.M.Nafasov. Methodological instructions for carrying out practical work in the subject "Operating Systems". 3. Silberschatz A., Galvin P. B., Gagne G. Operating System Concepts Essentials. – Wiley Publishing, 2013. 4. Chaudhary P., Gupta B. B., Singh A. K. Securing heterogeneous embedded devices against XSS attack in intelligent IoT system // Computers & Security. – 2022. – Vol. 118. – P. 102710. 5. Tanenbaum A. Modern Operating Systems. – Pearson Education, Inc., 2009.

	6. Stallings W. Operating Systems: Internals and Design Principles. – Pearson Education Limited, 2018.
Responsible Lecturer	Xolliyev Faxriddin Boxodirovich

GEOINFORMATION TECHNOLOGY

Field	Description
Module Title	GEOINFORMATION TECHNOLOGY
Module Code	GAT1704
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (30 hours Lecture, 30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	VII semester
Module objective	The objective of the Module Geoinformation Technology is to develop students' theoretical knowledge, practical skills, and scientific outlook regarding the structural organization of geographic information technologies. The Module examines the development trends of geoinformation hardware and software tools. Students study the application of geographic information technology data structures and models. The Module introduces the architecture and components of geographic information systems. Students develop practical skills in working with modern GIS software packages. The Module prepares students to apply geoinformation technologies to real-world spatial analysis and decision-making problems.
Module Learning Outcomes	Knowledge: LO1. Geoinformation Technology Architecture: The student will acquire theoretical knowledge of the components of geoinformation systems, SMP systems and modern GIS software packages (ArcGIS, SAGA, MapInfo). LO2. Geodata models: Will know vector and raster models, the structure of attribute tables and their relationship with spatial data.

	LO3. Cartographic fundamentals: Will have an understanding of geographic coordinates, topographic maps, concepts of scale and the principles of map visualisation. LO4. Technological processes: knows the theoretical stages of data collection, scanning, vectorisation and georeferencing. LO5. Working in a software environment: will have the skills to set up, configure and carry out practical tasks in ArcGIS/ArcMap/ArcCatalog. Skills: LO6. Analytical operations: can practically apply spatial queries, data format conversion and attribute calculations using Spatial Analysis and ArcToolbox. LO7. Mapping: Will have practical skills in developing electronic maps, compiling thematic layers and preparing them for printing. LO8. 3D Design: They will acquire the skills to enrich analytical projects by constructing 3D models of complex objects and visualising them. LO9. Technical Decision-Making: Will be competent in selecting the most suitable GAT system for given tasks and assessing its technical requirements (SMP, PostgreSQL/PostGIS). LO10. Data quality and analysis: Will have the competence to check the quality of geodatabase data, manage the relationships between spatial and attribute data, and edit errors. LO11. Presentation and communication: They will be qualified to professionally justify prepared projects (maps, models, analyses) through visual presentations and oral explanations. LO12. Innovative Application: Will be competent in adapting remote sensing data to their own field (e.g., telecommunications) and in operating an automated geographic analysis system. Competencies: LO13. Ability to analyze mathematical structures, logical systems, and computational processes using formal methods and critical reasoning. LO14. Ability to identify, formulate, and solve theoretical and practical problems using discrete mathematical techniques and logical methods.
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	LO15. Ability to present mathematical arguments, proofs, and analytical results clearly and effectively in written and oral forms. LO16. Ability to acquire new knowledge independently, conduct research, and apply modern information resources in professional activities. LO17. Ability to model real-world and computational problems using sets, relations, functions, and logical expressions.
Content	Lectures The Nature of the Geoinformation Technology Discipline and Its Areas of Application. The concept, objectives, and development of geoinformation technologies are studied. The role of geographic information systems in various fields, including environmental management, urban planning, transportation, agriculture, and telecommunications, is explored. The advantages and practical applications of geoinformation technologies are analyzed. Modern geospatial solutions and their importance in decision-making processes are examined. Software and Hardware of Geoinformation Technologies. The hardware and software components of geoinformation systems are studied. The functions and capabilities of GIS software packages are examined. Computer hardware requirements for geospatial data processing and analysis are explored. The interaction between software tools and hardware resources in GIS environments is discussed. Selection of Geoinformation Technology Systems. The criteria for selecting appropriate geoinformation technology systems are studied. Different GIS platforms and their functionalities are analyzed. System requirements, performance characteristics, and application areas are evaluated. Commercial and open-source GIS solutions are compared. Symmetric Multiprocessor Systems (SMP). The architecture and principles of symmetric multiprocessor systems (SMP) are studied. Parallel processing and resource sharing in SMP

environments are examined. The role of SMP systems in large-scale geospatial data processing is explored. The performance advantages and limitations of multiprocessor systems used in GIS are analyzed.

Geoinformation Technology Data Structures and Models. Spatial data structures and geospatial data models are studied. Vector and raster data representations are examined. The organization, storage, and management of geographic information are discussed. The advantages and limitations of different data models are analyzed.

Creating Databases in Geoinformation Technologies. The principles of database design and implementation in geoinformation systems are studied. Methods of organizing spatial and attribute data are examined. Database structures, relationships, and management techniques are discussed. Practical skills in creating and maintaining GIS databases are developed.

Geodatabases. PostgreSQL, PostGIS. Geodatabase concepts and architectures are studied. The fundamentals of PostgreSQL and PostGIS technologies are examined. Spatial database management and the storage of geographic information are discussed. Operations with spatial datasets are performed, developing practical skills in geodatabase administration.

Working with Attribute Data and Spatial Queries. Methods of managing attribute information in geoinformation systems are studied. The relationship between spatial and non-spatial data is examined. Attribute-based and spatial queries are performed. Query results are analyzed and applied in decision-making processes.

Working with Attribute Data and Spatial Queries. Advanced techniques for data filtering, selection, and analysis are studied. SQL-based queries are applied to geospatial datasets. Attribute and spatial analysis methods are combined to solve practical tasks involving geographic databases. Analytical and problem-solving skills in GIS environments are developed.

Data Entry Technologies and Data Sources.

Methods of data collection and acquisition in geoinformation systems are studied. Primary and secondary data sources are examined. Data input technologies, including scanning, digitizing, and remote sensing, are discussed. Data quality and reliability are evaluated.

Models for Data Editing and Summarization.

Techniques for editing and updating geospatial data are studied. Methods of data correction, validation, and generalization are examined. Summarization processes for geographic information are discussed. Editing models are applied to improve data accuracy and consistency.

Vector Data and Operations on It.

Vector data structures and their applications are studied. Operations such as buffering, overlay analysis, clipping, merging, and spatial editing are examined. Vector topology and spatial relationships are discussed. Practical tasks are performed using vector data analysis tools.

Schematic Mapping and Data Symbolology in Geoinformation Technologies.

Cartographic visualization principles and map design techniques are studied. Methods of symbolization and thematic mapping are examined. The visual representation of geographic information is discussed. Schematic maps are created following cartographic standards.

Geographic Coordinates, Topographic Maps and Their Scale.

Coordinate systems and geographic referencing methods are studied. The principles of topographic mapping and scale representation are examined. Map interpretation and coordinate transformations are discussed. Geographic coordinates are applied in practical GIS tasks.

Linking Data Obtained from Remote Sensing with Geographic Information Systems.

Remote sensing technologies and data acquisition methods are studied. Techniques for integrating satellite and aerial imagery into GIS environments are examined. Image processing and georeferencing procedures are discussed. Remote sensing data are analyzed for practical applications in spatial analysis.

Practical lessons

Installation and Configuration of the ArcGIS ArcINFO Software Suite. Study of Its Aims and Objectives. The purpose, architecture, and capabilities of the ArcGIS ArcINFO software package are studied. The installation and configuration process of ArcGIS components is performed. The main objectives and application areas of ArcGIS in geoinformation technologies are examined. Practical skills in preparing the software environment for GIS projects are developed.

The ArcCatalog System and How to Launch It. Creating a Data Geodatabase. The functions and interface of ArcCatalog are studied. Methods of organizing and managing geographic data are examined. Geodatabases are created and configured for storing spatial and attribute information. Practical skills in managing GIS resources and datasets are developed.

Generating the GIS Model Structure in ArcCatalog, Data Entry and Editing. The process of creating a geographic information system structure in ArcCatalog is studied. Methods of data entry, editing, and validation are examined. Spatial and attribute data are organized within GIS projects. Practical skills in creating and maintaining GIS datasets are developed.

Working in ArcMap and Linking ArcCatalog Applications to Build a Geographic Information System. The ArcMap working environment and its integration with ArcCatalog are studied. GIS projects are created using spatial and attribute data. Layers, maps, and data sources are managed. Practical skills in building and organizing geographic information systems are developed.

Establishing and Managing the Database of Geographic Information Systems. The principles of GIS database creation and administration are studied. Methods of storing, updating, and managing geographic information are examined. Database relationships and data integrity are discussed. Skills in maintaining geospatial databases efficiently are developed.

Learning the Principles of Digital Mapping in ArcMap, Its Structure and Operation. The fundamentals of digital cartography and map creation in ArcMap are studied. Map structures, layer management, and visualization techniques are examined. Map design principles and cartographic standards are discussed. Practical skills in creating and editing digital maps are developed.

Interlinking Geodatabase Data and Working with Tables in the ArcMap and ArcToolbox Software Modules. Methods of connecting spatial and attribute data within geodatabases are studied. Attribute tables and relational data structures are worked with. Data queries and updates are performed. Practical skills in managing and analyzing geospatial information are developed.

General Concepts of ArcToolbox Functional Modules in Geographic Information Systems and Their Technical, Software and Information Support. The structure and functionality of ArcToolbox modules are studied. Available tools for data processing, analysis, and transformation are examined. Technical and software requirements for GIS operations are discussed. Practical skills in using ArcToolbox for geospatial tasks are developed.

Retrieving, Processing and Editing Information from the Data Geodatabase in the ArcToolbox Software Module. Techniques for extracting and processing data from geodatabases are studied. Editing, filtering, and transformation operations are performed. ArcToolbox tools are applied for spatial analysis and data management. Skills in handling complex geospatial datasets are developed.

Modifying the Structural Organisation of Data Models, Formats and Content. Methods of transforming and reorganizing geospatial data models are studied. Data format conversion techniques and content management are examined. Interoperability between GIS software environments is discussed. Practical skills in adapting data for different applications are developed.

Attributes of Data and Their Representation in an Object. The role of attribute data in geographic

information systems is studied. Methods of storing and displaying object attributes are examined. Attribute tables and data relationships are managed. Skills in analyzing and visualizing attribute information are developed.

Transforming Data for Other Software, Three-Dimensional Raster Data, Composing Thematic Layers. Methods of exporting and converting GIS data for use in other software systems are studied. The principles of three-dimensional raster data representation are examined. Thematic layers are created and managed. Practical skills in data integration and visualization are developed.

Developing Electronic Maps in ArcMap. Developing Applications in ArcToolbox. Techniques for creating professional electronic maps in ArcMap are studied. ArcToolbox tools are used to automate and enhance GIS workflows. Practical mapping projects and geospatial applications are developed. Technical and analytical GIS skills are improved.

Conditional Symbols and Presentation and Print Preparation Features in Geodatabase Data Scheming. Cartographic symbolization and visualization methods are studied. Conditional symbols are applied to represent geographic information effectively. Map layout design, presentation techniques, and print preparation are examined. Professional cartographic communication skills are developed.

Creation of 3D Models of Important and Complex Objects, Key Strategies. The principles of three-dimensional modeling in GIS environments are studied. Three-dimensional representations of geographic objects and structures are created. Visualization techniques are applied to complex spatial data. Practical skills in 3D analysis and presentation of geospatial information are developed.

Self-study

Independent Study of Literature and Learning Materials.

Preparing for Practical Sessions and Writing a Report.

	Oral Commentary and Submission of Assigned Topics According to the Presentation Type. Preparation and Submission of Presentations.
Teaching Methods	The module is delivered through a combination of lectures, laboratory classes, practical demonstrations, project-based learning, and independent study activities. Teaching methods include classroom instruction and hands-on training in ArcGIS software environments, including ArcMap, ArcCatalog, and ArcToolbox. Geodatabase creation and management, spatial data processing and analysis, digital mapping, and GIS modeling are also employed. Case studies, problem-solving activities, group discussions, presentations, and the use of modern geoinformation technologies further support learning. Practical activities involve creating and managing geodatabases, editing spatial and attribute data, performing spatial queries, conducting geospatial analysis, developing thematic and electronic maps, integrating remote sensing data, and creating 3D geographic models.
Prerequisites	Database Computer Networks Programming 1,2 Data Structures and Algorithms
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, practical GIS exercises, seminar presentations, and the successful completion of independent study assignments, including geodatabase creation, spatial data collection and processing, map design tasks, georeferencing exercises, remote sensing data analysis, and GIS project development. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as the creation of thematic maps,

	geospatial database design, spatial analysis reports, GIS-based case studies, remote sensing applications, and presentations on contemporary geoinformation technologies. The final assessment is a written examination consisting of theoretical questions and practical problem-solving tasks designed to assess students' understanding of geoinformation technologies, GIS software environments, spatial databases, geospatial analysis methods, cartographic principles, and their ability to apply geoinformation technologies to real-world problems.
Recommended Literature	1. A. G. Karmanov, A. I. Knyshev, V. V. Yeliseeva. Geoinformation Systems for Territorial Management: A Textbook – St. Petersburg: ITMO University, 2015. – 121 pp. 2. E. G. Kapralov et al. Geoinformatics: A Textbook for University Students. Moscow: Publishing Centre "Akademia", 2005. — 480 pp. 3. DeMers, Michael N. Fundamentals of Geographic Information Systems. New Mexico State University. Translated from the English. M.: Data, 1999. – 506 pp. 4. J. Kh. Zhumanov. Geoinformation Technologies in Hydrogeology. Tashkent: GP "Hydrogeology and Engineering Institute", 2016. – 258 pp. 5. T.X. Boltayev, Q. Rahmonov, O.M. Akbarov. Scientific Foundations of the Geoinformation System. Textbook. T., 2015. – 275 pp. 6. T.A. Tsipileva. Geoinformation Systems: A Textbook. – Tomsk: Tomsk Inter-University Centre for Distance Education, 2004. – 162 pp. 7. V.I. Laikin, G.A. Uporov. Geoinformatics: A Textbook. Komsomolsk-on-Amur: AmGU Publishing House, 2010. – 162 pp. 8. E.Yu. Safarov et al. Cartography and Geovisualisation. Textbook. T. 2015. – 158 pp.
Responsible Lecturer	Abdullayev Sharof Abdug'afforovich, Senior Lecturer

USAGE OF THE UZBEK LANGUAGE IN THE FIELD

Field	Description
Module Title	USAGE OF THE UZBEK LANGUAGE IN THE FIELD
Module Code	O'SQ203
Credit Points (ECTS)	3 Credits
Workload	Total workload: 90 hours Contact hours: 30 hours (14 hours Lecture, 16 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	IV semester
Module objective	The objective of the Module Usage of the Uzbek Language in the Field is to enhance students' speech competence in the field in Uzbek. The Module develops skills in the effective use of specialized terminology in both oral and written professional communication. Students' abilities to express their acquired knowledge of the field in written and oral Uzbek are improved. The Module ensures the standardization of terminology, key concepts, and the formatting of documents used in the field. Skills in discourse styles are developed, particularly the use and practical application of the scientific and official styles within the field. The Module also improves students' written Uzbek language skills and enhances their literacy.
Module Learning Outcomes	Knowledge: LO1. Basic Principles and Structures of Speech: The student must know the purposes, structure, and types of professional speech, for example, understanding various communication formats such as presentations, lectures, and written letters. LO2. Rhetorical Devices and Strategies: Studying the rhetorical techniques and arguments needed for persuasive and effective expression in speech. LO3. Communication Theory: Knowledge of various communication theories, their application in professional discourse, and their effectiveness. Skills: LO4. Effective Speech Preparation and Delivery: Skill in preparing a speech in a timely manner and

	presenting the content in a systematic and engaging way. LO5. Proper Pronunciation and Vocal Techniques: The ability to use correct pronunciation, vocal management, and physical expressions effectively in speech. LO6. Oral Mimicry: The skill of adapting and effectively using facial expressions, gestures, and physical expression during speech. LO7. Application of Communication Strategies: The ability to apply appropriate communication strategies in various communicative situations (e.g., discussion, negotiation, decision-making). Competencies: LO8. Audience Analysis: Knowledge of how to understand and adapt to the needs, interests, and communication styles of different audience groups. LO9. Conflict Resolution: Skill in applying effective speech and communication methods to resolve conflicts and problems that arise in a professional environment.
Content	Lectures Objectives and Tasks of Teaching the Course "Application of Uzbek Language in the Field". The general purpose and educational significance of the course are examined. The main tasks aimed at developing students' professional language competence are studied. The role of the Uzbek language in specialized professional communication, along with course expectations, is considered. The Development of Language and Its Evolutionary Stages. The general concept of language development is examined. The main historical and evolutionary stages of language formation are studied. The factors influencing language change over time, along with their significance for modern linguistics, are considered. The Emergence and Development of Writing. The historical origins of writing systems are examined. The stages of development from early pictographic forms to modern alphabets are studied. The role of writing in preserving and transmitting language and culture, along with its evolution, is considered.

Lexical Layers of the Uzbek Language. The structure and classification of the Uzbek language's vocabulary are examined. The historical and functional layers of lexicon, including native and borrowed words, are studied. Their role in shaping the richness and diversity of the language, along with practical usage, is considered.

Language and Terminology. Formation of Terms. The relationship between general language and specialized terminology is examined. The principles and methods of term formation are studied. The role of terminology in professional and scientific communication, along with its standardization, is considered.

Text and Its Types. Microtext and Macrotext. The concept of text as a unit of communication is examined. The classification of texts into various types, including microtext and macrotext, is studied. Their structural and functional differences, along with practical applications, are considered.

Literary Speech and Its Styles. The concept and characteristics of literary speech are examined. The main functional styles of literary language are studied. Their distinguishing features and appropriate contexts of use, along with practical examples, are considered.

Practical lessons

Organization of Speech. The Beginning and End of Speech. The Communicative Qualities of Speech. The principles of structuring speech from beginning to end are examined. The methods of effectively opening and concluding a speech are studied. The essential communicative qualities that ensure clarity and impact, along with their practical application, are considered.

Speech and Literary Norms. Credible Speech and Television Show Speech. The concept of literary norms in shaping correct speech is examined. The characteristics of credible and persuasive speech are studied. The specific features of speech used in television broadcasting, along with its influence on public communication, are considered.

Writing an Initial Scientific Research Paper. Avoiding Plagiarism, Plagiarism, Citation, and

Referencing. The basic structure and requirements of an initial scientific research paper are examined. The concept of plagiarism and methods for avoiding it are studied. Proper techniques of citation and referencing, along with their importance for academic integrity, are considered.

Official-Business Style and Its Stylistic Features.

The concept and scope of the official-business style are examined. The distinctive stylistic features that characterize this functional style are studied. Its application in formal and administrative communication, along with practical examples, is considered.

Information and Data Documents. The purpose and classification of information and data documents are examined. The structural requirements and content standards for these documents are studied. Their role in facilitating accurate communication within organizations, along with practical examples, is considered.

Organizational Documents. The types and functions of organizational documents are examined. The structural and content requirements for drafting such documents are studied. Their role in regulating institutional activities and procedures, along with practical examples, is considered.

Order Documents. The purpose and legal significance of order documents are examined. The structural elements and drafting requirements for orders are studied. Their role in implementing decisions within organizations, along with practical examples, is considered.

Service Correspondence. Documents Related to Employee Activities. The concept and types of service correspondence are examined. The structure and requirements of documents related to employee activities are studied. Their role in maintaining formal communication and record-keeping within organizations, along with practical examples, is considered.

Self-study

	Speech Organization. The Beginning and End of a Speech. The Communicative Qualities of Speech. Speech and Literary Standards. Credible Speech and Television Show Speech. Writing an Initial Scientific Research Paper. Avoiding Plagiarism, Plagiarism, Citation, and Referencing. Official-Business Style and Its Stylistic Features. Information and Reporting Documents. Organizational Documents. Instructional Documents. Service Correspondence. Documents Related to Staff Activities.
Teaching Methods	Theoretical knowledge of the functional styles of the Uzbek literary language, the norms of official-business speech, and the linguistic foundations of professional communication is developed. The fundamentals of professional terminology in various fields, including legal, economic, technical, and humanitarian fields, and the rules for their correct and appropriate use are taught. Proficiency in creating official documents, including orders, statements, contracts, memorandums, and business letters, is developed in accordance with state standards and office work requirements. Issues of oral professional speech are addressed, developing skills for conducting meetings, negotiations, giving presentations, and speaking publicly in Uzbek. Skills in editing, reviewing, analyzing, and proofreading professional texts, as well as eliminating stylistic and grammatical errors, are developed. Competencies in the practical application of speech etiquette, business communication culture, and the effective use of the Uzbek language in intercultural professional dialogue are developed.
Prerequisites	Modern Uzbek Literary Language Stylistics Culture of Speech Official Language Proficiency
Exams and Assessment	Type of examination (written, oral, project) and its weight Students' knowledge is assessed through current control (20%), interim control (30%), and the final written exam (50%). Current control evaluates

	students' activity in class, participation in practical sessions, and successful completion of independent assignments. Independent assignments include compiling an analytical table on the lexical layers of the Uzbek language, analyzing texts in official and scientific styles, forming a dictionary of professional terms, completing practical assignments on the types of microtext and macrotext, and preparing a chronological scheme on the evolution of language and the stages of the emergence of writing. Interim control consists of written assignments, short in-class tests and surveys, and independent project work, including preparing a comparative analytical report on the functional styles of the Uzbek literary language, developing a proposal for standardizing the system of field-specific terms, preparing a video presentation on the topic of language and term formation, creating a practical guide project on text types, and writing a scientific essay on the historical stages of development of the Uzbek language. The final assessment is a written exam consisting of theoretical questions and practical assignments, aimed at assessing students' knowledge of the structure of the Uzbek language, its lexical layers, term formation, the fundamentals of text studies, literary speech and its styles, as well as their practical skills in analyzing and editing professional texts and creating field-specific documents. The timely and high-quality completion of all independent assignments is a mandatory condition for assessment within the course module.
Recommended Literature	1. Kadirova Z.Z. Application of the Uzbek Language in the Field. A Textbook. – Termez: "Globe Edit", 2024. – 110 p. 2. Korotkina I. B. Academic Writing: Process, Product. A textbook for universities. Moscow: Yurayt, 2021. 3. Kuvshinskaya Y.M., Zevakhina N.A., Akhapkina Y.E., Gordienko E.N. Academic Writing: From Research to Text. Textbook and Workbook for Universities. Moscow: Yurayt, 2022. 4. Korotkina I.B. A Model for Teaching Academic Writing. A Textbook for Universities. Moscow: Yurayt, 2021.

	5. N. Jiyanova, O. Mo'ydinova. Speech Culture. T.: "Ilm ziyo zakovat", 2019. 221 p. 6. Qo'ng'urov R., Begmatov E., Tojiyev Yo. Fundamentals of Speech Culture and Stylistics, T.: "Fan", 1992, 254 pp. 7. Husanov, N., Rasulov, Mo'ydinov, Q. The Culture of Speech and the Art of Oratory. T.: "IQTISOD-MOLIYA", 2006. 8. M. Akhmedova. Speech Culture and Professional Public Speaking. TATU, 2017, 114 p.
Responsible Lecturer	Zaynura Kadirova, PhD in Philology, Associate Professor.

PROFESSIONAL SPEECH CULTURE

Field	Description
Module Title	PROFESSIONAL SPEECH CULTURE
Module Code	KNM303
Credit Points (ECTS)	3 Credits
Workload	Total workload: 90 hours Contact hours: 30 hours (14 hours Lecture, 16 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	IV semester
Module objective	The objective of the Module Professional Speech Culture is to develop the written and oral communication culture, stylistics, and communication skills necessary for students to communicate successfully in their professional lives. This course enhances the knowledge gained during the educational process. The Module lays the foundation for students to find their place in their professional activities and become successful specialists. Students develop the ability to prepare and deliver effective professional speech. The Module also develops students' skills in professional etiquette and intercultural communication. It is expected that, during their future professional activities in their chosen field, students will be able to apply their speech-related knowledge, skills, and competencies in practice.

Module Learning Outcomes	Knowledge: LO1. Basic Principles and Structures of Speech: The student must know the purposes, structure, and types of professional speech, for example, understanding various communication formats such as presentations, lectures, and written letters. LO2. Grammatical and Stylistic Rules: Knowledge of proper grammatical structure, parts of speech, and the application of language style in a professional setting. LO3. Rhetorical Devices and Strategies: Study of rhetorical techniques and argumentation needed for persuasive and effective expression in speech. LO4. Communication Theory: Knowledge of various communication theories, their application in professional discourse, and their effectiveness. LO5. Audience Analysis: Knowledge of how to understand and adapt to the needs, interests, and communication styles of different audience groups. Skills: LO6. Effective Speech Preparation and Delivery: The skill of preparing a speech in a timely manner and presenting the content in a systematic and engaging way. LO7. Proper Pronunciation and Vocal Techniques: Skill in using correct pronunciation, voice management, and physical expressions effectively in speech. LO8. Speaker Mimicry: The skill of adapting and effectively using facial expressions, gestures, and physical expression during a speech. LO9. Application of Communication Strategies: The ability to apply appropriate communication strategies in various communicative situations (e.g., discussion, negotiation, decision-making). LO10. Conflict Resolution: The skill of applying effective speech and communication methods to resolve conflicts and problems that arise in a professional environment. Competencies: LO11. Skill in preparing and delivering effective speeches – the ability to plan a speech in advance
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	and present its content in an organized and engaging way for the audience. LO12. Proficiency in proper pronunciation and vocal techniques – the ability to adhere to phonetic norms in speech, control the voice, and coordinate physical expression with the speech. LO13. Ability to apply speech mime and physical expression – the ability to use gestures, facial expressions, and body movements in a way that is appropriate to the content. LO14. Communication and Conflict Management Competence – the ability to apply effective strategies in various communicative situations, reach agreements, and resolve problems through speech.
Content	Lectures Theoretical Foundations of Professional Speech Culture. The concept and significance of professional speech culture are examined. The theoretical principles underlying effective professional communication are studied. Its role in shaping a specialist's competence and reputation, along with its practical relevance, is considered. Rules and Standards of Communication in Professional Speech. The fundamental rules governing professional communication are examined. The established standards and norms of speech within professional contexts are studied. Their role in ensuring clarity, respect, and effectiveness in workplace interactions, along with practical examples, is considered. Professional Etiquette: Formal and Informal Communication. The concept of professional etiquette in various communication settings is examined. The distinguishing features between formal and informal communication are studied. Appropriate etiquette practices for different professional situations, along with their practical application, are considered. Speech Etiquette and Speech Activity. Oral Business Communication — The Etiquette of Communication. The principles of speech etiquette within professional speech activity are examined. The characteristics of oral business communication are

studied. The rules of etiquette governing effective and respectful oral exchanges, along with practical examples, are considered.

Analysis of the Characteristics and Types of Written and Spoken Speech. The distinguishing characteristics of written and spoken speech are examined. The various types and functions of each form of speech are studied. Their comparative features and appropriate contexts of use, along with practical examples, are considered.

Professional Writing: Documents and Their Written Structure. The principles of professional writing are examined. The structural requirements and formatting standards of professional documents are studied. Their role in ensuring clarity and formality in written communication, along with practical examples, is considered.

Creating Impactful Presentations Delivered with Confidence, Determination, and Skill. The principles of designing an impactful presentation are examined. Techniques for delivering presentations with confidence and skill are studied. The role of determination and effective delivery in engaging an audience, along with practical strategies, is considered.

Practical lessons

Developing Skills in Writing Formal Documents. The fundamental principles of writing formal documents are examined. Techniques for structuring and drafting various types of official documents are studied. Practical skills for ensuring clarity, accuracy, and appropriate formality in written communication, along with common exercises, are considered.

Email and Social Media Writing and Communication Etiquette. The principles of effective email and social media writing are examined. The etiquette rules governing digital communication are studied. Appropriate tone, structure, and professionalism in online correspondence, along with practical examples, are considered.

Reading and Presenting Oral Information. Techniques for reading and comprehending information effectively are examined. Methods of

organizing and presenting information orally are studied. The development of clarity and confidence in oral delivery, along with practical exercises, is considered.

Critical Thinking About Text. The concept and importance of critical thinking in text analysis are examined. Methods for evaluating the reliability, argumentation, and structure of written texts are studied. The application of critical thinking skills to improve comprehension and analysis, along with practical exercises, is considered.

Effective Transmission of Technical Information Through Graphs/Diagrams. The principles of presenting technical information visually are examined. Methods for designing clear and accurate graphs and diagrams are studied. Their role in enhancing understanding and communication of complex data, along with practical applications, are considered.

Developing Teamwork and Communication Skills. The importance of teamwork in professional and academic settings is examined. Strategies for fostering effective collaboration and communication within a team are studied. The role of interpersonal skills in achieving shared goals, along with practical exercises, is considered.

Speech Culture in Intercultural Communication. The concept of speech culture within intercultural contexts is examined. The challenges and considerations involved in communicating across different cultures are studied. Strategies for adapting speech and behavior to ensure respectful and effective intercultural interaction, along with practical examples, are considered.

International Communication and Cross-Cultural Speech Culture. The principles of international communication are examined. The characteristics of cross-cultural speech culture and its variations across regions are studied. The development of competencies necessary for successful communication in international and multicultural environments, along with practical applications, is considered.

	Self-study Speech Organization. The Beginning and End of a Speech. The Communicative Qualities of Speech. Speech and Literary Standards. Credible Speech and Television Show Speech. Writing an Initial Scientific Research Paper. Avoiding Plagiarism, Plagiarism, Citation, and Referencing. Official-Business Style and Its Stylistic Features. Information and Reporting Documents. Organizational Documents. Instructional Documents. Service Correspondence. Documents Related to Staff Activities.
Teaching Methods	To develop theoretical knowledge of the functional styles of the Uzbek literary language, the norms of official-business speech, and the linguistic foundations of professional communication. The fundamentals of professional terminology in various fields, including legal, economic, technical, and humanitarian fields, and the rules for their correct and appropriate use are taught. Proficiency in creating official documents, including orders, statements, contracts, memorandums, and business letters, is developed in accordance with state standards and office work requirements. Issues of oral professional speech are addressed, developing skills for conducting meetings, negotiations, giving presentations, and speaking publicly in Uzbek. Skills in editing, reviewing, analyzing, and proofreading professional texts, as well as eliminating stylistic and grammatical errors, are developed. Competencies in the practical application of speech etiquette, business communication culture, and the effective use of the Uzbek language in intercultural professional dialogue are developed.
Prerequisites	Modern Uzbek Literary Language Stylistics Culture of Speech Official Language Proficiency
Exams and Assessment	Type of examination (written, oral, project) and its weight

	Students' knowledge is assessed through current control (20%), interim control (30%), and the final written exam (50%). Current control evaluates students' activity in class, participation in practical sessions, and successful completion of independent assignments. Independent assignments include preparing an analytical overview of the theoretical principles of professional speech culture, compiling a comparative table of communication rules and standards used in professional contexts, analyzing examples of formal and informal professional etiquette, completing practical assignments on the etiquette of oral business communication, and preparing a comparative scheme on the distinguishing characteristics of written and spoken speech. Interim control consists of written assignments, short in-class tests and surveys, and independent project work, including preparing an analytical report on the role of speech culture in shaping a specialist's competence and reputation, developing a set of practical recommendations on communication etiquette in the workplace, preparing a video presentation demonstrating oral business communication skills, creating a sample professional document with proper written structure and formatting, and preparing and delivering an impactful presentation that demonstrates confidence, determination, and effective delivery skills. The final assessment is a written exam consisting of theoretical questions and practical assignments, aimed at assessing students' knowledge of the theoretical foundations of professional speech culture, the rules and standards of professional communication, formal and informal etiquette, the characteristics of written and spoken speech, as well as their practical skills in drafting professional documents, applying speech etiquette in oral business communication, and designing and delivering confident, well-structured presentations. The timely and high-quality completion of all independent assignments is a mandatory condition for assessment within the course module.
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Recommended Literature	1. Husanov B. Culture of Communication. Textbook for students of higher education institutions/Ministry of Higher and Secondary Specialized Education of the Republic of Uzbekistan. National University of Uzbekistan. – T.: "Ta'lim Nashriyoti", 2009. -156 p. 2. Jumaniyozov R. The Art of Speech. T.: Adolat, 2005. 3. Kortava T.V. Russian Language and Speech Culture. Moscow: Moscow Branch of the "Uchitel" Publishing House, 2018. 4. Trofimova O.V., Kupchik E.V. Fundamentals of Business Writing. Tyumen State University. – 4th ed., rev. – Moscow: Flinta, 2019. – 305 p. – URL: http://biblioclub.ru/ 5. Mirziyoyev, Sh. M. We will build our great future together with our brave and noble people. – Tashkent: "O'zbekiston" NMIU, 2017. – 488 p. 6. On the "Strategy of Actions" for the Further Development of the Republic of Uzbekistan, by the President of the Republic of Uzbekistan, Decree No. PF-4947 of February 7, 2017. 7. Decree of the President of the Republic of Uzbekistan No. PF-5850 of October 21, 2019, "On Measures to Fundamentally Increase the Prestige and Status of the Uzbek Language as the State Language" (National Legal Information Portal, 10/21/2019, No. 06/19/5712/3034). 8. Gafurov I., Boltayev M., Murodov M. Uzbek Speech Culture. – Tashkent: "Fan va Texnologiya", 2017. – 200 p. 9. Asqarov M. Speech Culture and Communicative Competence. – Tashkent: "Economics and Culture", 2018. – 148 p. 10. Nurmatov A. Professional Communication Ethics. – Tashkent: "O'zbekiston", 2020. – 172 p. 11. Hallsby, Atilla. Reading Rhetorical Theory: Speech, Representation & Power. University of Minnesota Libraries Publishing. https://open.lib.umn.edu/rhetoricaltheory/ 12. Hallsby, Atilla. The Rhetoric UnTextbook. https://the-untextbook.ghost.io/.
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	13. Jalolov X. Culture of Written and Oral Communication. – Tashkent: "Akademnashr", 2019. – 190 p. 14. Sattorov G. Fundamentals of Intercultural Communication. – Tashkent: "Sharq", 2021. – 160 p.
Responsible Lecturer	Amini Aziz Ahmad – Senior Lecturer, Department of Uzbek Language and Literature, (PhD)

PHYSICAL EDUCATION

Field	Description
Module Title	PHYSICAL EDUCATION
Module Code	JTA203
Credit Points (ECTS)	3 Credits
Workload	Total workload: 90 hours Contact hours: 30 hours (30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	IV semester
Module objective	The objective of the Module Physical Education is to develop students' theoretical knowledge and practical skills related to the physical culture system of Uzbekistan. The Module examines physical education, physical training, and children's sports in Uzbekistan. Physical development, physical preparation, sports training, and physical qualities are studied. Students are introduced to the three-stage sports competitions "Hope Sprouts," "Barkamol Avlod," and "Universiada," among others. The Module develops specialized theoretical knowledge and practical skills using modern tools and methods. Students develop the ability to organize mass health promotion events. The Module also prepares students for self-physical improvement and for teaching these skills independently.
Module Learning Outcomes	Knowledge: LO1. To provide students with the comprehensive knowledge to conduct training sessions in various sports, organize and analyze sports games, movement and ball-handling techniques in volleyball

	and handball, must have theoretical knowledge of offensive and defensive techniques and tactics. LO2. Must know methods for teaching and developing sports games, methods for organizing competitions, and ways to develop and train physical qualities. LO3. Knowledge acquired through oral responses and discussions during class must be applied in practice. LO4. Must know and be able to use the system of competition load assignments, and know how to organize and conduct the lesson and training process based on modern requirements. Skills: LO5. To provide students with comprehensive knowledge for conducting training sessions in various sports, organizing and analyzing sports games, and the techniques of movement and ball control in volleyball and handball, must have theoretical knowledge of offensive and defensive techniques and tactics. LO6. Must know methods for teaching and developing sports games, methods for organizing competitions, and ways to develop and train physical qualities. Competencies: LO7. The selection process can be carried out in collaboration with specialists from other disciplines closely related to sports (physiology, anatomy, sports medicine, massage, psychology, etc.). LO8. Is able to obtain rapid information and manage it effectively using modern technology during the training process.
Content	Practical lessons Assessment of Physical Fitness Level. The concept and importance of physical fitness assessment are introduced. Methods for evaluating strength, endurance, speed, flexibility, and coordination are studied, together with standard fitness tests and performance indicators. Assessment results are analyzed and interpreted. The role of physical fitness evaluation in promoting health and improving performance is discussed. Track and Field. Teaching the Standing Long Jump Technique. The fundamentals of the standing

long jump are introduced, together with proper body position during take-off and landing. Explosive strength and coordination are developed. Common technical errors and correction methods are studied. The standing long jump is practically performed.

Teaching Short-Distance Running Technique. The principles of sprint running are introduced, together with starting position and acceleration techniques. Proper running mechanics and finishing methods are studied. Exercises for improving speed and coordination are practiced. Performance is analyzed and technique is corrected.

Teaching Overhead and Underhand Passing and Receiving Techniques in Volleyball. The fundamentals of overhead and underhand passing are introduced, together with correct body position and hand placement. Passing and receiving techniques are practiced during game situations. Individual and pair drills are used for skill development. These techniques are practically applied in volleyball.

Teaching the Serve Technique in Volleyball. The principles of volleyball serving techniques are introduced, including underhand and overhand serve execution. Accuracy and consistency in serving are developed. Common technical mistakes and correction methods are studied. Practical serving exercises are performed.

Teaching Movement Techniques in Handball. The basic movement techniques in handball are introduced, including running, stopping, turning, and changing direction. Coordination and agility are developed. Practical movement drills are performed. These techniques are applied during play.

Teaching the Dribbling Technique in Handball. The principles of ball dribbling in handball are introduced, together with correct hand position and ball control. Dribbling while moving and changing direction is practiced. Individual skill development exercises are performed. These skills are practically applied in game situations.

Teaching the Movement Technique in Basketball. The fundamentals of movement without the ball in

basketball are introduced, including running, pivoting, stopping, and changing direction. Offensive and defensive movement techniques are studied. Coordination and agility exercises are practiced. These techniques are practically applied during gameplay.

Teaching Technical Methods of Ball Control in Basketball Play. The principles of basketball ball control are introduced, including dribbling, passing, catching, and protecting the ball. Individual technical drills are performed. Ball control is practiced under game conditions. Technical proficiency is developed.

Teaching Movement Techniques in Soccer. Training Game. The basic movement techniques used in soccer are introduced, including running, positioning, and changing direction. Coordination and agility are developed. Practical movement exercises are performed. These techniques are applied during training games.

Teaching Ball Carrying Technique in Soccer. Training Game. The principles of ball carrying and dribbling in soccer are introduced. Ball control is practiced while moving at different speeds, including direction changes and obstacle dribbling. Practical exercises for ball handling are performed. These skills are applied in training games.

Teaching the Technique of Striking the Ball in Soccer. Training Game. The techniques of kicking the ball using different parts of the foot are introduced, together with passing and shooting techniques. Accuracy and power are developed. Practical shooting and passing drills are performed. These techniques are applied during training games.

Rules of Badminton, Teaching Special Drills with the Shuttlecock. The official rules of badminton are studied, together with grip techniques and basic strokes. Shuttlecock control exercises are performed. Footwork and court movement drills are practiced. Badminton techniques are practically applied.

Rules of Table Tennis, Teaching Special Drills with Balls. The official rules of table tennis are studied, together with basic stroke and serving techniques. Ball control and rally exercises are

	practiced. Coordination and reaction training are performed. Playing skills are practically developed. Methodology for Organizing and Conducting Active Games. Practical Familiarization with the Game Content of "White Poplar or Blue Poplar," "Hunters and Ducks," and "Run and Carry the Ball." The principles of organizing and conducting active games are studied, together with the educational and health benefits of movement games. The rules and practical implementation of the games White Poplar or Blue Poplar, Hunters and Ducks, and Run and Carry the Ball are examined. Teamwork, coordination, and physical fitness are developed. Safety requirements and effective game management are discussed. Self-study Physical Fitness Assessment (BN). Track and Field. Teaching the Standing Long Jump Technique. Teaching Short-Distance Running Technique. Middle-Distance Running Technique. Teaching Running for 500–1000 Meters. Sports Games. Rules of Volleyball, Teaching Special Drills with Balls. Basketball Game Rules and Teaching Special Ball Drills. Rules of the Sport of Handball, Teaching Special Drills with Balls. Rules of the Game of Football, Teaching Special Drills with Balls. Rules of Badminton, Teaching Special Shuttlecock Drills. Rules of the Table Tennis Game, Teaching Special Drills with Balls. Teaching Movement Games. Gymnastics. Teaching Floor Exercises. Teaching General Developmental Exercises. Teaching Simple Acrobatic Exercises. Teaching Strength-Building Exercises on Gymnastic Equipment.
Teaching Methods	The module is delivered through interactive lectures, practical physical training sessions, demonstrations,

	guided practice, individual and group exercises, sports games, fitness assessment activities, and independent learning tasks. Students develop physical fitness, motor skills, and technical abilities through athletics, gymnastics, sports games, and movement-based activities. Teamwork and healthy lifestyle habits are also developed through these activities. Modern teaching methods, digital educational resources, multimedia materials, and sports equipment are used to enhance students' physical performance, theoretical knowledge, and practical competence. These methods further support students' lifelong commitment to physical activity and health promotion.
Prerequisites	General Biochemistry and Sports Biochemistry Track and Field Gymnastics National and Sports Games
Exams and Assessment	Type of examination (practical performance assessment) and its weight Student performance is assessed through continuous assessment (50%) and a final assessment (50%). The continuous assessment (50%) is conducted throughout the semester during practical classes. Students may earn up to 20 points by successfully completing the required 60-meter sprint performance standard. Assessment is based on the officially approved physical fitness standards for male and female students. Male students who complete the sprint in 9.0 seconds and female students in 10.0 seconds receive 18–20 points (90–100%). Male students who complete the sprint in 9.5 seconds and female students in 10.5 seconds receive 14–17.9 points (70–89.9%). Male students who complete the sprint in 10.0 seconds and female students in 11.5 seconds receive 12–13.9 points (60–69.9%). Male students who complete the sprint in 11.0 seconds and female students in 12.0 seconds receive up to 11.9 points (below 60%). In addition, students may earn up to 30 points for completing independent study assignments according

	to the module schedule. Independent study tasks may include preparing presentations on physical education and healthy lifestyles, writing reports, completing theoretical assignments, participating in sports-related projects, and carrying out other activities assigned by the instructor. All independent study assignments must be submitted within the specified deadlines. Assignments submitted after the deadline will not be accepted. Therefore, students may earn a maximum of 50 points in the continuous assessment. The final assessment (50%) is conducted after the completion of the course as a practical examination. Students are assessed by completing the required 100-meter sprint according to the approved physical fitness standards. Male students who complete the sprint in 13.5 seconds and female students in 17.0 seconds receive 45–50 points (90–100%). Male students who complete the sprint in 13.8 seconds and female students in 17.2 seconds receive 35–44.9 points (70–89.9%). Male students who complete the sprint in 14.0 seconds and female students in 17.5 seconds receive 30–34.9 points (60–69.9%). Male students who complete the sprint in 15.0 seconds and female students in 18.0 seconds receive up to 29.9 points (below 60%).
Recommended Literature	1. B.K. Amanturdiyev. The Game of Football. A Textbook. Termiz, "Surxon Nashr", 2021. 167 pages. 2. Q.Q.Xaqnazarov. Basketball: Sportivnaya podgotovka. O'quv qo'llanma. Termiz, "Surxon nashr", 2021. 132 bet. 3. B.K. Amanturdiyev. The Game of Football. Textbook. Termez, "Surhon Nashr", 2021. 167 pp. 4. Mirziyoyev, Sh.M. Strategy of the New Uzbekistan. – Tashkent: "Uzbekiston", 2021. 5. Law of the Republic of Uzbekistan "On Physical Culture and Sport" of September 4, 2015.
Responsible Lecturer	B.K. Amonturdiyev

PHYSICAL CULTURE

Field	Description
Module Title	PHYSICAL CULTURE
Module Code	JMA203
Credit Points (ECTS)	3 Credits
Workload	Total workload: 90 hours Contact hours: 30 hours (30 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	IV semester
Module objective	The objective of the Module Physical Culture is to develop students' theoretical knowledge and practical skills related to the physical culture system of Uzbekistan. The Module examines physical education, sport, and physical training in Uzbekistan, including children's sport. Physical development, physical preparation, sports training, and physical qualities are studied. Students are introduced to the three-stage sports competitions "Hope Sprouts," "Barkamol Avlod," and "Universiada," among others. The Module develops modern theoretical knowledge and practical skills and abilities, together with tools and methods that help to enhance them. Students develop the ability to organize mass health-promotion events. The Module also prepares students for self-physical improvement and for teaching these skills independently.
Module Learning Outcomes	Knowledge: LO1. To provide students with the comprehensive knowledge to conduct training sessions in various sports, organise and analyse sports games, movement and ball-handling techniques in volleyball and handball, must have theoretical knowledge of attacking and defensive techniques and tactics. LO2. Must know methods for teaching and developing sports games, organising competitions, and ways to develop and train physical qualities. LO3. The knowledge acquired through oral responses and discussions during lessons must be applied in practice.

	LO4. Must know and be able to use the system of competition loadings, and be able to organise and conduct the lesson and training process in accordance with modern requirements. Skills: LO5. To provide students with comprehensive knowledge for conducting training sessions in various sports, organising and analysing sports games, techniques for movement and ball control in volleyball and handball, must have theoretical knowledge of attacking and defensive techniques and tactics. LO6. Must know methods for teaching and developing sports games, organising competitions, and ways to develop and train physical qualities. Competencies: LO7. The selection process can be carried out in collaboration with specialists from disciplines closely related to sport (physiology, anatomy, sports medicine, massage, psychology, etc.). LO8. Is able to use modern technology to obtain information quickly and manage it effectively during the training process.
Content	Practical lessons Assessment of Physical Fitness Level. The concept and importance of physical fitness assessment are introduced. Methods for evaluating physical fitness components are studied, together with standard fitness tests and performance indicators. Assessment results are interpreted. The role of fitness evaluation in improving physical health is discussed. Athletics. Teaching the Technique of the Standing Long Jump. The fundamentals of the standing long jump are introduced, together with correct body position and movement phases. Techniques for take-off and landing are studied. Common errors and correction methods are examined. Long jump techniques are practically applied. Teaching Short-Distance Running Technique. The principles of sprint running are introduced, together with starting techniques and acceleration phases. Proper running posture and stride mechanics are studied. Finishing techniques and performance

improvement are practiced. Safety measures during sprint training are discussed.

Middle-Distance Running Technique. Teaching 500–1000 Metre Runs. The fundamentals of middle-distance running are introduced, together with running pace and energy management. Breathing techniques and endurance development are practiced. Running strategy and performance are analyzed. Practical training methods are applied.

Sports Games. Rules of Volleyball, Teaching Special Drills with Balls. The basic rules and objectives of volleyball are introduced. Fundamental technical skills and ball control exercises are studied. Individual and team drills are practiced, together with game tactics and cooperation. Volleyball techniques are practically applied.

Basketball: Rules of the Sport and Teaching Special Ball Drills. The rules and principles of basketball are introduced, together with ball handling, passing, dribbling, and shooting techniques. Individual and team practice drills are performed. Basic offensive and defensive strategies are studied. Performance is improved through practice.

Handball Rules and Special Ball Drills. The basic rules of handball are introduced, together with passing, catching, dribbling, and shooting techniques. Individual and team exercises are performed. Tactical principles of handball are studied. Game skills are practically applied.

Rules of Football, Teaching Special Ball Exercises. The rules and objectives of football are introduced, together with ball control, passing, dribbling, and shooting techniques. Individual skill development exercises are performed. Team coordination and tactical play are studied. Football skills are practically applied.

Rules of Badminton and Teaching Special Exercises with the Shuttlecock. The rules and objectives of badminton are introduced, together with grip techniques and basic strokes. Footwork and court movement are practiced. Shuttlecock control exercises are performed. Badminton techniques are practically applied.

	Table Tennis Rules, Teaching Special Drills with Balls. The rules and basic principles of table tennis are introduced, together with grip and stroke techniques. Ball control and rally exercises are practiced. Serving and receiving techniques are studied. Playing skills are practically improved. Teaching Games with Movement. The concept and educational importance of movement games are introduced. Movement-based activities are organized and implemented. Coordination, agility, and teamwork are developed, following safety requirements during games. Movement games are practically applied. Gymnastics. Teaching Floor Exercises. The fundamentals of floor gymnastics are introduced, together with basic body positions and movement techniques. Coordination, flexibility, and balance exercises are practiced. Safety requirements during floor exercises are discussed. Gymnastic routines are practically performed. Teaching General Developmental Exercises. The objectives and importance of general developmental exercises are introduced. Exercises for strength, flexibility, endurance, and coordination are practiced, together with their correct execution techniques. Balanced exercise routines are planned. These exercises are practically applied for overall fitness. Teaching Simple Acrobatic Exercises. The basic principles of acrobatic exercises are introduced, including techniques for rolls, balances, and simple acrobatic movements. Safety and injury prevention are discussed. Flexibility and coordination are developed. Acrobatic skills are practically performed. Teaching Strength-Developing Exercises on Gymnastic Apparatus. The purpose of strength training using gymnastic apparatus is introduced, together with techniques for performing exercises on different apparatus. Muscular strength and endurance are developed. Safety requirements and correct body mechanics are discussed. Strength-building exercises are practically applied. Self-study Physical Fitness Assessment (BN).
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	Athletics. Teaching the Standing Long Jump Technique. Teaching Short-Distance Running Technique. Middle-Distance Running Technique. Teaching 500–1000 Metre Runs. Sports Games. Rules of Volleyball, Teaching Special Drills with Balls. Rules of the Sport of Basketball and Teaching Specialised Ball Drills. Rules of the Sport of Handball, Teaching Special Ball Exercises. Rules of the Game of Football, Teaching Special Exercises with Balls. Rules of Badminton, Teaching Special Exercises with the Shuttlecock. Rules of Table Tennis and Teaching Special Drills with Balls. Teaching Movement Games. Gymnastics. Teaching Floor Exercises. Teaching General Developmental Exercises. Teaching Simple Acrobatic Exercises. Teaching Strength-Building Exercises on Gymnastic Apparatus.
Teaching Methods	The module is delivered through interactive lectures, practical physical training sessions, demonstrations, guided practice, individual and group exercises, sports games, fitness assessment activities, and independent learning tasks. Students develop physical fitness, motor skills, and technical abilities through athletics, gymnastics, sports games, and movement-based activities. Teamwork and healthy lifestyle habits are also developed through these activities. Modern teaching methods, digital educational resources, multimedia materials, and sports equipment are used to enhance students' physical performance, theoretical knowledge, and practical competence. These methods further support students' lifelong commitment to physical activity and health promotion.
Prerequisites	General Biochemistry and Sports Biochemistry Athletics Gymnastics

	National and Sports Games
Exams and Assessment	Type of examination (practical performance assessment) and its weight Student performance is assessed through continuous assessment (50%) and a final assessment (50%). The continuous assessment (50%) is conducted throughout the semester during practical classes. Students may earn up to 20 points by successfully completing the required 60-meter sprint performance standard. Assessment is based on the officially approved physical fitness standards for male and female students. Male students who complete the sprint in 9.0 seconds and female students in 10.0 seconds receive 18–20 points (90–100%). Male students who complete the sprint in 9.5 seconds and female students in 10.5 seconds receive 14–17.9 points (70–89.9%). Male students who complete the sprint in 10.0 seconds and female students in 11.5 seconds receive 12–13.9 points (60–69.9%). Male students who complete the sprint in 11.0 seconds and female students in 12.0 seconds receive up to 11.9 points (below 60%). In addition, students may earn up to 30 points for completing independent study assignments according to the module schedule. Independent study tasks may include preparing presentations on physical education and healthy lifestyles, writing reports, completing theoretical assignments, participating in sports-related projects, and carrying out other activities assigned by the instructor. All independent study assignments must be submitted within the specified deadlines. Assignments submitted after the deadline will not be accepted. Therefore, students may earn a maximum of 50 points in the continuous assessment. The final assessment (50%) is conducted after the completion of the course as a practical examination. Students are assessed by completing the required 100-meter sprint according to the approved physical fitness standards. Male students who complete the sprint in 13.5 seconds and female students in 17.0 seconds receive 45–50 points (90–100%). Male

	students who complete the sprint in 13.8 seconds and female students in 17.2 seconds receive 35–44.9 points (70–89.9%). Male students who complete the sprint in 14.0 seconds and female students in 17.5 seconds receive 30–34.9 points (60–69.9%). Male students who complete the sprint in 15.0 seconds and female students in 18.0 seconds receive up to 29.9 points (below 60%).
Recommended Literature	1. B.K. Amanturdiyev. The Game of Football. A Textbook. Termiz, "Surxon Nashr", 2021. 167 pages. 2. Q.Q.Xaqnazarov. Basketball: Sportivnaya podgotovka. O'quv qo'llanma. Termiz, "Surxon nashr", 2021. 132 bet. 3. B.K. Amanturdiyev. The Game of Football. A Textbook. Termez, "Surxon Nashr", 2021. 167 pp. 4. Mirziyoyev, Sh.M. The Strategy of the New Uzbekistan. – Tashkent: "Uzbekiston", 2021. 5. Law of the Republic of Uzbekistan "On Physical Education and Sport", 4 September 2015.
Responsible Lecturer	Shaymardanov Ravuf Narzulloyevich

ECOLOGY AND ENVIRONMENTAL PROTECTION

Field	Description
Module Title	ECOLOGY AND ENVIRONMENTAL PROTECTION
Module Code	EAM302
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 30 hours (14 hours Lecture, 16 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	V semester
Module objective	The objective of the Module Ecology and Environmental Protection is to provide students with the scientific and theoretical foundations of environmental protection and green development. The Module teaches students the scientific and theoretical principles underlying ecology and environmental protection. Students study the

	biosphere, natural resources, and the atmosphere, along with methods for their protection. The conservation of biological diversity is examined. The Module addresses issues of sustainable development and environmental education. Students develop the ability to analyze environmental problems and apply environmental legislation in practice.
Module Learning Outcomes	Knowledge: LO1. Gains an understanding of the fundamentals of the subject of "Ecology and Environmental Protection", and of the phenomena in the environment and the methods for their prevention. LO2. Is aware of the reforms being carried out in the field of ecology and environmental protection. LO3. Understands the role of society in ecology and environmental protection. LO4. They will be competent to describe methods for applying theoretical issues in ecology and environmental protection based on environmental legislation. Skills: LO5. They will have the skills to conserve and enhance biological resources that are important for humans. Competencies: LO6. Demonstrates a comprehensive understanding of ecological principles, environmental processes, biodiversity, natural resource conservation, and sustainable development. LO7. Identifies and analyzes local, regional, and global environmental problems using scientific and ecological approaches. LO8. Applies environmental legislation, ecological standards, and sustainable resource management principles in solving environmental issues.
Content	Lectures Subject Matter, Objectives, Aims and History of the Course "Ecology and Environmental Protection". The subject of ecology and environmental protection is introduced, together with its aims, objectives, and scope. The historical stages in the development of ecological science and the

contributions of major ecologists are examined. The role of ecology in understanding environmental processes and solving contemporary environmental problems is discussed.

Scientific and Theoretical Foundations of Nature Conservation. The fundamental principles and concepts of nature conservation are studied. Ecological laws governing the interaction between living organisms and the environment are examined. Scientific approaches to the sustainable use and protection of natural resources are discussed.

Concept of the Biosphere. The Doctrine of V.I. Vernadsky. The concept, structure, and functions of the biosphere are studied. Vernadsky's doctrine and its significance in ecological science are examined. The role of living organisms in maintaining ecological balance and global biogeochemical cycles is discussed.

Classification of Natural Resources and Protection of the Atmosphere. Natural resources are classified into renewable and non-renewable categories, and the importance of rational resource utilization is studied. Sources of atmospheric pollution and their environmental impacts are examined. Measures for air quality protection are discussed.

Water, Land Resources and Their Protection. The importance of water and land resources for sustainable development is studied. Major threats to water and soil quality, including pollution and degradation, are examined. Methods and strategies for the conservation and sustainable management of these resources are discussed.

Conservation of Biological Diversity. The concept and significance of biodiversity in maintaining ecosystem stability are studied. The causes and consequences of biodiversity loss are examined. National and international measures aimed at conserving biological diversity and endangered species are discussed.

Issues of Sustainable Development and Environmental Education. The principles and goals of sustainable development are studied. The relationship between environmental protection,

economic growth, and social well-being is examined. The role of environmental education and public awareness in promoting sustainable lifestyles is discussed.

Practical lessons

Modern Ecological Problems. Major environmental problems affecting the world today are analyzed. Pollution, climate change, resource depletion, and biodiversity loss are investigated. Possible solutions and environmental policies are discussed.

Interrelationships Between Nature and Society.

The interactions between human activities and natural ecosystems are studied. The environmental impacts of economic development are assessed. Sustainable approaches to human-nature relationships are evaluated.

Cycle of Matter and Energy in the Biosphere. The movement of matter and energy through ecosystems is investigated. Ecological cycles, including the water, carbon, nitrogen, and oxygen cycles, are analyzed. Their importance for environmental stability is examined.

Alternative Energy Sources and Their Environmental Safety. Renewable energy resources, such as solar, wind, hydro, and biomass energy, are studied. Their environmental benefits and limitations are evaluated. Their role in sustainable development is analyzed.

Ecological Problems of the Aral Sea and the Aral Sea Region. The causes and consequences of the Aral Sea environmental crisis are examined. The ecological, economic, and social impacts on the region are analyzed. Restoration and mitigation measures are discussed.

The Republic of Uzbekistan's Specially Protected Natural Areas. National parks, nature reserves, and other protected territories in Uzbekistan are studied. Their role in biodiversity conservation and environmental protection is analyzed. Management and protection strategies are evaluated.

Towards the Sustainable Development Goals. The United Nations Sustainable Development Goals (SDGs) are investigated. Environmental targets and

	indicators are analyzed. National and global efforts toward sustainable development are discussed. Environmental Protection, Ecological Culture, Education and Awareness-Raising Issues. Environmental education and ecological culture are studied. Methods for increasing public environmental awareness are analyzed. The role of education in promoting environmental responsibility is evaluated. Self-study The Development of Ecology in Uzbekistan. Ecological Problems and Ways to Prevent Them. Habitats of Living Organisms and Their Adaptations. Mechanisms by Which Various Environmental Factors Affect Living Organisms. Human Demographics and Environmental Protection. The Concept of the Noosphere and Environmental Protection. Mining, Chemical and Metallurgical Industries and Environmental Protection. Climate Change and Ecological Adaptation. Current State of Uzbekistan's Water Basins; Water Problems and Their Prevention. Salinization, Desertification Problems and Environmental Protection. The Role of Plants and Animals in the Biosphere; the Flora and Fauna of Uzbekistan. The Red Data Book of the Republic of Uzbekistan and Its Role in Environmental Protection. Protected Natural Areas in the Republic of Uzbekistan. Achieving the Goals of Sustainable Development. Issues of Developing Environmental Education in the Republic of Uzbekistan.
Teaching Methods	Knowledge of the scientific and theoretical foundations of ecology and environmental protection is formed. The biosphere, the theory of Vernadsky, and the classification of natural resources are taught. Methods for the protection of atmospheric air, water, and land resources are taught. Skills for the conservation of biological diversity and specially protected areas are developed. Competence in analyzing environmental problems, including island

	effects, salinization, and climate change, and their prevention methods, is developed. Competencies in sustainable development, environmental education, and the wise use of natural resources are developed.
Prerequisites	Environmental Protection and Green Development
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final assessment (50%). The continuous assessment evaluates students' classroom participation, practical activities, seminar discussions, presentations, and the successful completion of independent study assignments. These include analyses of environmental problems, climate change adaptation, biodiversity conservation, sustainable use of natural resources, water resource management, waste management and recycling technologies, renewable energy, environmental legislation, and the implementation of sustainable development principles. The midterm assessment consists of written assignments, short in-class quizzes, practical reports, presentations, and the evaluation of independent study projects. Students are assessed on their ability to analyze environmental issues, evaluate the impacts of climate change, explain biodiversity conservation strategies, assess environmental protection measures, interpret environmental legislation, compare sustainable development approaches, and propose practical solutions to environmental challenges. The final assessment may be conducted as a test, oral examination, or written examination. The test examination consists of multiple-choice questions covering all lecture topics of the module. The oral examination evaluates students' understanding of environmental concepts, sustainable development, environmental legislation, climate change adaptation, green energy, biodiversity conservation, and international environmental cooperation through discussion and questioning. The written examination

	consists of theoretical and analytical questions designed to assess students' knowledge of environmental protection, green development, natural resource management, environmental policy, ecological culture, and their ability to apply environmental principles to contemporary environmental problems. Successful completion of all practical activities and independent study assignments is required as an essential component of the overall module assessment.
Recommended Literature	1. X.X. Zokirov, M.B. Abramotov and others. Ecology and Environmental Protection (textbook). — 2025, 285 pp. 2. M. Abramotov, X. Zokirov. Ecology and Nature Conservation (textbook). — Tashkent: "Publishing High Future", 2024, 219 pp. 3. Ministry of Ecology, Environmental Protection and Climate Change of the Republic of Uzbekistan. National Report on the State of the Environment: Uzbekistan. — International Institute for Sustainable Development, 2023. 4. Mirziyoyev, S. M. We will build our great future together with our brave and noble people. — Tashkent: "Uzbekiston", 2017. 5. Michael Begon, Colin R. Townsend, John L. Harper. Ecology: from individuals to ecosystems. 4th ed. — 2006, 759 p. 6. Voluntary National Report of Uzbekistan on the implementation of the national goals and objectives for sustainable development up to 2030. — Tashkent, 2023. 7. Sattorov Z. Ecology. — Tashkent, 2018. 8. Xo'janazarov O., Yakubjonova Sh. Ecology and Nature Conservation. — Tashkent, 2018.
Responsible Lecturer	Mukhamadi Abramotov - Candidate of Biological Sciences, Associate Professor; Mukhriddin Qodirov - Teacher

ENVIRONMENTAL PROTECTION AND GREEN DEVELOPMENT

Field	Description
Module Title	ENVIRONMENTAL PROTECTION AND GREEN DEVELOPMENT
Module Code	AMYAR302
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 30 hours (14 hours Lecture, 16 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	V semester
Module objective	The objective of the Module Environmental Protection and Green Development is to provide students with the scientific and theoretical foundations of environmental protection and green development. The aim of the Module is to teach students the scientific and theoretical foundations of environmental protection and green development. Environmental problems and the factors that cause them are studied. Their variation in space and time is examined. Environmental assessment methods are introduced. The ecological principles of nature conservation are studied. The rational use of natural resources is examined.
Module Learning Outcomes	Knowledge: LO1. An understanding of the fundamentals of the subject "Environmental Protection and Green Development", and of environmental phenomena and their prevention methods. LO2. Reforms being carried out in environmental protection and green development. LO3. The role of society in environmental protection and green development. LO4. Must have the skills to describe the application methods of theoretical issues in environmental protection and green development based on environmental laws. Skills: LO5. For environmental protection and green development, they should possess skills such as

	conserving and nurturing the environment, and preserving, safeguarding and increasing the biological resources that are important for the people in the environment. Competencies: LO6. They will be competent to draw independent conclusions and make decisions on environmental issues. LO7. They will acquire the competence to carry out practical activities for environmental protection and the promotion of ecological culture.
Content	Lectures Subject, Purpose and Objectives of the Discipline of Environmental Protection and Green Development. The subject, objectives, and scope of environmental protection and green development are introduced. The importance of sustainable resource management and environmental responsibility is studied. The role of green development in achieving economic growth and environmental sustainability is discussed. The Nature, Consequences and Ways of Solving Environmental Problems. Environmental Culture. The major causes and consequences of environmental degradation are studied. Pollution, ecosystem disturbances, and resource depletion are analyzed. The importance of environmental culture and public awareness in protecting the environment is discussed. Adapting to the Impacts of Climate Change and Ways to Use Natural Resources Wisely. Climate change, its environmental and socio-economic impacts, and adaptation strategies are studied. The sustainable management of natural resources is analyzed. Resource efficiency and environmental resilience are evaluated. Conservation of Biodiversity. The concept and importance of biodiversity for ecosystem stability are studied. Threats to biological diversity and endangered species are analyzed. National and international biodiversity conservation strategies are evaluated.

	Sustainable Development Goals. The United Nations Sustainable Development Goals (SDGs) are introduced. The environmental, economic, and social dimensions of sustainable development are analyzed. Global and national efforts to achieve sustainability are evaluated. "Green Energy" as a Factor of Sustainable Development. Renewable energy resources and green technologies are studied. The environmental and economic benefits of clean energy are analyzed. The role of green energy in reducing greenhouse gas emissions is evaluated. Environmental Protection Issues in Implementing the "Uzbekistan-2030" Strategy. The environmental priorities within the Uzbekistan-2030 Strategy are studied. Green reforms, sustainable development initiatives, and environmental policies are analyzed. Their contribution to environmental protection and national development is evaluated. Practical lessons Transition to a Green Economy: Theory, Practice and Opportunities in Uzbekistan. Green economy principles and their implementation in Uzbekistan are analyzed. Sustainable production and consumption practices are evaluated. Opportunities for environmentally friendly economic development are discussed. Climate Change and Ecological Adaptation. Climate change impacts on ecosystems and society are investigated. Adaptation and mitigation measures are analyzed. Practical approaches to climate resilience are evaluated. Concept for Enhancing the Ecological Culture of the Population in Uzbekistan up to 2030. National strategies for improving environmental awareness and ecological culture are studied. Educational initiatives and public participation are analyzed. Their effectiveness in promoting environmental responsibility is evaluated. Effective Use of Water Resources. Sustainable water resource management techniques are analyzed. Water conservation, efficient irrigation, and pollution prevention measures are investigated.
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	Water management practices in Uzbekistan are evaluated. The Problem of Waste, Recycling and Zero-Waste Technologies. Waste generation, classification, and environmental impacts are studied. Recycling methods and circular economy principles are analyzed. Zero-waste technologies and sustainable waste management are evaluated. Sustainable Development Goals in Uzbekistan. Uzbekistan's progress toward achieving the Sustainable Development Goals is investigated. Environmental indicators and national development strategies are analyzed. Implementation challenges and achievements are evaluated. Green Energy: Development and Prospects of Renewable Energy Sources. Solar, wind, hydro, and biomass energy technologies are studied. Renewable energy development in Uzbekistan is analyzed. Future prospects for sustainable energy production are evaluated. International Cooperation in Environmental Protection. International environmental organizations, agreements, and conventions are analyzed. Global environmental cooperation and sustainable development initiatives are studied. Uzbekistan's participation in international environmental programs is evaluated. Self-study The Development of Ecology in Uzbekistan. Environmental Problems and Ways to Prevent Them. Habitats of Living Organisms and the Adaptations of Organisms to Them. Mechanisms of Various Environmental Factors Affecting Living Organisms. Human Demographics and Environmental Protection. The Concept of the Noosphere and Environmental Protection. Mining, Chemical and Metallurgical Industries and Environmental Protection. Climate Change and Ecological Adaptation. Biodiversity Conservation and Sustainable Use of Natural Resources.
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	Water Resources Management and Water Pollution Control. Air Pollution, Its Sources and Prevention Measures. Waste Management and Recycling Technologies. Renewable Energy Sources and Environmental Sustainability. Environmental Monitoring and Ecological Assessment Methods. Environmental Legislation and International Environmental Agreements.
Teaching Methods	Knowledge of the scientific and theoretical foundations of ecology and environmental protection is formed. The biosphere, the theory of Vernadsky, and the classification of natural resources are taught. Methods for the protection of atmospheric air, water, and land resources are taught. Skills for the conservation of biological diversity and specially protected areas are developed. Competence in analyzing environmental problems, including island effects, salinization, and climate change, and their prevention methods, is developed. Competencies in sustainable development, environmental education, and the wise use of natural resources are developed.
Prerequisites	Ecology and Environmental Protection
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final assessment (50%). The continuous assessment evaluates students' classroom participation, practical activities, seminar discussions, presentations, and the successful completion of independent study assignments. These include analyses of environmental problems, climate change adaptation, biodiversity conservation, sustainable use of natural resources, water resource management, waste management and recycling technologies, renewable energy, environmental legislation, and the implementation of sustainable development principles. The midterm assessment consists of written assignments, short in-class quizzes, practical

	reports, presentations, and the evaluation of independent study projects. Students are assessed on their ability to analyze environmental issues, evaluate the impacts of climate change, explain biodiversity conservation strategies, assess environmental protection measures, interpret environmental legislation, compare sustainable development approaches, and propose practical solutions to environmental challenges. The final assessment may be conducted as a test, oral examination, or written examination. The test examination consists of multiple-choice questions covering all lecture topics of the module. The oral examination evaluates students' understanding of environmental concepts, sustainable development, environmental legislation, climate change adaptation, green energy, biodiversity conservation, and international environmental cooperation through discussion and questioning. The written examination consists of theoretical and analytical questions designed to assess students' knowledge of environmental protection, green development, natural resource management, environmental policy, ecological culture, and their ability to apply environmental principles to contemporary environmental problems. Successful completion of all practical activities and independent study assignments is required as an essential component of the overall module assessment.
Recommended Literature	1. M. Abramotov, X. Zokirov. Ecology and Nature Conservation. Textbook. – Tashkent, PUBLISHING HIGH FUTURE OK, 2024. – 219 pp. 2. S.N. Khashimova. "Green Economy". Tashkent. "Ma'rifat". 2024. Textbook. 3. A.V. Vakhobov, Sh.Kh. Khajibakiev. "Green Economy". Tashkent: "Universitet", 2020. – 262 pp. 4. Nigmatov. Theoretical Foundations of Ecology. Tashkent, 2014. 260 pages. 5. Odum Y. Fundamentals of Ecology. Translation of the English edition. M.: Mir, 1975, 740 pp. 6. Mustafoev S., O'roqov S., Suvonov P. General Ecology. Tashkent, 2006.

	7. Tursunov X.T., Rakhimova T.U. Ecology: A Textbook. Tashkent, 2006.
Responsible Lecturer	Mukhamadi Abramotov - Candidate of Biological Sciences, Associate Professor; Mukhriddin Qodirov - Teacher

FUNDAMENTALS OF ROBOTICS

Field	Description
Module Title	FUNDAMENTALS OF ROBOTICS
Module Code	RBA305
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 76 hours (44 hours Lecture, 32 hours Practical lessons) Self-Assignments: 74 hours
Frequency/Duration	V semester
Module objective	The aim of the Module Fundamentals of Robotics is to provide students with theoretical and practical knowledge of the structure, operating principles, and application areas of robotic systems. Through this course, students are taught the fundamentals of designing and controlling the mechanical components of robots. The electronic and software components of robots are also studied. Skills in working with sensors, actuators, and microcontrollers are developed. Students develop competencies in utilising modern robotics technologies. The Module prepares students to innovatively solve practical problems using robotics technologies.
Module Learning Outcomes	Knowledge: LO1. Knows the fundamental concepts, principles, and development trends of robotics. LO2. Can explain the structure, components and operating mechanisms of robots. LO3. Understands the functions and operating principles of sensors, actuators, microcontrollers and control systems. LO4. Has knowledge of programming methods and control algorithms used in robotic systems.

	LO5. Is aware of the potential applications of robotics technologies in industry, service, medicine, education and other fields. LO6. Master the theoretical foundations of the processes for designing, modelling and testing robotic devices. Skills: LO7. Will have practical skills in assembling, configuring and commissioning robotic devices. LO8. Can connect sensors and actuators to microcontrollers and programme them to control their operation. LO9. Can develop and implement algorithms for controlling robot motion. LO10. Acquires skills in modelling, testing and analysing the results of robotic systems. LO11. Can effectively use modern software tools and technologies in creating robotics projects. LO12. They will be qualified to develop, improve and apply robotic solutions to practical problems in real-world settings. Competencies: LO13. Can carry out professional activities in the design, development and implementation of robotic systems. LO14. They will have the competence to assess the technical condition of robots, diagnose faults, and identify and rectify any malfunctions that arise. LO15. They can create and manage automated systems based on microcontrollers, sensors and control modules.
Content	Lectures Introduction to Robotics. The Subject Matter, Aims and Objectives of the Science of Robotics. The concept of robotics, its objectives, scope, and significance in modern technology are studied. The role of robotics in industry, science, education, and everyday life is examined. The interdisciplinary nature of robotics and its relationship with mechanics, electronics, computer science, and artificial intelligence are explored. Classification of Robots and Fields of Application. Different classifications of robots based on structure,

mobility, functionality, and application areas are studied. The use of robots in manufacturing, healthcare, agriculture, transportation, education, military systems, and service industries is examined. The advantages and limitations of various robotic systems are analyzed.

History of Robotics and Modern Trends. The historical development of robotics, from early mechanical devices to intelligent autonomous systems, is explored. Major milestones in robotics development are studied. Current trends, innovations, and future directions in robotic technologies are analyzed.

Structure and Main Components of Robots. The architecture of robotic systems and their main components are studied. Mechanical structures, sensors, actuators, controllers, and power systems are examined. The interaction between these components in performing robotic tasks is discussed.

Fundamentals of Mechanics in Robotics. The basic principles of mechanics used in robotic systems are studied. Force, motion, torque, balance, and mechanical transmission systems are examined. The role of mechanical engineering principles in robot design and operation is discussed.

Robot Kinematics and Types of Motion. Robot movement and kinematic models are studied. Coordinate systems, degrees of freedom, position, velocity, and trajectory planning are examined. Different types of robotic motion and movement control are analyzed.

Robot Manipulators and Their Design Characteristics. The structure and operation of robot manipulators are studied. Robotic arms, joints, links, and end effectors are examined. Design principles and applications of robotic manipulators in industrial environments are discussed.

Drive Mechanisms and Transmissions in Robots. Mechanical transmission systems used in robots are studied. Gears, belts, chains, and transmission mechanisms are examined. Their role in converting and transmitting motion and power is discussed.

Types of Actuators and Drivers and Their Operating Principles. Different types of actuators, including electric, pneumatic, and hydraulic systems, are studied. Motor drivers and control circuits are examined. Actuator performance characteristics and applications are discussed.

The Role of Electric Motors and Servo Mechanisms in Robotics. DC motors, stepper motors, and servo motors are studied. Their construction, operating principles, and control methods are examined. Applications of motor systems in robotic projects are analyzed.

General Overview of Sensors and Detectors. Sensor technologies used in robotics are studied. The functions and classifications of sensors are examined. Data acquisition and environmental monitoring through sensor systems are discussed.

Use of Distance, Temperature and Light Sensors in Robots. The operation and applications of ultrasonic, infrared, temperature, and light sensors are studied. Sensors are integrated into robotic systems. Sensor data is analyzed for automation and decision-making processes.

Fundamentals of Vision Systems and Image Processing in Robots. Computer vision concepts and image processing techniques are studied. The methods by which robots detect, recognize, and interpret visual information are examined. The role of cameras and vision systems in intelligent robotics is discussed.

Control Fundamentals of Robotic Systems. The principles of robot control systems are studied. Open-loop and closed-loop control methods are examined. Stability, accuracy, and efficiency in robotic operations are discussed.

Feedback and Automatic Control Systems in Robots. Feedback mechanisms and automatic control principles are studied. The interaction between sensors and controllers in improving robot performance is examined. Practical examples of automated robotic systems are analyzed.

Fundamentals of Microprocessors and Microcontrollers. The architecture and operation of

microprocessors and microcontrollers are studied. Their role in robot control and embedded systems is examined. Practical applications in robotics projects are explored.

The Arduino Platform and Its Applications in Robotics. The Arduino platform is studied, together with its architecture, programming environment, and hardware components. The use of Arduino for sensor integration and motor control is examined. Applications of Arduino in robotic automation are discussed.

Fundamentals of Robot Programming. Programming concepts used in robotics are studied. Variables, control structures, functions, and programming logic are examined. Simple robotic control programs are developed.

Creating Algorithms and Software for Robots. Algorithm design and software development for robotic systems are studied. Problem-solving techniques and decision-making processes are examined. Algorithms for autonomous robot behavior are developed.

Mobile Robots and Autonomous Navigation Systems. Mobile robot platforms and navigation techniques are studied. Localization, mapping, path planning, and obstacle avoidance methods are examined. Autonomous movement and intelligent navigation systems are discussed.

Industrial Robots and Automation of Manufacturing Processes. Industrial robotics and automated production systems are explored. Robotic applications in manufacturing, assembly, quality control, and logistics are studied. The impact of automation on modern industry is discussed.

Prospective Directions in Artificial Intelligence and Robotics. The integration of artificial intelligence and robotics is studied. Machine learning, intelligent agents, collaborative robots, and autonomous systems are examined. Future trends and innovations in intelligent robotics are analyzed.

Practical lessons

Introduction to the Arduino Platform and Software Environment Installation. The Arduino IDE is

installed and configured. The Arduino board structure and development environment are explored. The workflow of creating and uploading programs is examined.

Creating and Uploading the First Program on the Arduino Device. A simple Arduino program is written and uploaded. The program structure, including the setup() and loop() functions, is examined. Program execution is tested on the hardware.

Working with Analogue and Digital Input/Output Ports. Digital and analog pins are studied. Input signals are read, and output devices are controlled. Practical skills in hardware interfacing are developed.

Working with the Serial Monitor and Data Exchange Processes. The Serial Monitor is used for debugging and communication. Data is sent and received between Arduino and a computer. Sensor readings and system performance are monitored.

Working with Variables, Arithmetic Operations and Performing Programmatic Calculations. Variables and data types are used in Arduino programming. Arithmetic operations and calculations are performed. Problem-solving skills are developed through programming exercises.

Creating Control Algorithms Using Conditional Operators (if, else). Decision-making logic is implemented in programs. Conditional statements are used to control system behavior. Intelligent responses to sensor inputs are developed.

Connecting LEDs and Controlling Them Based on an Algorithm. LED circuits are built, and LED control is programmed. Digital outputs and timing functions are examined. Simple lighting control systems are created.

Creating Various Lighting Effects Using Multiple LEDs. LED patterns and visual effects are designed. Sequential and dynamic lighting systems are programmed. Creativity and programming skills are developed.

Generate a PWM Signal Using the AnalogWrite() Function. Pulse-width modulation techniques are studied. Brightness and power levels are controlled

	using PWM signals. PWM is applied in practical electronics projects. Controlling an RGB LED and Producing Various Colours. RGB LEDs are connected and programmed. Colors are mixed using PWM control. Interactive lighting applications are developed. Processing Input Signals Using a Button. Button states are read, and user input is detected. Control functions are implemented based on button interactions. Interactive electronic systems are developed. Reading and Analysing Analogue Signals Using a Potentiometer. Potentiometers are used as analog input devices. Sensor values are read and analyzed. Analog control techniques are applied in projects. Detection and Control of Light Levels Using a Photoresistor. Ambient light levels are measured using photoresistors. Automatic lighting control systems are created. Sensor-based automation projects are developed. Generating Sound and Musical Signals Using a Buzzer. Buzzers are programmed to generate tones and melodies. Frequency and timing control are examined. Audio feedback systems are developed. Connecting and Programming a Servo Motor. Servo motors are controlled using Arduino. Position control and movement programming are studied. Servo mechanisms are applied in robotic projects. Create a Simple Robotics Project Using Sensors and Actuators. Sensors, actuators, and control algorithms are integrated into a complete robotic system. A simple robotics project is designed and implemented. Practical robotics and programming skills are demonstrated. Self-study Preparing for Practical Sessions and Completing Assignments. Independent Study of Literature and Learning Materials on Computer Software. Performing Practical Tasks on Working with and Upgrading Computer Components.
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	Study of Scientific Articles, Analytical Materials and Sources. Preparation of Mini-Projects Using Software Tools. Preparation of Presentations, Reports and Independent Work.
Teaching Methods	The module is delivered through a combination of lectures, laboratory classes, practical demonstrations, project-based learning, and independent study activities. Teaching methods include classroom instruction, interactive discussions, and hands-on training with Arduino platforms. Sensor integration, actuator control, robotics programming, and algorithm development are also taught. Circuit design, problem-solving exercises, project-based learning, teamwork activities, case studies, and presentations further support learning, together with the use of modern robotics and automation technologies. Practical activities involve programming microcontrollers, connecting and testing electronic components, working with sensors and actuators, controlling LEDs, buzzers, and servo motors, developing robotic algorithms, building simple robotic systems, and implementing automation projects using Arduino.
Prerequisites	Programming 1,2 Foundations of Artificial Intelligence Data Structures and Algorithms
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, practical robotics exercises, programming assignments, seminar presentations, and the successful completion of independent study tasks, including Arduino programming activities, sensor integration exercises, electronic circuit assembly, servo motor control projects, and mini robotics projects. The midterm assessment consists of written assignments, short in-class quizzes, laboratory evaluations, and the assessment of independent study projects, such as robot control

	algorithm development, sensor-based automation systems, Arduino-based applications, robotics project reports, technical presentations, and analyses of modern robotic technologies. The final assessment is a written examination consisting of theoretical questions and practical problem-solving tasks designed to assess students' understanding of robotics fundamentals, robotic systems, sensors and actuators, microcontrollers, Arduino programming, automation technologies, intelligent systems, and their ability to apply engineering and programming principles to robotic applications.
Recommended Literature	1. G.Y.Bo'ronova – Fundamentals of Virtual Robotics. Textbook. Buxoro 2023. 2. R.O. Sultanov, K.K. Matyakubov – Fundamentals of Robotics. Textbook. Chirchiq 2023. 3. A. Aripov, M. Muxitdinov. Fundamentals of Robotics. – Tashkent: A textbook for higher education institutions of the Republic of Uzbekistan, 2022. 4. Sh. Nazirov, B. Axmedov. Microcontrollers and their practical application. – Tashkent: Fan va Texnologiya, 2021. 5. A. Abduqodirov. Fundamentals of Programming on the Arduino Platform. – Tashkent: Innovatsion Rivojlanish Publishing House, 2023. 6. O. To'rayev, R. Karimov. Automation and Control Systems. – Tashkent: Tafakkur Bo'stoni, 2020. 7. Introduction to Robotics: Mechanics and Control – John J. Craig. Pearson, 2018.
Responsible Lecturer	Tursunov Farxod Bakhodir o'g'li

COMPUTER SOFTWARE

Field	Description
Module Title	COMPUTER SOFTWARE
Module Code	KMT305
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 76 hours (44 hours Lecture, 32 hours Practical lessons) Self-Assignments: 74 hours

Frequency/Duration	V semester
Module objective	The objective of the Module Computer Software is to develop students' deep understanding of the hardware and software components of computer systems. The Module examines the structure of computer hardware and the methods for their effective management. During the course, students master the correct use of modern operating systems and software tools. Students acquire practical skills in configuring computer hardware. Diagnostic skills are also developed. Students learn methods for optimally utilising computer hardware. The course prepares qualified specialists capable of solving practical problems that arise in the field of information technology.
Module Learning Outcomes	Knowledge: LO1. The student will have knowledge of the hardware and software architecture of computer systems and their principles of interaction. LO2. Understands the functions and mechanisms of the main components of a computer (processor, memory, storage devices, etc.). LO3. Gains theoretical knowledge of the main functions and capabilities of modern operating systems. LO4. Has knowledge of the types of software and the processes for installing and configuring them. LO5. They will have an understanding of the main problems that arise in computer systems and the methods for diagnosing them. Skills: LO6. Understands the principles of security in information technology and the rules for the effective use of computer resources. LO7. Demonstrates practical skills in correctly assembling, connecting and powering up computer hardware components. LO8. Performs the practical installation, configuration and effective use of operating systems. LO9. Independently carries out the processes of installing, updating and configuring software.

	LO10. Has the skills to diagnose and resolve faults that occur in computer systems. LO11. Applies practices for the effective use and optimisation of system resources (CPU, memory, disk). LO12. Ensures the safe use of computer and network resources in accordance with information security policies. Competencies: LO13. Competence to analyse and evaluate the hardware and software of computer systems at a professional level. LO14. Competence in the effective selection, implementation and management of modern operating systems and software tools. LO15. Competence in the technical maintenance and modernisation of computer hardware and its components.
Content	Lectures History and Development of the Computer. The history of computer technology, the evolution of computing devices, and major milestones in computer development are studied. The contributions of pioneering scientists and inventors are explored. The impact of computers on society, science, industry, and everyday life is examined. Generations of Computers and Their Characteristics. The different generations of computers and their technological advancements are studied. Vacuum tubes, transistors, integrated circuits, microprocessors, and modern computing technologies are examined. The characteristics, advantages, and limitations of each generation are compared. Computer Architecture and Its Main Components. The structure and organization of computer systems are studied. The central processing unit, memory, storage devices, input/output units, and system buses are examined. The interaction of hardware components in performing computing tasks is discussed. Operating Principles of Computer Systems. The fundamental principles of computer operation are

studied. Data processing, instruction execution, machine cycles, and communication between hardware components are examined. The workflow of modern computer systems is analyzed.

Motherboard and Its Functional Tasks. The motherboard is studied as the central component of a computer system. Its structure, ports, slots, chipsets, and communication functions are examined. The role of the motherboard in coordinating hardware operations is discussed.

Microprocessor and Its Types. Microprocessors, their architecture, and classification are studied. Different processor families and generations are examined. Processor specifications and performance characteristics are discussed.

Microprocessor's Working Mechanism and Performance. The execution of instructions and processing of data by microprocessors are studied. Clock speed, cores, cache memory, and performance indicators are examined. Factors affecting processor efficiency are analyzed.

Types of Memory (RAM, ROM, Cache, etc.). Primary memory technologies and their functions are studied. RAM, ROM, cache memory, and virtual memory are examined. Their role in system performance and data storage is discussed.

Structure and Operating Principle of Memory Devices. Memory architecture and data storage mechanisms are studied. The storage, retrieval, and transfer of information by memory devices are examined. Memory hierarchy and performance optimization are explored.

Data Storage Devices (HDD, SSD and Others). Different storage technologies, including hard disk drives, solid-state drives, and external storage devices, are studied. Storage capacities, speeds, and reliability are compared. Data management and storage solutions are discussed.

Graphics Cards and Their Functions. The architecture and operation of graphics processing units are studied. Graphics rendering, video processing, and display technologies are examined.

	The role of graphics cards in computing applications is discussed. Computer Cooling Systems and Their Importance. Thermal management in computer systems is studied. Cooling methods, including air cooling, liquid cooling, and heat dissipation technologies, are examined. The importance of temperature control for system stability is discussed. Computer Peripherals. Input, output, and peripheral devices used in computer systems are studied. Keyboards, mice, printers, scanners, monitors, and external devices are examined. Their functions and applications are discussed. The Concept of Computer Software. The definition, purpose, and importance of software in computer systems are studied. The relationship between hardware and software is examined. The role of software in solving practical problems is explored. Types of Software (System, Application, Utility). Different categories of software and their functions are studied. Operating systems, application software, and utility programs are examined. Their roles in computer operations and user productivity are discussed. Operating Systems and Their Main Functions. The architecture and functions of operating systems are studied. Process management, memory management, file systems, and user interfaces are examined. The role of operating systems in computer resource management is discussed. Fundamentals of Software Installation and Configuration. Methods of software installation, setup, and maintenance are studied. System requirements, software updates, and configuration processes are examined. Practical skills in software deployment are developed. Fundamentals of Computer Networks. Computer network concepts, architectures, and communication protocols are studied. Network devices, topologies, and data transmission methods are examined. The fundamentals of network connectivity are discussed. Internet and Network Services. Internet technologies and network-based services are studied.
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	Web services, email systems, cloud technologies, and online communication tools are examined. The role of the internet in modern computing is discussed. Computer Security Fundamentals. The principles of computer security and information protection are studied. Threats, vulnerabilities, malware, and security policies are examined. Methods of protecting computer systems and data are discussed. Methods for Carrying Out Security Tasks on a Computer. Practical techniques for securing computer systems are studied. Antivirus software, firewalls, encryption, authentication, and access control are examined. Skills in implementing security measures are developed. Technical Maintenance and Diagnostics of Computer Systems. Preventive maintenance and troubleshooting techniques are studied. Methods for diagnosing hardware and software problems are examined. Practical skills in maintaining and repairing computer systems are developed. Practical lessons Identifying Computer Components and the Basics of Assembling Them. Computer hardware components are identified, and the principles of computer assembly are studied. Component compatibility is examined. Installation procedures are practiced. Learning About and Connecting the Components of the Motherboard. Motherboard components and their functions are studied. The installation of hardware devices to the motherboard is practiced. Connections between devices and the motherboard are verified. Installing the Microprocessor and Determining Its Parameters. Processors are installed, and their specifications are identified. Processor performance is verified. Configuration settings are examined. Installation and Testing of RAM Modules. Memory modules are installed, and system memory performance is tested. Memory compatibility is examined. Troubleshooting techniques are applied. Connecting and Configuring Permanent Storage Devices (HDD/SSD). Storage devices are installed
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and configured. Partitions are managed. Storage performance is verified.

Installation of the Graphics Card and Cooling System. Graphics cards and cooling solutions are installed. Graphics performance is tested. System temperatures are monitored.

Operating System Installation (Windows/Linux). Operating systems are installed, and initial system configuration is performed. Installation procedures are studied. System setup techniques are practiced.

Installing and Updating Drivers. Hardware drivers are installed, and system compatibility is maintained. Driver updates are performed. Their importance for system stability is discussed.

Software Installation and Configuration. Application software is installed, and system settings are configured. Software resources are managed. User preferences are set.

Optimising System Settings. System parameters are adjusted to improve performance and efficiency. System resources are monitored. Operations are optimized based on the results.

Creating and Managing User Accounts. User accounts are created, and permissions are assigned. Access rights are managed. Account security and administration are examined.

Configure and Verify Network Settings. Local network settings are configured, and connectivity is tested. IP addressing is examined. Network communication is studied.

Configure and Diagnose the Internet Connection. Internet access is set up, and connectivity problems are troubleshooted. Network performance is analyzed. Network reliability is assessed.

Installing and Using Antivirus Software. Antivirus programs are installed, and malware detection is performed. Security settings are managed. System protection is configured.

Data Protection and Backup. Data backup procedures and recovery methods are implemented. The importance of data security is examined. Disaster recovery principles are studied.

	Detecting and Diagnosing Computer Malfunctions. Hardware and software faults are identified. Troubleshooting techniques are applied. Diagnostic tools are used to resolve system issues. Self-study Preparing for Practical Sessions and Completing Assignments. Independent Study of Literature and Learning Materials on Computer Software. Performing Practical Tasks on Working with and Upgrading Computer Components. Study of Scientific Articles, Analytical Materials and Sources. Preparation of Mini-Projects Using Software Tools. Preparation of Presentations, Reports and Independent Work.
Teaching Methods	The module is delivered through a combination of lectures, laboratory classes, practical demonstrations, hands-on training, project-based learning, and independent study activities. Teaching methods include classroom instruction, interactive discussions, and hardware assembly workshops. Software installation and configuration exercises, operating system administration, and network configuration activities are also employed. Troubleshooting sessions, case studies, problem-solving tasks, group work, and presentations further support learning, together with the use of modern computer technologies.
Prerequisites	Cyber Security Fundamentals Computer Networks Data Structures and Algorithms
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, practical exercises, seminar presentations, and the successful completion of independent study tasks, including computer

	hardware identification, computer assembly activities, operating system installation, software configuration, network setup exercises, computer security tasks, and system maintenance activities. The midterm assessment consists of written assignments, short in-class quizzes, laboratory evaluations, and the assessment of independent study projects, such as computer architecture analysis, hardware performance evaluation, operating system configuration reports, computer network setup projects, cybersecurity case studies, troubleshooting exercises, and presentations on modern computer technologies. The final assessment is a written examination consisting of theoretical questions and practical problem-solving tasks designed to assess students' understanding of computer architecture, hardware components, software systems, operating systems, computer networks, information security, technical maintenance procedures, and their ability to apply technical knowledge to real-world computing environments.
Recommended Literature	1. I.B. Aminov – Computer Hardware and Software. SamDU Publishing House. 2020. 2. M.E. Mamarajabov, Sh.Sh. Adinayev, R.A. Razzakov - Computer Software. 2017. 374 pages. 3. J. Glen Brookshear, Dennis Brylow – Computer Science. 12th Global Edition. 4. Easttom C. Computer security fundamentals. – Pearson IT Certification, 2019, 447 p. 5. I. Englander – Computer Hardware, System Software and Networking. 4th edition. 678 p. 6. Markov A. S., Barabanov A. V., Dorofeev A. V., Tsirlov V. L. Seven Safe Information Technologies / ed. by A. S. Markov. – M.: DMK Press, 2017. – 224 pp. 7. Yablonsky S.V. Introduction to Discrete Mathematics. – M.: Vyshaya Shkola, 2003.
Responsible Lecturer	Tursunov Farxod Bakhodir o'g'li

FUNDAMENTALS OF CRYPTOGRAPHY

Field	Description
Module Title	FUNDAMENTALS OF CRYPTOGRAPHY
Module Code	KGA305
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 74 hours (44 hours Lecture, 30 hours Practical lessons) Self-Assignments: 76 hours
Frequency/Duration	V semester
Module objective	The objective of the Module Fundamentals of Cryptography is to develop students' theoretical knowledge and practical skills in the theoretical foundations of cryptography, symmetric and asymmetric encryption methods, hash functions and the operating principles of digital signatures. The Module traces the historical development of cryptography from classical substitution and transposition ciphers to modern computer-based cryptographic systems. It examines symmetric-key algorithms, including stream and block ciphers such as the Advanced Encryption Standard, together with public-key systems such as RSA and the Diffie-Hellman key-exchange algorithm. Students study cryptographic hash functions, Message Authentication Codes and the principles underlying digital signatures, public-key infrastructure and SSL/TLS protocols. The Module further addresses key management and the cryptographic key lifecycle, authentication and identification technologies, and the requirements of cryptographically secure random number generators. Applications of cryptography in email security, messengers, blockchain, e-commerce and e-government systems are considered, together with an introduction to post-quantum cryptography. Students thereby develop the competence to understand, evaluate and apply cryptographic tools in order to ensure the security of information systems.
Module Learning Outcomes	Knowledge:

	LO1. Develops the necessary knowledge and skills in the field of cryptography. LO2. Develops practical skills in creating programs using modern programming languages. LO3. Acquires initial knowledge of network security and its protection. LO4. Develops practical skills in working with databases and Big Data. Skills: LO5. Becomes familiar with software related to cryptography. LO6. Studies the theoretical foundations of using personal computers in management processes and the methods for applying them in various branches and sectors of the national economy. LO7. Learns to work with database management systems and Big Data. LO8. Acquires the necessary knowledge and skills related to information security. Competencies: LO9. Encodes and decodes texts and files using various encryption algorithms. LO10. Understands the mathematical foundations of cryptographic algorithms. LO11. Uses cryptographic tools to ensure the security of information systems.
Content	Lectures Introduction to Cryptography and Basic Concepts of Information Security. The purpose, scope, and fundamental principles of cryptography are introduced. The concepts of confidentiality, integrity, availability, authentication, authorization, and non-repudiation are examined. The role of cryptographic methods in protecting information systems and communication channels is also considered. Historical Development of Cryptography. The principal stages in the development of cryptography are studied, from ancient substitution techniques to modern computer-based cryptographic systems. The influence of military and diplomatic developments on encryption methods is considered. The influence of scientific and technological developments on encryption methods is also considered.

Fundamentals of Cryptography and Cryptanalysis. The basic elements of cryptographic systems, including plaintext, ciphertext, encryption algorithms, decryption algorithms, and cryptographic keys, are examined. The objectives and general methods of cryptanalysis are discussed, together with security assumptions and attack models. Criteria for evaluating the strength of cryptographic systems are also discussed.

Classical Encryption Methods: Caesar and Vigenere Ciphers. The operating principles of the Caesar substitution cipher and the Vigenère polyalphabetic cipher are studied. Encryption and decryption procedures, together with key selection and frequency analysis, are examined through illustrative examples. The principal security weaknesses of classical ciphers are also examined.

Symmetric and Public-Key Cryptosystems. Symmetric-key and asymmetric-key encryption systems are compared according to their architecture and key distribution requirements. Their performance, scalability, and security properties are also compared. Their respective roles in data encryption, secure communication, authentication, and hybrid cryptographic systems are analyzed.

The Concept and Operating Principle of Stream Cipher Algorithms. Stream ciphers are studied as encryption algorithms that process plaintext one bit or byte at a time. Keystream generation, synchronization, and initialization vectors are explained. Encryption and decryption procedures, together with the security requirements of stream cipher systems, are also explained.

The Concept and Operating Principle of Block Cipher Algorithms. Block ciphers are examined as algorithms that encrypt fixed-size blocks of data using a secret key. Their internal structures, together with substitution and permutation operations, are discussed. Round functions, key schedules, and principal modes of operation are also discussed.

Introduction to the Advanced Encryption Standard (AES) Algorithm. The structure and operating principles of the AES algorithm are studied.

Key sizes, block size, and round transformations, including SubBytes, ShiftRows, MixColumns, and AddRoundKey, are explained. The role of AES in modern data-protection systems is also explained.

Cryptographic Hash Functions and Their Applications. Cryptographic hash functions are introduced as algorithms that transform input data of arbitrary length into fixed-length hash values. Preimage resistance, second-preimage resistance, and collision resistance are considered. Data-integrity verification, password protection, digital signatures, and blockchain applications are also considered.

The Concept and Operating Principle of Digital Signatures. Digital signatures are studied as cryptographic mechanisms that provide authentication, integrity, and non-repudiation. The processes of signature generation and signature verification are explained. Document hashing and the use of public and private keys are also explained.

Fundamentals of Public-Key Cryptography. The theoretical principles of asymmetric cryptography are examined. Public and private key pairs, one-way mathematical functions, encryption, decryption, digital signatures, and secure key distribution are studied. Their advantages and computational limitations are also considered.

Diffie-Hellman Key Exchange Algorithm. The Diffie–Hellman algorithm is studied as a method for establishing a shared secret over an insecure communication channel. Its mathematical foundations, key-generation process, and shared-secret calculation are explained. The importance of authentication in preventing intermediary attacks is also explained.

Fundamentals of the RSA Algorithm. The mathematical foundations and operating principles of RSA are studied. Key generation, public and private key formation, encryption, decryption, and digital-signature applications are examined. The importance of secure padding and appropriate key lengths is also considered.

PKI, Digital Certificates, and SSL/TLS Protocols. Public Key Infrastructure, digital certificates,

certificate authorities, registration authorities, and certificate-validation processes are studied. The role of SSL/TLS protocols in authenticating communication parties and establishing secure sessions is explained. The role of SSL/TLS protocols in protecting data transmitted over networks is also explained.

Random Number Generators and Their Role in Cryptography. The importance of random and pseudorandom numbers in key generation, initialization vectors, nonces, salts, and cryptographic protocols is examined. Requirements for cryptographically secure random number generators, entropy sources, and unpredictability are discussed. Common risks associated with weak randomness are also discussed.

Message Authentication Codes (MAC). Message Authentication Codes are studied as mechanisms for verifying data integrity and source authenticity using a shared secret key. The operating principles of HMAC and block-cipher-based MAC constructions, together with their applications, are explained. Their differences from hash functions and digital signatures are also explained.

Key Management and the Key Lifecycle. The complete lifecycle of cryptographic keys, including generation, distribution, storage, and activation, is examined. The lifecycle stages of use, rotation, backup, revocation, archival, and destruction are also examined. Organizational policies and technical controls required to protect keys from unauthorized access and loss are discussed.

Authentication and Identification Technologies. Identification, authentication, authorization, and access-control concepts are studied. Password-based authentication, one-time passwords, cryptographic tokens, biometric methods, certificates, and multi-factor authentication are compared. Their security, usability, and application areas are analyzed.

Email Security and Cryptographic Protection. Security threats affecting electronic mail, including interception, impersonation, and phishing, are

examined. Message modification and malicious attachments are also examined as threats. Encryption, digital signatures, certificate-based protection, secure mail protocols, and standards such as S/MIME and OpenPGP are considered.

Cryptography in Messengers and Mobile Applications. The use of cryptographic mechanisms in instant messaging and mobile software is studied. End-to-end encryption, session-key establishment, and device authentication are discussed. Secure data storage, key verification, backup protection, and risks associated with metadata and compromised endpoints are also discussed.

Fundamentals of Blockchain and Post-Quantum Cryptography. The cryptographic foundations of blockchain systems, including hash functions, digital signatures, and linked data blocks, are introduced. Consensus support and transaction integrity in blockchain systems are also introduced. The potential impact of quantum computing on current public-key algorithms and the objectives of post-quantum cryptographic methods are examined.

Application of Cryptography in E-Commerce and E-Government Systems. The role of cryptography in protecting electronic payments, online transactions, and digital identities is studied. Its role in protecting official documents, public services, and inter-organizational data exchange is also studied. Secure communication, digital signatures, certificates, authentication, confidentiality, integrity, legal validity, and trust-management mechanisms are considered.

Practical lessons

Identifying and Analyzing Everyday Information Systems Where Encryption is Used. Students identify information systems and digital services that use encryption in everyday activities, including online banking, messaging applications, and electronic commerce. Cloud storage, mobile communications, and government services are also identified as examples. The purpose of encryption, the types of protected data, and the security mechanisms used in these systems are analyzed.

Analysis of Practical Cases for Ensuring Data Confidentiality and Integrity. Practical cases involving the protection of confidential data and the prevention of unauthorized modification are examined. Students analyze the use of encryption, cryptographic hash functions, and message authentication codes. The use of access controls and digital signatures to maintain data confidentiality and integrity is also analyzed.

The Role of Cryptography in Digital Transformation Processes. The importance of cryptography in the digitalization of public services and business processes is studied. Its importance in the digitalization of education, healthcare, finance, and electronic document management is also studied. Students evaluate how cryptographic technologies support secure digital identities, trusted transactions, protected communication, and reliable data exchange.

Comparison and Evaluation of Information Protection Methods. Different information-protection methods are compared according to their purpose, effectiveness, and implementation complexity. Their performance and application area are also compared. Encryption, hashing, digital signatures, authentication, access control, data backup, and network protection mechanisms are evaluated through practical examples.

Comparison of Modern Authentication Methods. Password-based authentication, one-time passwords, and biometric authentication are compared. Security tokens, digital certificates, and passwordless authentication methods are also compared. Students assess their security, usability, reliability, cost, and suitability for different information systems.

Study of Multi-Factor Authentication (MFA) Technologies. The operating principles and components of multi-factor authentication are examined. Students analyze combinations of knowledge, possession, and biometric factors. Students also evaluate how MFA reduces the risks

associated with stolen passwords and unauthorized account access.

Analysis of Website Security Certificates.

Students examine digital certificates used by websites and identify certificate owners, issuing authorities, and validity periods. Public keys, domain information, and signature algorithms are also identified. The role of certificates in HTTPS connections, server authentication, encrypted communication, and user trust is analyzed.

Application of Digital Signature Technologies in the Republic of Uzbekistan.

The use of digital signatures in electronic government and electronic document management in the Republic of Uzbekistan is studied. Their use in public services, taxation, banking, and business processes is also studied. Students analyze their role in confirming identity, ensuring document integrity, establishing authorship, and providing legal validity to electronic documents.

Analysis of the Use of Cryptographic Tools in Open-Source and Proprietary Software.

Cryptographic capabilities implemented in open-source and proprietary software are compared. Students examine differences in transparency, licensing, documentation, and update mechanisms. Differences in security audits, algorithm support, compatibility, and user control are also examined.

Study of Security Issues in Electronic Document Management Systems.

Security requirements for creating, transmitting, and storing electronic documents are examined. Security requirements for approving and archiving electronic documents are also examined. Students analyze authentication, authorization, encryption, digital signatures, access logging, version control, backup, and protection against unauthorized modification.

Study of Information Security Problems in IoT Devices.

Security risks associated with Internet of Things devices are studied, including weak authentication and insecure communication. Outdated software, limited computing resources, and insufficient key protection are also studied as risk factors. Cryptographic measures for device

authentication, secure data transmission, firmware verification, and protected updates are evaluated.

Analysis of the Influence of the Human Factor in Social Engineering Attacks. The role of human behaviour in phishing, impersonation, manipulation, and other social-engineering threats is analyzed from a defensive perspective. Students examine common risk factors associated with these threats. Students also develop awareness, verification, access-control, incident-reporting, and security-training measures to reduce these risks.

Comparison of the Application of Cryptographic Tools in the Public and Business Sectors. The use of encryption, digital signatures, and certificates in public institutions and private organizations is compared. The use of secure authentication and key-management systems in these sectors is also compared. Differences in legal requirements, information sensitivity, infrastructure, operational needs, and risk-management approaches are analyzed.

Study of the Role and Importance of Cryptography in Modern Cybercrime. The relationship between cryptography and contemporary cybercrime is examined from information-security and prevention perspectives. Students analyze how cryptographic technologies protect systems and evidence. Students also consider challenges such as encrypted malicious traffic, unauthorized data encryption, and the need for effective incident response.

Independent Research on New Technologies and Trends in the Field of Cryptography. Students conduct independent research on emerging cryptographic technologies, standards, and application areas. Topics include post-quantum cryptography, lightweight cryptography, privacy-enhancing technologies, secure multiparty computation, homomorphic encryption, and blockchain security. Research findings are presented in the form of a report, presentation, or analytical project.

Self-study

	Basic Concepts of Cryptography, History of Development, and Classical Encryption Methods. Fundamentals of Symmetric and Asymmetric Cryptography and Security Analysis. Public-Key Cryptography and the RSA Algorithm: Mathematical Foundations. Public-Key Cryptography and the RSA Algorithm: Illustrative Calculations. Public-Key Cryptography and the RSA Algorithm: Practical Applications. Hash Functions and Data Integrity. Electronic Digital Signatures and Authentication Mechanisms. PKI Systems and SSL/TLS Protocols. Key Management and Random Number Generators. Authentication Systems and Password Security. Cryptographic Protocols and Practical Analysis. Practical Cryptography: Email, Mobile, and Government Systems. Open-Source Cryptographic Tools and Web Security. Blockchain and Cryptocurrency Security. Trends and Prospects in Modern Cryptography.
Teaching Methods	The Module is delivered through lectures, practical sessions, problem-based and case-based learning, analytical activities, presentations, independent research, and mini-projects. Students examine classical and modern encryption methods, symmetric and asymmetric cryptosystems, hash functions, digital signatures, authentication mechanisms, cryptographic protocols, digital certificates, and key-management processes. Practical activities include analyzing real-world applications of cryptography, evaluating information-protection methods, examining SSL/TLS certificates, comparing authentication technologies, and studying cryptographic tools used in web, mobile, e-government, and electronic document systems. Teacher-guided and student independent learning, scientific literature, electronic resources, reports, presentations, and practical assignments further develop students' analytical, research, information-security, and cryptographic problem-solving skills.

Prerequisites	Computer Networks Network Security Fundamentals of Cybersecurity
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and final assessment (50%). The continuous assessment evaluates participation in 15 practical sessions and completion of six Teacher-Guided Independent Work assignments, with a maximum of 8 points awarded for practical participation and 12 points allocated to independent assignments, analytical work, reports, presentations, and mini-projects. The midterm assessment consists of a theoretical component worth 12 points and independent work worth 18 points; the theoretical component may be conducted as a test, oral examination, or written work, with a written assessment comprising three theoretical and two practical questions or a test of 24 questions worth 0.5 points each. The final assessment may be conducted as a written examination consisting of five questions worth 10 points each, or a test consisting of 25 questions worth 2 points each. All required independent study assignments must be submitted before the final assessment.
Recommended Literature	1. Katz, J., & Lindell, Y. Introduction to Modern Cryptography. 3rd ed. Chapman & Hall/CRC, 2021. 2. Boneh, D., & Shoup, V. A Graduate Course in Applied Cryptography. Version 0.6, 2020. 3. Menezes, A. J., van Oorschot, P. C., & Vanstone, S. A. Handbook of Applied Cryptography. CRC Press, 1996. 4. Paar, C., & Pelzl, J. Understanding Cryptography: A Textbook for Students and Practitioners. Springer, 2010. 5. Stallings, W. Cryptography and Network Security: Principles and Practice. 8th ed. Pearson, 2023. 6. Ryabko, B. Ya., & Fionov, A. N. Cryptographic Methods of Information Protection. Moscow: Goryachaya Liniya–Telekom, 2017.

	7. Moldovyan, A. A., & Moldovyan, N. A. Cryptography. Saint Petersburg: BHV-Petersburg, 2004. 8. Domarev, V. V. Security of Information Technologies: A Systematic Approach. Kyiv: DiaSoft, 2004. 9. Romanets, Yu. V., Timofeev, P. A., & Shangin, V. F. Information Protection in Computer Systems and Networks. Moscow: Radio i Svyaz, 2018. 10. Khoroshko, V. A., & Chekatkov, A. A. Methods and Tools for Information Protection. Kyiv: Junior, 2003.
Responsible Lecturer	Shaymanov Farxod Xushbakovich, Department of Computer and Software Engineering, Termiz State University.

PHP PROGRAMMING LANGUAGE

Field	Description
Module Title	PHP PROGRAMMING LANGUAGE
Module Code	PHPDT305
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (34 hours Lecture, 26 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	V semester
Module objective	The aim of the Module PHP Programming Language is to teach students the PHP programming language. Students develop projects using the programming language. These projects are analyzed. Students learn to create problem-oriented programs using PHP. The Module also develops students' skills in working with the Yii2 framework. Students gain the ability to design and deploy real web applications using PHP and Yii2.
Module Learning Outcomes	Knowledge: LO1. To develop knowledge and skills in the PHP programming language and to learn the necessary software tools.

LO2. Study the principles of the PHP programming language, its main features, and analyse its advantages.

LO3. Have an understanding of the history of the PHP programming language and its development.

LO4. Know how to work with databases using the ActiveRecord ORM in Yii2.

LO5. Has knowledge of Yii2's security components (input validation, CSRF, RBAC).

LO6. Theoretically understands how to create, validate and submit forms in Yii2.

LO7. Learns how to work with modules, widgets and extensions in Yii2.

LO8. Will have a basic understanding of creating and configuring a REST API.

Skills:

LO9. Knows how to create and configure a Yii2 project via Composer.

LO10. Creates CRUD functions for working with the database quickly using Yii2.

LO11. Skills in creating forms, validation and data storage are developed.

LO12. Manages user permissions using RBAC or Access Control.

LO13. Integrates Ajax and Pjax technologies with Yii2.

LO14. Uses Yii2's logging, error and debugging tools.

LO15. Creates a RESTful API and can prepare responses in JSON format.

Competencies:

LO16. They will have the competence to independently design and launch real web applications using PHP and the Yii2 framework.

LO17. Acquires skills in structuring and modularising large-scale software projects based on the MVC architecture.

LO18. Develops professional database skills - migration, optimisation, and creating complex queries via ActiveRecord ORM.

LO19. Develops the competence to apply professional approaches and solutions for ensuring web application security (CSRF, XSS, SQL Injection, RBAC).

	LO20. Create RESTful APIs, integrate with external services, and perform professional-level data exchange in JSON format.
Content	Lectures Introduction to Web Programming - Client-Server Architecture, HTTP, Requests and Responses. The basic principles of web programming are introduced. Client-server interaction and HTTP communication are examined. The processing of requests and responses is studied. Introduction to the PHP Language - Syntax, Variables, Data Types, Operators. The fundamental syntax of PHP is studied, together with variable declaration and data types. Constants are examined. Arithmetic, comparison, and logical operators are studied. Conditional Statements and Loops - if, else, switch, for, while, foreach. Conditional structures, including if, else, and switch statements, are examined for controlling program execution and making decisions. Loop statements, including for, while, and foreach, are studied for repeatedly processing data. Their application in program control flow is discussed. Working with Arrays and Strings – PHP Functions, Manipulation. Indexed and associative arrays, together with multidimensional arrays, are studied. String manipulation is examined. Built-in PHP functions for searching, sorting, formatting, and modifying data are discussed. Functions and Scopes – Creating Functions, Parameters, Global Variables. The principles of creating and calling functions are explained, together with passing parameters and returning values. Local, global, and static variable scopes are examined. Their management within PHP programs is discussed. Forms and Superglobal Variables – \$_GET, \$_POST, \$_SESSION, \$_COOKIE. HTML form data processing is studied, together with the use of PHP superglobal variables, including \$_GET, \$_POST, \$_SESSION, and \$_COOKIE. Sessions are examined for storing user-related information.

Cookies are examined for managing user-related information across requests.

PHP OOP Basics – Class, Object, Inheritance, Interfaces. The foundations of object-oriented programming in PHP are introduced, including classes, objects, properties, and methods. Inheritance and encapsulation are examined. Interfaces are studied.

Working with Files – Reading/Writing, Form via File Upload. Methods for creating, reading, and writing files are studied. File management techniques are examined. Secure file-upload processing through web forms is also studied.

Working with MySQL – PDO, CRUD Queries, SELECT/INSERT. PHP-MySQL database connectivity using PDO is studied. CRUD operations are examined. Secure SELECT, INSERT, UPDATE, and DELETE queries are executed.

MVC and the Concept of a Framework – Architecture, Why a Framework is Needed. The Model-View-Controller (MVC) architecture and the purpose of web frameworks are introduced. Their role in organizing code and improving maintainability is explained. Their contribution to security and application development efficiency is discussed.

Installing and Configuring Yii2 – via Composer, Basic/Advanced Templates. The installation and initial configuration of Yii2 using Composer are studied. The structure and purpose of the Basic and Advanced project templates are examined. Their respective application areas are compared.

Yii2 Project Structure – Folders, Configurations, Components. The main folders and configuration files of a Yii2 project are examined. Application components and dependencies are studied. Their roles in organizing and managing a web application are explained.

Controllers and Routing – Actions, URL Management. The creation of controllers and actions is studied, together with request processing. Route definition and URL management are examined. Methods for directing users to appropriate application pages are discussed.

Model and ActiveRecord – Connecting to a MySQL Table, Validation Rules. Yii2 models and ActiveRecord are studied for interacting with MySQL database tables. Database connections and data retrieval are examined. Record management and validation rules are explained.

Forms and Widgets – Creating a Form, Contact Form, Widgets. Yii2 tools for creating and processing web forms are examined. Form validation and contact forms are studied. Data input fields and reusable interface widgets are discussed.

Authorisation and Authentication – Login, Logout, RBAC Concept. User authentication and authorisation mechanisms are studied, including login and logout functions. Session management and access control are examined. The basic principles of Role-Based Access Control (RBAC) are introduced.

Project Deployment – Deploying the Project to the Server/Hosting. The procedures for preparing and deploying a Yii2 application to a web server or hosting environment are studied. Server configuration, dependencies, and database settings are examined. Security and deployment testing are considered.

Practical lessons

Environment Configuration – OpenServer/XAMPP, phpinfo(), Checking the Environment. The local web development environment is installed and configured using OpenServer or XAMPP. PHP settings and server status are verified through the phpinfo() function. Extensions and environment compatibility are checked.

PHP Fundamentals – Variables, Simple Mathematical Operations. Basic PHP syntax and variable declaration are practiced. Data types and arithmetic operators are applied. Simple programming tasks and mathematical calculations are performed.

Forms and Arrays – \$_POST, \$_GET, Working with Arrays. HTML forms are created and processed using the \$_POST and \$_GET superglobal variables. Indexed and associative arrays are used to store and retrieve submitted data. Submitted data is manipulated using array functions.

Session and Cookies – Working with Sessions via Authorisation. PHP sessions and cookies are applied to store user information. Authentication states are managed using these mechanisms. Basic authorisation, session creation, verification, and termination processes are implemented.

Working with MySQL – Connecting to the Database and Retrieving Data via PDO. A PHP application is connected to a MySQL database using PDO. Secure queries are executed to retrieve and process database records. The retrieved records are displayed.

Creating a Simple Blog (PHP) – Adding and Displaying Posts. A basic PHP blog application is developed with functions for adding and storing posts. Functions for retrieving and displaying posts are implemented. Form processing and database interaction are integrated into the application.

Yii2 Installation – Composer, Project Setup. The Yii2 framework is installed using Composer. A new project is created and configured. The basic application structure and initial settings are examined.

Controllers and Routing – Controller, Action and URLs. Yii2 controllers and actions are created to process user requests. Routes are configured to connect application pages with the appropriate controller actions. URL rules are defined for navigation.

Creating CRUD with Gii – Automatic Model, Form and CRUD. Yii2 Gii is used to automatically generate models and forms. Controllers and CRUD interfaces are generated. The generated code is examined and adapted to meet application requirements.

Form and Validation – Application (Form) and Data Validation. Yii2 forms are created using models and form widgets. Validation rules are defined to check user input. Submitted data is verified to ensure that it meets application requirements.

Authentication – Login/Logout Function. User authentication functionality is implemented in Yii2. Login and logout functions are developed, together with password verification. Session management and access control processes are practiced.

	File Uploads – Uploading an Image via the UploadForm Model. Image-upload functionality is developed using the Yii2 UploadForm model. File validation and secure storage are implemented. Filename management and uploaded-image processing are practiced. Mini-project – Final Independent Project: a Blog, a Catalogue, etc. A complete web application, such as a blog, product catalogue, or similar system, is developed. The project integrates database operations, forms, and validation. Authentication, file uploads, and Yii2 architectural principles are also applied. Self-study History, Development and Role of the PHP Programming Language in Web Development. PHP Syntax, Data Types, Operators and Conditional Statements. Loops and Arrays - PHP Functions and Manipulation. Functions, Scopes and Superglobal Variables. OOP Fundamentals in PHP - Class, Object, Inheritance and Interfaces. Working with Files and Connecting to MySQL via PDO. Introduction to MVC Architecture and the Yii2 Framework. Creating and Configuring a Yii2 Project via Composer. Creating Controllers, Routing and Managing URLs. Creating a Model, ActiveRecord ORM and Working with the Database. Creating View Files, Configuring Layout and Design Fundamentals. Automatically Generate CRUD Actions Using the Gii Generator. Creating Forms, Validation and Data Storage. Session, Cookies and User Registration. Authentication in Yii2, Login/Logout Functionality.
Teaching Methods	The module is delivered through lectures, practical programming sessions, problem-based learning, code analysis, demonstrations, independent assignments, presentations, and project-based

	activities. Students develop web applications using PHP, MySQL, PDO, and the Yii2 framework, create MVC components, forms, validation rules, CRUD operations, authentication and RBAC systems, and apply security mechanisms against CSRF, XSS, and SQL injection. Practical work also includes REST API development, integration with external services, version control through Git, testing, debugging, and project deployment to a server. Teacher-guided and student independent learning, technical documentation, electronic resources, reports, and mini-projects further develop students' programming, analytical, database, web security, software design, and deployment skills.
Prerequisites	Programming 1,2 Database Data Structures and Algorithms
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and final assessment (50%). The continuous assessment evaluates students' preparation for and participation in practical sessions, completion of programming tasks, and submission of independent assignments, reports, presentations, and mini-projects. A maximum of 20 points may be earned during the semester. The midterm assessment is worth 30 points. Up to 20 points are awarded for a test or written assessment covering the theoretical and practical topics of the course, while up to 10 points are awarded for the completion and submission of independent study assignments. The final assessment may be conducted as a test or written examination. The test consists of 25 questions worth 2 points each, while the written examination consists of five questions worth 10 points each. Questions may be selected from lecture, practical, and independent study topics. Students must submit all required assignments and achieve at least 60 points overall to successfully complete the module.

Recommended Literature	1. Welling, L., & Thomson, L. PHP and MySQL Web Development. 5th ed. Addison-Wesley, 2017. 2. Grodzenskiy, Ya. S. Information Security. Moscow: RG-Press, 2020. 144 p. 3. Safronov, M. Web Application Development with Yii 2 and PHP. Packt Publishing, 2016. 4. Nixon, R. Learning PHP, MySQL & JavaScript. 6th ed. O'Reilly Media, 2021. 5. Way, J. PHP Objects, Patterns, and Practice. 6th ed. Apress, 2019. 6. Yii Software LLC. Yii 2.0 Official Guide. Available at: https://www.yiiframework.com/doc/guide/2.0/en 7. The PHP Group. PHP Official Documentation. Available at: https://www.php.net/manual/en/ 8. Harris, S. CISSP All-in-One Exam Guide. Guide for CISSP examination preparation, 2011. 876 p. 9. Masalkov, A. S. Features of Cybercrime: Tools for Information Attacks and Protection. Moscow: DMC Press, 2018. 10. Yii Software LLC. Yii2 GitHub Repository. Available at: https://github.com/yiisoft/yii2 11. Decree of the President of the Republic of Uzbekistan dated 13 December 2018 on the development and use of digital information resources in public administration.
Responsible Lecturer	Tursunov Farxod Baxodir o'g'li, Department of Computer and Software Engineering, Termiz State University.

BIG DATA AND DATA ANALYSIS

Field	Description
Module Title	BIG DATA AND DATA ANALYSIS
Module Code	BDMT305
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 76 hours (42 hours Lecture, 34 hours Practical lessons) Self-Assignments: 74 hours
Frequency/Duration	VI semester

Module objective	The objective of the Module Big Data and Data Analysis is to prepare students to work with large volumes of data. The Module equips students with theoretical knowledge and practical skills to store, process, analyse, and interpret large volumes of data using modern technologies. During the course, students become familiar with big data infrastructure. Parallel computing and data warehousing are studied. Data streams and analytical models are examined. Visualisation techniques are introduced. The fundamentals of machine learning are also studied.
Module Learning Outcomes	Knowledge: LO1. Explain the concept of Big Data, its types and characteristics. LO2. Knows the main components of large-scale data infrastructure (Hadoop, Spark, etc.). LO3. Theoretically understands the stages of the data analysis process (collection, cleansing, analysis, visualisation). LO4. Understands the theoretical foundations of parallel and distributed computing methods. LO5. Understands the differences and methods for working with structured and unstructured data. LO6. They will have knowledge of the principles of working with databases and data warehouses. Skills: LO7. They can carry out practical tasks related to collecting, organising and processing large volumes of data. LO8. Acquires the skill to perform data analysis using the Python or R programming languages. LO9. Can perform initial configurations when working with Hadoop, Spark or other Big Data platforms. LO10. Uses data cleaning, anomaly detection and visualisation techniques. LO11. Can apply an analytical approach to data using statistical methods. LO12. Draws conclusions on decision-making based on the results of the analysis.

	Competencies: LO13. Applies Big Data technologies and data analysis methods effectively in professional practice. LO14. Utilises technologies for collecting, storing, processing and managing large volumes of data. LO15. Analyses data and substantiates the results using statistical analysis, data mining and machine learning methods. LO16. Applies big data platforms (Hadoop, Spark, etc.) and modern analytical tools in practical projects. LO17. Visualises data, interprets analytical results and prepares analytical reports for management decision-making. LO18. They will independently explore new technologies and analytical platforms in the field of Big Data and will be competent in implementing them in scientific and professional activities.
Content	Lectures Introduction to Big Data Technologies. The concept of Big Data is introduced. The 5V principle of Big Data is examined. Differences between Big Data and traditional data are discussed. Big Data Ecosystem: Infrastructure and Architecture. Distributed systems and NoSQL databases are studied. The MapReduce processing model is examined. Real-time processing techniques are discussed. Data Types and Their Modelling. Structured data is studied. Semi-structured data is examined. Unstructured data and their modelling approaches are discussed. Storage Technologies for Large-Scale Data. The Hadoop Distributed File System (HDFS) is studied. Its architecture and fault-tolerance mechanisms are examined. Cloud storage technologies for large-scale data are discussed. Hadoop Architecture and Components. The NameNode and DataNode components of Hadoop are studied. YARN resource management is examined. The MapReduce mechanism is discussed. Apache Spark Technology. Spark Core and Resilient Distributed Datasets (RDD) are studied.

DataFrames and Spark SQL are examined. The MLlib machine learning library is discussed.

Data Collection and Integration. The ETL (Extract, Transform, Load) process is studied. Apache Kafka and Flume are examined for data streaming and collection. Sqoop is discussed for data transfer between relational databases and Hadoop.

Data Cleansing and Preparation. Methods for handling missing values are studied. Noise removal techniques are examined. Data transformation and normalisation are discussed.

Fundamentals of Statistical Analysis. Descriptive statistics are studied. Dispersion and correlation are examined. The p-value and its role in statistical significance are discussed.

Data Visualisation. Visual analysis techniques are studied. Graphical tools such as Tableau and Power BI are examined. Python visualization libraries, including Matplotlib and Seaborn, are discussed.

Working with Databases and Warehouses. OLTP and OLAP systems are compared. Data warehouses are studied. Data lakes are discussed.

SQL and NoSQL in Big Data Processing. MongoDB and Cassandra, as NoSQL databases, are studied. Hive is examined for data warehousing on Hadoop. Pig is discussed for data flow processing.

Machine Learning Fundamentals. Regression algorithms are studied. Classification algorithms are examined. Clustering and other core machine learning algorithms are discussed.

Data Stream Processing. Apache Storm is studied for real-time stream processing. Apache Flink is examined. Real-time analytics techniques are discussed.

Big Data Security and Privacy. Data governance principles are studied. Encryption techniques are examined. Access control mechanisms are discussed.

Application of Big Data Analytics in Practical Fields. Applications of Big Data analytics in healthcare and finance are studied. Applications in marketing and transport are examined. Applications in IoT systems are discussed.

	Managing and Evaluating Big Data Projects. Agile methodology for managing Big Data projects is studied. Key performance indicators (KPIs) and evaluation metrics are examined. Real-world Big Data project analysis is discussed. Cloud Big Data Platforms. Cloud Big Data platforms are studied, including AWS, Azure, and GCP services. Scaling techniques for these platforms are examined. Cost optimisation strategies using services such as EMR, Databricks, and BigQuery are discussed. Graph Data Analysis. Social network and relationship analysis techniques are studied. Neo4j and Spark GraphX are examined as graph analysis tools. The PageRank algorithm is discussed. Recommendation Systems and Big Data Algorithms. Collaborative and content-based filtering techniques are studied. LSH and MinHash similarity search methods are examined. Association rules and market basket analysis are discussed. Scalable ML and Deep Learning. Spark MLlib is studied for scalable machine learning. Integration with TensorFlow and PyTorch is examined. Model deployment and the fundamentals of MLOps are discussed. Practical lessons Introduction to XAMPP and phpMyAdmin. XAMPP is installed. The Apache and MySQL servers are started. A new database is created via the phpMyAdmin interface. Creating a Database and Performing Simple Queries. A database is created using the CREATE DATABASE statement. Tables are created using the CREATE TABLE statement. Data is inserted and retrieved using INSERT INTO and SELECT FROM queries via phpMyAdmin. Importing Data from a CSV File via phpMyAdmin. A prepared CSV dataset file is uploaded to a table via the phpMyAdmin Import menu. The imported data format is verified. Data integrity is checked after import. Designing the Database Based on the ER Diagram. Table relationships are created based on
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an entity-relationship (ER) diagram. Tables are linked using FOREIGN KEY constraints. The ER diagram is viewed using phpMyAdmin's ER Designer.

Analysis of Table Joins Using the JOIN Operators. INNER JOIN operations are performed on two or more tables. LEFT JOIN operations are practiced. RIGHT JOIN operations are performed and analyzed.

Analysis with Aggregate Functions and GROUP BY. Data is summarized using the COUNT(), SUM(), and AVG() functions. The MIN() and MAX() functions are applied. These functions are used in conjunction with GROUP BY to analyze grouped data.

Filtering and Sorting Data. Data is filtered using the WHERE clause. Pattern matching is performed using the LIKE operator. Data is sorted using ORDER BY.

Data Cleansing: Identifying NULL and Redundant Data. NULL values are identified using the IS NULL operator. Duplicate records are identified using the DISTINCT operator. Duplicate or empty data is deleted or updated.

Analytical Queries on Categorical Data. Categorical values are grouped using CASE or IF statements. Data types are converted using the CONVERT() function. Data types are also converted using the CAST() function.

Working with Time Data. Time-related data is analyzed using the DATE and NOW() functions. The CURDATE() and DATEDIFF() functions are applied. Time data is formatted using the DATE_FORMAT() function.

MySQL JSON Functions (MySQL 5.7+). Columns of the JSON data type are created. Data is extracted from JSON columns using the JSON_EXTRACT() function. JSON objects and arrays are created using the JSON_OBJECT() and JSON_ARRAY() functions.

Connecting to a MySQL Database via PHP (mysqli or PDO). A connection to the MySQL database is established using mysqli_connect(). A query is executed via a PHP file using mysqli_query(). The connection process is also examined using PDO.

Dynamically Displaying a Data Table Using PHP and HTML. The result of a SELECT query is retrieved using PHP. The retrieved data is displayed in an

	HTML table. A while loop is used to iterate through and display the data dynamically. Input Data via a User Interface Using PHP. An HTML form is created for user data input. The form data is sent to a PHP script via the POST method. The data is written to the database using an INSERT INTO query. Aggregate Analysis and Simple Visualisation Using PHP. Aggregate results, such as COUNT() and SUM(), are calculated in PHP. The results are formatted for display. The results are displayed in the user interface using HTML. Completing CRUD: UPDATE and DELETE Operations. An existing record is selected by its ID via an HTML form. The record is edited using an UPDATE query. The record is deleted using a DELETE query, with both actions executed using PHP. Secure Queries and Session-Based Login (Final Mini-Project). Queries are secured against SQL Injection using Prepared Statements, including mysqli_prepare and PDO with bind_param. A login and logout feature is implemented using \$_SESSION. The application is finalized as a complete mini-project. Self-study History of the Development of Big Data Technologies and Their Main Directions. Large-Scale Data Storage Technologies: HDFS and Its Alternatives. Analytical Comparison of Hadoop and Spark Technologies. ETL (Extract, Transform, Load) Process and Its Analytical Significance. Managing Large Volumes of Data in MySQL and PostgreSQL Databases. Analysis of CSV and JSON Data Types: Storage, Import, and Export Methods. Preparing Analytical Reports Using Aggregate Functions. Ensuring Data Security and Privacy in Big Data Systems.
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	Real-Time Data Streaming Using Apache Kafka, Flume, and Sqoop. Time Series Analysis: Practical Work with Temporal Data in SQL. The Life Cycle and Stages of a Big Data Project. OLAP and OLTP Systems: Differences, Advantages, and Applications. Visualization Tools for Analyzing Large-Scale Data. The Role of Machine Learning Models in Big Data Analysis. Working with Data via a Web Interface: PHP and MySQL Integration. Statistical Methods for Data Cleansing. The Concepts of Data Lake and Data Warehouse. Application of Big Data Analysis by Industry. Introduction and Prospects of Big Data Technologies in Uzbekistan. Popular Open Data Sets Used in Big Data Analysis. Analytical Comparisons of SQL and NoSQL Systems. Data Mining and Big Data: Concepts, Methods, and Interrelationships. Data Stream Analysis and Real-Time Monitoring Systems. Integration of IoT (Internet of Things) and Big Data. Methods and Metrics for Ensuring Data Quality. Advantages of Cloud Technologies in Working with Big Data. Application of Data Analysis Results in Decision-Making (Data-Driven Decision Making). The Most Popular Programming Languages Used in Big Data Projects. The Formation and Management of a Big Data Strategy in Large Enterprises. Preparation of a Presentation and a Mini-Project on Independent Learning Topics.
Teaching Methods	Knowledge of the theoretical foundations of Big Data technologies and their practical application areas is developed. Teaching methods for collecting, storing, managing, and processing large volumes of data are applied. Skills in using statistical and intelligent data analysis methods are developed. Proficiency in data processing using Big Data platforms and programming tools is developed. Skills in data

	visualisation, interpreting analytical results, and preparing reports are developed. Competencies in problem analysis, the resolution of scientific and practical issues, and decision-making based on Big Data are developed.
Prerequisites	Database Data Structures and Algorithms Discrete Structures
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, practical database exercises, seminar presentations, and the successful completion of independent study assignments. These assignments include SQL query development, data preprocessing and cleansing tasks, CSV and JSON data analysis, Hadoop and Spark practice activities, data visualization exercises, and short reports on current Big Data technologies and applications. The midterm assessment consists of written assignments, short in-class quizzes, practical laboratory evaluations, and the assessment of independent study projects. Typical projects include designing an ETL workflow, comparative analysis of SQL and NoSQL databases, implementation of Big Data storage solutions, development of analytical dashboards, data mining case studies, and presentations on Big Data applications in business, healthcare, finance, or smart city systems. The final assessment is a written examination consisting of theoretical questions, analytical case studies, SQL-based problem-solving tasks, and data interpretation exercises designed to assess students' understanding of Big Data technologies, distributed data processing, database management, cloud computing, data analytics, and machine learning applications. The examination also evaluates students' ability to apply Big Data methods and analytical techniques to real-world data-driven

	problems. Successful completion of all laboratory work, practical assignments, and independent study projects is required as an essential component of the module assessment.
Recommended Literature	1. Damji J.S., Wenig B., Das T., Lee D. Learning Spark, 2nd Edition. O'Reilly / Databricks, 2020. 2. James G., Witten D., Hastie T., Tibshirani R. An Introduction to Statistical Learning (2nd ed., R/Python). Springer, 2021. 3. Aliqulov S.A., Shukurov I.S., Kushatov R.X. et al. Big Data (textbook). Uzbekistan. 4. Big Data and Data Analysis (a teaching and methodological manual). Bukhara State University, 2021. 5. Problems in Big Data and its Intelligent Analysis. Modern Science and Research: Problems and Solutions, 2025.
Responsible Lecturer	Dilshod Abdualimov, son of Abdujabbor, teacher

R PROGRAMMING LANGUAGE

Field	Description
Module Title	R PROGRAMMING LANGUAGE
Module Code	RDT305
Credit Points (ECTS)	6 Credits
Workload	Total workload: 180 hours Contact hours: 76 hours (42 hours Lecture, 34 hours Practical lessons) Self-Assignments: 104 hours
Frequency/Duration	VI semester
Module objective	The objective of the Module R Programming Language is to develop students' theoretical knowledge and practical skills in data processing, statistical analysis, and visualisation using the R programming language. During the course, students learn the syntax of the R language. Data types and structures, including vectors, lists, data frames, and factors, are studied. Control flow operators and functions are examined. Object-oriented

	programming using the S3, S4, and R6 systems is introduced. The tidyverse ecosystem, including dplyr, tidyr, and ggplot2, is studied. Data import and export, statistical analysis, and linear regression are examined. Reproducible reporting using R Markdown is also studied.
Module Learning Outcomes	Knowledge: LO1. Knows the fundamental concepts and syntax of the R programming language. LO2. Understands how to work with variables and data structures (vector, matrix, list, data frame, factor). LO3. Understands the principles of operation of conditional (if/else) and looping (for/while/repeat) statements. LO4. Theoretically understands the processes of data import/export, cleaning and transformation. LO5. Is familiar with R's capabilities for statistical functions, linear regression, and statistical analysis. LO6. Understands the fundamentals of visualisation using graphical modules (plot, ggplot2). Skills: LO7. Writes and runs standalone code in the R environment. LO8. Reads and saves data in the correct format (CSV, Excel, TXT, JSON). LO9. Works with and manipulates complex data structures (lists, data frames). LO10. Performs statistical calculations independently (mean, variance, correlation, t-test, regression). LO11. Creates graphical charts (histogram, line graph, bar chart, scatter plot). LO12. Uses the dplyr, tidyr and ggplot2 libraries to clean, group and visualise data. Competencies: LO13. Applies data processing and analysis methods using the R programming language effectively in professional practice. LO14. Works with data structures such as vectors, lists, data frames and factors to import, export and transform data. LO15. Draws conclusions using statistical analysis, correlation and linear regression methods.

	LO16. Cleans, groups and visualises data using the tidyverse (dplyr, tidyr, ggplot2) ecosystem. LO17. Writes user functions, modularises code, and prepares reproducible reports with R Markdown. LO18. Independently learns and applies modern tools and packages in the field of Data Science to professional practice.
Content	Lectures The R Programming Language and Programming Environment. The purpose of the R language is introduced. The RStudio interface, including the console, script editor, and workspace, is examined. Key packages such as ggplot2, dplyr, and tidyr are discussed. Variables in R. Variables in R are created using assignment operators. Naming conventions for variables are studied. Assigning values to variables is discussed. Data Types in R. The numeric, integer, logical, character, and complex data types are studied. The class() and typeof() functions are examined. Converting between data types is discussed. Operators in R. Arithmetic, comparison, and logical operators are studied. String operations are examined. The remainder operator %% and integer division operator %/% are discussed. Branching Operators. Conditional operators if, else if, and else are studied. Their use in controlling program flow is examined. The ifelse() function for vectorized conditions is discussed. Loop Operators. The for, while, and repeat loops are studied. The break and next statements are examined. Automation of repetitive tasks using loops is discussed. Strings in R. The nchar() and substr() functions are studied. The toupper() and tolower() functions are examined. The grep(), sub(), and gsub() functions are discussed. Lists in R. Lists are studied as objects capable of storing different data types. Creating lists using list() is examined. Accessing list elements by index and by name is discussed.

Vectors in R. Vectors are created using the `c()` function. Indexing of vectors is studied. Vectorisation and logical indexing are discussed.

Functions in R. Functions are defined using the `function()` keyword. Parameters and the `return()` statement are studied. Anonymous functions are discussed.

Standard Libraries. The `stats`, `graphics`, `utils`, and `datasets` libraries are studied. The `library()` function is examined. The `install.packages()` function is discussed.

Modularisation. Breaking code into modular parts is studied. The `source()` function is examined. Package structure is discussed.

Classes and Objects. Object-oriented programming in R is studied. The S3, S4, and R6 systems are examined. Classes, objects, and methods are discussed.

The Concept of a Class. Inheritance in R's object-oriented systems is studied. Overriding methods in S3 and S4 is examined. Extending methods in R6 is discussed.

Working with Graphical Modules. The `plot()`, `hist()`, and `barplot()` functions are studied. The `ggplot2`, `lattice`, and `plotly` packages are examined. Customising and saving graphics are discussed.

Data Frames and Data Structures. Data frames are created using `data.frame()` and `tibble`. Indexing, filtering, and sorting of data frames are studied. The factor data type is discussed.

Importing and Exporting Data. Data is imported and exported in CSV, Excel, TXT, and JSON formats. The `read.csv()` function is studied. The `readr`, `readxl`, and `jsonlite` packages are discussed.

Transformation with Tidyverse and dplyr. Tidy data principles are studied. The `dplyr` verbs `filter`, `select`, `mutate`, `summarise`, and `group_by` are examined. The `tidyr` package and the pipe operator are discussed.

Visualisation with ggplot2. The grammar of graphics, including `aes` and `geom`, is studied. Scatter plots, bar charts, histograms, and box plots are examined. Faceting and themes are discussed.

	Statistical Analysis and Linear Regression. Descriptive statistics and correlation are studied. The t-test and chi-square test are examined. Regression and model evaluation using <code>lm()</code> are discussed. Reproducible Reporting with R Markdown. R Markdown and knitr are studied. Combining text, code, and results is examined. Generating HTML, PDF, and Word reports is discussed. Practical lessons Installing and Configuring the R Environment. The R and RStudio environments are installed and configured. Work with the console and scripts is practiced. Files are saved within the RStudio environment. Working with Variables. Values of various types are stored in variables. Variable outputs are displayed. Stored values are modified. Data Types and Detection. Data types are analyzed using the <code>class()</code> function. The <code>typeof()</code> function is applied. The <code>is.numeric()</code> function is used to detect numeric data. Working with Operators. Arithmetic operations are performed. Logical operations are performed. Expression results are analyzed. Conditional Operators. Algorithms are created using if statements. The else statement is applied. The <code>ifelse()</code> function is used for vectorized conditions. Loop Operators. Sequences are generated using the for loop. Sequences are generated using the while loop. Sums and factorials are computed using the repeat loop. Working with Strings. Text is processed using the <code>paste()</code> function. The <code>substr()</code> and <code>nchar()</code> functions are applied. The <code>grep()</code> function is used for pattern matching. Working with Lists. Lists are created. List elements are accessed. List elements are modified. Vectors and Statistical Analysis. Basic statistics are computed using the <code>c()</code> function. The <code>mean()</code> function is applied. The <code>sd()</code> and <code>summary()</code> functions are used.
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	User Functions. Parametric functions are written using function(). These functions are called with different arguments. Function outputs are analyzed. Standard Libraries. The dplyr package is used for data manipulation. The ggplot2 package is used for visualization. The tidyr package is used for data tidying. Modularisation. Code is split into separate files. These files are called using the source() function. Modular code organization is practiced. Object-Oriented Programming. Objects are created using the S3 class system. Objects are created using the S4 class system. Objects are created using the R6 class system. Inheritance. Interclass relationships are examined. Inheritance is implemented using R6Class. Class hierarchies are analyzed. Visualisation. Plots are created using the plot() function. Plots are created using the barplot() and hist() functions. Plots are created using the ggplot() function. Data Frames and Import/Export. Data is read from CSV files. Data is read from Excel files. Data is saved and exported in JSON format. dplyr/tidyr and R Markdown. Data is cleaned and transformed using dplyr and tidyr. An R Markdown report is prepared. The final report combines code, results, and text. Self-study History and Development Stages of the R Programming Language. Getting Started in the RStudio Environment: Files, Console, and Scripts. Variables and Their Types: Naming and Assigning Values. Basic Data Types: Numeric, Character, Logical, Factor. Vectors in R: Creation, Indexing, and Manipulation. Data Transformation Using the dplyr Package. Converting to 'Long' and 'Wide' Formats with tidyr. Basic Statistical Analyses: Mean, Median, Dispersion. Calculating Correlation and Covariance.
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	Building and Evaluating Linear Regression Models. Data Clustering (K-Means). Random Numbers and Probability Distributions. T-Test, Chi-Square, and Other Statistical Tests. Installing and Updating Packages (install.packages, update). Data Cleaning and Filtering (na.omit, is.na, filter). Working with R Markdown Documents and Preparing Reports. Working with APIs in R: JSON and RESTful Services. Operations on Matrices and Arrays. Working with the List Data Structure. The Concept of a Data Frame and the data.frame() Function. Importing and Exporting Data (CSV, Excel, TXT, JSON). Conditional Statements: if, else, ifelse. Looping Statements: for, while, repeat. Writing and Calling User Functions. Recursive Functions in R. Working with Strings in R (stringr, substr, nchar). Factor Types and Working with Them. apply(), lapply(), sapply(): Functional Programming. Plotting in R: plot(), hist(), boxplot(). Visualization Using the ggplot2 Library.
Teaching Methods	Knowledge of the syntax, data types and structures (vector, list, data frame, factor) of the R programming language is developed. Skills in control statements (conditional, looping) and writing user functions are developed. Methods for data import/export, cleaning and transformation (tidyverse: dplyr, tidyr) are taught. Proficiency in applying statistical analysis, correlation and linear regression methods is developed. Skills in data visualisation using ggplot2 and other graphical tools are developed. Competencies in object-oriented programming (S3/S4/R6), code modularisation, and reproducible reporting with R Markdown are developed.
Prerequisites	Programming 1,2 Probability and Statistics Database

Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory performance, practical programming activities, seminar presentations, and the successful completion of independent study assignments. These assignments include writing R scripts, data preprocessing and visualization, statistical analysis, creating functions, working with data frames, and using R packages such as ggplot2, dplyr, and tidyr for data analysis and reporting. The midterm assessment consists of written assignments, short in-class quizzes, practical programming tasks, and the evaluation of independent study projects. Typical projects include developing statistical data analysis applications, creating interactive data visualizations, performing regression and classification analyses, implementing data mining techniques, and preparing presentations on real-world applications of the R programming language in business, healthcare, finance, and scientific research. The final assessment is a written examination consisting of theoretical questions, programming exercises, statistical problem-solving tasks, and data analysis case studies designed to assess students' understanding of the R programming language, statistical computing, data visualization, data manipulation, and analytical methods. The examination also evaluates students' ability to apply R programming techniques to solve practical data analysis problems and develop reproducible analytical workflows. Successful completion of all laboratory work, programming assignments, and independent study projects is required as an essential component of the module assessment.
Recommended Literature	1. Abduvaliyev I.A., Turg'unov S.I. Statistical Analysis and Visualisation: R Programming Fundamentals. — Tashkent: TATU, 2022.

	2. Kabirov A.K., Umarov I. Practical Statistics and the R Programming Language. — Tashkent: Iqtisodiyot, 2023. 3. Wickham H., Çetinkaya-Rundel M., Grolemund G. R for Data Science. 2nd ed. — O'Reilly, 2023. 4. Davies T.M. The Book of R: A First Course in Programming and Statistics. — No Starch Press, 2016. 5. Matloff N. The Art of R Programming. — No Starch Press, 2011. 6. Abdurahmonov M.M. Analysis of Economic Indicators in the R Programming Language. — Samarkand, 2021. 7. Irizarry R.A. Introduction to Data Science: Data Analysis and Prediction Algorithms with R. — 2024 (open book). 8. Maindonald J., Braun W.J. Data Analysis and Graphics Using R. — Cambridge University Press, 2010.
Responsible Lecturer	Dilshod Abdualimov, teacher

FUNDAMENTALS OF THE DJANGO FRAMEWORK

Field	Description
Module Title	FUNDAMENTALS OF THE DJANGO FRAMEWORK
Module Code	DFA305
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 74 hours (42 hours Lecture, 32 hours Practical lessons) Self-Assignments: 76 hours
Frequency/Duration	VI semester
Module objective	The objective of the Module Fundamentals of the Django Framework is to develop modern programming skills through the Django framework for creating robust, secure, and scalable web applications. Students gain theoretical knowledge and practical skills in developing dynamic web applications using Django architecture. Database

	management within Django is studied. User authentication mechanisms are examined. Project deployment procedures are discussed. Students also gain proficiency in building complete, production-ready web applications using the Django framework.
Module Learning Outcomes	Knowledge: LO1. Understand core concepts of Django framework: Model, View, Template, and URL dispatcher components. LO2. Know the MTV (Model-Template-View) architecture of Django applications. LO3. Understand project and application creation process in Django. LO4. Know the purpose of key Django files: settings.py, urls.py, views.py, models.py, admin.py. LO5. Understand ORM (Object-Relational Mapping) concept and database work in Django. LO6. Know templates and static files role in creating user interface. Skills: LO7. Configure admin panel, register models, perform data entry in Django. LO8. Communicate interactively with users through Django Forms. LO9. Understand and implement authentication and authorization system (login, logout, permissions). LO10. Create and debug code using AI tools responsibly. Competencies: LO11. Build complete web applications using Django framework. LO12. Use generative AI tools responsibly in scientific and professional activities. LO13. Independently explore new AI platforms and implement them in professional practice.
Content	Lectures Introduction to the Django Framework. The Django framework and its role in web development are defined. Django's basic architecture and working principles are examined. Django is compared with other frameworks, and its historical development and current usage are discussed.

	Creating a Django Project and Exploring Structure. The project creation process is studied. The project directory structure is examined. Configuration files, development server setup, and environment preparation are discussed. Creating a Django Application. The App and Project concepts are studied. The application creation procedure is examined. Registering and structuring applications for scalable projects is discussed. Django Settings (settings.py) and Important Parameters. The structure and purpose of the settings.py file are studied. Configurable parameters, database configuration, and security settings are examined. Static and media file configuration is discussed. Django URL Routing (urls.py) Basics. The URL dispatcher and routing mechanism are studied. Pattern matching and the path and re_path functions are examined. Connecting views through URLs, using URL namespaces, and include functions are discussed. Django View Functions and Their Tasks. The concept and functionality of views are studied. Function-based view creation and request processing are examined. GET and POST handling, redirects, and response types are discussed. Django Template System and HTML Integration. The purpose and structure of the template system are studied. Variables, tags, template inheritance, and the block system are examined. Template filters, custom tags, and the relationship between views and templates are discussed. Working with Static Files (CSS, JavaScript, Images). Static file organization and serving are studied. CSS and JavaScript integration, and image handling and optimization, are examined. Static file collection for production and best practices are discussed. Creating a Model and Database Connection. Model definition in Django is studied. Field types, constraints, and model relationships, including one-to-one, one-to-many, and many-to-many, are
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examined. Model Meta options and database schema creation are discussed.

Django ORM (Object-Relational Mapping) Basics. The ORM concept and its benefits are studied. Query creation without SQL, and model methods and QuerySet operations, are examined. Filtering, limiting results, aggregation, and annotation are discussed.

Using the Django Admin Panel. Admin interface setup and its purpose are studied. Model registration and customizing the admin display are examined. Admin permissions, user management, admin actions, and filters are discussed.

Django Forms: Simple and ModelForms. Form creation and field definition are studied. Form validation and cleaning, and ModelForm advantages, are examined. Form rendering in templates and form error handling are discussed.

Performing CRUD Operations (Create, Read, Update, Delete). Create and read operations are studied. Update and delete operations are examined. Transaction management is discussed.

User Authentication and Registration. The authentication system is studied. The user model, authentication backends, and registration form are examined. Password hashing, security, and email verification systems are discussed.

User Rights Management (login, logout, permissions). Login and logout functionality are studied. Session management and permission and group systems are examined. Decorator-based access control and custom permission creation are discussed.

Django Class-Based Views (CBV) Basics. The advantages of class-based views over function-based views are studied. Generic views, detail views, and list views are examined. Mixins for extending functionality and template name specification are discussed.

Introduction to Django REST Framework (DRF). REST API concepts and principles are studied. Serializers, APIView, and ViewSets are examined. Request and response handling and token authentication are discussed.

	Creating an API and Working with Serializers. API endpoint creation is studied. Serializer definition, validation, and many-to-many serialization are examined. Nested serializers and custom serializer methods are discussed. Uploading Files and Images in Django (Media Files). Media file handling is studied. File fields in models and file upload forms are examined. Image processing, storage, and the differences between static and media files are discussed. Basics of Deploying Django to Heroku Platform. Heroku platform setup is studied. Procfile creation and environment variables are examined. Database configuration on Heroku, logging, and monitoring are discussed. Preparing and Deploying Django to Production Environment. The production checklist and requirements are studied. Security settings for production, and static and media file handling, are examined. Load balancing, scaling, and performance optimization are discussed. Practical lessons Creating and Launching a Django Project. Django is installed. A project is created using django-admin. The directory structure is examined, and the development server is run and tested. Creating and Registering Django Application. An application is created using the startapp command. Application files and their purpose are examined. The application is registered in settings and integrated into the project. Exploring and Configuring Django Settings File. The settings file is examined. Database configuration and installed apps setup are practiced. Middleware configuration and static and media directories are configured. Working with URLs: Creating Routes. URL patterns are created. Views are mapped to URLs. Path converters, regular expressions, URL naming, and URL reversal in templates are practiced. Creating Simple Pages with View Functions. Function-based views are implemented. Request and response handling and template usage are practiced.
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	Context data passing and dynamic content generation are performed. Generating HTML Pages Using Django Templates. Templates are created and organized. Template tags, filters, and variable interpolation are practiced. Conditional statements and loops are used within templates. Creating Page Design with Static Files and Base Template. A base template is set up. Template inheritance and CSS/JavaScript integration are practiced. Static file loading and design consistency are ensured. Creating a Model and Database Migration. Model fields and relationships are defined. Migrations are created and executed. The database schema is verified and model updates are applied. Using Django Admin Panel and Model Registration. The admin site is accessed. Models are registered in the admin panel. The admin interface, list display, and search and filter configuration are customized. CRUD Operations via Django ORM. Objects are created using Django ORM. Objects are retrieved and queried. Object attributes are updated, objects are deleted, and batch operations are performed. Creating Simple Forms and User Input. Form classes and field types are defined. Forms are rendered in templates. Form submission handling and error display and validation are implemented. Working with Forms Using ModelForm. ModelForms are created from models. Fields are selected and customized. Form saving, validation, related field handling, and multi-step forms are practiced. Creating CRUD Functions. List and detail views are implemented. Create and update views with forms are developed. Delete views with confirmation are implemented. User Registration and Login. The user model is used for registration. A registration form and login form are created. User creation, password hashing, and successful login responses are implemented.
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	User Logout and Rights Restriction. Logout functionality is implemented. Login-required decorators and permission checking are applied. Access control and redirection for unauthorized users are implemented. Adding Filters and Search Functionality. Query filtering and conditions are implemented. Search fields and filter combinations are developed. Performance optimization and results pagination are applied. Self-study Django Framework History and Development. Django MVT Architecture Analysis. Database Migration Processes in Django. Django Template System Features. Security Mechanisms in Django Projects. Django Forms Advantages and Challenges. File and Image Handling in Django. Email Functionality in Django. User Roles and Permissions Management. AJAX and JavaScript Integration. Bootstrap and Django Integration. Django REST Framework Fundamentals. Pagination and Search Functionality. Django Deployment Procedures. Learning Platform Creation Project.
Teaching Methods	Interactive lectures with practical examples are conducted. Hands-on practical sessions with coding exercises are held. Project-based learning is applied using real Django projects. Code review, debugging practice, and pair programming sessions are conducted. Independent project development is encouraged. AI tools are used for code generation and debugging, supplemented by online resources and documentation study.
Prerequisites	Python Programming Fundamentals Database Management Systems Fundamentals of Web Technologies
Exams and Assessment	Type of examination (written, oral, project) and its weight

	Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation in practical sessions, coding ability during hands-on work, and the completion of teacher-guided independent work assignments, including Django project components, coding reports, and mini-projects on web application development. The midterm assessment consists of a written test covering theoretical knowledge of the Django framework, MVT architecture, ORM concepts, and authentication systems, together with independent work projects evaluating practical skills in Django application development, model creation, and form handling. The final examination is a written examination with problems requiring students to create Django applications, implement specific features, debug code, and explain architectural decisions. The assessment measures theoretical understanding, coding skills, and the ability to apply the Django framework to practical projects.
Recommended Literature	1. Holovaty A., Kaplan-Moss J. The Definitive Guide to Django. Apress, 2009. 2. Freeman E., Freeman E. Head First Design Patterns. O'Reilly, 2004. 3. Martin R.C. Clean Code. Prentice Hall, 2008. 4. Django Official Documentation. djangoproject.com/documentation
Responsible Lecturer	Ikromov Husniddin Abdivoidovich, Trainee teacher, Department of Computer and Software Engineering

ARDUINO PROGRAMMING LANGUAGE

Field	Description
Module Title	ARDUINO PROGRAMMING LANGUAGE
Module Code	ADT305
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 74 hours (42 hours Lecture, 32 hours Practical lessons)

	Self-Assignments: 76 hours
Frequency/Duration	VI semester
Module objective	The objective of the Module Arduino Programming Language is to develop theoretical knowledge and practical skills in Arduino microcontroller programming. Students learn to develop C/C++ programs in the Arduino IDE. Working with sensors and actuators is studied. Control algorithms are developed. Robotics projects are examined. IoT projects for automated and intelligent systems are developed.
Module Learning Outcomes	Knowledge: LO1. Explain the structure, capabilities, and operating principles of the Arduino platform. LO2. Install, configure, and effectively use Arduino IDE programming environment. LO3. Write programs using Arduino programming language syntax and basic statements. LO4. Work with variables, arrays, functions, and Arduino libraries. LO5. Understand data types and their characteristics in Arduino. LO6. Understand digital and analog input/output port operations. LO7. Receive and process data from various sensors. LO8. Control devices such as LEDs, relays, servo motors, and displays. Skills: LO9. Develop control algorithms using conditional and looping statements. LO10. Assemble, wire, and troubleshoot electronic circuits. LO11. Use data exchange protocols between Arduino and external modules. LO12. Perform testing, optimization, and debugging of Arduino programs. Competencies: LO13. Develop practical Arduino-based projects for IoT and automation systems. LO14. Create modern smart devices and embedded systems using Arduino.

	LO15. Apply Arduino technologies to solve real-world automation and control problems.
Content	Lectures Introduction to Arduino Platform and Capabilities. The Arduino platform is defined, and its role in embedded systems is explained. Its capabilities, advantages, and historical development are examined. Current applications of Arduino in industry and education are discussed. Types and Technical Specifications of Arduino Boards. The Arduino Uno, Mega, Nano, and Pro boards are studied. Their technical specifications, memory, and processing capabilities are compared. Pin configurations and the selection of an appropriate board for projects are discussed. Working with Arduino IDE Programming Environment. Arduino IDE installation and setup are studied. The user interface, sketch creation, and library management are examined. Code compilation, uploading, and the use of the serial monitor for debugging are discussed. Fundamentals of Arduino Programming Language (C/C++ Elements). Arduino programming language is studied as a subset of C/C++. Data types, declarations, operators, and expressions are examined. Program structure, code formatting, and standard Arduino functions are discussed. Variables, Data Types, and Operators. Integer, floating-point, character, and boolean data types are studied. Variable declaration, initialization, scope, and lifetime are examined. Arithmetic, logical, and bitwise operators, along with operator precedence, are discussed. Condition and Loop Operators. If-else conditional statements and switch-case statements are studied. While, do-while, and for loops are examined. Break and continue statements, along with nested conditions and loops, are discussed. Working with Functions and Libraries. Function definition and calling are studied. Parameter passing, return values, and function scope are examined. Arduino standard library functions, external libraries, and custom function creation are discussed.

Digital Input/Output Ports in Arduino. GPIO concepts and digital pin modes are studied. Reading digital inputs with `digitalRead()` and writing digital outputs with `digitalWrite()` are examined. Pull-up and pull-down resistors and debouncing techniques are discussed.

Analog Signals and ADC/DAC Concepts. Analog signal characteristics are studied. Analog-to-digital and digital-to-analog conversion principles are examined. Analog input reading with `analogRead()` and analog output using PWM are discussed.

Controlling LEDs, Buttons, and Displays. LED connection and control are studied through the blink program and its variations. Button interfacing and input reading are examined. LCD and OLED display types and character display programming are discussed.

Working with Timing and Interrupts. The `delay()`, `millis()`, and `micros()` timing functions are studied. Timer interrupts and Interrupt Service Routines (ISR) are examined. External interrupts and time-based programming patterns are discussed.

Basic Principles of Working with Sensors. Sensor types and classifications are studied. Analog and digital sensor outputs are examined. Sensor calibration, signal conditioning, and data reading are discussed.

Programming Temperature, Humidity, and Light Sensors. DHT11 and DHT22 temperature and humidity sensors are studied. LDR light sensors and DS18B20 digital temperature sensors are examined. Sensor data interpretation and multiple sensor integration are discussed.

Servo and Stepper Motor Control. Servo motor principles and PWM signal generation are studied. The Servo library is examined. Stepper motor types, motor drivers, and speed and direction control are discussed.

Working with LCD and OLED Displays. LCD 16x2 display connection and character programming are studied. OLED display technology and its advantages are examined. Graphics, custom characters, and display formatting are discussed.

Serial Communication Technologies (UART, I2C, SPI). UART serial communication is studied. The I2C protocol and its master-slave architecture are examined. SPI communication and protocol selection criteria are discussed.

Wireless Communication Modules (Bluetooth, Wi-Fi). The HC-05 Bluetooth module setup and serial communication are studied. The ESP8266 Wi-Fi module integration is examined. Wireless network communication and remote control are discussed.

Internet of Things (IoT) and Arduino. IoT concepts and architecture are studied. Cloud platform integration with Arduino is examined. Data transmission to cloud services and remote monitoring and control are discussed.

Creating Automation Systems Based on Arduino. Home automation concepts are studied. Sensor-based automation logic and multi-sensor system integration are examined. Event-driven programming and practical automation scenarios are discussed.

Using Arduino in Robotics Projects. Robot platform design and construction are studied. Motor control and sensor-based navigation are examined. Obstacle detection, avoidance, and robot communication are discussed.

Development and Defense of Practical Arduino-Based Project. Project planning, design, and component selection are studied. Circuit building, testing, and code development are examined. Project documentation, presentation, and demonstration are discussed.

Practical lessons

Installing Arduino IDE and Creating First Program (Blink). Arduino IDE is installed, and the board and port are selected. A first sketch is created, compiled, and uploaded. LED blinking is observed via the serial monitor.

Working with Digital Input/Output Ports. Digital output is configured using `digitalWrite()`. Digital input is configured using `digitalRead()`. LEDs and buttons are connected, and button states are read.

Creating Control Programs Using LEDs and Buttons. A button-controlled LED program is created.

Multiple LED sequences and patterns are implemented. Debouncing and user interaction programming are practiced.

Using Conditional and Iteration Operators. If-else and switch-case statements are practiced. While loop implementation is performed. For loop variations and loop termination conditions are practiced.

Controlling LED Brightness Using PWM Signal. PWM principles are reviewed. Brightness control is implemented using `analogWrite()`. Potentiometer input and LED fade effects are practiced.

Working with Analog Input Ports and Potentiometers. Analog pins are configured. Analog values are read using `analogRead()` and output to the serial monitor. Potentiometer voltage division and sensor value interpretation are practiced.

Creating and Using Functions and Libraries. Custom functions are defined with parameters and return values. Libraries are installed and used. Code is organized using functions.

Working with DHT11/DHT22 Sensors. DHT11/DHT22 sensors are connected and configured. The corresponding library is installed and used. Temperature and humidity data are read, interpreted, and displayed on the serial monitor.

Creating Automatic Control System Using Light Sensor. An LDR sensor is connected. Analog values are read and mapped. Threshold-based automatic light control is implemented and tested.

Displaying Data on LCD 16x2 Display. The LCD 16x2 display is connected and initialized. Characters and numbers are displayed and formatted. Real-time data display is implemented.

Controlling Servo Motor Using Arduino. The Servo library is installed. The servo motor is connected and powered. Angle control commands and smooth servo movement are practiced.

Working with Ultrasonic Distance Sensor (HC-SR04). The HC-SR04 ultrasonic sensor is connected via its trigger and echo pins. Distance measurements are calculated and output to the serial monitor. Object detection and range-finding applications are implemented.

	Data Exchange via Bluetooth Module (HC-05). The HC-05 Bluetooth module is connected. Serial communication is configured for data transmission and reception. Remote control is implemented via a mobile app. Creating IoT Project Using Wi-Fi Module (ESP8266). The ESP8266 module is set up and configured. A Wi-Fi connection is established, and HTTP requests are sent. Data is uploaded to a cloud service in real time. Establishing Communication Between Arduino and Mobile Applications. Bluetooth serial communication is reviewed. A mobile app is installed and used for command transmission. Data is monitored and controlled from the mobile device. Developing and Testing Final Mini-Project. Project requirements are defined, and components are assembled. Code is developed for the final mini-project. System integration, testing, documentation, and demonstration are completed. Self-study History of Arduino Platform and Development Stages. Technical Specifications of Arduino Boards. Data Types and Their Characteristics in Arduino. Arduino Libraries and How to Use Them. PWM Technology and Its Practical Application. Designing an LED Lighting System Based on Arduino. Operating Principles of Temperature and Humidity Sensors. Analysis of Light and Motion Sensors. Applications of Servo and Stepper Motors. Features and Differences of LCD and OLED Displays. Learning UART, I2C, and SPI Protocols. Bluetooth Technology and Integration with Arduino. Operating Principle and Application of Wi-Fi Modules. Development Prospects of IoT Technologies. Creating Arduino-Based Smart Home Systems.
Teaching Methods	Interactive lectures with practical demonstrations are conducted. Hands-on practical sessions with circuit building are held. Project-based learning is applied using real Arduino projects. Troubleshooting and debugging practice, along with pair programming and

	peer review, are conducted. Independent project development is encouraged. Simulators, simulation software, online resources, and official documentation study are used.
Prerequisites	Fundamentals of Programming and Algorithms Computer Architecture and Microprocessor Fundamentals Fundamentals of Electronics and Circuit Design
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation in practical sessions, circuit building and programming ability during hands-on work, and the completion of teacher-guided independent work assignments, including Arduino project components, circuit diagrams, code reports, and mini-projects on sensor integration and control systems. The midterm assessment consists of a written test covering theoretical knowledge of the Arduino platform, C/C++ programming basics, sensor operation, and communication protocols, together with independent work projects evaluating practical skills in Arduino program development, circuit design, and sensor interfacing. The final examination is a written examination with problems requiring students to design Arduino systems, write control programs, debug circuits, and explain design decisions. The assessment measures theoretical understanding, coding skills, and the ability to apply the Arduino platform to practical embedded systems.
Recommended Literature	1. Massimo Banzi. Getting Started with Arduino. O'Reilly Media, 2011. 2. Michael Margolis. Arduino Cookbook: Recipes to Begin, Expand, and Enhance Your Projects. O'Reilly Media, 2012. 3. Simon Monk. Programming Arduino: Getting Started with Sketches. McGraw Hill, 2011.

	4. Brian Evans. Beginning Arduino Programming. Apress, 2011.
Responsible Lecturer	Ikromov Husniddin Abdivoidovich, Department of Computer and Software Engineering, Termez State University

MOBILE PROGRAMMING LANGUAGE (FLUTTER)

Field	Description
Module Title	MOBILE PROGRAMMING LANGUAGE (FLUTTER)
Module Code	MDT305
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 60 hours (34 hours Lecture, 26 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	VI semester
Module objective	The objective of the Module Mobile Programming Language (Flutter) is to provide a scientifically grounded understanding of the fundamental principles, tools, and technologies used in modern cross-platform mobile application development. The Module examines the Flutter framework and the Dart programming language. Widget-based user interface development, application architecture, and state management are studied. Navigation, user interaction, data storage, and networking are examined. Integration with mobile device features is also studied. The Module aims to develop students' ability to design, develop, test, debug, and deploy efficient and user-friendly mobile applications for different platforms. The Module fosters students' programming, analytical, and problem-solving skills. It provides the knowledge and practical competencies required for the creation and effective utilization of cross-platform mobile applications using Flutter.
Module Learning Outcomes	Knowledge: LO1. Knows the basic concepts, capabilities, and application areas of the Flutter framework and the Dart programming language.

	LO2. Understands the key stages of the mobile application development process, architecture, and life cycle. LO3. Has knowledge of Flutter widgets, user interface elements, and their design principles. LO4. Knows how to work with data in mobile applications and how to use local and remote databases. Skills: LO5. They will acquire the skills to create and customize mobile applications using Flutter and Dart programming tools. LO6. Can design and implement user interfaces for Android and iOS platforms. LO7. Can apply methods for data entry, processing, storage, and management in mobile applications. LO8. Has the skill to develop mobile applications integrated with REST APIs and other external services. Competencies: LO9. Will have the competence to design and develop mobile applications for various industries using modern mobile development technologies. LO10. Demonstrates the competence to apply modern software development methodologies and standards when creating mobile applications. LO11. Will have the competence to integrate mobile applications with databases, web services, and external information systems. LO12. Acquires professional competencies in participating in team software projects, managing project documentation, and testing software products. LO13. Performs professional activities to ensure the security, reliability, and efficiency of mobile applications and to implement them in practical environments.
Content	Lectures Introduction to the Course of Mobile Application Development and Trends in the Development of Mobile Technologies. The field of mobile application development and its role in modern information technologies are introduced. The history and evolution of mobile devices and operating systems,

along with the main types of mobile applications, are examined. Current trends, including wearable devices, the Internet of Things, cloud technologies, artificial intelligence, and cross-platform development, are discussed. The general stages of mobile application development, from idea and design to deployment and maintenance, are also covered.

Introduction to the Dart Programming Language and Program Structure. The Dart programming language, its history, and its features are introduced. The basic principles of Dart and its role in Flutter application development are examined. The structure of a Dart program, including the main() function, statements, and comments, is discussed. The compilation and execution of Dart programs and the development environment are also covered.

Variables and Data Types in the Dart Programming Language. Variables, constants, and literals in Dart are studied. The declaration, initialization, and modification of variables are examined. The basic data types, including int, double, String, and bool, and the var, final, and const keywords, are discussed. Type inference, type conversion, and the principles of null safety in Dart are also covered.

Operators and Expressions in the Dart Programming Language. The concept and classification of operators in Dart are studied. Arithmetic, comparison, logical, and bitwise operators are examined. Operator precedence, associativity, and the construction of expressions are discussed. The conditional operator and null-aware operators are also covered.

Control Operators (Conditional and Loop Operators) in the Dart Programming Language. The principles of controlling program execution flow are studied. Conditional statements, including if, else, and switch, are examined. Loop structures, including for, while, and do-while, along with break and continue statements, are discussed. Nested control structures and their application to program logic are also covered.

Functions, List, Set, and Map Collections in the Dart Programming Language. Function declaration, parameters, return values, and anonymous functions in Dart are studied. Higher-order functions are examined. The List, Set, and Map collection types are discussed, including adding, modifying, deleting, and searching for elements. Iteration through collections and common collection methods are also covered.

Object-Oriented Programming in the Dart Programming Language (Class and Object). The fundamental concepts of object-oriented programming in Dart are studied. Classes, objects, fields, methods, and constructors are examined. Default, named, and factory constructors, along with encapsulation and access control, are discussed. Instance and static members and their use in designing reusable software components are also covered.

Inheritance, Polymorphism, and Abstraction in the Dart Programming Language. Inheritance and the creation of relationships between parent and child classes in Dart are studied. Method overriding and polymorphism are examined. Abstract classes, abstract methods, and interfaces are discussed. The extends, implements, and with mechanisms are also covered.

Working with Packages, Sharing, and Importing Libraries. The concept of packages and libraries in the Dart ecosystem is studied. The organization of reusable source code into libraries and packages is examined. Importing standard, external, and user-defined libraries, and managing project dependencies, are discussed.

Basics of Working with Sensors and Touch-Enabled Mobile Devices. The hardware capabilities of modern mobile devices are introduced. Touch screens and touch events are examined. Sensor types, including accelerometers, gyroscopes, proximity sensors, and GPS, are discussed. Processing user gestures and accessing device hardware through software interfaces are also covered.

Introduction to the Flutter Framework and Working with the Development Environment. The Flutter framework, its architecture, and its advantages are introduced. Its role in cross-platform mobile application development is examined. Installation and configuration of the Flutter SDK, and working with Android Studio and Visual Studio Code, are discussed. Creating, running, and debugging Flutter projects are also covered.

Flutter App Structure and Working with Widgets. The general structure and architecture of Flutter applications are studied. Widgets are examined as the fundamental building blocks of Flutter user interfaces. Stateless and Stateful widgets, widget trees, and the build() method are discussed. Basic built-in widgets and the composition of complex interfaces are also covered.

Designing Layouts and User Interfaces in Flutter. The fundamental principles of designing user interfaces for mobile applications are studied. Layout organization using Row, Column, Container, and other layout widgets is examined. Alignment, spacing, sizing, and positioning of interface elements are discussed. Responsive and adaptive design principles are also covered.

Working with Text Widgets: Text, ElevatedButton, TextButton, and IconButton. The Text widget is studied for displaying and formatting text. Interactive buttons, including ElevatedButton, TextButton, and IconButton, are examined. Button properties, styles, icons, and callback events are discussed.

Working with Lists: ListView, ListTile, and GridView. The principles of displaying collections of data in Flutter applications are studied. Scrollable lists are created using ListView, and individual items are designed using ListTile. Grid-based layouts are created using GridView. The efficient rendering of static and dynamically generated lists is also discussed.

Navigation and Data Exchange Between Screens in Flutter. The principles of multi-screen application development in Flutter are studied. Navigation between screens using the Navigator mechanism and

named routes is examined. Passing data between screens and returning results are discussed. Managing the navigation stack is also covered.

Form Elements, Data Entry, and Validation in Flutter. Forms and user data entry in Flutter applications are studied. The Form and TextFormField widgets, along with text input controllers, are examined. Form validation rules and error messages are discussed. Processing submitted data and implementing reliable data entry forms are also covered.

Practical lessons

Mobile Application Development Environment and Flutter SDK Installation. The environment for mobile application development is set up. The Flutter SDK, Dart SDK, Android Studio, and Visual Studio Code are installed and configured. The correctness of the development environment is verified using Flutter diagnostic tools. A test project is created and run to resolve configuration issues.

Creating Simple Programs in the Dart Programming Language. Simple Dart programs are created and executed. The main() function, statements, and basic program structure are used. Programs performing simple calculations and displaying results are written. Basic programming skills are developed through simple computational tasks.

Working with Variables and Data Types in Dart. Variables are declared, initialized, and modified in Dart programs. Basic data types, including integers, strings, and Boolean values, are used. The var, final, and const keywords are applied. Type conversion and null safety principles are practiced.

Using Operators and Control Structures in the Dart Programming Language. Arithmetic, comparison, logical, and conditional operators are applied in Dart programs. Programs are developed using if, else, and switch statements. Repetitive operations are implemented using for, while, and do-while loops. Practical computational problems are solved by combining operators and control structures.

Working with Functions and Collections (List, Set, Map) in Dart. Functions are created and used in Dart programs, including parameters and return values. List, Set, and Map collections are created, modified, and processed. Collection methods and iteration techniques are applied to process groups of data. Reusable program components are developed using functions and collections.

Object-Oriented Programming Basics: Class and Object. Classes and objects are created in Dart programs. Fields, methods, and constructors are defined and initialized. Encapsulation is applied to organize program code using object-oriented principles. Simple class-based applications are developed.

Adding Packages and External Libraries to the Project. Packages and external libraries are used in Dart and Flutter projects. Project dependencies are managed using the project configuration file. External packages are searched for, added, configured, and imported into the project.

Creating a Flutter App and Working with Basic Widgets. A new Flutter project is created and its basic structure is examined. The application is run on an emulator or physical device. The MaterialApp, Scaffold, AppBar, Container, and Text widgets are used. The widget tree and the build() method are applied to create a simple functional application.

Designing a User Interface in Flutter Using Layout Elements. Mobile application interfaces are designed using Flutter layout widgets. Row, Column, Container, Center, Expanded, and Flexible elements are applied. The position, size, alignment, and spacing of interface components are controlled. A complete responsive user interface is developed based on a given design.

Working with Text, Icons, and Buttons. Text is displayed and formatted using the Text widget. Icons are added and configured using the Icon and IconButton widgets. Interactive interfaces are created using ElevatedButton and TextButton. Styles, callbacks, and event handling are applied to develop an interactive interface.

	Displaying Data Using ListView, ListTile, and GridView. Structured data is displayed using list and grid components in Flutter. Scrollable lists are created using ListView, and individual items are designed using ListTile. Grid-based interfaces are created using GridView. List and grid elements are generated dynamically from data collections. Navigation and Passing Data Between Screens in Flutter. Multi-screen Flutter applications are created. Navigation between screens is implemented using the Navigator and routes. Data is passed from one screen to another, and results are returned between screens. Named routes and navigation stacks are used in a multi-page application. Form Elements and User Data Validation. Forms are created for entering and processing user data in Flutter. The Form and TextFormField widgets, along with input controllers, are used. Data validation rules are implemented, and error messages are displayed. A complete data-entry form, such as a registration or login form, is developed. Self-study Preparing for Practical Sessions and Completing Assignments. Independent Study of Literature and Educational Materials on Mobile Programming. Performing Practical Tasks for Creating and Improving Programs. Study of Scientific Articles, Analytical Materials, and Sources. Developing Applications for Mobile Devices. Preparation of Presentations, Reports, and Independent Work.
Teaching Methods	The course is delivered through a combination of theoretical instruction, interactive lectures, practical programming activities, problem-based and project-based learning, demonstration, live coding, and independent study. Interactive lectures explain the fundamental concepts of Dart programming and Flutter architecture, including widgets, layouts, navigation, and user interface design. Demonstration and live coding are used to show the step-by-step

	process of creating, debugging, and testing Dart programs and Flutter applications. Practical and laboratory activities develop programming skills through implementing variables, data types, functions, collections, and object-oriented concepts, as well as building user interfaces and navigation. Collaborative and pair programming activities promote teamwork, code review, and joint problem-solving. Continuous formative assessment is conducted through practical assignments, programming exercises, and project presentations.
Prerequisites	Programming Fundamentals Object-Oriented Programming Data Structures and Algorithms Computer Organization Operating Systems
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, practical performance, computer-based laboratory activities, seminar presentations, and the successful completion of independent study assignments, including Dart programming exercises, Flutter widget and user interface development, layout design, navigation between screens, and form creation with data validation. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as a Dart programming project, a Flutter user interface design project, and a form-based application with user input validation. The final assessment is a written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of mobile programming principles, the Dart programming language, Flutter framework architecture, widgets, navigation, forms, and mobile application development, as well as their ability to apply theoretical principles to practical problems. Successful completion of all independent study

	assignments is required as an essential component of the module assessment.
Recommended Literature	1. O. K. Qodirov, Z. M. Murodov. Android Operating System Mobile Application Development. Tashkent University of Information Technologies Publishing House. 2. A. E. G'ulomov. Fundamentals of Mobile Application Development. Fan and Technology. 3. Sh. M. Usmonov. Mobile Technologies and Programming Fundamentals. TATU Publishing House. 4. Jakob Iversen, Michael Eierman. Learning Mobile App Development. New York, 2014, 464 pp. 5. Shane Conder, Lauren Darcey. Android™ Wireless Application Development. USA, 2011, 138 pp. 6. L.V. Priskaya. Developing Mobile Applications in the Android Studio Environment: A Textbook. Southern Federal University, 2019. — 123 pp.
Responsible Lecturer	Tursunov Farxod Bakhodir o'g'li

COMPUTER GRAPHICS

Field	Description
Module Title	COMPUTER GRAPHICS
Module Code	KGF305
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 60 hours (34 hours Lecture, 26 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	VI semester
Module objective	The objective of the Module Computer Graphics is to develop students' theoretical knowledge and practical skills in the principles of two- and three-dimensional computer graphics. Image formation, processing, and visualization are studied. Geometric transformations in the plane and in space are examined. Rasterization and rendering algorithms are studied. Lighting, shading, and texturing techniques are examined.

	Modern graphics programming technologies, including OpenGL and GLSL, are studied.
Module Learning Outcomes	Knowledge: LO1. Can describe the fundamentals of raster, vector, and fractal graphics, their types, and areas of application. LO2. Explain color models (RGB, CMYK, HSV, CIE XYZ/Lab) and the fundamentals of colorimetry. LO3. Can explain geometric transformations in the plane and in space, as well as the principles of homogeneous coordinates. LO4. Can explain the theoretical foundations of projection methods, spline curves and surfaces, and polygonal meshes. LO5. Can describe rasterization, clipping, and hidden-line and hidden-surface removal algorithms. LO6. Can explain lighting and shading models, rendering techniques, and the fundamentals of OpenGL/GLSL. Skills: LO7. Works with raster and vector graphic formats and represents and converts colors in various models. LO8. Programmatically performs geometric transformations and projections in the plane and in space. LO9. Constructs fractal images, spline curves, and surfaces using control points. LO10. Programs rasterization, filling, and clipping algorithms. LO11. Applies techniques for removing hidden surfaces, lighting, and shading. LO12. Creates three-dimensional scenes and rendering effects using OpenGL and GLSL shaders. Competencies: LO13. Effectively uses modern graphical software tools and libraries in professional activities. LO14. Models three-dimensional scenes and implements rendering processes. LO15. Applies graphic algorithms in practical projects and software systems.

	LO16. Independently studies modern graphics APIs (Vulkan, DirectX) and real-time graphics technologies.
Content	Lectures Introduction to the Field of Computer Graphics: Concept, Purpose, Application Areas, Development History, and Technical Tools. The fundamental concepts and objectives of computer graphics are introduced. The major application areas and historical development of the field are examined. The technical tools used for creating, processing, and visualizing graphical information are discussed. Raster Graphics and Image Formats. Pixel, Resolution, Color Depth, Encoding and Compression Methods (BMP, JPEG, PNG, GIF, TIFF). The structure of raster images and the concepts of pixels, resolution, and color depth are studied. Image encoding and compression methods are examined. The characteristics and applications of the BMP, JPEG, PNG, GIF, and TIFF formats are discussed. Vector Graphics. Vector Objects, Their Mathematical Representation, Formats (SVG, EPS, AI, CDR), and Rasterization. The principles of vector graphics and the mathematical representation of vector objects are examined. The main vector formats, including SVG, EPS, AI, and CDR, are studied. The process of converting vector graphics into raster images is discussed. Fractal Graphics. Fractal Geometry, Self-Similarity, Mandelbrot and Julia Sets, L-Systems. Fractal geometry and the principle of self-similarity are introduced. The mathematical and graphical properties of the Mandelbrot and Julia sets are studied. The use of L-systems for generating complex graphical structures is discussed. Fundamentals of Color Models and Colorimetry. Additive and Subtractive Models (RGB, CMYK), HSV/HSL, CIE XYZ and Lab, Color Management. The fundamental principles of color models and colorimetry are studied. Additive and subtractive color models, including RGB and CMYK, and the HSV/HSL, CIE XYZ, and Lab color spaces are

examined. Color management techniques are discussed.

Geometric Transformations in the Plane. Affine Coordinates, Translation, Scaling, Rotation, Reflection, Composite Transformations. The mathematical foundations of two-dimensional geometric transformations are studied. Translation, scaling, rotation, and reflection using affine coordinates are examined. Composite transformations are discussed.

Geometric Transformations in Space. Three-Dimensional Coordinates, Transformations in Space, and Inverse Transformations. Geometric transformations in three-dimensional space are examined. Three-dimensional coordinate systems and spatial translation, scaling, and rotation are studied. Inverse transformations for modeling and manipulating 3D objects are discussed.

Projections and Viewing Parameters. Parallel (Orthographic, Axonometric, Oblique) and Perspective Projections, the Viewing Pyramid. Methods for representing three-dimensional objects on a two-dimensional plane are studied. Parallel, orthographic, axonometric, oblique, and perspective projections are examined. Viewing parameters and the viewing pyramid are discussed.

Spline Curves. Hermite, Bézier, B-spline, and NURBS Curves, Control Points and Smoothness Conditions. Spline-based methods for modeling complex curves are examined. Hermite, Bézier, B-spline, and NURBS curves and their control points are studied. Interpolation, approximation principles, and smoothness conditions are discussed.

Spline Surfaces and Polygonal Meshes. Bézier and B-spline Surfaces, the Concept of a Mesh, Mesh Storage, and Tessellation. The mathematical and computational representation of spline surfaces and polygonal meshes is studied. Bézier and B-spline surfaces and mesh structures are examined. Mesh storage methods and tessellation techniques are discussed.

Rasterization Algorithms. Rasterization of Line Segments (Bresenham, DDA) and Circles, Fill

Algorithms, Anti-Aliasing. Algorithms for converting geometric primitives into raster images are studied. The Bresenham and DDA algorithms for line and circle rasterization are examined. Filling algorithms and anti-aliasing techniques are discussed.

Clipping and Removal of Hidden Lines/Surfaces. Cohen–Sutherland, Sutherland–Hodgman, Z-buffer, Back-Face Culling. Methods for removing parts of graphical objects outside the viewing area are examined. The Cohen-Sutherland and Sutherland-Hodgman clipping algorithms are studied. The Z-buffer technique and back-face culling for eliminating hidden lines and surfaces are discussed.

Lighting and Shading Models. Light Sources, Phong Lighting Model, Flat/Gouraud/Phong Shading. The principles of lighting and shading in computer graphics are introduced. Different types of light sources and the Phong lighting model are examined. Flat, Gouraud, and Phong shading techniques are discussed.

Texturing and Creating Realistic Images. Texture Mapping, UV Coordinates, Mip-Mapping, Bump and Shadow Maps. Techniques for applying textures to three-dimensional objects and creating realistic images are studied. Texture mapping and UV coordinates are examined. Mip-mapping, bump mapping, and shadow mapping methods are discussed.

Rendering Techniques: Ray Tracing and Global Illumination. Ray Tracing, Reflection and Refraction, PBR Concepts. Modern rendering techniques for producing realistic computer-generated images are examined. Ray tracing and the reflection and refraction of light are studied. Global illumination and the fundamental concepts of Physically Based Rendering are discussed.

Fundamentals of the OpenGL Graphics Library. OpenGL Architecture, the Graphics Pipeline, Primitives, Buffers, and Transformations. The fundamental concepts and architecture of the OpenGL graphics library are introduced. The graphics pipeline and graphical primitives are examined.

Buffers and geometric transformations implemented using OpenGL are discussed.

Modern Graphics Programming and Shaders. Vertex and Fragment Shaders, the GLSL Language, GPUs, and Modern Graphics APIs (Vulkan, DirectX). Modern graphics programming technologies and shader-based rendering are examined. Vertex and fragment shaders and the GLSL programming language are studied. GPU computing and modern graphics APIs, such as Vulkan and DirectX, are discussed.

Practical lessons

Working with Raster and Vector Graphics Formats and Editing Images. Raster and vector graphics formats are worked with. Basic image editing techniques are applied. Images are opened, modified, transformed, and saved in different graphical formats.

Representing and Converting Colors in the RGB, CMYK, and HSV Color Models. Colors are represented using the RGB, CMYK, and HSV color models. Color values are converted between different models. The differences between additive and subtractive color systems are analyzed.

Programmatically Constructing Fractal Images (Mandelbrot and Julia Sets). Fractal images are generated programmatically. Algorithms for constructing Mandelbrot and Julia sets are implemented. The influence of mathematical parameters on the resulting graphical patterns is investigated.

Programming Geometric Transformations (Translation, Scaling, Rotation) in the Plane. Two-dimensional geometric transformations are implemented. Translation, scaling, and rotation operations are programmed. These operations are applied to graphical objects using appropriate mathematical transformations.

Performing Transformations in Space Using Homogeneous Coordinates and Matrices. Three-dimensional geometric transformations using homogeneous coordinates are implemented. Transformation matrices are applied. Spatial

translation, scaling, and rotation operations for 3D objects are practiced.

Constructing Parallel and Perspective Projections and Adjusting Viewing Parameters. Parallel and perspective projections for three-dimensional objects are constructed. Viewing parameters are adjusted. The effects of different projection methods on the representation of 3D scenes are analyzed.

Constructing Bezier and B-spline Curves by Control Points. Bézier and B-spline curves are constructed using control points. The position of control points is varied. The effects on the shape, smoothness, and geometric properties of the resulting curves are investigated.

Rasterizing Line Segments and Circles Using Bresenham and DDA Algorithms. Rasterization algorithms for line segments and circles are implemented. The Bresenham and DDA algorithms are applied. Their effectiveness in converting geometric primitives into raster images is analyzed.

Programming Polygon Filling and Clipping Algorithms. Algorithms for filling polygonal regions are implemented. Clipping algorithms for graphical objects are developed. Programs for processing polygons and limiting graphical elements to a specified viewing region are created.

Removing Invisible Surfaces Using Z-buffer and Back-Face Culling Methods. Hidden-surface removal in three-dimensional computer graphics is studied. The Z-buffer method is implemented. The back-face culling method is analyzed to determine the visible surfaces of 3D objects.

Application of the Phong Lighting Model and Gouraud/Phong Shading Techniques. Lighting and shading models are applied to graphical objects. The Phong lighting model is implemented. Flat, Gouraud, and Phong shading techniques are applied to create realistic lighting and surface effects.

Creating Three-Dimensional Scenes and Geometric Objects Using OpenGL. Three-dimensional geometric objects and graphical scenes are created using the OpenGL library. Graphical primitives and transformations are applied. Viewing

	parameters and the basic OpenGL graphics pipeline are used. Implementing Simple Rendering Effects Using Vertex and Fragment Shaders (GLSL). Basic rendering effects are implemented using vertex and fragment shaders. The GLSL language is used to control the processing of geometric data. Vertices, fragments, and visual properties of graphical objects are controlled through shader programming. Self-study Theoretical Foundations of Computer Graphics and Modern Development Trends. Comparative Analysis of Raster and Vector Graphics and Image Formats. Color Models and Colorimetry: Comparative Analysis of RGB, CMYK, HSV, and Lab Color Spaces. Fractal Geometry and Algorithms for Constructing Fractals. Practical Application of Geometric Transformations in the Plane and Space. Configuration of Projections and Virtual Camera Systems Methodology. The Role of Spline Curves and Surfaces in Design and Modeling. Polygonal Meshes and Methods for Tessellating Surfaces. Comparative Analysis of Rasterization and Anti-Aliasing Algorithms. Algorithms for Removing Invisible Lines and Surfaces (Z-buffer, Back-Face Culling). Lighting and Shading Models: Phong, Gouraud, and Flat Shading. Texturing, Bump, and Shadow Mapping Technologies. Ray Tracing and Physically Based Rendering (PBR). Graphics Programming with OpenGL/GLSL and Using Shaders. Modern Graphics APIs and Real-Time Graphics Trends.
Teaching Methods	The module is delivered through interactive lectures, problem-based and inquiry-based learning, practical exercises, computer simulations, and project-based

	activities. Interactive lectures introduce the fundamental concepts of computer graphics, including raster and vector graphics, color models, geometric transformations, and rendering. Problem-based learning enables students to analyze graphical problems and apply mathematical and algorithmic methods. Practical exercises focus on image processing, geometric transformations, rasterization, lighting and shading, texturing, and 3D scene creation using OpenGL. Computer simulations and graphics programming environments are used to visualize graphical processes and analyze algorithm results. Project-based learning and independent study develop students' analytical, creative, and technical skills through individual and group assignments, presentations, and mini-projects.
Prerequisites	Fundamentals of Informatics Programming Fundamentals Digital Logic and Electronics Operating Systems
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, practical performance, computer-based laboratory activities, seminar presentations, and the successful completion of independent study assignments, including raster and vector graphics editing, color model conversion and analysis, fractal image generation, geometric transformations, projection and virtual camera configuration, spline curve and surface construction, rasterization and anti-aliasing algorithms, hidden-surface removal, lighting and shading models, texturing and mapping techniques, and OpenGL/GLSL graphics programming. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as a comparative study of raster and vector graphics, an analysis of color models, a fractal geometry research project, a

	study of geometric transformations and projections, an analysis of spline curves and surfaces, a comparative study of rasterization and anti-aliasing algorithms, a study of Z-buffer and back-face culling methods, an analysis of lighting and shading models, and a research project on ray tracing, PBR, OpenGL/GLSL, or modern graphics APIs. The final assessment is a written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of computer graphics fundamentals, geometric transformations, rasterization, hidden-surface removal, lighting and shading, texturing, rendering techniques, and OpenGL/GLSL programming, as well as their ability to apply theoretical principles to practical and analytical problems. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Marschner S., Shirley P. Fundamentals of Computer Graphics. 5th Edition. — A K Peters / CRC Press, 2021. — 716 p. 2. Pharr M., Jakob W., Humphreys G. Physically Based Rendering: From Theory to Implementation. 4th Edition. — MIT Press, 2023. — 1280 p. (Full text: https://pbr-book.org/4ed) 3. Eck D. J. Introduction to Computer Graphics. Version 1.4. — Hobart and William Smith Colleges, 2023. — 541 p. (Free PDF: https://math.hws.edu/eck/cs424/downloads/graphics_book.pdf) 4. Nikulin E. A. Computer Graphics: Models and Algorithms. A Textbook. 2nd ed. — St. Petersburg: Lan', 2022. — 708 pp. 5. Romanyuk O. N., Romanyuk O. V., Chekhmestruk R. Yu. Computer Graphics. — Vinnytsia: VNTU, 2023. 6. Hughes J. F., van Dam A., McGuire M. et al. Computer Graphics: Principles and Practice. 3rd Edition. — Addison-Wesley, 2014. — 1264 p. 7. https://www.opengl.org — Official website of the OpenGL graphics library.

	8. https://learnopengl.com — Practical tutorials on OpenGL.
Responsible Lecturer	Mengnarov Adham Ergashevich

NETWORK SECURITY

Field	Description
Module Title	NETWORK SECURITY
Module Code	TX305
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 60 hours (34 hours Lecture, 26 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	VI semester
Module objective	The objective of the Module Network Security is to develop students' knowledge, skills, and competencies required to understand the theoretical foundations of information security in modern computer networks. Network threats are identified and classified. Methods for protecting network infrastructure are studied. Firewalls, VPNs, and IDS/IPS systems are configured and monitored. Network security policies are developed. The Module also develops students' ability to apply modern cybersecurity principles and international best practices to real-world network security challenges.
Module Learning Outcomes	Knowledge: LO1. Knows the fundamental concepts, principles, and objectives of network security, including the CIA triad, security models, modern cyber threats, and risk management approaches. LO2. Understands the architecture of the OSI and TCP/IP models, security challenges at each network layer, and the vulnerabilities of common network protocols such as ARP, DNS, DHCP, HTTP, and wireless communication protocols. LO3. Knows the operating principles, configuration methods, and practical applications of network security technologies, including firewalls, IDS/IPS,

	VPNs, network segmentation, authentication mechanisms, and Zero Trust architecture. LO4. Understands network traffic monitoring, log management, incident response procedures, digital forensics, penetration testing principles, cloud network security, IoT security, and current cybersecurity standards and best practices. LO5. Knows modern cybersecurity frameworks, security policies, international standards, and emerging technologies applied in protecting enterprise network infrastructures and distributed computing environments. Skills: LO6. Is able to analyze network traffic, identify communication protocols, detect suspicious activities, and investigate security incidents using modern network analysis tools such as Wireshark, Nmap, and Nessus. LO7. Possesses practical skills in configuring and managing firewalls, VPN services, IDS/IPS systems, wireless network security mechanisms, and secure network infrastructures using industry-standard technologies. LO8. Is able to identify network vulnerabilities, perform basic penetration testing, analyze attack scenarios, and recommend appropriate mitigation strategies for protecting network resources. LO9. Possesses practical skills in implementing network segmentation, authentication systems, access control mechanisms, security monitoring solutions, and security policy enforcement within enterprise networks. LO10. Is able to evaluate network security risks, prepare technical reports, document security incidents, and develop practical recommendations for improving organizational cybersecurity. Competencies: LO11. Possesses the competence to design, implement, and manage secure computer network infrastructures by applying modern cybersecurity principles, security technologies, and international best practices.
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	LO12. Possesses the competence to select appropriate security mechanisms, authentication protocols, encryption technologies, monitoring tools, and defensive strategies according to organizational security requirements. LO13. Possesses the competence to analyze cybersecurity incidents, perform vulnerability assessments, support incident response activities, and contribute to maintaining the confidentiality, integrity, and availability of information systems. LO14. Possesses the competence to evaluate the effectiveness of network security solutions, optimize security configurations, conduct security audits, and ensure compliance with organizational policies and cybersecurity standards. LO15. Possesses the competence to prepare professional technical documentation, security reports, audit findings, and incident response documentation while effectively communicating cybersecurity issues to both technical and non-technical audiences. LO16. Possesses the competence to work independently and collaboratively in cybersecurity projects, demonstrating ethical responsibility, professional conduct, critical thinking, and effective decision-making in protecting digital infrastructures. LO17. Possesses the competence to continuously develop professional knowledge, follow emerging trends in cybersecurity, artificial intelligence, cloud security, IoT security, and network defense technologies, and apply innovative approaches to solving modern network security challenges.
Content	Lectures Introduction to Network Security: The CIA Triad Model and Security Strategies. Fundamental concepts and objectives of network security are studied. The principles of confidentiality, integrity, and availability, known as the CIA Triad, are examined as the foundation of information security. Common security threats, risk management concepts, and defense-in-depth strategies are discussed, along with the role of cybersecurity in protecting modern computer networks.

OSI and TCP/IP Models: Security Issues at Each Layer. The architecture of the OSI and TCP/IP networking models and their role in secure communication are studied. Security vulnerabilities associated with each network layer, including the physical, data link, network, transport, and application layers, are examined. Security mechanisms, protocols, and best practices for protecting network communications are discussed.

Network Protocol Security: Analysis of ARP, DNS and DHCP Vulnerabilities. The fundamental operation of the ARP, DNS, and DHCP protocols is studied. Protocol vulnerabilities, including ARP spoofing, DNS cache poisoning, and DHCP starvation, are examined. Techniques for detecting, preventing, and mitigating protocol-based security threats are discussed.

Classification of Network Attacks: Active and Passive Attacks, DoS/DDoS Attacks. The classification of cyberattacks targeting computer networks is studied. Active and passive attacks, denial-of-service and distributed denial-of-service attacks, and man-in-the-middle attacks are examined. Countermeasures and defense strategies against network attacks are discussed.

Network Scanning and Vulnerability Identification: Nmap and Nessus Tools. The principles of network reconnaissance and vulnerability assessment are studied. Network discovery, port scanning, and service identification using Nmap are examined. Vulnerability scanning and security auditing using Nessus, along with ethical considerations in penetration testing, are discussed.

Authentication and Authorisation: 802.1X, RADIUS and LDAP Protocols. The concepts of authentication, authorization, and accounting are studied. User identity verification mechanisms and access control models are examined. The operation of the IEEE 802.1X, RADIUS, and LDAP protocols, along with multi-factor authentication, is discussed.

Firewalls: Types, Operating Principles and Configuration. The fundamental principles of firewall technologies are studied. Packet filtering, stateful

inspection, proxy firewalls, and next-generation firewalls are examined. Firewall deployment strategies and rule configuration for protecting enterprise networks are discussed.

IDS and IPS Systems: Mechanisms for Detecting and Preventing Attacks. Intrusion Detection Systems and Intrusion Prevention Systems are studied. Signature-based and anomaly-based detection techniques are examined. Deployment architectures, event analysis, and the integration of IDS/IPS solutions into network security infrastructures are discussed.

VPN Technologies: IPsec, SSL/TLS, OpenVPN and WireGuard Protocols. Virtual Private Network technologies for secure remote communication are studied. Encryption, tunneling protocols, and authentication mechanisms are examined. IPsec, SSL/TLS VPN, OpenVPN, and WireGuard are comparatively analyzed and discussed.

Network Segmentation: VLAN, DMZ and Zero Trust Architecture. The concepts and benefits of network segmentation are studied. VLANs, demilitarized zones, and network isolation techniques are examined. The principles of Zero Trust Architecture and least-privilege access are discussed.

Wireless Network Security: WPA2/WPA3, Evil Twin and Rogue AP Attacks. The fundamentals of wireless network security are studied. Wireless encryption standards, including WPA2 and WPA3, are examined. Common wireless attacks, such as Evil Twin and Rogue Access Point attacks, are discussed.

Web Application Security: OWASP Top 10, XSS, SQL Injection and CSRF. The security principles for web-based applications are studied. The OWASP Top 10 security risks and their impact on web systems are examined. Cross-Site Scripting, SQL Injection, and Cross-Site Request Forgery, along with secure coding practices, are discussed.

Network Traffic Monitoring: SNMP, NetFlow and SIEM Systems. Techniques for monitoring and analyzing network traffic are studied. Network management using SNMP and traffic analysis using NetFlow are examined. Centralized log collection

using SIEM systems and continuous security monitoring are discussed.

Log Analysis and the Incident Response Process.

Methods of security log collection, storage, and analysis are studied. Incident detection, investigation, containment, and recovery are examined. Digital forensic principles and the role of incident response teams are discussed.

Cloud Network Security: AWS, Azure and Google Cloud Security Models.

Security principles in cloud computing environments are studied. Shared responsibility models, identity and access management, and virtual network security are examined across AWS, Azure, and Google Cloud Platform. Best practices for securing cloud infrastructures are discussed.

Network Security of IoT Devices: Challenges and Solutions.

Security challenges associated with Internet of Things devices and networks are studied. Device authentication, secure communication protocols, and firmware protection are examined. Botnet threats and modern approaches to securing IoT ecosystems are discussed.

Network Security Policy, Audit and Modern Trends.

The development and implementation of organizational network security policies are studied. Security governance, compliance requirements, and security audits are examined. Emerging trends, including artificial intelligence, Zero Trust security, and quantum-resistant cryptography, are discussed.

Practical lessons

Analysing Network Traffic and Identifying Protocols Using Wireshark.

Network traffic is captured and analyzed using Wireshark. Communication protocols within the captured traffic are identified. Packet-level details are examined to understand network behavior.

Network Scanning and Identifying Open Ports Using Nmap.

Network scanning is performed using Nmap. Open ports and active services are identified. Results are analyzed to assess the network's exposure to potential threats.

Simulating and Detecting ARP Spoofing and DNS Poisoning Attacks. ARP spoofing attacks are simulated in a controlled environment. DNS poisoning attacks are also simulated. Detection techniques are applied to identify these attacks.

Modelling a DoS Attack and Testing Protection Methods. A denial-of-service attack is modeled in a test environment. The impact of the attack on network resources is observed. Protection methods are tested to mitigate the attack.

Analysing Phishing Emails and Applying OSINT Methods. Phishing emails are analyzed to identify malicious indicators. Open-source intelligence (OSINT) methods are applied to gather relevant information. Findings are used to assess the credibility and risk of suspicious communications.

Configuring Firewall Rules: Working with iptables and pfSense. Firewall rules are configured using iptables. Firewall policies are also configured using pfSense. Rule sets are tested to verify correct traffic filtering.

Installing the Snort IDS System and Configuring Rules. The Snort intrusion detection system is installed. Detection rules are configured within Snort. The system is tested to verify its ability to detect suspicious network activity.

Configuring a VPN Connection: OpenVPN Server and Client Configuration. An OpenVPN server is configured. The corresponding OpenVPN client is configured to connect to the server. The VPN connection is tested to verify secure data transmission.

Configuring VLAN and Network Segmentation. VLANs are configured to segment the network. Network segmentation is implemented to isolate traffic between different network zones. The configuration is tested to verify proper isolation and connectivity.

Testing Wireless Network Security: WPA2 Handshake Analysis. Wireless network security is tested in a controlled environment. The WPA2 handshake process is captured and analyzed.

	Findings are used to assess the strength of the wireless security configuration. Introduction to SIEM Systems: Log Analysis with ELK Stack. SIEM systems are introduced as tools for centralized security monitoring. Logs are collected and analyzed using the ELK Stack. Patterns and anomalies within the logs are identified. Identifying Web Application Vulnerabilities: Working with Burp Suite. Web application vulnerabilities are identified using Burp Suite. Requests and responses are intercepted and analyzed. Potential security weaknesses are documented and evaluated. Final Practical Project: Designing a Secure Network Architecture. A secure network architecture is designed as a final practical project. Appropriate security mechanisms and technologies are selected and integrated. The designed architecture is documented and presented. Self-study Theoretical Foundations of Information and Network Security. Network Protocol Vulnerabilities and Spoofing Attacks. Network Attacks and Methods for Their Detection. Network Protection Tools and Segmentation. Network Monitoring and Security Analysis. VPN and Zero Trust Security Model. Wireless Network Security. Web Application Security and OWASP Vulnerabilities. Incident Response and Digital Forensics. Cloud Computing Environment Security. IoT and Industrial Network Security. Fundamentals of Penetration Testing and Ethical Hacking. Security Policy and Social Engineering Attacks.
Teaching Methods	The module is delivered through interactive lectures, practical laboratory sessions, simulation-based learning, case study analysis, problem-based learning, and cybersecurity workshops. Students develop practical competencies by configuring secure network infrastructures, analyzing network protocols, and performing vulnerability assessments. Firewall,

	VPN, IDS/IPS, and wireless security solutions are implemented using modern cybersecurity tools and platforms. Laboratory activities emphasize network traffic analysis, penetration testing, log analysis, incident response, and cloud security through hands-on exercises. Independent learning, research-based assignments, and collaborative discussions further strengthen students' analytical thinking and technical problem-solving skills.
Prerequisites	mathematics, physics, informatics
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final assessment (50%). The continuous assessment evaluates students' classroom participation, laboratory performance, practical cybersecurity exercises, network configuration tasks, and the successful completion of independent study assignments, including the analysis of network protocols, identification of security vulnerabilities, firewall configuration, and implementation of authentication mechanisms and secure VPN connections. The midterm assessment consists of written assignments, laboratory exercises, practical configuration tasks, quizzes, and the evaluation of project-based activities, including network scanning and vulnerability assessment using tools such as Nmap and Nessus, IDS/IPS configuration, network segmentation, and wireless network security. The final assessment is conducted as a written and practical examination consisting of theoretical questions, network security analysis tasks, configuration exercises, and scenario-based problem-solving activities covering network security principles, authentication and authorization mechanisms, firewall technologies, VPN solutions, intrusion detection and prevention systems, cloud and IoT security, and incident response procedures. Successful completion of all independent study assignments is required as an essential component of the module assessment.

Recommended Literature	1. Kurose, J. F. & Ross, K. W. Computer Networking: A Top-Down Approach. Pearson Education, 5th–8th editions. 2. Tanenbaum, A. S. & Wetherall, D. J. Computer Networks. Pearson Education, 5th edition. 3. Stallings, W. Network Security Essentials: Applications and Standards. Pearson, 6th edition, 2021. 4. Stallings, W. Cryptography and Network Security: Principles and Practice. Pearson, 6th edition, 2014. 5. Bishop, M. Computer Security: Art and Science. Addison-Wesley, 2nd edition. 6. Schneier, B. Secrets and Lies: Digital Security in a Networked World. Wiley, 2000. 7. Menezes, A. J., van Oorschot, P. C., Vanstone, S. A. Handbook of Applied Cryptography. CRC Press, 1996. 8. Stinson, D. R. Cryptography: Theory and Practice. CRC Press, 3rd edition.
Responsible Lecturer	Shaymanov Farxod, teacher.

COMPUTER VISION

Field	Description
Module Title	COMPUTER VISION
Module Code	KK305
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 60 hours (34 hours Lecture, 26 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	VI semester
Module objective	The objective of the Module Computer Vision is to develop students' theoretical knowledge and practical skills in computer vision, with a focus on automatically extracting meaningful information from digital images and videos. During the course, students are introduced to image processing and feature detection and matching. Multi-view geometry, camera modelling, and stereo and motion analysis are studied. Students become familiar with state-of-the-

	art methods based on deep learning, including CNNs, object detection, segmentation, generative models, and vision transformers. These methods are applied in practice using Python, OpenCV, and PyTorch tools.
Module Learning Outcomes	Knowledge: LO1. Knows the fundamental concepts, principles, and applications of computer vision, including image formation, image acquisition, digital image representation, and image processing techniques. LO2. Understands image enhancement methods, feature extraction and matching algorithms, camera models, multi-view geometry, stereo vision, and motion analysis techniques used in computer vision systems. LO3. Knows traditional and deep learning-based computer vision methods, including convolutional neural networks (CNNs), object detection, image segmentation, image classification, object tracking, and generative models. LO4. Understands the principles of camera calibration, optical flow, homography, Structure from Motion (SfM), Simultaneous Localization and Mapping (SLAM), and three-dimensional computer vision techniques. LO5. Knows modern computer vision frameworks, libraries, and development environments, including OpenCV, PyTorch, deep learning architectures, Vision Transformers (ViT), and vision-language models used in intelligent visual systems. Skills: LO6. Is able to acquire, process, enhance, and analyze digital images using modern computer vision techniques and software development tools. LO7. Possesses practical skills in implementing feature detection and matching algorithms, camera calibration, stereo vision, image stitching, motion analysis, and object tracking using Python and OpenCV. LO8. Is able to design, train, evaluate, and optimize deep learning models for image classification, object detection, semantic segmentation, and instance

	segmentation using modern machine learning frameworks. LO9. Possesses practical skills in evaluating computer vision algorithms, interpreting experimental results, visualizing data, and comparing the performance of different computer vision models. LO10. Is able to develop, test, debug, document, and present computer vision applications that solve practical problems in areas such as healthcare, industry, robotics, autonomous systems, security, and intelligent transportation. Competencies: LO11. Possesses the competence to analyze real-world visual information, identify computer vision problems, and design appropriate algorithmic solutions using classical and deep learning approaches. LO12. Possesses the competence to select appropriate image processing techniques, feature extraction methods, neural network architectures, and computer vision frameworks according to specific application requirements. LO13. Possesses the competence to develop reliable, scalable, and efficient computer vision systems by integrating image processing, machine learning, and artificial intelligence techniques. LO14. Possesses the competence to critically evaluate computer vision models, optimize algorithm performance, validate experimental results, and ensure the reliability and accuracy of intelligent visual systems. LO15. Possesses the competence to prepare technical reports, research papers, and professional presentations, effectively communicating computer vision concepts, methodologies, and project outcomes to academic and professional audiences. LO16. Possesses the competence to work independently and collaboratively in multidisciplinary teams while applying ethical principles, professional responsibility, and modern software engineering practices in computer vision projects. LO17. Possesses the competence to engage in lifelong learning, follow emerging developments in
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	computer vision, artificial intelligence, and deep learning, and continuously improve professional knowledge and practical skills in intelligent visual computing.
Content	Lectures Introduction to Computer Vision and Image Formation. The fundamental concepts and applications of computer vision in modern intelligent systems are introduced. Its role in robotics, healthcare, autonomous vehicles, and industrial automation is examined. The principles of light propagation, optics, image formation, and image acquisition are discussed. Fundamentals of Digital Image Processing. Basic techniques of digital image processing, including point operations and image histograms, are studied. Linear filtering and convolution are examined. Image enhancement methods, such as Gaussian smoothing, median filtering, and sharpening, are discussed. Edge Detection and Frequency Analysis. Edge detection using gradient-based methods, the Sobel operator, and the Canny edge detector is studied. Frequency-domain image analysis using the two-dimensional Fourier Transform is examined. Multi-scale image representation through Gaussian and Laplacian pyramids is discussed. Feature Detection and Descriptors. Distinctive image features are detected using Harris corner detection. Scale-invariant feature extraction techniques, including SIFT, SURF, and ORB, are examined. The Histogram of Oriented Gradients (HOG) descriptor and its role in object recognition and image matching are discussed. Feature Matching and RANSAC. Methods for matching local image descriptors between multiple images are studied. Distance metrics and feature correspondence techniques are examined. Outlier removal and robust parameter estimation using the RANSAC algorithm are discussed. Image Transformations and Homography. Geometric image transformations, including affine and projective transformations, are studied. Homography estimation and perspective correction

are examined. Image stitching techniques for panorama generation are discussed.

Camera Model and Calibration. The pinhole camera model and image projection principles are studied. Internal and external camera parameters are examined. Camera calibration techniques based on Zhang's method are discussed.

Multi-view Geometry and Stereo. The fundamentals of epipolar geometry, including fundamental and essential matrices, are studied. Stereo correspondence algorithms for depth estimation are examined. The construction of depth maps and three-dimensional scene reconstruction using stereo vision are discussed.

Structure from Motion and Optical Flow. The principles of Structure from Motion for reconstructing three-dimensional scenes from multiple images are studied. Optical flow estimation using the Lucas-Kanade and Horn-Schunck methods is examined. Motion analysis and visual tracking in dynamic environments are discussed.

Image Segmentation. Image segmentation techniques, including thresholding, region growing, and k-means clustering, are studied. Graph-based segmentation and watershed algorithms are examined. The partitioning of images into meaningful regions for object recognition and medical image analysis is discussed.

Image Classification: kNN and Linear Classifiers. The fundamentals of image classification using traditional machine learning methods are studied. Data-driven approaches, including k-nearest neighbour and linear classifiers, are examined. Softmax classification, loss functions, and performance evaluation of image classification models are discussed.

Neural Networks and Backpropagation. Artificial neural networks, perceptrons, and multilayer neural network architectures are studied. Activation functions, forward propagation, and backpropagation algorithms are examined. Optimisation methods, including Stochastic Gradient Descent and Adam, are discussed.

Convolutional Neural Networks (CNN). The architecture and operation of Convolutional Neural Networks are studied. Convolutional layers, pooling operations, and fully connected layers are examined. The advantages of CNNs for image analysis and their applications in computer vision are discussed.

Training CNNs and Architectures. Techniques for training deep convolutional neural networks are studied. Batch normalization, dropout, and regularization are examined. Well-known CNN architectures, including AlexNet, VGG, and ResNet, are comparatively analyzed.

Object Detection and Semantic Segmentation. Modern object detection algorithms, including the R-CNN family, YOLO, and SSD, are studied. Semantic and instance segmentation using U-Net and Mask R-CNN are examined. Applications of object localisation and scene understanding in intelligent vision systems are discussed.

Generative Models and Vision Transformers. Generative deep learning models, including Autoencoders, Variational Autoencoders, and Generative Adversarial Networks, are studied. Diffusion models are also examined. The fundamentals of Vision Transformers and their application to large-scale image understanding are discussed.

3D Vision and Vision–Language Models. Three-dimensional computer vision techniques for object representation and scene reconstruction, including Neural Radiance Fields, are studied. The integration of vision and language through multimodal models, including CLIP, is examined. Emerging trends in intelligent visual computing and multimodal artificial intelligence are discussed.

Practical lessons

Setting Up the Environment and Working with Images. Python and OpenCV are installed and configured. Images are read and displayed using OpenCV. Images are saved in different formats.

Image Filtering. Image filtering is performed using convolution. Images are smoothed using Gaussian

and median filters. Image sharpening techniques are applied.

Edge Detection. Edge detection is performed using the Sobel operator. Edge detection is also performed using the Canny detector. The effect of threshold values on detection results is studied.

Feature Detection and Matching. Keypoints are detected using SIFT and ORB algorithms. Feature descriptors are extracted for the detected keypoints. Descriptor matching is performed between images.

Panorama Creation. Images are stitched together to create a panorama. The RANSAC algorithm is used for robust feature matching. Homography is applied to align and blend the images.

Camera Calibration. A chessboard calibration pattern is used for camera calibration. Camera parameters are determined from multiple calibration images. The calibration results are evaluated for accuracy.

Stereo and Depth. A stereo image pair is processed for depth estimation. A disparity map is calculated from the stereo pair. The resulting depth map is analyzed and visualized.

Optical Flow and Tracking. Optical flow is estimated using the Lucas-Kanade method. Motion vectors are computed between consecutive frames. Object tracking is implemented based on the estimated optical flow.

Image Segmentation. Image segmentation is performed using the k-means clustering algorithm. Segmentation is also performed using the watershed algorithm. The results of both methods are compared and analyzed.

Classification with CNN. A convolutional neural network is built using PyTorch or Keras. The network is trained on the CIFAR-10 dataset. The trained model's classification performance is evaluated.

Transfer Learning. A pre-trained ResNet model is adapted for a new classification task. The model is fine-tuned using transfer learning techniques. The performance of the adapted model is evaluated on the new task.

	Object Detection. Objects are detected in images using the YOLO algorithm. Object detection is also performed using Faster R-CNN. The detection results are compared and analyzed. Final Project. A real-world computer vision problem is selected. An appropriate solution is designed and implemented. The results are analyzed and presented as a final project. Self-study History of Computer Vision and Its Application Areas. Colour Spaces (RGB, HSV, Lab) and Image Formats. Histogram and Contrast Enhancement (Equalisation). Spatial and Frequency Filtering. Morphological Operations (Dilatation, Erosion). Comparison of SIFT, SURF and ORB Descriptors. HOG and Traditional Features. RANSAC and Robust Estimation Methods. Homography and Image Stitching (Panorama). Camera Model and Calibration (Zhang Method). Epipolar Geometry and Stereo Matching. Structure from Motion (SfM) and SLAM Fundamentals. Optical Flow Methods (Lucas–Kanade, Horn–Schunck). Object Tracking (Kalman Filter, Mean Shift). The Comparison of CNN Architectures (AlexNet, VGG, ResNet). Object Detection Methods (R-CNN, YOLO, SSD). Semantic and Instance Segmentation (U-Net, Mask R-CNN). Generative Models (GAN, VAE, Diffusion). Vision Transformers and Self-Supervised Learning (ViT, CLIP, DINO). 3D Vision, NeRF and Vision–Language Models.
Teaching Methods	The module is delivered through interactive lectures, practical laboratory sessions, computer-based experiments, problem-based learning, project-based activities, and guided independent study. Students develop practical competencies by applying digital image processing techniques, feature extraction algorithms, camera calibration, and image segmentation. Object detection and deep learning

	methods are implemented using modern computer vision libraries, such as OpenCV, TensorFlow, and PyTorch. Laboratory sessions emphasize image analysis, algorithm implementation, model training, and performance evaluation for real-world problems. Independent learning, scientific literature review, and research-oriented projects further enhance students' analytical thinking and computational skills.
Prerequisites	Programming 1,2 Linear Algebra Signal and Image Processing
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final assessment (50%). The continuous assessment evaluates students' classroom participation, laboratory performance, programming exercises, image processing assignments, and the successful completion of independent study tasks, including the implementation of digital image processing techniques, feature extraction algorithms, image transformations, camera calibration, stereo vision, and image segmentation. The midterm assessment consists of written assignments, practical laboratory exercises, programming tasks, quizzes, and the evaluation of project-based activities, including the development of computer vision applications using OpenCV and deep learning frameworks, feature detection and matching, image classification, object detection, and segmentation. The final assessment is conducted as a written and practical examination consisting of theoretical questions, algorithm design tasks, image analysis problems, and computer-based programming exercises covering computer vision principles, digital image processing, geometric vision, deep learning architectures, object detection, semantic segmentation, three-dimensional vision, and vision-language models. Successful completion of all

	independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Szeliski R. Computer Vision: Algorithms and Applications. 2nd ed. — Springer, 2022. szeliski.org/Book 2. Prince S.J.D. Computer Vision: Models, Learning, and Inference. — Cambridge University Press, 2012. computervisionmodels.com 3. Forsyth D.A., Ponce J. Computer Vision: A Modern Approach. 2nd ed. — Pearson, 2012. 4. Goodfellow I., Bengio Y., Courville A. Deep Learning. — MIT Press, 2016. deeplearningbook.org 5. Gonzalez R.C., Woods R.E. Digital Image Processing. 4th ed. — Pearson, 2018.
Responsible Lecturer	Abdualimov Dilshod, teacher.

CLOUD TECHNOLOGIES

Field	Description
Module Title	CLOUD TECHNOLOGIES
Module Code	BT304
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (34 hours Lecture, 26 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	VI semester
Module objective	The objective of the Module Cloud Technologies is to develop students' theoretical and practical skills in designing cloud computing models. Virtualization technologies are studied. Cloud technology architectures are examined. Developing and managing software on cloud platforms is practiced. Cloud security, monitoring, and billing systems are studied. Modern cloud service ecosystems, including Google, Amazon, and Yandex platforms, are examined.
Module Learning Outcomes	Knowledge: LO1. Cloud Architecture Principles: Differentiates between the fundamental models of cloud computing

	(IaaS, PaaS, SaaS) and their technological layers (Frontend, Backend, Network), and understands their role in business. LO2. Cloud services ecosystem: Theoretically grasps the architecture, business opportunities, and market position of the Google (Disk, Workspace), Amazon (AWS), and Yandex platforms. LO3. Virtualization and Platform Management: Understands the operating mechanisms of virtualization technologies (VMs and containers), cloud databases (RDS, NoSQL), and network technologies (VPN, SDN, APIs). LO4. Security and financial models: Understands the operating principles of authentication, encryption, monitoring (Grafana, CloudWatch), and billing systems in a cloud environment. Skills: LO5. Working with cloud platforms: will have practical skills in creating digital resources with Google cloud tools (Docs, Forms, Sheets, Drive) and organizing online classes and conferences. LO6. Managing business processes in the cloud: using Amazon and Yandex services, you can manage and configure business operations (commerce, image processing, data analysis) in practice. LO7. Integration and visualization: Works with cloud maps and can connect external services (weather data, payment systems) to its own applications and systems via API. LO8. CI/CD and Automation: Will have skills in deploying software on cloud platforms and applying continuous integration and delivery (CI/CD) processes using tools such as Docker and GitHub Actions. Competencies: LO9. Will have the competence to independently design educational or business systems using cloud technology tools (Google, AWS, Yandex) and ensure their performance. LO10. Secure and efficient resource management: It will have the ability to select strategies for ensuring data security in a cloud environment, monitoring
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	resource consumption, and effectively managing billing systems. LO11. Technological Integration and Innovation: Quickly adopts new cloud services (AI, ML, serverless) and emerging trends (edge computing, quantum cloud), and can apply them to existing engineering solutions. LO12. Team and remote collaboration: Will be qualified to use cloud platforms for teamwork, implement remote management, and professionally organize a digital collaboration environment.
Content	Lectures Cloud Technologies: Concept, Key Ideas, and Historical Development. The fundamental principles of cloud computing, including on-demand self-service, broad network access, and rapid elasticity, are studied. Resource pooling and measured service are examined. The evolution from grid computing and utility computing to modern scalable distributed cloud architectures is discussed. Cloud Technology Architecture: User, Service, and Technological Layers. The structural organization of cloud ecosystems across user interface, service delivery, and infrastructure layers is studied. Orchestration mechanisms and middleware abstractions are examined. Virtualization management and API-driven system interactions are discussed. Types of Cloud Services: IaaS, PaaS, SaaS. Infrastructure as a Service, Platform as a Service, and Software as a Service delivery models are comparatively evaluated. The allocation of administrative control and shared responsibility models are examined. Resource abstraction and software stack management between providers and consumers are discussed. Private, Public, and Hybrid Clouds: Comparison and Areas of Application. Public, private, hybrid, and multi-cloud deployment models are studied. Security posture, compliance requirements, and operational costs are examined. Latency constraints and dynamic workload burstability across enterprise infrastructure are discussed.

Virtualization Technology and Its Role in Cloud Computing. The core mechanisms of hardware abstraction and hypervisors, including Type-1 and Type-2, are studied. Virtual machine management is examined. Hardware-assisted virtualization and containerization fundamentals are discussed.

Cloud Storage Services and Data Warehouse Management. The architecture of cloud storage modalities, including block, file, and object storage, is studied. Scalable data warehousing and distributed object persistence are examined. Storage tiering strategies and data replication models are discussed.

Network Technologies and Connectivity in Cloud Computing. Software-Defined Networking, Virtual Private Clouds, and subnetting are studied. Dynamic routing and load balancing configurations are examined. Network Function Virtualization and VPN gateways are discussed.

Database as a Service (DBaaS) in the Cloud. The architecture of managed relational and non-relational database solutions in cloud environments is studied. Provisioning, automated scaling, and multi-region replication are examined. Fault tolerance, backup mechanisms, and high-availability database clusters are discussed.

Cloud Infrastructure Management and Automation. Infrastructure as Code paradigms using tools such as Terraform and CloudFormation are studied. Automated resource provisioning and configuration management are examined. State synchronization and immutable infrastructure practices are discussed.

Developing and Deploying Applications on Cloud Platforms (Amazon AWS, Microsoft Azure, Google Cloud). The architectural comparison of major cloud service provider ecosystems, including AWS, Azure, and GCP, is studied. Deployment methodologies and SDK integration are examined. Identity management and serverless runtime execution are discussed.

DevOps Practices in Cloud Computing: CI/CD, Docker, and GitHub Actions. Continuous Integration and Continuous Deployment pipelines operating within cloud platforms are studied.

Containerization using Docker is examined. Automated testing and version control integration via GitHub Actions are discussed.

Security and Privacy Issues, and Approaches to Their Resolution. Identity and Access Management and zero-trust network architectures are studied. Encryption at rest and in transit are examined. Key management services and compliance frameworks, including GDPR and ISO/IEC 27001, are discussed.

Resource Monitoring and Reporting in Cloud Computing. Infrastructure telemetry and real-time metrics collection are studied. Centralized log aggregation and dynamic alerting systems are examined. Observability frameworks and performance benchmarking are discussed.

Billing Systems and Pricing Models for Cloud Services. FinOps frameworks and economic models, including pay-as-you-go and reserved instances, are studied. Spot instances and sustained-use discounts are examined. Cost allocation tags and resource optimization strategies are discussed.

Machine Learning and Artificial Intelligence Services Used in Cloud Computing. Cloud-hosted AI and ML platforms and managed model training pipelines are studied. Pre-built cognitive services are examined. Distributed model inference and MLOps practices are discussed.

The Future of Cloud Technologies: Edge Computing, Serverless, Quantum Cloud. Next-generation distributed paradigms, including edge computing and event-driven serverless architectures, are studied. Cloud-accessible quantum computing processing units are examined. Low-latency edge-to-cloud integration is discussed.

Course Review and Emerging Trends in Cloud Computing. A comprehensive synthesis of cloud architecture, security governance, and DevOps methodologies is presented. Emerging trends such as multi-cloud orchestration and sovereign clouds are examined. Sustainable green cloud computing and autonomous infrastructure management are discussed.

Practical lessons

Working with Modern Clouds. Popular cloud platforms are initially set up and navigated. User accounts, storage quotas, and synchronized folders are configured. Basic access control policies across cloud ecosystems are applied.

Getting to Know Google Drive Features and Creating Online Lessons. The Google Workspace cloud infrastructure is explored. Shared drives, folder hierarchies, and dynamic access permissions are configured. Cloud-based instructional content is distributed for online lessons.

Working with Spreadsheets and Word Processors in Google Drive. Real-time collaborative documents are created using Google Docs and Google Sheets. Data processing functions and concurrent version tracking are practiced. Comment threads and revision history management are used.

Options for Storing Images and Organizing Lessons in the Google Cloud. Media assets are managed using Google Photos and Google Drive storage. Visual digital assets are organized with metadata tagging. Public and private sharing configurations are applied to embed media into cloud lesson workflows.

Creating Test Forms in Google Drive. Interactive surveys and assessment modules are designed using Google Forms. Automated grading rules and data validation constraints are set up. Response collection pipelines linked to Google Sheets are configured.

Organizing Online Conferences and Meetings on Google Drive. Cloud scheduling and video conferencing are integrated using Google Calendar and Google Meet. Secure meeting rooms and participant access privileges are configured. Screen-sharing protocols and automated cloud recording storage are set up.

Processes for Creating Presentations and Sites in Google Drive. Cloud-hosted interactive presentations are designed using Google Slides. Web portals are constructed using Google Sites. Embedded cloud components and custom domain mapping are integrated.

Introduction to the Amazon Cloud and Business Opportunities. The AWS Management Console and core infrastructure services are navigated. Cloud-based enterprise solutions and pay-as-you-go economic models are explored. Foundational IaaS and PaaS architecture capabilities are examined.

Sales Processes on Amazon.com. Cloud-enabled e-commerce mechanisms and digital marketplace platforms are analyzed. Inventory management API integration is examined. Automated order processing and cloud backend transaction management are practiced.

Yandex.com and Its Features. The Yandex Cloud ecosystem components and regional infrastructure services are reviewed. Storage buckets and compute instances are configured. Administrative control dashboards are used.

Editing Images and Getting Weather Information on Yandex. API-driven cloud web services and microservices are utilized. RESTful APIs for real-time weather data retrieval are integrated. Cloud-based media processing tools for image manipulation are applied.

Working with Maps in the Cloud. Cloud-based spatial location services are integrated using interactive Mapping APIs. Geolocation tracking and dynamic marker placement are configured. Spatial data rendering in web applications is practiced.

Payments via Clouds and the Wallet Features They Provide. Cloud payment gateways and digital wallet integrations are practically evaluated. Sandbox API endpoints for e-commerce payment processing are implemented. Tokenization and cloud security protocols are applied.

Self-study

Cloud Technology Capabilities.
 Key Features of Cloud Computing and Their Applications in Real Life.
 Cloud Models.
 SaaS, PaaS, and IaaS Service Models: Differences and Application Areas.

	The Concept of Cloud Computing and Its Comparison with Traditional Computing Models. Virtualization Technologies and the Operating Principle of Hypervisors. Comparison of Public, Private, and Hybrid Cloud Types. Storing Data in the Cloud: An Analysis of Object Storage, Block Storage, and File Storage Models. Amazon AWS Services and Their Advantages. Microsoft Azure Platform and Its Key Capabilities. Comparison of Google Cloud Platform Services and Providers. Features of the IBM Cloud Platform. Amazon Cloud Features. Creating Forms and Tests in Google Drive, and Designing Surveys. Yandex Disk Features. Creating a Site on Google Drive. Google Services. Using Yandex Music and Wallet. Amazon Services. Image Processing in Yandex. Modern Network Devices: HUB, Switch, WiFi, Modems. Cloud Technologies in E-Government Systems. Opportunities for Using Big Data Technologies in a Cloud Environment. Cloud Data Backup and Recovery Methods. The Importance of SDN and CDN Technologies in the Cloud. Application of Cloud Technologies in Education. Legal Foundations for Ensuring Privacy in Cloud Computing (for Example, GDPR). The Economic Efficiency of Cloud Computing: An Analysis of the Pay-as-You-Go Model. Security Issues in Cloud Technologies and Their Mitigation Methods. Future Development Prospects of Cloud Technologies: Edge and Fog Computing Approaches.
Teaching Methods	The course employs a blend of problem-based learning and hands-on cloud environment training to bridge theoretical infrastructure concepts with practical solutions. Interactive workshops in AWS,

	Azure, and Google Workspace allow students to deploy virtual servers, manage cloud storage, and construct automated CI/CD pipelines. Collaborative group projects and peer reviews enhance critical thinking, cloud architecture optimization, and technical communication skills. Practical scenario-based assignments involving database management, network security configuration, and API integrations enable students to independently design, deploy, and manage scalable cloud solutions.
Prerequisites	Programming 1,2 Database Computer Networks Cyber Security Fundamentals
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment reflects the student's theoretical knowledge, practical skills in configuring cloud infrastructure and services, ability to work with cloud storage, manage digital resources, integrate web APIs, adhere to cloud security principles, and present technical findings logically and coherently. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as a cloud service model comparative report, a hypervisor virtualization analysis presentation, an object/block storage analysis, an AWS/Azure ecosystem review, and practical Google Workspace/Yandex integration projects. The final assessment is a written examination consisting of theoretical concepts, architectural analysis tasks, and practical cloud deployment problems designed to assess students' understanding of cloud service models, virtualized networking, database-as-a-service, cloud security, billing mechanisms, and future trends such as edge computing and serverless architectures. Successful completion of all independent study assignments is

	required as an essential component of the module assessment.
Recommended Literature	1. Sandeep Bhowmik. Cloud Computing. Cambridge University Press & Assessment 2017. – 434 p. 2. Borko Furht, Armando Escalante. Handbook of Cloud Computing. Florida, Atlantic University. 2010. 3. Nayan B. Ruparelia. Cloud Computing. 2016 Massachusetts Institute of Technology. – 277 p. 4. Andrew Tanenbaum, Nick Fimstër, David Weatherall. Computer Networks. 6th ed. – St. Petersburg: Piter, 2023. – 784 p. 5. D.V. Druzhinin. High-Performance Computing and Cloud Technologies: A Textbook. – Tomsk: Tomsk State University Press, 2020. – 94 p. 6. T.E. Delov. Cloud Technologies. Textbook. 2020. – 217 p.
Responsible Lecturer	S.H.Norqobilov, teacher.

DATABASE MANAGEMENT SYSTEMS

Field	Description
Module Title	DATABASE MANAGEMENT SYSTEMS
Module Code	MBBT305
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (34 hours Lecture, 26 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	VI semester
Module objective	The objective of the Module Database Management Systems is to form and develop the knowledge, skills, and competencies that a student must acquire in their professional field for practical application. This programme includes the fundamental theoretical and practical knowledge that every professional must master, regardless of their area of practice. During the course, students study the fundamental principles of database management systems. Database design, SQL querying, and database administration are studied. Database security and integrity mechanisms

	are examined. Modern DBMS platforms, including MySQL, PostgreSQL, and Oracle, are studied.
Module Learning Outcomes	Knowledge: LO1. To provide information about database management systems, to demonstrate the role of information technology in the lives of individuals and in the development of society, and to reveal the capabilities of database management systems. LO2. To develop practical skills in managing computer databases and working on the Internet. LO3. To develop an understanding of when and for what purposes database management systems are used. LO4. To develop knowledge of the application and development of database management. LO5. Familiarisation with methods for ensuring information security in the context of reworking database management systems. Skills: LO6. Studies the theoretical foundations of using personal computers in database management system processes and the methods for applying them to various data structure domains. LO7. Studies effective methods of using database management systems. LO8. Studies the development trends of databases and their characteristics. LO9. It consists of forming concepts about when and for what purposes to use a database, and providing knowledge on how to generate data. LO10. They will implement methods to ensure security when working with the database and familiarise themselves with its functional capabilities. Competencies: LO11. Designing and creating databases. LO12. Managing and analysing data using SQL queries. LO13. Ensuring database integrity, security, and protection. LO14. Working with modern DBMSs (MySQL, PostgreSQL, Oracle, MS SQL Server, etc.).

	LO15. Optimisation and administration of databases for information systems.
Content	Lectures Introduction. Database Management. The core principles of database management systems and data abstraction levels are studied. The ANSI-SPARC three-schema architecture is examined. Traditional file processing systems are compared with modern DBMS solutions in terms of data independence and redundancy control. Connecting to Data in INTERNAL Mode. The low-level physical storage representation and internal schema mapping are studied. Access paths and storage engines are examined. Disk block allocation strategies and buffer management mechanisms are discussed. Database Administrator Applications. Administrative tools and utilities used by database administrators are studied. Operational oversight of database instances and user privilege management are examined. System resource allocation, performance tuning, and health auditing are discussed. Indexing: Tree-Based and Hash-Based Methods, Learning Indexes from Data. The theoretical foundations of search structures, including B/B+ trees and ISAM, are studied. Static and extendible hash tables are examined. Index search efficiency, range queries, and modern ML-learned indexing techniques are discussed. Query Evaluation: Theory, Sorting and Processing, Selection, Projection, Aggregation. The mathematical evaluation of relational algebra queries is studied. Physical operator execution pipelines are examined. External merge-sort algorithms and hash-based versus sort-based aggregation strategies are discussed. Data Blocks, Extensions and Segments. The logical and physical storage hierarchy within relational databases is studied. Physical data blocks and contiguous block extents are examined. Logical storage segments for tables, indexes, and rollback logs are discussed.

Transactions. The fundamental concepts of transaction management and the ACID properties are studied. Concurrency control algorithms, including two-phase locking, are examined. Serializability and write-ahead logging protocols are discussed.

Ensuring the Security of Database Management Systems. Role-Based Access Control and Discretionary Access Control mechanisms are studied. Data encryption at rest and in transit is examined. SQL injection prevention and security auditing logs are discussed.

Query Optimisation. Cost-based and rule-based query optimization strategies are studied. The construction and transformation of relational query trees are examined. Join order selection algorithms and join execution mechanics are discussed.

Big Data and NoSQL Systems. The limitations of traditional relational databases under massive scale and the CAP theorem are studied. Document-oriented, key-value, and wide-column database models are examined. Graph databases for unstructured and semi-structured Big Data storage are discussed.

Database Integrity. The enforcement of semantic integrity constraints across relational structures is studied. Domain constraints and entity integrity are examined. Referential integrity, dynamic assertions, and declarative check conditions are discussed.

Creating the Database. Physical schema initialization and table space allocation are studied. System catalog bootstrapping is examined. Structural definition using Data Definition Language and initial storage sizing are discussed.

Database Management. The operational lifecycle administration of databases is studied. Dynamic instance re-configuration and storage auto-extension are examined. Active connection pool management and runtime performance monitoring are discussed.

Working with Database Management Systems. Interactive and programmatic interactions with RDBMS platforms are studied. Data Manipulation Language commands and embedded SQL execution

are examined. Server-side cursors and application developer database connectors are discussed.

Complex Queries and Metadata. Advanced SQL constructs, including window functions and common table expressions, are studied. Set operations and analytical processing are examined. Querying system data dictionaries and metadata catalogs is discussed.

Packages: Using Standard Packages, Creating User Packages. Modular procedural database programming using server-side packages is studied. Encapsulation of stored procedures and user-defined functions is examined. Cursor variables and package initialization logic are discussed.

Working with Complex Queries: Qualifying with Subqueries and Using Dynamic Queries. Advanced relational query design using scalar, correlated, and nested subqueries is studied. Quantified predicates, including EXISTS and IN, are examined. The construction and execution of dynamic SQL statements are discussed.

Practical lessons

Designing a Relational Database. Conceptual, logical, and physical database design phases are implemented. Entity-Relationship and Enhanced Entity-Relationship modeling are practiced. Normalization rules, from First through Third Normal Form, are applied to eliminate data redundancy.

Database System Architectures, Row and Column Stores, OLTP and OLAP, In-Memory Database Systems. Row-oriented and column-oriented database engines are benchmarked. Workload profiles for transactional and analytical execution pathways are configured. In-memory storage tuning is practiced.

Storage: Secondary Storage Devices. Page layout specifications and slotted-page architectures are analyzed. Disk-I/O access overhead is evaluated. Buffer pool hit ratios and dirty page flushing mechanisms are practically assessed.

Working with Database Objects: Creating Indexes, References and Constraints. Tables, primary and foreign key relationships, and check constraints are created and maintained. Sparse,

dense, and composite indexes are configured. Partitioned indexes are created to optimize query execution paths.

Query Optimisation: Cardinality Estimation, Cost-Based Optimisation, Dynamic Programming, Rule-Based Optimisation. Execution plans are analyzed using EXPLAIN ANALYZE tools. Cardinality estimation statistics and cost formula calculations are evaluated. Rule-based and cost-based optimizer decisions on complex join networks are compared.

Working with the Database Using Standard Packages, Creating and Dropping Packages. Modular database programming is practiced. Related procedures, functions, and cursors are packaged into reusable schema components. Package lifecycle management is applied.

Working with Complex Queries: Multi-Step Queries, Dynamic SQL, Joining Multiple Tables. Multi-stage SQL execution pipelines are designed using Common Table Expressions and temporary tables. Dynamic SQL strings are constructed for runtime filter execution. Multiple tables are joined in complex query topologies.

Using Conditions, Types and Loops of the PL/SQL Procedural Extension Language; Handling Exceptions. Conditional branching and composite data types are implemented in server-side scripts. Explicit loops are used for iterative processing. Custom and predefined exception-handling blocks are implemented for transaction resilience.

Transaction Management: ACID Properties, Parallelism Control, Locking and Multi-Versioning Protocols, Recovery. Transaction isolation levels, including Read Committed and Serializable, are analyzed. Concurrency phenomena and lock escalation are simulated. Multi-Version Concurrency Control and write-ahead log crash recovery are examined.

Distributed Database Systems: Parallel Query Evaluation, Distributed Transaction Processing. Distributed database architectures and horizontal sharding are explored. Partition key selection is

practiced. Distributed two-phase commit protocols and parallel join strategies are analyzed.

Database Operators and Query Processing.

Relational operators are executed and performance-profiled. Physical join implementations, including Nested Loop, Hash Join, and Sort-Merge Join, are benchmarked. Aggregate operators are evaluated under varying memory constraints.

Working with Relations and Sequences, Sorting Data, Indexing.

Database relations and auto-incrementing sequences are created and managed. Complex multi-column ORDER BY clauses are used for querying. Composite multi-attribute indexes are built to support covering queries.

Working with Stored Procedures and Functions, Creating Triggers.

Server-side stored procedures and scalar/table-valued functions are developed. Encapsulated business logic is implemented. BEFORE, AFTER, and INSTEAD OF triggers are designed for automated auditing and referential integrity.

Self-study

The Relational Model of a Database and Its Main Concepts (Domain, Attribute, Tuple).

Designing a Database Using ER Diagrams and Converting Them into a Relational Schema.

Stages of Relational Database Normalisation (1NF, 2NF, 3NF and BCNF).

Creating Complex Queries and Using SQL Statements in Microsoft Access.

Optimising Indexing and Search Processes in a Database (B-Trees and Hash Indexes).

Transactions in a Database and Their ACID Properties.

Data Security and User Rights Management (Grant/Revoke Mechanisms).

Backup and Recovery Strategies.

Comparative Analysis of MySQL and PostgreSQL DBMSs: Advantages and Disadvantages.

NoSQL Databases: Types and Their Differences from Relational DBMSs.

	Object-Oriented Database Management Systems (ODBMS) and Their Applications. Distributed Database Architecture and Data Replication. Cloud Databases: Firebase, Amazon RDS and Google Cloud Spanner Technologies. Data Warehouse and OLAP Technologies for Working with Big Data. Integrating Databases with Modern Web Applications: ORM Technologies.
Teaching Methods	This course incorporates a blend of active problem-based learning, hands-on database lab exercises, and practical engineering projects to connect theoretical database foundations with software engineering applications. Interactive laboratory sessions provide direct experience in ER/EER schema modeling, complex SQL query design, database indexing, and stored procedure execution. Collaborative team projects focus on real-world database development, multi-table schema normalization, and ORM integration into modern web applications. Case study analyses and comparative research projects involving transaction management, query optimization, and distributed NoSQL systems are also conducted.
Prerequisites	Database Big Data and Data Analysis Foundations of Artificial Intelligence
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment reflects the student's theoretical knowledge in database engineering, practical skills in relational database design, proficiency in writing complex SQL queries and PL/SQL procedural blocks, implementation of security and access controls, and clear presentation of technical database solutions. The midterm assessment consists of written assignments, short in-class database quizzes, and the evaluation of independent study projects, such as

	ER diagram conversion models, normalisation case studies, SQL code generation and testing reports, index complexity and performance evaluations, and transaction ACID properties analysis. The final assessment is a written examination consisting of theoretical concepts, database schema design tasks, query execution plan optimizations, and practical database programming problems designed to assess students' mastery of core DBMS principles, concurrency control, distributed and cloud databases, NoSQL paradigms, and ORM integrations. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Murodova G.B. Database Management Systems. Practical manual. Durdona Publishing House, Bukhara, 2024. 148 pp. 2. Yury Bekarevich, Nina Pushkina. Self-teaching guide to Microsoft Access 2013. St. Petersburg: BHV-St. Petersburg, 2014. — 464 pp. 3. Ergashev B.T., Abidov K.Z. Database Management Systems. Tashkent: ILM ZIYO Publishing House, 2020. — 306 pages. 4. T.S.Gaipnazarov, D.Y.Irgasheva. Data Warehouse. Textbook. — Tashkent: "National Society of Uzbek Philosophers" Publishing House, 2017. — 224 pages. 5. Sh.Nazirov, A.Ne'matov, R.Qobulov, N.Mardonova. Database. Textbook. T.: "Sharq", 2007. — 200 pp. 6. S.Atajonova, B.Abdullayev. Creation of a Database System. Textbook. Omadbek Print Number One, 2024.
Responsible Lecturer	M.K.Ibragimova, teacher.

CRYPTOGRAPHY AND NETWORK SECURITY

Field	Description
Module Title	CRYPTOGRAPHY AND NETWORK SECURITY
Module Code	KTA405
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 60 hours (34 hours Lecture, 26 hours Practical lessons) Self-Assignments: 60 hours
Frequency/Duration	VII semester
Module objective	The objective of the Module Cryptography and Network Security is to enable students to design a comprehensive security architecture for modern information systems. Cryptographic protocols are applied in a network environment. Integrated cryptography and network security are implemented through hands-on learning. High-level professional competencies in managing the PKI infrastructure are developed. Students independently conduct a cybersecurity audit methodology.
Module Learning Outcomes	Knowledge: LO1. Can explain the integrated approach of cryptography and network security, and their combined role in modern information systems. LO2. Can explain the working mechanisms and security features of complex cryptographic protocols such as TLS 1.3, SSH, and IPsec IKEv2. LO3. Describe the lifecycle of the PKI infrastructure, certificate management and cryptographic keys. LO4. Can explain modern cryptographic attacks and post-quantum cryptography strategies. Skills: LO5. Configure cryptographic protocols in a complex network environment and verify their security. LO6. Installs the PKI infrastructure, implements the CA configuration and manages certificates. LO7. Applies penetration testing and cryptographic audit methods to identify cryptographic weaknesses. Competencies:

	LO8. Designs a comprehensive security architecture that integrates cryptography and network security in complex information systems. LO9. Makes sound decisions on applying the principles of Zero Trust and post-quantum cryptography to practical projects. LO10. Demonstrates the ability to independently manage cybersecurity audit and monitoring systems and to respond professionally to security incidents.
Content	Lectures An Integrated Approach to Cryptography and Network Security. The combined perspective needed to secure modern information systems is introduced. How cryptographic primitives support network-level security controls is explained. The layered relationship between data protection and network defense is discussed. The TLS 1.3 Protocol: New Features, the Handshake Mechanism and the 0-RTT Mode. The streamlined handshake process introduced in TLS 1.3 is explained. The removal of legacy cryptographic algorithms and improved forward secrecy are discussed. The 0-RTT mode and its trade-offs between performance and replay attack risk are covered. IPsec IKEv2: Enhanced Authentication and Key Management. The role of IKEv2 in establishing secure IPsec tunnels is explained. Improvements in authentication methods and mobility support over IKEv1 are discussed. Key exchange and security association management within IPsec are covered. SSH Protocol: Complex Configuration and Tunneling Techniques. The architecture and authentication mechanisms of the SSH protocol are reviewed. Advanced configuration options for secure remote access are explained. SSH tunneling techniques for secure data forwarding are discussed. DTLS and QUIC Protocols: Secure Communication over UDP. How DTLS adapts TLS security features for datagram-based communication is explained. QUIC is introduced as a modern transport protocol combining security and performance. Use cases where UDP-based secure

protocols are preferred over TCP-based ones are discussed.

Cryptographic APIs and Libraries: OpenSSL, libsodium, Bouncy Castle. The purpose and capabilities of widely used cryptographic libraries are reviewed. Design philosophies and use cases for OpenSSL, libsodium, and Bouncy Castle are compared. Best practices for integrating cryptographic libraries into applications securely are discussed.

PKI Architecture: Root CA, Intermediate CA and Certificate Chain. The hierarchical trust model underlying public key infrastructure is explained. The roles of root and intermediate certificate authorities are described. How certificate chains establish trust between endpoints is discussed.

Certificate Management: Life Cycle, CRL and OCSP Mechanisms. The stages of certificate issuance, renewal, and expiration are covered. The purpose and operation of Certificate Revocation Lists are explained. The Online Certificate Status Protocol as a real-time revocation-checking mechanism is discussed.

Hardware Security Module (HSM): Key Storage and Cryptographic Operations. The role of HSMs in securely storing and managing cryptographic keys is explained. How HSMs perform cryptographic operations without exposing key material is discussed. Typical deployment scenarios for HSMs in enterprise environments are reviewed.

Code Signing and Document Signing: Programme and Document Authentication. How digital signatures verify the authenticity and integrity of software and documents is explained. The certificate-based trust model underlying code signing is discussed. Practical applications in software distribution and legal documentation are reviewed.

Cryptographic Vulnerability Analysis: Side-Channel, Timing and Padding Oracle Attacks. How side-channel attacks exploit physical implementation characteristics is explained. Timing attacks and their exploitation of processing time variations are

discussed. Padding oracle attacks and their impact on cryptographic protocol security are covered.

Zero Trust Architecture: Microsegmentation and Cryptographic Identification. The core principles of the Zero Trust security model are introduced. Microsegmentation as a method of limiting lateral movement within networks is explained. The role of cryptographic identity verification in Zero Trust implementations is discussed.

Quantum Threats: Shor and Grover Algorithms, Impact on Cryptography. How Shor's algorithm threatens widely used public-key cryptosystems is explained. The impact of Grover's algorithm on symmetric cryptography key strength is discussed. The broader implications of quantum computing for current cryptographic standards are reviewed.

Post-Quantum Cryptography: NIST PQC Standards (CRYSTALS-Kyber, Dilithium). The NIST post-quantum cryptography standardization effort is introduced. The design principles behind CRYSTALS-Kyber for key encapsulation are explained. CRYSTALS-Dilithium is discussed as a post-quantum digital signature scheme.

Cybersecurity Audit Methodology: NIST CSF, ISO 27001, PCI DSS. The structure and purpose of the NIST Cybersecurity Framework are reviewed. The certification and control requirements of ISO 27001 are explained. PCI DSS compliance requirements for payment card data security are discussed.

Threat Intelligence and Cryptographic Monitoring Systems. The role of threat intelligence in proactive security defense is explained. Methods for monitoring cryptographic systems for anomalies and misuse are discussed. The integration of threat intelligence feeds into security operations is reviewed.

Future Directions for the Development of Cryptography and Network Security. Emerging trends shaping the future of cryptographic research are discussed. The evolving relationship between AI, automation, and security operations is explored. Long-term challenges facing cryptography in an increasingly connected world are reviewed.

Practical lessons

TLS 1.3 Configuration: nginx/Apache Server Setup and Testing. A web server is configured to support TLS 1.3 protocol settings. Certificates are generated and installed for secure server operation. The configuration is tested using tools such as OpenSSL and browser diagnostics.

IPsec IKEv2 Tunnel Configuration: Practical Work with strongSwan. strongSwan is installed and configured to establish an IPsec VPN tunnel. IKEv2 authentication and key exchange parameters are configured. Tunnel establishment and secure data transmission are verified.

SSH Tunnel and Port Forwarding: Complex Configuration Scenarios. Local, remote, and dynamic SSH port forwarding scenarios are set up. SSH key-based authentication for secure access is configured. Tunneled connections are tested to verify correct traffic routing.

Testing the DTLS Protocol: OpenSSL DTLS Client-Server Implementation. A basic DTLS client-server communication is implemented using OpenSSL. Certificates and cipher suites for the DTLS session are configured. Captured traffic is analyzed to verify secure UDP-based communication.

Working with the Cryptographic API: The Python cryptography Library. The Python cryptography library is used to perform encryption and decryption operations. Key generation, hashing, and digital signature functions are implemented. The correctness and security of the implemented cryptographic operations are tested.

Creating a Root CA and Intermediate CA: A Complete PKI Setup with OpenSSL. A Root CA and subordinate Intermediate CA are generated using OpenSSL. End-entity certificates signed by the Intermediate CA are issued. The complete certificate chain of trust is verified.

Certificate Revocation: CRL and OCSP Responder Configuration. A Certificate Revocation List is configured for issued certificates. An OCSP responder is set up for real-time certificate status checking. Revocation checking from a client application is tested.

Code Signing: Implementing the Process of Signing and Verifying Software. A sample software package is signed using a code-signing certificate. The digital signature is verified on the client side before execution. The security guarantees provided by the signing process are discussed.

Detecting Cryptographic Vulnerabilities: Padding Oracle and Timing Attack Analysis. A padding oracle vulnerability is simulated in a sample cryptographic implementation. Timing variations are measured to identify potential timing attack vectors. Mitigation strategies for the identified vulnerabilities are proposed.

Configuring and Testing a Zero Trust Network Architecture. Basic microsegmentation is implemented within a test network environment. Identity-based access controls following Zero Trust principles are configured. Unauthorized lateral movement is tested and verified to be prevented.

Testing Post-Quantum Algorithms: Working with the liboqs Library. The liboqs library is installed and configured for post-quantum cryptographic testing. Key encapsulation is implemented using a post-quantum algorithm such as CRYSTALS-Kyber. Performance characteristics are compared against classical cryptographic algorithms.

Cyber Security Audit: System Assessment Based on the NIST CSF. A security assessment of a sample system is conducted using the NIST Cybersecurity Framework. Gaps between current practices and framework recommendations are identified. Findings are documented and a remediation plan is proposed.

Final Comprehensive Project: Integrating Cryptography and Network Security. A security solution integrating multiple cryptographic protocols is designed and implemented. PKI, secure protocols, and monitoring are applied within a unified project scenario. The design choices made throughout the project are presented and defended.

Self-study

Information Entropy and Random Number Generator. One-Time Pad Model, the Concept of Padding.

	Stream and Block Ciphers. Key Length Security and Public/Private Key. Brute-Force Attack. Fundamentals of Cryptanalysis. PKI System. Hash Collision. Certificate Authority. Digital Certificate. RSA Weaknesses. SHA Algorithms. HMAC Fundamentals. Application of Digital Signature. TCP/IP Security. VPN Technology. Types of Firewalls. IDS/IPS Systems. The Role of NAT in Security. Secure Email Systems. Man-in-the-Middle Attack. Replay Attack. Fundamentals of SQL Injection. Fundamentals of XSS Attacks. Malicious Software. Social Engineering Techniques.
Teaching Methods	The module is delivered through interactive lectures, problem-based and case-study learning, laboratory practicals, and hands-on protocol configuration exercises. Computer simulations, seminars, group discussions, and project-based activities are also employed. Students configure and test cryptographic protocols, PKI systems, and network security mechanisms using tools such as OpenSSL, strongSwan, and the Python cryptography library. Technical, analytical, and problem-solving skills are developed through these practical exercises. Independent learning and digital educational resources further promote critical thinking and professional communication.
Prerequisites	Foundations of Cryptography Network Security Cyber Security Fundamentals Foundations of Artificial Intelligence
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory and practical session performance, protocol configuration exercises, seminar presentations, and the successful completion of

	independent study assignments, including entropy and random number analysis, PKI system exercises, cryptanalysis problem sets, and reports on network security attacks and countermeasures. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as a PKI infrastructure deployment report, a cryptographic vulnerability analysis, a presentation on a modern cryptographic protocol, a Zero Trust architecture case study, and an essay on post-quantum cryptography. The final assessment is a written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of cryptography and network security, analytical skills, and their ability to apply cryptographic principles and protocols to practical scenarios. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Katz J., Lindell Y. Introduction to Modern Cryptography. 3rd Edition. Chapman & Hall/CRC, 2021. 2. Boneh D., Shoup V. A Graduate Course in Applied Cryptography. Version 0.6, 2020. 3. Menezes A., van Oorschot P., Vanstone S. Handbook of Applied Cryptography. CRC Press, 1996. 4. Paar C., Pelzl J. Understanding Cryptography: A Textbook for Students and Practitioners. Springer, 2010. 5. Stallings W. Cryptography and Network Security: Principles and Practice. 8th Edition. Pearson, 2023. 6. Ryabko B. Ya., Fionov A. N. Cryptographic Methods of Information Protection. Moscow: Goryachaya Linia – Telekom, 2017. 7. Moldovan A. A., Moldovan N. A. Cryptography. Saint Petersburg: BHV-St. Petersburg, 2004. 8. Domarev V. V. Information Technology Security: A Systemic Approach. Kyiv: DIASoft, 2004.

	9. Romanets Yu. V., Timofeyev P. A., Shankin V. F. Protection of Information in Computer Systems and Networks. Moscow: Radio and Communication, 2018. 10. Khorosko V. A., Chekatkov A. A. Methods and Means of Information Protection. Kyiv: Yuniop, 2003.
Responsible Lecturer	Shaymanov Farxod Xushbakovich.

MATHEMATICAL MODELING OF NONLINEAR PROCESSES

Field	Description
Module Title	MATHEMATICAL MODELING OF NONLINEAR PROCESSES
Module Code	CHJMM405
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 60 hours (34 hours Lecture, 26 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	VII semester
Module objective	The aim of the Module Mathematical Modeling of Nonlinear Processes is to equip students with theoretical knowledge and practical skills in the mathematical modelling of nonlinear processes. Students formulate mathematical models of complex nonlinear dynamic systems. Their behaviour is analysed. Computer-based simulations are performed. Computational results are interpreted. Modern mathematical methods and software tools are applied to solve practical engineering, scientific, and socio-economic problems.
Module Learning Outcomes	Knowledge: LO1. Knows the theoretical foundations, principles and areas of application of mathematical modelling of non-linear processes. LO2. Understands nonlinear differential equations, dynamic systems and the methods for constructing their mathematical models. LO3. Is familiar with the fundamental concepts of stability, equilibrium states, bifurcation and chaotic processes in nonlinear systems.

	LO4. Is familiar with the theoretical foundations of methods for analysing mathematical models, numerical solution and computer simulation. LO5. Is familiar with the capabilities and tools for modelling and analysing nonlinear processes in MATLAB, Python and other modern computational environments. Skills: LO6. They will be able to analyse real processes and systems, construct their nonlinear mathematical models and determine their parameters. LO7. They can solve nonlinear differential equations and dynamical systems using analytical and numerical methods and interpret the results. LO8. Can implement mathematical models on a computer and conduct simulation experiments using MATLAB, Python or other modern software tools. LO9. Has the skills to analyse and evaluate the stability, dynamic behaviour and control characteristics of nonlinear systems. LO10. Can visualise modelling results, analyse them, draw scientifically grounded conclusions and apply them to solving practical problems. Competencies: LO11. They will have the competence to mathematically model complex nonlinear processes and systems encountered in various fields and to develop appropriate models for them. LO12. They can carry out professional activities involving the analysis, forecasting and optimisation of the dynamic characteristics of processes based on mathematical models. LO13. Using modern computing technologies and software packages, they can independently organise modelling, simulation and numerical experiments. LO14. They will have the competence to solve practical problems arising in technical, economic, ecological and information systems using mathematical modelling methods. LO15. Can analyse the obtained results on a scientific basis, present them in graphical, tabular and report form, and apply them in decision-making processes.
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	LO16. Using interdisciplinary knowledge, they can develop and implement modelling projects integrated with artificial intelligence, data analysis and digital technologies. LO17. They will have the competence to conduct research in the field of mathematical modelling, to explore new methods and to apply innovative approaches in their professional practice.
Content	Lectures Introduction to the Mathematical Modelling of Nonlinear Processes. Application Areas in Computer Engineering. The scope and objectives of studying nonlinear process modelling are introduced. Foundational concepts distinguishing nonlinear from linear phenomena are discussed. Typical applications of nonlinear modelling within computer engineering are reviewed. Linear and Non-Linear Models. The Role of Mathematical Modelling in Software Systems. The structural and behavioral differences between linear and nonlinear models are compared. How mathematical modelling supports the design and analysis of software systems is explained. Criteria for selecting an appropriate modelling approach for a given problem are discussed. Non-Linear Functions and Their Analysis Using Computer Graphics. The properties and classification of common nonlinear functions are reviewed. How computer graphics tools aid in visualizing function behavior is demonstrated. Techniques for interpreting graphical representations of nonlinear relationships are discussed. Algorithm Complexity and Non-Linear Dependencies. How algorithm complexity is characterized using nonlinear growth functions is explained. Common complexity classes and their practical implications are discussed. Methods for analyzing and comparing algorithm efficiency are reviewed. Nonlinear Algebraic Equations and Their Modelling. The structure and classification of nonlinear algebraic equations are introduced. Approaches for formulating such equations from real-

world problems are explained. The challenges nonlinear equations pose compared to linear counterparts are discussed.

Numerical Algorithms for Solving Nonlinear Equations. Iterative numerical methods, such as bisection, Newton's method, and fixed-point iteration, are reviewed. Convergence criteria and conditions for method reliability are discussed. The computational efficiency of different numerical approaches is compared.

Modelling of Iterative Processes and Recursive Algorithms. The mathematical basis of iterative and recursive computation is explained. How recursive algorithms can be modelled as nonlinear processes is discussed. Convergence and stability considerations for iterative modelling are reviewed.

Nonlinear Systems of Equations and Their Solution on a Computer. Methods for representing systems of nonlinear equations computationally are introduced. Numerical techniques for solving such systems, including Newton-Raphson for systems, are discussed. Practical considerations for implementation and convergence are reviewed.

Modelling Traffic and Load Processes in Computer Networks. Mathematical models used to represent network traffic behavior are introduced. Queuing theory concepts applied to load and congestion analysis are discussed. How nonlinear effects arise in network traffic dynamics is reviewed.

Mathematical Models of Server Systems and Service Processes. Modelling approaches for server request handling and service processes are explained. Key performance metrics derived from such models are discussed. The application of queuing and stochastic models to server systems is reviewed.

Modelling and Analysis of Data Flows. Methods for representing and analyzing data flow within computing systems are introduced. Techniques for identifying bottlenecks and inefficiencies in data flow are discussed. Practical tools used for data flow modelling and analysis are reviewed.

Modelling Dynamic Systems Based on Differential Equations. How differential equations represent the evolution of dynamic systems over time is explained. Common formulations used in engineering and computing contexts are discussed. Solution approaches and their interpretation for system behavior are reviewed.

Fundamentals of Mathematical Modelling in the Python Programming Language. Python is introduced as a tool for implementing mathematical models. Core programming constructs relevant to modelling tasks are reviewed. Best practices for structuring modelling code in Python are discussed.

Computing Non-Linear Models Using the NumPy and SciPy Libraries. The core functionalities of NumPy and SciPy for numerical computation are explained. How these libraries support solving and analyzing nonlinear models is demonstrated. Practical examples of applying these tools to modelling problems are reviewed.

Mathematical Modelling of the Performance of Computer Systems. Modelling techniques for evaluating computer system performance are introduced. Key performance indicators and their mathematical representation are discussed. Methods for predicting system behavior under varying workloads are reviewed.

Nonlinear Models in Artificial Intelligence and Machine Learning. The role of nonlinear functions in machine learning models is explained. How nonlinearity enables complex pattern recognition in AI systems is discussed. Examples of nonlinear modelling within neural networks and related methods are reviewed.

Modern Problems and Prospects of Mathematical Modelling in the Field of IT Services. Current challenges facing mathematical modelling in IT service contexts are discussed. Emerging trends and technologies shaping future modelling approaches are explored. The evolving role of modelling in supporting IT service innovation is reviewed.

Practical lessons

Installation of the Python Environment. Working with Jupyter Notebook and Mathematical Computing Packages. Python is installed and a working development environment is configured. Jupyter Notebook is set up for interactive computation. Core mathematical packages, such as NumPy, SciPy, and Matplotlib, are installed and tested.

Plotting and Analysing Linear and Non-Linear Functions Using Python. Python code is written to generate plots of linear and nonlinear functions. Key features of function graphs, including extrema and asymptotic behavior, are analyzed. The visual characteristics distinguishing linear from nonlinear functions are compared.

Experimental Analysis of the Time Complexity of Algorithms. Sample algorithms are implemented and their execution time is measured empirically. Runtime data is plotted against input size to observe complexity trends. Experimental results are compared with theoretical complexity expectations.

Graphical Solution of Nonlinear Algebraic Equations. Nonlinear equations are plotted to visually identify approximate root locations. Graphical solutions are verified through numerical substitution. The limitations of graphical methods for precise root-finding are discussed.

Implementing the Bisection and Successive Approximation Methods for Dividing a Segment into Two Equal Parts in Python. The bisection method is coded to iteratively narrow down root intervals. Successive approximation techniques for equation solving are implemented. Convergence speed and accuracy between the two methods are compared.

Programming the Simple Iteration and Newton Methods. The simple iteration method for solving nonlinear equations is implemented. Newton's method is coded and tested on sample functions. Convergence behavior and computational efficiency of both methods are compared.

Solving a System of Nonlinear Equations Using Numerical Methods. A system of nonlinear

equations is formulated for numerical solution. A suitable numerical method, such as Newton-Raphson for systems, is implemented. Solution accuracy and convergence behavior are verified and analyzed.

Creating and Analysing a Simple Mathematical Model of Network Traffic. A basic mathematical model representing network traffic behavior is built. The model is implemented in Python and sample simulations are run. Model output is analyzed to identify traffic patterns and congestion points.

Building and Software Implementation of a Server Queue System Model. A queuing model representing server request handling is developed. The model is implemented computationally to simulate service processes. Key performance metrics, such as wait time and queue length, are analyzed.

Modelling and Visualising Changes in Data Flow. A model representing data flow within a computing system is constructed. Visualization tools are implemented to display data flow changes over time. Visualized results are interpreted to identify flow irregularities or bottlenecks.

Solving Differential Equations Using the Euler Method and Analysing Them by Means of Graphs. The Euler method is implemented to numerically solve differential equations. Graphs illustrating the solution's behavior over time are generated. Accuracy and stability considerations of the numerical solution are analyzed.

Computing Nonlinear Models Using the NumPy and SciPy Libraries. Nonlinear model computations are implemented using NumPy and SciPy functions. Library-based numerical solvers are applied to sample modelling problems. Custom-coded solutions are compared with library-based implementations.

Creating a Mathematical Model of a Computer Network, Server or IT Service Process and Its Software Analysis. A mathematical model for a chosen real-world system is selected and formulated. The model is implemented in Python and simulation experiments are conducted. Results are analyzed and

	presented, with conclusions drawn relevant to the chosen system. Self-study Subject Matter and Objectives of the Discipline of Mathematical Modelling of Nonlinear Processes. The Concept of a Mathematical Model and the Stages of Modelling. Linear and Nonlinear Models: Similarities and Differences. Nonlinear Functions and Their Main Properties. Analysis of Nonlinear Functions Using Computer Graphics. Fundamentals of Mathematical Model Identification. Modelling Errors and Assessment of Model Adequacy. Capabilities of MATLAB and Python Software Environments for Mathematical Modelling. The Concept and Classification of Nonlinear Differential Equations. First-Order Nonlinear Differential Equations. Numerical Methods for Solving Nonlinear Differential Equations. Mathematical Models. Equilibrium Points and Methods for Their Determination. Mathematical Models of Oscillatory Processes. The Predator–Prey (Lotka–Volterra) Mathematical Model. Analysis of the SIR Epidemiological Model. The Concept and Types of Stability. Lyapunov Stability Theory. Methods for Constructing Lyapunov Functions. Fundamental Concepts of Bifurcation Theory. Hopf Bifurcation and Periodic Solutions. Chaotic Processes and Deterministic Chaos. The Lorenz System and Its Dynamic Behaviour. Nonlinear Mathematical Models of Technical Systems. Nonlinear Processes in Electrical and Electronic Systems. Mathematical Modelling of Mechanical Systems.
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	Development of Nonlinear Mathematical Models of Economic Processes. Optimisation Problems and Nonlinear Optimisation Methods. Application of Artificial Intelligence Methods in Mathematical Modelling. Development of a Mathematical Model for a Real-World Process.
Teaching Methods	The module is delivered through interactive lectures, problem-based and inquiry-based learning, and computational laboratory sessions. Software-based simulations, seminars, group discussions, and project-based activities are also employed. Students construct and analyze nonlinear mathematical models using Python, MATLAB, and specialized numerical libraries such as NumPy and SciPy. Analytical, computational, and research skills are developed through these activities. Independent learning and digital educational resources further promote critical thinking and scientific communication.
Prerequisites	Calculus Differential Equations Linear Algebra Programming
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation, laboratory and computational session performance, programming exercises, seminar presentations, and the successful completion of independent study assignments, including nonlinear function analysis, numerical method implementation, dynamic system simulations, and stability analysis reports. The midterm assessment consists of written assignments, short in-class quizzes, and the evaluation of independent study projects, such as a report analyzing the Predator–Prey (Lotka–Volterra) model, a bifurcation analysis case study, a presentation on a real-world nonlinear modelling application, and an

	essay on chaotic dynamics. The final assessment is a written examination consisting of theoretical questions and problem-solving tasks designed to assess students' understanding of mathematical modelling of nonlinear processes, analytical skills, and their ability to apply modelling methods to practical problems. Successful completion of all independent study assignments is required as an essential component of the module assessment.
Recommended Literature	1. Isralov M.I. Computational Methods. Part 1. – Tashkent: O'qituvchi, 2003. – 450 pp. 2. Israyelov M.I. Computational Methods. Part 2. – Tashkent: O'qituvchi, 2008. – 340 pp. 3. S.S.Irisqulov, K.D. Ismanova, M. Olimov, A. Imomov. Numerical methods and algorithms. – N.: "Namangan" Publishing House, 2013, 244 pp. 4. Kalitkin N.N., Alshina E.A. Numerical Methods: in 2 vols. Numerical Analysis: textbook for students of higher professional education institutions. - M.: Publishing Centre "Academy", 2013. Vol. 1. - 304 pp. Vol. 2. - 304 pp. 5. Giordano F.R., Weir M.D., Fox W.P. A First Course in Mathematical Modeling. – 5th Edition. – Boston: Cengage Learning, 2014. 6. Strogatz S.H. Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry and Engineering. – 2nd Edition. – Boulder: Westview Press, 2015. 7. Hirsch M.W., Smale S., Devaney R.L. Differential Equations, Dynamical Systems and an Introduction to Chaos. – 3rd Edition. – Academic Press, 2013. 8. Chapra S.C., Canale R.P. Numerical Methods for Engineers. – 8th Edition. – McGraw-Hill Education, 2020. 9. Richard L. Burden and J. Douglas Faires. Numerical Analysis. 9th Edition. Youngstown State University Press, 2011. – 895 p. 10. L. Ridgway Scott. Numerical Analysis. Princeton University Press, 2011. – 342 p.

	11. Ismatullayev G.P., Pulatov S.I., Fayazov Q.S. Handbook of Numerical Methods. – Tashkent, University, 2006. – 140 pp. 12. Bakvalov N. S., Zhidkov N. P., Kobelkov G. M. Numerical Methods. – Moscow: Bionom Publishing House, Laboratory of Knowledge, 2011. – 640 pp.
Responsible Lecturer	Xolliyev Faxriddin Boxodirovich, Doctor of Philosophy (PhD) in Physical and Mathematical Sciences.

APPLICATION SOFTWARE PACKAGES

Field	Description
Module Title	APPLICATION SOFTWARE PACKAGES
Module Code	ADP405
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 60 hours (34 hours Lecture, 26 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	VII semester
Module objective	The objective of the Module Application Software Packages is to provide theoretical knowledge and practical skills in modern application software packages. Students learn to effectively use word processors for document creation. Spreadsheets are used for data analysis. Presentation software is used for professional communication. Databases are studied for organizing information. Cloud technologies are used for collaboration and document management.
Module Learning Outcomes	Knowledge: LO1. Know types, functions and application areas of practical software packages. LO2. Master creation, editing and formatting of documents using word processors. LO3. Master use of spreadsheets including formulas and functions. LO4. Apply data analysis and visual presentation methods.

	LO5. Understand basic principles of databases and apply them in practice. LO6. Develop skills in solving professional problems using software packages. Skills: LO7. Create, format and manage text documents professionally. LO8. Develop spreadsheets with complex formulas and data analysis. LO9. Create presentations with design and multimedia elements. LO10. Design and manage databases with queries and reports. LO11. Use cloud technologies for collaboration and document management. LO12. Apply information security principles in software use. Competencies: LO13. Apply modern software packages effectively in professional activities. LO14. Organize, manage and analyze digital information. LO15. Demonstrate competence in improving productivity and communication. LO16. Follow digital ethics and information security principles.
Content	Lectures Introduction to Practical Software Packages. Software packages are defined and classified according to their types and functions. Modern software applications and their uses are examined. The relationship between operating systems and applications, along with development trends in software, is discussed. Interrelationship Between Operating Systems and Applications. Operating system functions and roles are studied. Application installation, configuration, and system resource requirements are examined. Compatibility considerations and performance optimization are discussed. Word Processors and Their Features. Microsoft Word's overview and capabilities are introduced. Text document creation, editing, and formatting options are

examined. Document structure, organization, and advanced features such as templates are discussed.

Technologies for Creating, Formatting and Editing Text Documents. Text input and editing procedures are studied. Font and paragraph formatting, along with page setup and printing, are examined. Headers, footers, page numbers, and document protection and sharing are discussed.

Electronic Spreadsheets and Their Functions. Microsoft Excel's overview and capabilities are introduced. Spreadsheet structure, navigation, and cell data entry are examined. Row and column operations, along with data types and formatting, are discussed.

Organizing Formulas, Functions and Calculations. Formula syntax and operators are studied. Built-in functions for calculations, including statistical and mathematical functions, are examined. Logical and text functions, along with error handling in formulas, are discussed.

Methods for Data Analysis and Creating Charts. Data sorting and filtering techniques are studied. Subtotals, summaries, and pivot table creation are examined. Chart types, selection, formatting, and analysis are discussed.

Presentation Programs and Their Features. Microsoft PowerPoint's overview and capabilities are introduced. Slide creation, design, layout, and master slides are examined. Text and content addition, along with animation and transitions, are discussed.

Professional Presentation Design and Delivery. Design principles and best practices for presentations are studied. Color schemes, fonts, visual hierarchy, and balance are examined. Audience engagement techniques and presentation delivery skills are discussed.

Database Concept and Management Systems. Database fundamentals and structure are studied. Database models, relationships, and DBMS software features are examined. Data organization principles and query and reporting techniques are discussed.

Creating Databases in Microsoft Access. The Access interface and its components are introduced.

Table creation, design, and field types and properties are examined. Primary keys, indexes, and data entry and validation are discussed.

Graphic Editors and Multimedia Software. Image editing basics and tools are studied. Graphics file formats and multimedia creation and editing are examined. Audio and video processing, along with design software applications, are discussed.

Information Security and Data Protection. Data security principles and practices are studied. Password protection, encryption, and user access controls are examined. Backup and recovery procedures, along with malware protection, are discussed.

Cloud Technologies and Collaboration. Google Workspace and Microsoft 365 are introduced. Cloud storage, sharing, and online document collaboration are examined. Real-time editing, comments, and version control are discussed.

Network Security Fundamentals. Network security concepts are studied. Firewalls, access control, and user authentication methods are examined. Data encryption techniques and security policies and procedures are discussed.

Practical lessons

Installing and Configuring Operating Systems and Applications. Operating systems and applications are installed following standard procedures. System and software configurations are set up. Hardware compatibility is verified and installation issues are troubleshoot.

Creating and Formatting Text Documents in Word. Text documents are created in Microsoft Word. Text is input, edited, and formatted at the paragraph level. Documents are saved, organized, and managed using file management techniques.

Working with Tables, Images and SmartArt Objects. Tables are inserted and formatted within documents. Images are inserted and positioned, and SmartArt diagrams are created. Object formatting, properties, and layout management are practiced.

Creating Styles, Tables of Contents and Links. Styles are defined and applied to documents. An

automatic table of contents is created and updated. Cross-references, bookmarks, and document navigation are configured.

Creating Spreadsheet and Entering Data. A spreadsheet is created and set up. Data is entered using appropriate techniques and validated. Cells are formatted and organized using standard methods.

Using Formulas and Standard Functions. Basic formulas are created and standard functions are inserted. Formulas are copied, modified, and combined into complex expressions. Formula errors are checked and corrected.

Sorting, Filtering and Conditional Formatting. Data is sorted using various procedures and options. Filters are applied based on specified criteria, and conditional formatting rules are set. AutoFilter is used to control data visibility.

Creating Charts and Graphs. Appropriate chart types are selected and created from data. Axes, legends, titles, and labels are configured. Charts are formatted for clarity and presentation.

Creating Presentations in PowerPoint. A presentation is created in PowerPoint. Slides are added, laid out, and designed. Content is inserted and an appropriate theme is selected.

Working with Design, Animation and Transitions. Animation effects are selected and timed appropriately. Slide transitions are applied and master slides are edited. Custom animations and interaction buttons are configured.

Creating Databases in Access. A database is created and set up in Microsoft Access. Tables are designed with defined relationships. Field properties, constraints, and primary keys are configured.

Creating Queries, Forms and Reports. Queries are created with specified criteria. Forms are designed with appropriate controls, and reports are generated and formatted. Data is retrieved, exported, and printed.

Collaborative Work in Cloud Services. Cloud accounts are set up and files are uploaded and shared. Collaborative editing is practiced with

	comments and version history. Integration between different cloud services is configured. Self-study History of Practical Software Package Development. Features of Modern Operating Systems. Types and Classification of Software. Fundamental Functions of Microsoft Word. Electronic Document Management Systems. Modern Text Document Formatting Methods. Technology for Automatic Table of Contents. Role of Microsoft Excel in Practical Problems. Using Mathematical and Logical Functions. Performing Financial Calculations. Using Pivot Tables for Data Analysis. Importance of Charts and Graphs. Rules for Creating Effective Presentations. Principles of Presentation Design. Role of Multimedia in Presentations.
Teaching Methods	Interactive lectures with software demonstrations are conducted. Hands-on practical sessions with real-world examples and problem-based learning scenarios are held. Project-based activities with actual applications are carried out. Peer collaboration and group work are encouraged. Independent exploration of advanced features is supported through sample files, templates, and practice with diverse document types.
Prerequisites	Computer Literacy and Information Technology Fundamentals Fundamentals of Programming and Algorithms Fundamentals of Operating Systems
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation in practical sessions, practical proficiency with software tools during hands-on work, and the completion of teacher-guided independent work assignments, including formatted documents, spreadsheet projects,

	presentations, database designs, and mini-projects using various applications. The midterm assessment consists of a written test covering theoretical knowledge of software packages, file management, formatting techniques, formulas and functions, database concepts, and cloud technologies, together with independent work projects evaluating practical skills in document creation, data analysis, and presentation design. The final examination is a practical examination requiring students to create documents, build spreadsheets with calculations, design presentations, create database structures, and demonstrate proficiency with multiple software packages. The assessment measures practical skills and the ability to apply tools effectively.
Recommended Literature	1. Joan Lambert. Microsoft Office 2021 Step by Step. Microsoft Press, 2021. 2. Timothy J. O'Leary, Linda O'Leary. Computing Essentials 2023. McGraw-Hill Education, 2023. 3. Brian K. Williams, Stacey Sawyer. Using Information Technology. McGraw-Hill Education. 4. Microsoft Office Online Documentation. Microsoft Learn, https://learn.microsoft.com
Responsible Lecturer	Ikromov Husniddin Abdivoidovich, Department of Computer and Software Engineering, Termez State University

ADMINISTRATIVE MANAGEMENT OF COMPUTER NETWORKS

Field	Description
Module Title	ADMINISTRATIVE MANAGEMENT OF COMPUTER NETWORKS
Module Code	KTMB405
Credit Points (ECTS)	5 Credits
Workload	Total workload: 150 hours Contact hours: 60 hours (34 hours Lecture, 26 hours Practical lessons) Self-Assignments: 90 hours
Frequency/Duration	VII semester
Module objective	The objective of the Module Administrative Management of Computer Networks is to provide

	theoretical and practical knowledge on organizing, configuring, managing, and securing computer networks. Students learn to manage LANs, WANs, and corporate networks. Network devices are configured. Network services are installed. Network performance is monitored. Network security is ensured through modern administration practices.
Module Learning Outcomes	Knowledge: LO1. Understand main types, architecture and operating principles of computer networks. LO2. Know network devices and their functions: routers, switches, servers. LO3. Understand IPv4 and IPv6 addressing systems and subnetting. LO4. Know routing fundamentals and routing protocols. LO5. Understand network services: DHCP, DNS, NAT. LO6. Know network security principles, firewalls and VPN technologies. Skills: LO7. Design and plan network infrastructure. LO8. Configure and manage network devices. LO9. Install and manage network services. LO10. Manage users and groups in network environments. LO11. Implement network security measures. LO12. Monitor and diagnose network problems. Competencies: LO13. Demonstrate systems thinking and analytical skills. LO14. Design secure network infrastructure. LO15. Make strategic security decisions. LO16. Optimize network efficiency and reliability. LO17. Troubleshoot and resolve network issues. LO18. Manage corporate network environments effectively.
Content	Lectures Introduction to Computer Networks and Basic Concepts. The definition and importance of computer networks are introduced. Network types, including LANs, WANs, and MANs, and their classifications are

examined. The historical development of networks and their impact on organizations and society are discussed.

OSI and TCP/IP Network Models. The layers and functions of the OSI model are studied. The structure of the TCP/IP model and the protocol stack are examined. The two models are compared, and data flow through the layers and practical applications are discussed.

Network Devices and Their Functions. Routers and their routing functions are studied. Switches, switching operations, and servers and their services are examined. Network interface cards and device interconnection are discussed.

IPv4 Addressing System. The structure and notation of IP addresses are studied. Class-based addressing, including Classes A through E, and default masks are examined. Address ranges, limitations of IPv4, and the transition to IPv6 are discussed.

IPv6 Addressing System. The IPv6 address format and length are studied. Address types, including unicast, multicast, and anycast, are examined. Advantages over IPv4, migration strategies, and dual-stack implementations are discussed.

Subnetting and Network Segmentation. The concept and advantages of subnetting are studied. Calculation methods, subnet masks, and CIDR notation are examined. Variable length subnetting and network design considerations are discussed.

Routing Fundamentals. Routing definitions and routing tables are studied. The routing decision process, including directly connected and default routes, is examined. Static versus dynamic routing and routing metrics are discussed.

Switching Technologies and VLAN. Switch operation and switching tables are studied. The VLAN concept, segmentation, and configuration are examined. Inter-VLAN communication, trunk ports, and the VTP protocol are discussed.

DHCP and DNS Services. The DHCP protocol, its operation, and server configuration are studied. IP address allocation and leasing are examined. DNS

fundamentals, the resolution process, and DNS records and zones are discussed.

Network Operating Systems. Windows Server architecture and functions are studied. Linux networking fundamentals are examined. Server roles, services, installation, configuration, and networking capabilities are discussed.

Managing Users and Groups. User account concepts are studied. Group creation, management, access control, and permissions are examined. Group policies, user authentication methods, and Active Directory basics are discussed.

Network Security Fundamentals. Security threats and vulnerabilities are studied. The principles of confidentiality, integrity, and availability are examined. Defense mechanisms, security policies, risk assessment, and security architecture are discussed.

Firewalls and Access Control Systems. Firewall types and functions are studied. Packet filtering, stateful inspection, and application-level gateways are examined. ACL configuration and NAT functionality for security are discussed.

Network Monitoring and Diagnostic Tools. Monitoring concepts, objectives, and network performance metrics are studied. Monitoring tools, including ping, traceroute, and netstat, are examined. The SNMP protocol, MIBs, and syslog log analysis are discussed.

Network Fault Detection and Troubleshooting. Common network problems and troubleshooting methodology are studied. Documentation and tracking practices are examined. Packet sniffing with Wireshark, performance optimization, and proactive maintenance are discussed.

VPN and Remote Access Technologies. VPN concepts and benefits are studied. Tunneling protocols, including PPTP, L2TP, and IPsec, and SSL/TLS VPNs are examined. Remote access solutions and their configuration and security considerations are discussed.

Management and Automation of Corporate Networks. Network management frameworks, including SNMP and NETCONF, are studied.

Configuration management and automation tools and scripts are examined. Orchestration platforms and security automation are discussed.

Practical lessons

Designing Network Topologies in Cisco Packet Tracer. Network topologies are designed using Cisco Packet Tracer. Device placement, connectivity, and cable selection are practiced. Simulation models are created, tested, and developed into practical scenarios.

Practical Analysis of OSI and TCP/IP Models. The OSI and TCP/IP models are analyzed layer by layer. Protocols are identified at each layer and data transformation is examined. Packet structure is examined and the two models are compared.

Connecting and Configuring Network Devices. Switches and routers are configured with basic settings. Device communication and configuration are verified. CLI basics and management access are practiced.

IPv4 Addressing and IP Configuration. IP addresses are assigned using static and DHCP configuration. Address assignments are verified and tested. Configuration procedures are documented.

Practical Subnetting Calculations. Subnet masks are calculated for given network requirements. Address ranges and host numbers are determined. Network design exercises are solved through practical problem solving.

Configuring Static Routing. Static routes are created and default routes are configured. Routing tables are examined for multiple route configurations. The configured routes are tested and verified.

Configuring Dynamic Routing Protocols. RIP and OSPF protocols are configured and compared. Neighbor discovery and convergence behavior are observed. Routing metrics are analyzed for each protocol.

Creating VLANs and Segmentation. VLANs are created and named, and ports are assigned to them. Trunk configuration and inter-VLAN routing are implemented. VLAN isolation and communication are tested.

	Configuring DHCP Server. A DHCP scope and IP pool are configured. Lease times and DHCP options are set. Client connectivity and address allocation are tested and verified. Setting Up DNS Server. DNS zones are created and records, including A, MX, and CNAME, are configured. Primary and secondary zones are set up. DNS resolution and recursive queries are tested. Managing Users in Server Environments. User accounts are created and configured. Password policies and group assignments are applied. Permissions, home directories, and user profiles are managed. Configuring Network Security. Firewall rules and access control lists are created. Port filtering and protocol-specific rules are configured. NAT is configured and security policies are implemented. Network Monitoring and Diagnostics. Network diagnostic tools, including ping, traceroute, and ipconfig, are used. Network traffic is captured and analyzed using Wireshark. Performance is monitored and problems are identified. Self-study Network Theory and Evolution. Network Technologies and Ethernet. Addressing Systems IPv4 and IPv6. Fundamentals of Routing and Switching. Network Services DHCP, DNS, NAT. Security Technologies: Firewall, VPN, Cryptography. Wireless Networks and Monitoring. Server Network Services. Modern Architectures: Cloud and Corporate.
Teaching Methods	Interactive lectures with network diagrams and scenarios are conducted. Hands-on practical sessions with Cisco Packet Tracer and real equipment are held. Lab experiments with actual network devices, network simulation, and modeling are practiced. Troubleshooting case studies and group projects are conducted. Independent research on emerging technologies and professional tool demonstrations are also included.

Prerequisites	Computer Networks Computer Organization
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation in practical sessions, network configuration and troubleshooting ability during hands-on work, and the completion of teacher-guided independent work assignments, including network designs, configuration reports, analysis documents, and mini-projects on network administration and security. The midterm assessment consists of a written test covering theoretical knowledge of network models, addressing systems, routing, protocols, and security concepts, together with independent work projects evaluating practical skills in network design, device configuration, service implementation, and problem diagnosis. The final examination is a written and practical examination requiring students to design networks, configure devices, implement services, diagnose problems, and explain design decisions. The assessment measures theoretical understanding, practical skills, and the ability to manage network environments effectively.
Recommended Literature	1. Cisco Certified Network Associate (CCNA) Training Book. Ciscopress, Latest Edition. 2. Kurose J., Ross K. Computer Networking: A Top-Down Approach. Pearson. 3. Jill West. Network+ Guide to Networks. McGraw-Hill. 4. Darril Gibson. CompTIA Security+: Get Certified Get Ahead. Sybex.
Responsible Lecturer	Abdullayev Mamayusup Kasimovich, Department of Computer and Software Engineering, Termez State University

ARTIFICIAL INTELLIGENCE IN CYBERSECURITY

Field	Description
Module Title	ARTIFICIAL INTELLIGENCE IN CYBERSECURITY
Module Code	KXSI404
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 46 hours (26 hours Lecture, 20 hours Practical lessons) Self-Assignments: 74 hours
Frequency/Duration	VII semester
Module objective	The objective of the Module Artificial Intelligence in Cybersecurity is to develop theoretical knowledge and practical skills in applying artificial intelligence and machine learning to cybersecurity. Students learn threat detection using AI-based systems. Network monitoring and log analysis are studied. Anomaly detection techniques are examined. Intrusion detection using AI-based systems is practiced. Machine learning models are implemented and evaluated for security applications.
Module Learning Outcomes	Knowledge: LO1. Explain fundamental concepts of cybersecurity and artificial intelligence and their interrelationship. LO2. Describe main concepts of machine learning: classification, clustering, anomaly detection. LO3. Explain main types of cybersecurity threats, attack models and defence mechanisms. LO4. Explain structure of security data and fundamentals of data processing for AI models. LO5. Understand anomaly detection, intrusion detection systems (IDS) and SIEM systems. LO6. Describe network traffic analysis and log file analysis techniques. Skills: LO7. Classify cybersecurity data using machine learning models. LO8. Analyze network traffic and system logs to identify suspicious patterns. LO9. Develop conceptual models of AI-based intrusion detection systems.

	LO10. Detect spam and phishing using text classification methods. LO11. Identify and classify malware using feature-based approaches. LO12. Implement and evaluate simple machine learning models for security. Competencies: LO13. Use AI tools responsibly in cybersecurity and professional activities. LO14. Make informed decisions on analyzing cybersecurity incidents with AI support. LO15. Understand capabilities and limitations of AI-based security systems. LO16. Apply AI techniques to real-world cybersecurity problems and challenges.
Content	Lectures Fundamentals of Cybersecurity and Artificial Intelligence. The definition and importance of cybersecurity are introduced. The basics of AI and machine learning are examined. The interrelationship between AI and security, along with current trends and challenges, is discussed. Concept of Artificial Intelligence and Its Types. AI is defined and classified into narrow and general AI. Reactive, limited memory, and theory of mind AI types are examined. AI applications in security and the distinction between machine and human intelligence are discussed. Fundamentals of Machine Learning. Supervised learning, including classification and regression, is studied. Unsupervised learning, including clustering and dimensionality reduction, and reinforcement learning basics are examined. Training and testing data and model evaluation metrics are discussed. Cybersecurity Threats and Attack Models. The threat landscape and vulnerability analysis are studied. Attack types, including malware, ransomware, and phishing, and their attack vectors are examined. Defense mechanisms, threat modeling frameworks, and risk assessment are discussed. Dataset Concept for Data and Security. Data collection sources and data labeling and annotation are studied. Feature engineering for security and

dataset preparation and cleaning are examined. Class imbalance issues and privacy considerations are discussed.

Fundamentals of Anomaly Detection. Anomaly definitions and types are studied. Statistical and machine learning approaches to anomaly detection are examined. Outlier detection techniques, applications in network monitoring, and performance evaluation are discussed.

Fundamentals of Intrusion Detection Systems. IDS types and architectures are studied. Signature-based, anomaly-based, and hybrid detection approaches are examined. Rule engines and alert generation and management are discussed.

Fundamentals of Spam and Phishing Detection. Spam characteristics and evolution are studied. Phishing tactics, indicators, and text classification methods are examined. Email filtering techniques, user behavior analysis, and detection challenges are discussed.

Methods for Detecting Malicious Software. Malware types and behaviors are studied. Static and dynamic analysis approaches, including sandboxing, are examined. Heuristic-based and behavior-based detection, along with polymorphic malware challenges, are discussed.

Analyzing Network Traffic Using AI. Network traffic characteristics and flow-based analysis are studied. Packet inspection and traffic classification methods are examined. Anomaly detection in traffic and protocol analysis are discussed.

Security Log Analysis. Log sources, types, aggregation, and parsing are studied. Correlation and analysis techniques are examined. Anomaly detection in logs, alert tuning, and forensics applications are discussed.

Introduction to SIEM Systems. SIEM architecture and components are studied. Data collection, normalization, and rule creation and management are examined. Alerting mechanisms, dashboards, reporting, and integration considerations are discussed.

Natural Language Processing and Security. NLP fundamentals and techniques are studied. Text preprocessing, feature extraction, and sentiment analysis in threat detection are examined. Named entity recognition and cybersecurity knowledge extraction are discussed.

Practical lessons

Creating a Simple Spam Filter. A text classification approach is used to build a spam filter. Features are extracted from emails and a training dataset is prepared. A model is selected, trained, and evaluated for performance.

Analyzing Phishing Emails. Phishing indicators are identified in sample emails. Email headers, URLs, and content are analyzed. Sender verification methods are applied and classification accuracy is assessed.

Data Preparation for AI Models. Data is loaded and explored, and missing values are handled. Data is normalized, scaled, and relevant features are selected. A train-test split and validation strategy are implemented.

Building a Machine Learning Model. A classification model is selected and its hyperparameters are tuned. The training procedure is monitored and convergence is assessed. Cross-validation is executed and performance metrics are calculated.

Anomaly Detection Model. Anomalies are defined and labeled using an unsupervised learning approach. Isolation Forest or Local Outlier Factor methods are implemented. Thresholds are set and the model's performance is evaluated in a practical application.

Malware Classification. Malware features are extracted and feature vectors are created. A classification model is trained for binary and multiclass classification. The model is evaluated in a real-world application.

Intrusion Detection System Simulation. Network traffic is simulated or a dataset is used to develop an IDS model. Normal and attack behavior are distinguished and alerts are generated. Performance

	metrics, including TPR and FPR, are calculated and the system is tuned. Model Accuracy Evaluation. Precision, recall, and F1-score are calculated for the model. A confusion matrix is analyzed and an ROC curve is generated. The AUC is computed, the threshold is optimized, and results are reported. Log File Analysis. Log files are ingested and parsed. Patterns and anomalies are identified through correlation analysis. Alerts are prioritized and an incident timeline is created. Network Traffic Visualization. Traffic flow visualization tools are used to analyze network data. Protocol distribution and traffic patterns are analyzed. Anomalies are visualized to support capacity planning and troubleshooting. Self-study History, Development and Role of AI in Cybersecurity. Types of Machine Learning and Main Approaches. Fundamentals of Neural Networks and Feature Engineering. Cyber Threat Intelligence and Threat Modeling. Security Operations Centers and SIEM Systems. Botnets, Fraud and User Behavior Detection. Zero-Day Attacks and Endpoint Protection. AI and Cloud Security. Adversarial Attacks and Threats Against AI Models. AI Safety Risks and Explainable AI (XAI). Security Automation and Incident Response. Malware Evolution and Threat Hunting. Network Segmentation, Privacy-Preserving Learning and AI Governance. AI Application in Cryptography. Quantum Computing, Blockchain and Future AI-Cybersecurity Technologies.
Teaching Methods	Interactive lectures with cybersecurity case studies are conducted. Hands-on practical sessions with machine learning tools and datasets are held. Threat analysis, incident simulations, and security data analysis exercises are practiced. Log examination, log correlation, and network traffic analysis using Wireshark are conducted. AI tool demonstrations and

	collaborative problem-solving on security challenges are also included.
Prerequisites	Computer Networks Programming 1,2 Foundations of Artificial Intelligence Fundamentals of Cybersecurity
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation in practical sessions, practical proficiency with AI and security tools during hands-on work, and the completion of teacher-guided independent work assignments, including security analysis reports, model implementation code, threat detection exercises, and mini-projects on AI-based cybersecurity applications. The midterm assessment consists of a written test covering theoretical knowledge of machine learning, cybersecurity threats, AI-based detection methods, and analysis techniques, together with independent work projects evaluating practical skills in data preparation, model development, threat analysis, and log examination. The final examination is a written and practical examination requiring students to develop security solutions using AI, analyze security data, classify threats, detect anomalies, and explain design decisions. The assessment measures theoretical understanding, practical AI-security skills, and the ability to apply AI in cybersecurity defense.
Recommended Literature	1. Russell S.J., Norvig P. Artificial Intelligence: A Modern Approach. Prentice Hall, 2010. 2. Bishop C.M. Pattern Recognition and Machine Learning. Springer, 2006. 3. Stallings W. Cryptography and Network Security: Principles and Practice. Prentice Hall, 2011. 4. Baratov J.R. et al. Fundamentals of Cybersecurity, Parts 1 & 2. Ilm nuri print, 2024.

	5. Madraximov Sh.F. Artificial Intelligence Systems. Metodist Publishing House, 2023.
Responsible Lecturer	Shaymanov Farxod Xushbakovich, Department of Computer and Software Engineering, Termez State University

OBJECT-ORIENTED PROGRAMMING (OOP)

Field	Description
Module Title	OBJECT-ORIENTED PROGRAMMING (OOP)
Module Code	OYUD404
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 46 hours (26 hours Lecture, 20 hours Practical lessons) Self-Assignments: 74 hours
Frequency/Duration	VII semester
Module objective	The objective of the Module Object-Oriented Programming is to develop theoretical knowledge and practical skills in creating high-quality, extensible, and reusable software using object-oriented programming principles in Java. Students learn classes and objects. Encapsulation, inheritance, and polymorphism are studied. Abstraction mechanisms are examined. Design patterns and SOLID principles are applied. Unit testing is practiced to ensure code quality.
Module Learning Outcomes	Knowledge: LO1. Understand fundamental OOP concepts: class, object, message and abstraction. LO2. Understand data protection principles through encapsulation and access modifiers. LO3. Understand inheritance structure, superclasses and class hierarchies. LO4. Understand polymorphism principle: method overriding and dynamic binding. LO5. Understand abstract classes, interfaces and generics for flexible design. LO6. Understand exception handling, collections framework and Object methods. Skills: LO7. Program classes and objects in Java.

	LO8. Apply encapsulation, inheritance and polymorphism in practical code. LO9. Write flexible code using abstract classes, interfaces and generics. LO10. Handle exceptions correctly and work with collections. LO11. Create UML diagrams and apply design patterns. LO12. Write unit tests using JUnit and ensure code quality. Competencies: LO13. Apply OOP principles effectively in professional software development. LO14. Write high-quality, reusable code based on OOP principles. LO15. Use abstraction mechanisms to create flexible software solutions. LO16. Apply UML modeling and design patterns in real-world projects. LO17. Write and maintain unit tests using JUnit framework. LO18. Independently explore and apply modern software tools and technologies.
Content	Lectures Introduction to Object-Oriented Programming. The OOP paradigm is compared with procedural programming. The concepts of class and object, message passing, and method calls are examined. Creating and using objects in Java and the benefits of the OOP approach are discussed. Encapsulation and Access Modifiers. Data hiding principles are studied. Private, protected, and public access modifiers, along with getter and setter methods, are examined. Information hiding, encapsulation best practices, and field protection strategies are discussed. Constructors, Methods and Class Members. Constructor definition, invocation, and overloading are studied. The this keyword and static fields and methods are examined. Class versus instance members and initialization ordering are discussed. Inheritance Fundamentals. Class hierarchy and the inheritance mechanism are studied. The superclass-

subclass relationship and the super keyword are examined. Single and multi-level inheritance, their advantages and limitations, and method resolution are discussed.

Polymorphism and Dynamic Binding. Method overriding concepts and rules are studied. Static and dynamic polymorphism and dynamic method dispatch are examined. Binding time considerations, runtime type checking, and polymorphic behavior are discussed.

Abstract Classes and Interfaces. The abstraction principle and abstract classes are studied. Interface definition, multiple interface implementation, and default methods are examined. Abstract method declaration and the comparison between abstract classes and interfaces are discussed.

Generics and Parameterized Types. Generic class definition and usage are studied. Type parameters, type arguments, bounded type parameters, and wildcards are examined. Generic methods, type safety benefits, and type erasure are discussed.

Exception Handling Fundamentals. The exception hierarchy and exception types are studied. Checked versus unchecked exceptions and the try-catch-finally structure are examined. Multiple catch clauses, custom exceptions, and exception propagation are discussed.

Collections Framework and Object Methods. The List, Set, and Map collection interfaces are studied. ArrayList, LinkedList, HashSet, and iterator usage are examined. HashMap and TreeMap implementations, sorting collections, and the Object class methods equals, hashCode, and toString are discussed.

UML and Object-Oriented Modeling. Class diagrams and relationships are studied. Sequence diagrams and interaction modeling are examined. The object-oriented analysis and design process, CRC cards, and UML notation are discussed.

Design Principles: SOLID. The single responsibility and open-closed principles are studied. The Liskov substitution and interface segregation principles are examined. The dependency inversion principle and design for change and reusability are discussed.

Design Patterns Introduction. The factory pattern for object creation and the singleton pattern are studied. The adapter and decorator structural patterns are examined. The observer and strategy patterns, along with pattern selection criteria, are discussed.

Unit Testing and Concurrency Fundamentals. Unit testing with the JUnit framework and test-driven development are studied. Threads and thread creation are examined. Synchronization, thread safety, race conditions, and deadlock issues are discussed.

Practical lessons

Setting Up Environment and Creating First Class. The development environment is configured using IntelliJ IDEA or Eclipse. Java classes and objects are created. Methods are invoked with parameters, and output is generated and tested.

Encapsulation Practice. Private fields are declared and protected. Getter and setter methods are implemented with appropriate access modifiers. Data validation is added to setters to demonstrate encapsulation benefits.

Constructors and Method Overloading. Multiple constructors are implemented with different parameter lists. Method overloading and signature differences are practiced. Constructor chaining is implemented using the this keyword.

Building Class Hierarchies with Inheritance. Superclasses and subclasses are created to form class hierarchies. The super keyword is used to access parent methods. Constructor chaining and method overriding in child classes are practiced.

Implementing Polymorphism. Method overriding is implemented and tested. Dynamic binding is observed through polymorphic array usage. Casting and type checking with the instanceof operator are practiced in inheritance hierarchies.

Abstract Classes and Interfaces Practice. Abstract classes are created and used. Interfaces are implemented multiple times, including default methods. Contracts are defined through interfaces, and the choice between abstract classes and interfaces is practiced.

	Working with Generics and Collections. Generic classes are designed and implemented. List, set, and map operations, including adding, removing, and iterating, are practiced. Collections are sorted using custom comparators. Exception Handling in Practice. Try-catch blocks are implemented for exception handling. Multiple exceptions are handled and finally blocks are used for resource cleanup. Custom exception classes are created within an exception hierarchy. Design Pattern Implementation. The factory pattern is implemented with a code example. The observer and strategy patterns are implemented for event handling and algorithm selection. The singleton pattern is used, and pattern benefits and trade-offs are discussed. Unit Testing with JUnit and Final Project. Unit tests are written for classes using JUnit assertions and test methods. Test data is prepared for the test cases. A mini-project is developed with a complete OOP design, including testing and documentation. Self-study History of OOP and Distinction from Procedural Programming. Principles of Abstraction and Data Hiding. Encapsulation and Access Modifiers. Constructors, This and Static Members. Types of Inheritance and Super Keyword. Method Overriding and Overloading. Polymorphism: Static and Dynamic Binding. Abstract Classes Versus Interfaces. Generics and Parametric Types. Collections Framework Deep Dive. Exception Hierarchy and Error Handling. Object Class Methods and Advanced Concepts. SOLID Principles and Clean Code. UML Modeling and Design Documentation. Unit Testing, TDD and Concurrency Fundamentals.
Teaching Methods	Interactive lectures with code examples and demonstrations are conducted. Hands-on practical sessions are held in an IDE environment with live coding demonstrations. Object-oriented design

	exercises and UML modeling workshops are practiced. Design pattern implementation activities, code review, and peer feedback are conducted. Unit testing practice with JUnit and group discussions on design decisions are also included.
Prerequisites	Programming 1,2 Data Structures and Algorithms Discrete Structures
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final written examination (50%). The continuous assessment evaluates classroom participation in practical sessions, coding ability and software design during hands-on work, and the completion of teacher-guided independent work assignments, including Java programs, design documents, UML diagrams, unit tests, and mini-projects demonstrating OOP principles. The midterm assessment consists of a written test covering theoretical knowledge of OOP concepts, design principles, patterns, and exception handling, together with independent work projects evaluating practical skills in class design, inheritance hierarchies, interface implementation, and collection usage. The final examination is a written and practical examination requiring students to design classes, implement inheritance and polymorphism, apply design patterns, create UML models, write unit tests, and explain design decisions. The assessment measures theoretical understanding and practical OOP programming skills.
Recommended Literature	1. Bloch J. Effective Java. 3rd edition. Addison-Wesley, 2018. 2. Horstmann C.S. Core Java, Volume I — Fundamentals. 12th edition. Pearson, 2021. 3. Larman C. Applying UML and Patterns: An Introduction to OO Analysis and Design. 3rd edition. Prentice Hall, 2004.

	4. Gamma E., Helm R., Johnson R., Vlissides J. Design Patterns: Elements of Reusable OO Software. Addison-Wesley, 1994.
Responsible Lecturer	Dilshod Abdualimov, Department of Computer and Software Engineering, Termez State University

INTRODUCTION TO MACHINE LEARNING

Field	Description
Module Title	INTRODUCTION TO MACHINE LEARNING
Module Code	MO'K404
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 44 hours (24 hours Lecture, 20 hours Practical lessons) Self-Assignments: 76 hours
Frequency/Duration	VII semester
Module objective	The objective of the Module Introduction to Machine Learning is to develop students' knowledge and skills in the theoretical foundations of machine learning, its core algorithms, and their application to solving practical problems. During the course, students study linear and logistic regression. Classification and regularisation techniques are examined. Optimisation methods are studied. Modern models based on artificial neural networks and deep learning, including CNN, RNN, and autoencoders, are introduced. These methods are applied in practice using Matlab, Python, and TensorFlow/Keras tools.
Module Learning Outcomes	Knowledge: LO1. Knows the main concepts, types and stages of the training process in machine learning. LO2. Knows the main concepts, types and stages of the training process in machine learning. LO3. Understands the principles of linear and logistic regression and the working of the gradient descent method. LO4. Understands the concepts of classification tasks, regularisation and the training dataset. LO5. Understands the operating principles of artificial neural networks and multi-layer networks.

	LO6. Understands convolutional and recurrent neural networks (CNN, RNN) and deep learning architectures. Skills: LO7. Performs operations on vectors and matrices using Matlab/Python tools. LO8. Builds linear and logistic regression models and programmes the gradient descent method. LO9. Solves classification problems and evaluates model accuracy (confusion matrix). LO10. Builds and trains artificial neural networks using TensorFlow/Keras tools. LO11. It solves image recognition tasks using a convolutional neural network (CNN). LO12. The instructor prepares the training set, trains the model and evaluates the results. Competencies: LO13. Applies machine learning methods effectively in professional practice. LO14. Uses data processing, model building and evaluation technologies. LO15. Builds and applies neural networks and deep learning models in real-world projects. LO16. Independently develops machine learning projects using Matlab/Python and TensorFlow/Keras. LO17. Visualises, interprets, and presents model results.
Content	Lectures Introduction to Machine Learning and Basic Concepts. The fundamental concepts, objectives, and applications of machine learning are introduced. The relationship between artificial intelligence, machine learning, and data science is studied. The role of machine learning in solving real-world problems is analyzed. Types of Machine Learning and Instrumental Tools. Supervised, unsupervised, semi-supervised, and reinforcement learning methods are studied. The strengths and limitations of different learning paradigms are analyzed. Commonly used machine learning tools and software libraries are introduced. Linear Regression with One and Multiple Variables. Linear regression models for predictive

analysis are studied. Simple and multiple linear regression techniques are analyzed. Model performance is evaluated using statistical measures.

Linear Algebra for Machine Learning. Vectors, matrices, matrix operations, and linear transformations are studied. The mathematical foundations required for machine learning algorithms are analyzed. The application of linear algebra to data representation and model computation is discussed.

Logistic Regression. Logistic regression for binary classification problems is introduced. Probability estimation, decision boundaries, and model interpretation are analyzed. Logistic regression performance is evaluated using classification metrics.

Classification Problems in Machine Learning. Classification techniques and algorithms are studied. Binary and multi-class classification problems are analyzed. Model accuracy, precision, recall, and other performance measures are evaluated.

Regularisation and the Training Dataset. Overfitting and underfitting problems in machine learning are studied. Regularisation techniques and dataset preparation methods are analyzed. Strategies for improving model generalization are evaluated.

Gradient Descent and Optimisation Methods. Optimization algorithms used in machine learning are studied. Gradient descent, stochastic gradient descent, and advanced optimization methods are analyzed. Convergence and model training efficiency are evaluated.

Introduction to Artificial Neural Networks. The structure and operation of artificial neural networks are studied. Neurons, activation functions, and network architecture are analyzed. Neural network applications in intelligent systems are evaluated.

Multi-layer Neural Networks and Back-propagation. Deep neural network architectures and hidden layers are studied. The back-propagation algorithm and weight optimization are analyzed. Model learning and training performance are evaluated.

Convolutional and Recurrent Neural Networks. Convolutional Neural Networks for image processing

are studied. Recurrent Neural Networks for sequential data analysis are examined. Their architectures and practical applications are analyzed.

Deep Learning Architectures and Modern Tools. Modern deep learning frameworks and architectures are introduced. TensorFlow, PyTorch, and Keras for developing intelligent applications are analyzed. Deep learning solutions in computer vision, natural language processing, and data analytics are evaluated.

Practical lessons

Introduction to Machine Learning, Familiarisation with Basic Concepts and Algorithms. Machine learning concepts and workflow are practically introduced. Basic algorithms are explored using real datasets. Model training and prediction processes are analyzed.

Types of Machine Learning and Instrumental Tools. Supervised, unsupervised, and reinforcement learning methods are practically compared. Python libraries used for machine learning are explored. Development environments for machine learning are examined.

Programming Linear Regression. Simple and multiple linear regression models are implemented. Prediction accuracy is analyzed. Regression results are visualized using programming tools.

Linear Algebra Operations. Vector and matrix operations required for machine learning are practically implemented. Linear algebra techniques are applied using programming environments. Computational results are analyzed.

Logistic Regression: Model Building and Optimisation. Logistic regression models for binary classification are developed. Probability estimation and decision boundaries are analyzed. Optimization techniques are applied to improve model performance.

Classification Problems in Machine Learning. Classification algorithms are implemented and evaluated. Classification performance is analyzed using confusion matrices. Evaluation metrics are calculated and interpreted.

	Regularisation and the Training Set. Regularisation techniques and dataset preparation methods are practically applied. Overfitting and underfitting are investigated. Model generalization is evaluated. Building a Simple Neural Network. A basic artificial neural network is designed and implemented. Network architecture and activation functions are analyzed. Prediction performance is evaluated. Building a Multi-layer Neural Network and Training with Back-propagation. A multi-layer neural network is implemented using back-propagation. Training accuracy and learning curves are evaluated. Model optimization techniques are applied. Image Recognition Using a Convolutional Neural Network (CNN). A CNN model is developed for image classification. Image recognition performance is analyzed. Experimental results are presented. Self-study History of Machine Learning and Its Application Areas. Types of Machine Learning and Steps of the Training Process. Working with Instrumental Tools (Octave/Matlab/Python). Linear Algebra Skills for Machine Learning. Linear Regression with One and Multiple Variables. Logistic Regression and Classification Problems. The Training Dataset and Its Preparation. Evaluating Classification Performance (Confusion Matrix). Softmax Regression and Multi-Class Classification. Regularisation and Structured Linear Models. Gradient Descent and Stochastic Gradient Descent. Methods for Optimising Model Training. Artificial Neural Network Model and Perceptron. Multi-layer Neural Networks and Back-propagation. Convolutional Neural Networks (CNN) and Image Analysis.
Teaching Methods	To form knowledge of the scientific and theoretical foundations of ecology and environmental protection. Teaching about the Biosphere, the theory of

	Vernadsky, and the classification of natural resources. Teaching methods for the protection of atmospheric air, water and land resources. Developing skills for the conservation of biological diversity and specially protected areas. Develop competence in analysing environmental problems (island effects, salinisation, climate change) and their prevention methods. Developing competencies in sustainable development, environmental education and the wise use of natural resources.
Prerequisites	Programming 1,2 Linear Algebra Probability and Statistics
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final assessment (50%). The continuous assessment evaluates students' classroom participation, practical laboratory activities, programming exercises, presentations, and the successful completion of independent study assignments. These include the implementation of machine learning algorithms, linear regression and logistic regression models, linear algebra programming tasks, neural network construction, image classification using convolutional neural networks, and the analysis of machine learning techniques using modern software tools such as Python, MATLAB, or Octave. The midterm assessment consists of written assignments, short in-class quizzes, practical programming tasks, laboratory reports, presentations, and the evaluation of independent study projects. Students are assessed on their ability to develop and evaluate machine learning models, apply optimization algorithms, prepare and preprocess datasets, analyze classification performance, implement regression techniques, compare machine learning algorithms, and explain the operation of artificial neural networks and deep learning models. The final assessment may be conducted as either a written examination or a test

	examination. The written examination consists of theoretical questions, analytical problems, and practical machine learning tasks covering the main topics of the module, including machine learning fundamentals, supervised and unsupervised learning, linear and logistic regression, linear algebra, classification algorithms, regularisation techniques, optimization methods, artificial neural networks, deep learning architectures, convolutional neural networks, and modern machine learning frameworks. The test examination consists of multiple-choice questions designed to evaluate students' knowledge of machine learning concepts, mathematical foundations, model development, optimization techniques, neural network architectures, deep learning methods, and the practical application of machine learning algorithms to real-world problems. Successful completion of all practical laboratory exercises and independent study assignments is required as an essential component of the overall module assessment.
Recommended Literature	1. Author, A. A., & Author, B. B. (2022). Machine learning with PyTorch and Scikit-learn. Packt Publishing. 2. Alpaydin, E. (2010). Introduction to machine learning (2nd ed.). The MIT Press. 3. Kelleher, J. D., Mac Namee, B., & D'Arcy, A. (2020). Fundamentals of Machine Learning for Predictive Data Analytics (2nd ed.). MIT Press. https://sciarium.com/file/514719/ 4. Burkov, A. (2020). The Hundred-Page Machine Learning Book. Andriy Burkov. 5. Zollanvari, A. (2023). Machine Learning with Python. Springer. https://doi.org/10.1007/978-3-031-33342-2
Responsible Lecturer	Mengnarov Adham Ergashevich

DIGITAL HARDWARE AND SOFTWARE

Field	Description
Module Title	DIGITAL HARDWARE AND SOFTWARE
Module Code	RIT404
Credit Points (ECTS)	4 Credits
Workload	Total workload: 120 hours Contact hours: 44 hours (24 hours Lecture, 20 hours Practical lessons) Self-Assignments: 76 hours
Frequency/Duration	VII semester
Module objective	The objective of the Module Digital Hardware and Software is to provide students with knowledge of modern digital devices, microprocessor systems, and computer hardware and software architecture. The operating principles of these systems are studied. Students learn the design, development, and practical application of digital devices. The main directions of modern information and communication technology development are examined. The integration of hardware and software tools is studied. Methods for modelling and managing digital systems are introduced.
Module Learning Outcomes	Knowledge: LO1. Knows the basic concepts and development trends of digital technology and software. LO2. Understand methods of representing information in digital form and number systems. LO3. Know the laws of logical algebra and logical functions. LO4. Understand the functions of triggers, registers, counters and memory devices. LO5. Knowledge of types of software, operating systems and the functions of application programmes. LO6. Must be familiar with modern methods of designing and modelling digital systems. Skills: LO7. Be able to analyse and design logic circuits. LO8. Be able to create schematics for combinational devices.

	LO9. Be able to carry out practical tasks with microprocessor systems. LO10. To develop algorithms and be able to express them in the form of a programme. LO11. Must be able to work with and analyse technical and software documentation. Competencies: LO12. Be able to create simple automated systems based on the integration of hardware and software tools. LO13. Be competent in designing, testing and improving digital devices and software products. LO14. Develop the competence to develop technical solutions using modern programming and modelling tools. LO15. Must be competent in the effective use of information and communication technologies in professional activities, teamwork, and independent learning.
Content	Lectures Introduction to Digital Hardware and Software. Key Concepts and Development Trends. The concepts of digital hardware and software, their functions, and areas of application are introduced. The evolution of digital technologies and modern computing systems is studied. Current trends in digital transformation and intelligent systems are analyzed. Numbering Systems and Methods of Number Encoding. Binary, octal, decimal, and hexadecimal number systems are studied. Number conversion methods and binary arithmetic are analyzed. Information encoding techniques used in digital systems are examined. Boolean Algebra Elements and Logical Functions. Boolean algebra, logical operations, and logical expressions are introduced. Boolean laws and their simplification techniques are studied. Logical functions used in digital circuit design are analyzed. Logical Elements and Their Implementation in Integrated Circuits. Basic logic gates, including AND, OR, NOT, NAND, NOR, XOR, and XNOR, are studied. Their implementation in integrated circuits is

analyzed. Logic gate applications in digital electronics are evaluated.

Combinational Digital Devices: Encoder, Decoder, Multiplexer and Demultiplexer. Combinational logic circuits and their operating principles are studied. Encoders, decoders, multiplexers, and demultiplexers are analyzed. Their applications in digital communication and computer systems are evaluated.

Sequential Digital Devices: Triggers and Registers. Sequential logic circuits, flip-flops, and registers are studied. Memory elements, timing diagrams, and data storage operations are analyzed. Register applications in digital systems are evaluated.

Counters, Memory Devices and Digital Automata. Digital counters, RAM, ROM, and finite-state machines are studied. Counting methods, memory organization, and digital automata are analyzed. Their role in computer architecture and control systems is evaluated.

Architecture of Microprocessors and Microcontrollers. Microprocessor and microcontroller architecture is introduced. CPU organization, instruction execution, and memory structure are analyzed. Embedded system applications are evaluated.

Computer Hardware and Interface Devices. Computer hardware components and peripheral devices are studied. Communication interfaces, input/output systems, and hardware integration are analyzed. Modern interface technologies are evaluated.

System and Application Software. System software, operating systems, utilities, and application software are studied. Software functions and their interaction with hardware are analyzed. Software environments for digital systems are evaluated.

Fundamentals of Algorithms and Programming Languages. Algorithms, flowcharts, and programming logic are introduced. Programming language classifications and software development principles are studied. Algorithm efficiency and problem-solving techniques are analyzed.

Hardware-Software Systems, IoT Technologies and Promising Digital Solutions. Embedded systems, Internet of Things (IoT), and hardware-software integration are studied. Smart devices, cloud connectivity, and intelligent digital technologies are analyzed. Future trends in digital innovation are evaluated.

Practical lessons

Performing Operations on Numbers in Various Number Systems. Conversion between number systems and arithmetic operations are practically performed. Binary calculations and encoding methods are analyzed. Numerical accuracy is verified.

Constructing Logical Functions and Creating Truth Tables. Logical expressions are developed and truth tables are constructed. Logical operations are analyzed. Boolean functions are verified.

Minimising Logical Functions Using Karnaugh Maps. Karnaugh maps are applied for simplifying Boolean expressions. Optimized logic circuits are analyzed. Circuit efficiency is evaluated.

Designing and Analysing Circuits Based on Logic Gates. Digital circuits are constructed and simulated using logic gates. Circuit operation is analyzed. Logical outputs are verified.

Designing Devices Based on Multiplexers and Decoders. Combinational circuits are implemented using multiplexers, demultiplexers, encoders, and decoders. Data selection and routing techniques are analyzed. Circuit functionality is verified through simulation.

Modelling the Operation of Triggers and Registers. Flip-flops and register circuits are simulated. Timing behavior and data storage are analyzed. Sequential operations are evaluated.

Designing and Verifying Counter Circuits. Synchronous and asynchronous counters are developed and tested. Counting sequences are analyzed. Circuit performance is evaluated.

Working with Microprocessor Instructions and Creating Simple Programmes. Basic microprocessor instructions are programmed. Simple assembly or embedded applications are developed.

	Instruction execution and processor operation are analyzed. Developing Software to Control Hardware Devices. Software applications for controlling digital hardware are designed. Programming is integrated with microcontrollers and peripheral devices. Hardware control functionality is tested. Simulating Digital Devices and Analysing the Results. Digital systems are simulated using specialized software. Functional behavior and timing diagrams are analyzed. System performance is evaluated. Self-study History of the Development of Digital Technology and Its Current State. Characteristics of Analogue and Digital Signals. Methods of Encoding and Decoding Information. Laws of Boolean Algebra and Their Practical Application. Comparison of TTL and CMOS Technologies. Industrial Applications of Combinational Circuits. Types and Functions of Triggers. The Role of Registers in Microprocessor Systems. Principle of Operation and Applications of Counters. Features of RAM and ROM. Generations and Development Stages of Microprocessors. Modern Microcontroller Platforms (Arduino, ESP32 and Others). Functions and Types of Operating Systems. Classification of Programming Languages. Software Development for Embedded Systems. Fundamentals of Internet of Things (IoT) Technologies. Integration of Artificial Intelligence and Digital Devices. Issues of Information Security in Digital Systems. Testing Methods for Hardware and Software Tools. Digital Transformation and Industry 4.0 Technologies.
Teaching Methods	The theoretical foundations of digital technology, including methods of representing and processing numerical information, are studied. Logical functions

	and algebraic laws, and methods for the analysis and synthesis of digital circuits, are covered. The operating principles of various logic elements, triggers, registers, counters, multiplexers, decoders, and other digital devices are studied. The structure of microprocessors and microcontrollers, their instruction sets, memory organisation, and mechanisms for interacting with peripheral devices are also examined.
Prerequisites	Computer Architecture Programming 1,2 Computer Organisation
Exams and Assessment	Type of examination (written, oral, project) and its weight Student performance is assessed through continuous assessment (20%), midterm assessment (30%), and a final assessment (50%). The continuous assessment evaluates students' classroom participation, practical laboratory activities, programming exercises, digital circuit design and simulation tasks, presentations, and the successful completion of independent study assignments. These include number system conversions, Boolean algebra and logic function design, Karnaugh map simplification, implementation of combinational and sequential circuits, programming microprocessors and microcontrollers, hardware-software integration, digital device simulation, and the analysis of Internet of Things (IoT) technologies and modern digital systems. The midterm assessment consists of written assignments, short in-class quizzes, practical laboratory tasks, programming exercises, presentations, and the evaluation of independent study projects. Students are assessed on their ability to analyze digital logic circuits, perform number system operations, design combinational and sequential digital devices, apply Boolean algebra, develop simple embedded software, evaluate microprocessor and microcontroller architectures, compare digital technologies, and explain the operation of hardware-software systems and IoT applications. The final assessment may be conducted

	as either a written examination or a test examination. The written examination consists of theoretical questions, analytical problems, and practical tasks covering the main topics of the module, including number systems, Boolean algebra, logic gates, combinational and sequential circuits, registers, counters, memory devices, microprocessors, microcontrollers, computer hardware, system and application software, programming fundamentals, embedded systems, Internet of Things (IoT), digital transformation, and information security. The test examination consists of multiple-choice questions designed to evaluate students' knowledge of digital hardware and software, digital circuit design, microprocessor and microcontroller systems, programming concepts, IoT technologies, embedded systems, and the practical application of modern digital technologies. Successful completion of all practical laboratory exercises and independent study assignments is required as an essential component of the overall module assessment.
Recommended Literature	1. M. Morris Mano, Michael D. Ciletti. Digital Design. 6th Edition. Boston: Pearson Education, 2018. 720 p. 2. Thomas L. Floyd. Digital Fundamentals. 11th Edition. Boston: Pearson Education, 2015. 928 p. 3. Ronald J. Tocci, Neal S. Widmer, Gregory L. Moss. Digital Systems: Principles and Applications. 12th Edition. Boston: Pearson Education, 2017. 960 p. 4. Barry B. Brey. The Intel Microprocessors: Architecture, Programming, and Interfacing. 9th Edition. Boston: Pearson Education, 2019. 960 p. 5. Paul Deitel, Harvey Deitel. C++ How to Program. 10th Edition. Hoboken, NJ: Pearson Education, 2017. 992 p. 6. Stephen Brown, Zvonko Vranesic. Fundamentals of Digital Logic with Verilog Design. 3rd Edition. New York: McGraw-Hill Education, 2014. 984 p. 7. Charles H. Roth Jr., Larry L. Kinney. Fundamentals of Logic Design. 7th Edition. Boston: Cengage Learning, 2014. 832 p.
Responsible Lecturer	Ibragimova Mohigul Komiljonovna